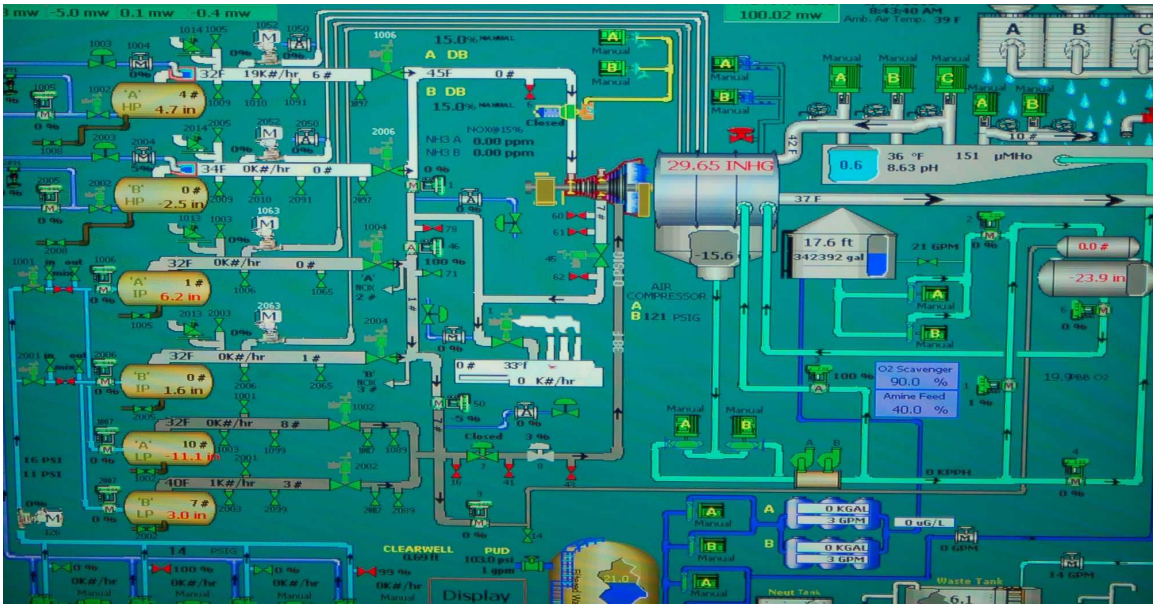
Lessons In Industrial Instrumentation



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Version 2.33 (development) – Last update 9 October 2024

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Revision history[1]

- Version 0.1 – July 2008 to September 2008 (initial development)
- Version 0.2 – released September 29, 2008 for Fall quarter student use (SafeCreative registration code `0810111072182`)
- Version 0.4 – released January 12, 2009 for Winter quarter student use (SafeCreative registration code `0901122394919`)
- Version 0.6 – released April 21, 2009 for public use (SafeCreative registration code `0904213106101`)
- Version 0.8 – released September 8, 2009 for public use (SafeCreative registration code `0909094400980`)
- Version 1.0 – released September 28, 2009 for public use (SafeCreative registration code `0909284601913`)
- Version 1.2 – released January 12, 2010 for public use (SafeCreative registration code `1001125303242`)
- Version 1.4 – released April 11, 2010 for public use (SafeCreative registration code `1004125976416`)
- Version 1.6 – released May 4, 2010 for public use (SafeCreative registration code `1005046194859`)
- Version 1.8 – released June 1, 2010 for public use (SafeCreative registration code `1006016477958`)
- Version 1.10 – released June 28, 2010 for public use (SafeCreative registration code `1006286691573`)
- Version 1.12 – released September 20, 2010 for public use (SafeCreative registration code `1009217397803`)

[1] Version numbers ending in odd digits are developmental (e.g. 0.7, 1.23, 4.5), with only the latest revision made accessible to the public. Version numbers ending in even digits (e.g. 0.6, 1.0, 2.14) are considered "public-release" and will be archived. Version numbers beginning with zero (e.g. 0.1, 0.2, etc.) represent early editions that were substantially incomplete.

Chapter 19

Continuous pressure measurement

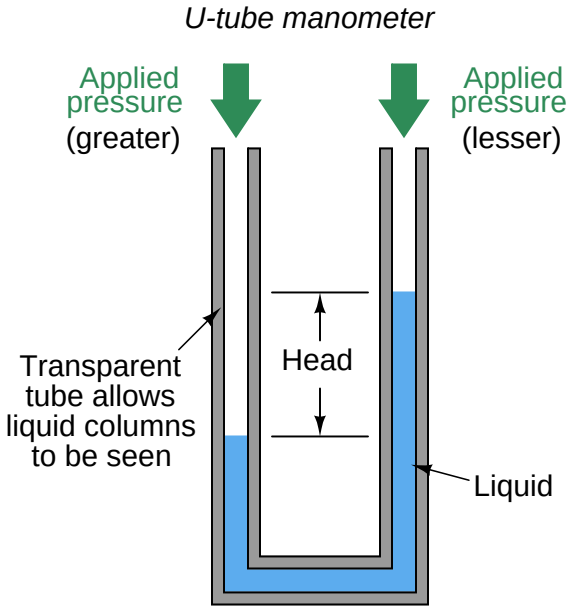
In many ways, pressure is the primary variable for a wide range of process measurements. Many types of industrial measurements are actually inferred from pressure, such as:

- Flow (measuring the pressure dropped across a restriction)
- Liquid level (measuring the pressure created by a vertical liquid column)
- Liquid density (measuring the pressure difference across a fixed-height liquid column)
- Weight (hydraulic load cell)

Even temperature may be inferred from pressure measurement, as in the case of a fluid-filled chamber where fluid pressure and fluid temperature are directly related. As such, pressure is a very important quantity to measure, and measure accurately. This section describes different technologies for the measurement of pressure.

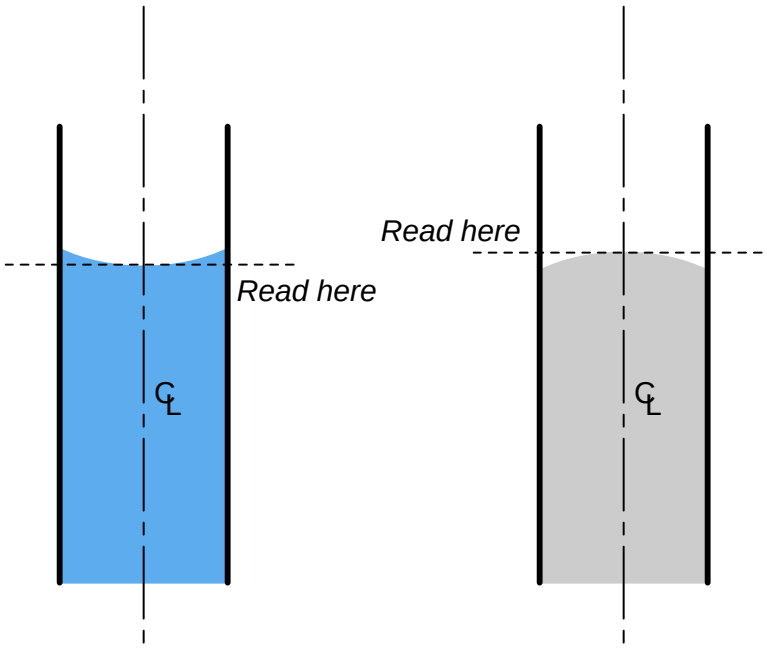
19.1 Manometers

A very simple device used to measure pressure is the *manometer*: a fluid-filled tube where an applied gas pressure causes the fluid height to shift proportionately. This is why pressure is often measured in units of liquid height (e.g. inches of water, inches of mercury). As you can see, a manometer is fundamentally an instrument of *differential* pressure measurement, indicating the difference between two pressures by a shift in liquid column height:

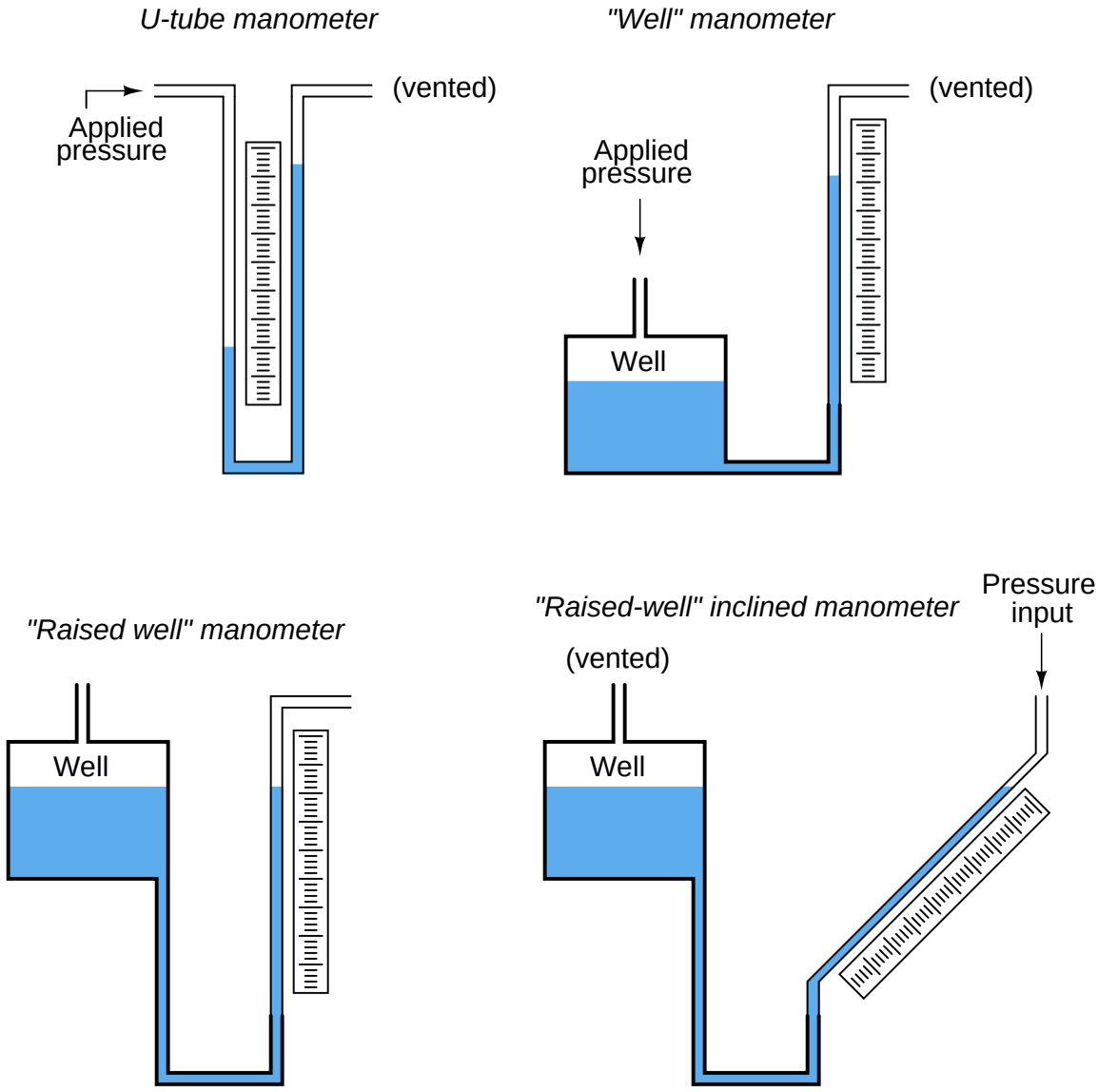


Of course, it is entirely acceptable to simply vent one tube of a manometer and use it as a *gauge* pressure instrument, comparing the applied pressure at one tube against atmospheric pressure in the other.

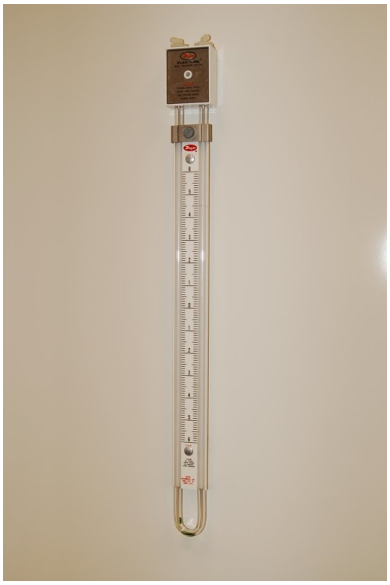
Liquid column height in a manometer should always be interpreted at the centerline of the liquid column, regardless of the shape of the liquid's meniscus (the curved air/liquid interface):



Manometers come in a variety of forms, the most common being the *U-tube*, *well* (sometimes called a *cistern*), *raised well*, and *inclined*:



U-tube manometers are very inexpensive, and are generally made from clear plastic (see the left-hand photo). Cistern-style manometers are the norm for calibration bench work, and are typically constructed from metal cisterns and glass tubes (see the right-hand photo):





Inclined manometers are used to measure very low pressures, owing to their exceptional sensitivity (note the fractional scale for inches of water column in the following photograph, extending from 0 to 1.5 inches on the scale, reading left to right):



Note that venting one side of a manometer is standard practice when using it as a *gauge pressure* indicator (responding to pressure in excess of atmospheric). Both pressure ports will be used if the manometer is applied to the measurement of differential pressure, just as in the case of the U-tube manometer first shown in this section. Absolute pressure may also be measured by a manometer, if one of the pressure ports connects to a sealed vacuum chamber. This is how a *mercury barometer* is constructed for the measurement of absolute ambient air pressure: by sealing off one side of a manometer and removing all the air in that side, such that the applied (atmospheric) pressure is always compared against a vacuum.

Manometers incorporating a "well" have the advantage of single-point reading: one need only compare the height of *one* liquid column, not the difference in height between *two* liquid columns. The cross-sectional area of the liquid column in the well is so much greater than that within the transparent manometer tube that the change in height within the well is usually negligible. In cases where the difference is significant, the spacing between divisions on the manometer scale may be skewed to compensate[1].

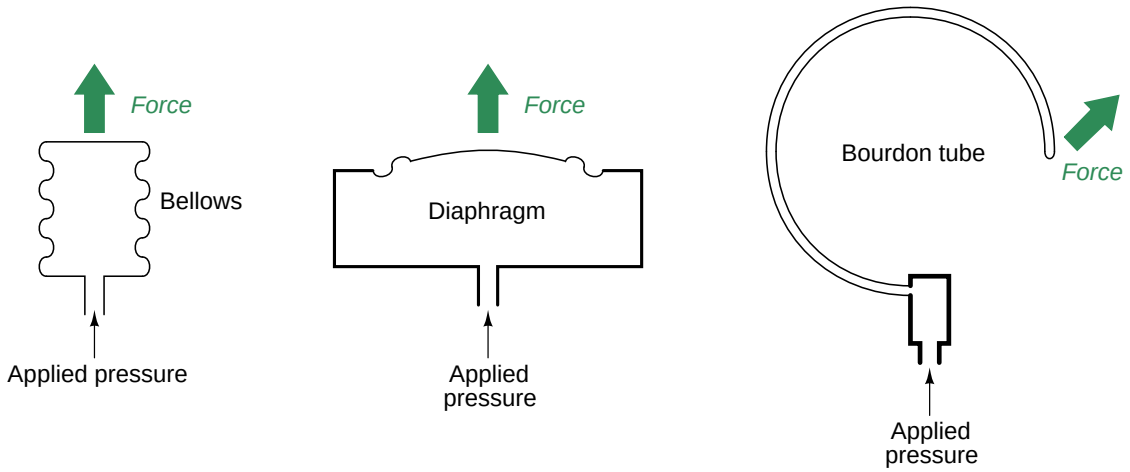
Inclined manometers enjoy the advantage of increased sensitivity. Since manometers fundamentally operate on the principle of pressure balanced by liquid height, and this liquid height is always measured parallel to the line of gravitational pull (perfectly vertical), inclining the manometer tube means that liquid must travel farther along the tube to generate the same change in (purely)

[1] If you are having difficulty understanding this concept, imagine a simple U-tube manometer where one of the tubes is opaque, and therefore one of the two liquid columns cannot be seen. In order to be able to measure pressure just by looking at one liquid column height, we would have to make a custom scale where every inch of height registered as *two* inches of water column pressure, because for each inch of height change in the liquid column we can see, the liquid column we can't see also changes by an inch. A scale custom-made for a well-type manometer is just the same concept, only without such dramatic skewing of scales.

vertical height than it would in a vertical manometer tube. Thus, an inclined manometer tube causes an amplification in liquid motion for a given amount of pressure change, allowing measurements of greater resolution.

19.2 Mechanical pressure elements

Mechanical pressure-sensing elements include the *bellows*, the *diaphragm*, and the *bourdon tube*. Each of these devices converts a fluid pressure into a force. If unrestrained, the natural elastic properties of the element will produce a motion proportional to the applied pressure.



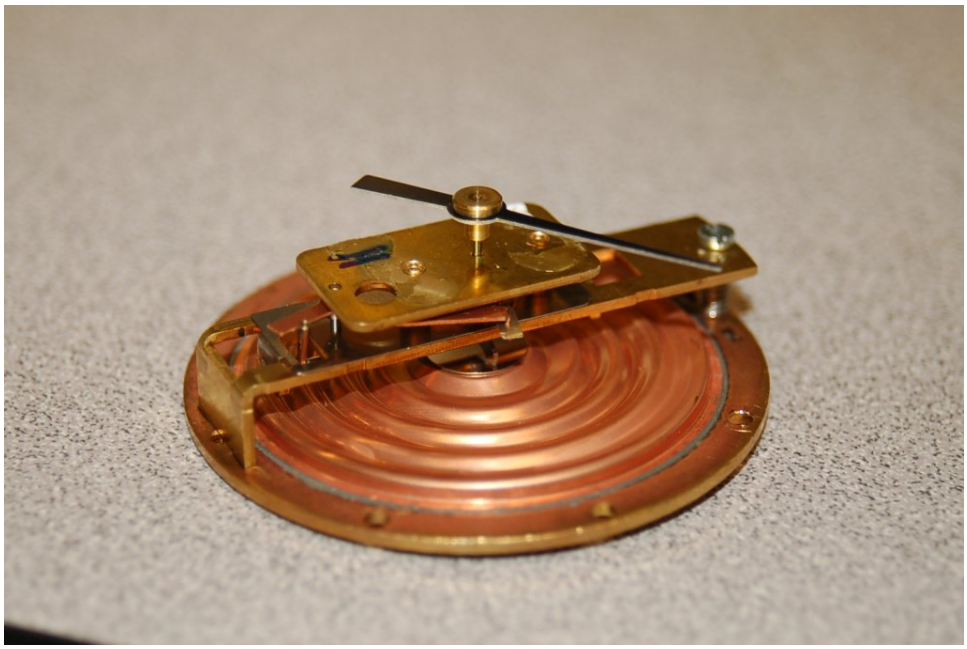
Bellows resemble an accordion constructed from metal instead of fabric. Increasing pressure inside a bellows unit causes it to elongate. A photograph of a bellows is shown here:



A diaphragm is nothing more than a thin disk of material which bows outward under the influence of a fluid pressure. Many diaphragms are constructed from metal, which gives them spring-like qualities. Some diaphragms are intentionally constructed out of materials with little strength, such that there is negligible spring effect. These are called *slack diaphragms*, and they are used in

conjunction with external mechanisms (e.g. springs) producing the necessary restraining force to prevent damage from applied pressure.

The following photograph shows the mechanism of a small pressure gauge using a brass diaphragm as the sensing element:



As pressure is applied to the rear of the diaphragm, it distends upward (away from the table on which it rests as shown in the photograph), causing a small shaft to twist in response. This twisting motion is transferred to a lever which pulls on a tiny link chain wrapped around the pointer shaft, causing it to rotate and move the pointer needle around the gauge scale. Both the needle and scale on this gauge mechanism have been removed for easier viewing of diaphragm and mechanism.

Bourdon tubes are made of spring-like metal alloys bent into a circular shape. Under the influence of internal pressure, a bourdon tube "tries" to straighten out into its original shape before being bent at the time of manufacture.

Most pressure gauges use a bourdon tube as their pressure-sensing element. Most pressure transmitters use a diaphragm as their pressure-sensing element. Bourdon tubes may be made in *spiral* or *helical* forms for greater motion (and therefore greater gauge resolution).

The Bourdon tube pressure element is a very robust and time-tested design. An illustration taken from page 471 of volume 1 of *Cassier's Magazine* published in the year 1891 shows a typical C-shaped bourdon tube pressure gauge mechanism complete with gears and pointing needle:

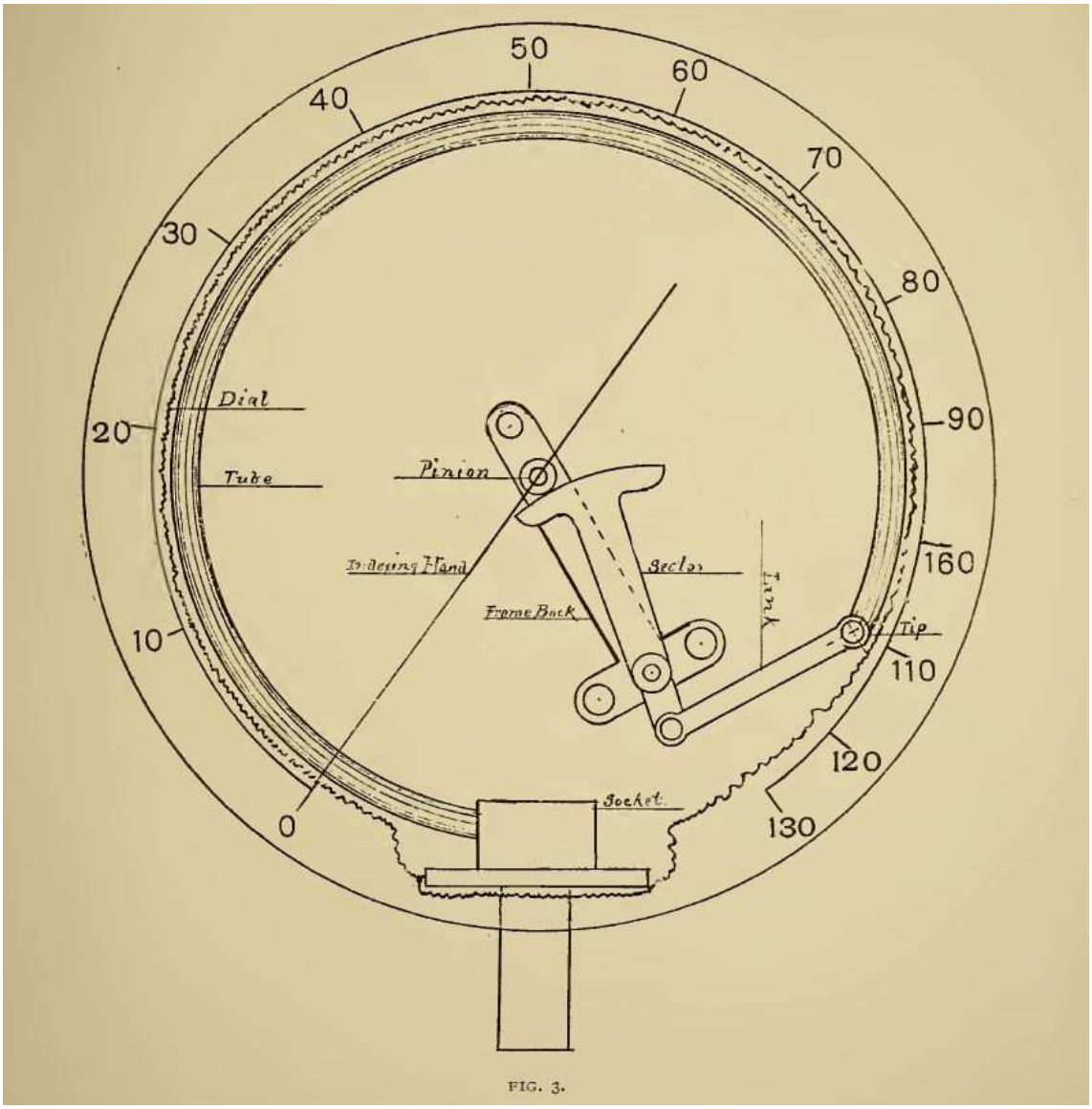


FIG. 3.

Looking closely at the labeled components of this mechanism, we see a circular "pinion" touching a curved "sector". Both of these are *gears* meshing with one another, but as is typical with mechanical drawings the individual teeth of the meshing gears are not shown.

It is a useful mental exercise to imagine the components of this gauge moving under the influence

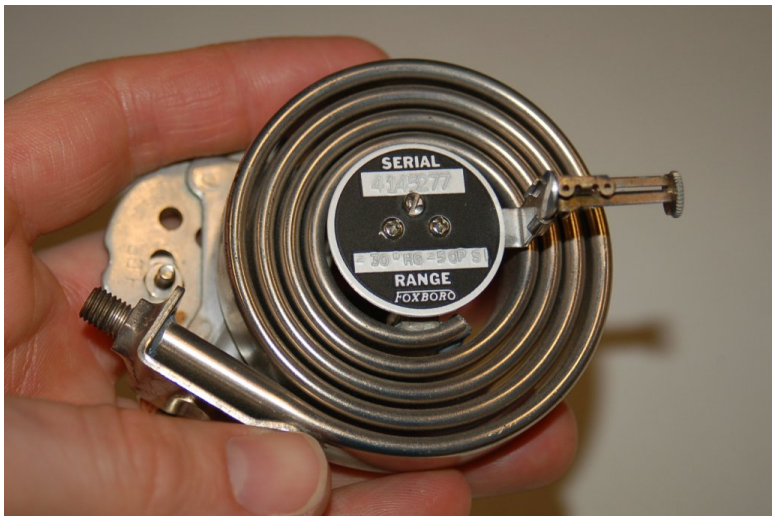
of a rising fluid pressure. The bourdon tube will straighten, resulting in its tip extending outward from center (up and right) as its socket remains stationary (anchored to the gauge body). This pulls on the link, which in turn causes the sector gear to rotate counter-clockwise on its bearing axis (that axis anchored on the backing plate of the gauge). This causes the pinion gear to rotate clockwise, driving the needle (the "indexing hand") clockwise as well, so that the needle's tip rises up the numerical scale printed on the gauge face.

A photograph of a C-tube pressure gauge mechanism (taken from the rear of the gauge, behind the pointer and scale) reveals its mechanical workings:



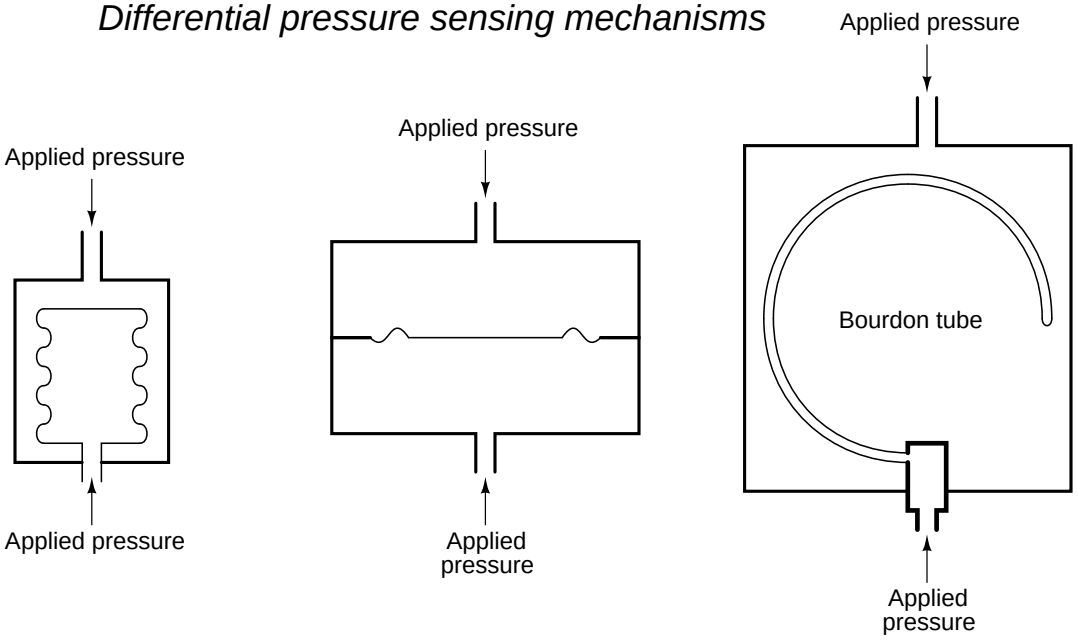
The dark, C-shaped tube is the bourdon tube sensing element, while the shiny metal parts are the linkage, lever, and gear assembly.

The next photograph shows a *spiral* bourdon tube, designed to produce a wider range of motion than a C-tube bourdon:



It should be noted that bellows, diaphragms, and bourdon tubes alike may all be used to measure differential and/or absolute pressure in addition to gauge pressure. All that is needed for these other functionalities is to subject the *other* side of each pressure-sensing element to either another applied pressure (in the case of differential measurement) or to a vacuum chamber (in the case of absolute pressure measurement).

This next set of illustrations shows how bellows, diaphragms, and bourdon tubes may be used as differential pressure-sensing elements:



The challenge in doing this, of course, is how to extract the mechanical motion of the pressure-sensing element to an external mechanism (such as a pointer) while maintaining a good pressure seal. In gauge pressure mechanisms, this is no problem because one side of the pressure-sensing element must be exposed to atmospheric pressure anyway, and so that side is always available for mechanical connection.

A differential pressure gauge is shown in the next photograph. The two pressure ports are clearly evident on either side of the gauge:



19.3 Electrical pressure elements

Several different technologies exist for the conversion of fluid pressure into an electrical signal response. These technologies form the basis of electronic *pressure transmitters*: devices designed to measure fluid pressure and transmit that information via electrical signals such as the 4-20 mA analog standard, or in digital form such as HART or FOUNDATION Fieldbus.

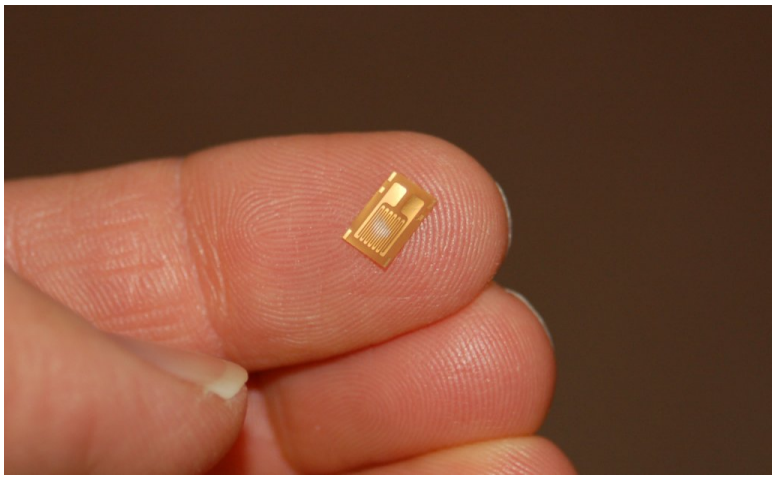
A brief survey of electronic pressure transmitters in contemporary[2] use reveals a diverse representation of electrical pressure-sensing elements:

Manufacturer	Model	Pressure sensor technology
ABB/Bailey	PTSD	Differential reluctance
ABB/Bailey	PTSP	Piezoresistive (strain gauge)
Foxboro	IDP10	Piezoresistive (strain gauge)
Honeywell	ST3000	Piezoresistive (strain gauge)
Rosemount	1151	Differential capacitance
Rosemount	3051	Differential capacitance
Rosemount	3095	Differential capacitance
Yokogawa	EJX series	Mechanical resonance

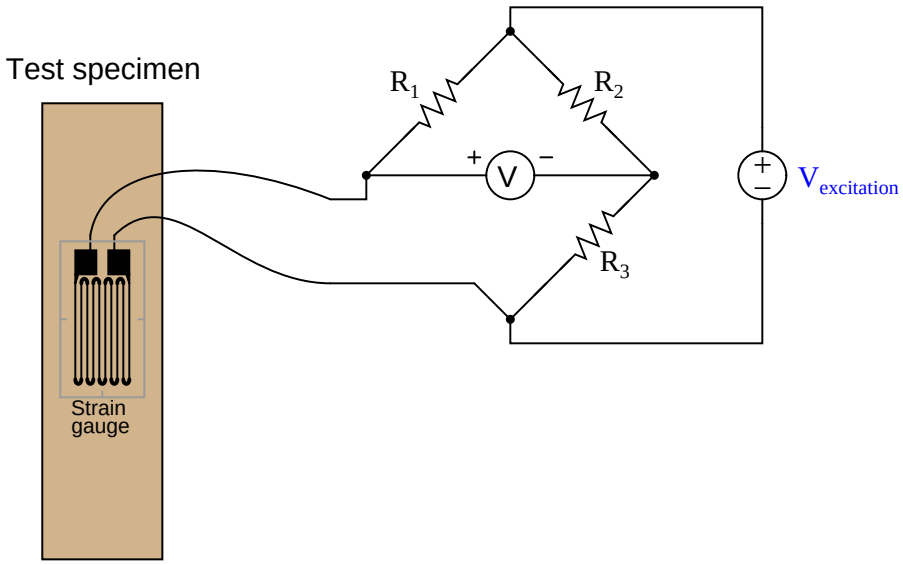
[2] As of this writing, 2008.

19.3.1 Piezoresistive (strain gauge) sensors

Piezoresistive means "pressure-sensitive resistance," or a resistance that changes value with applied pressure. The *strain gauge* is a classic example of a piezoresistive element, a typical strain gauge element shown here on the tip of my finger:



In order to be practical, a strain gauge must be glued (*bonded*) on to a larger specimen capable of withstanding an applied force (stress):



As the test specimen is stretched or compressed by the application of force, the conductors of the strain gauge are similarly deformed. Electrical resistance of any conductor is proportional to the ratio of length over cross-sectional area ($R \propto \frac{l}{A}$), which means that tensile deformation (stretching)

will increase electrical resistance by simultaneously increasing length and decreasing cross-sectional area while compressive deformation (squishing) will decrease electrical resistance by simultaneously decreasing length and increasing cross-sectional area.

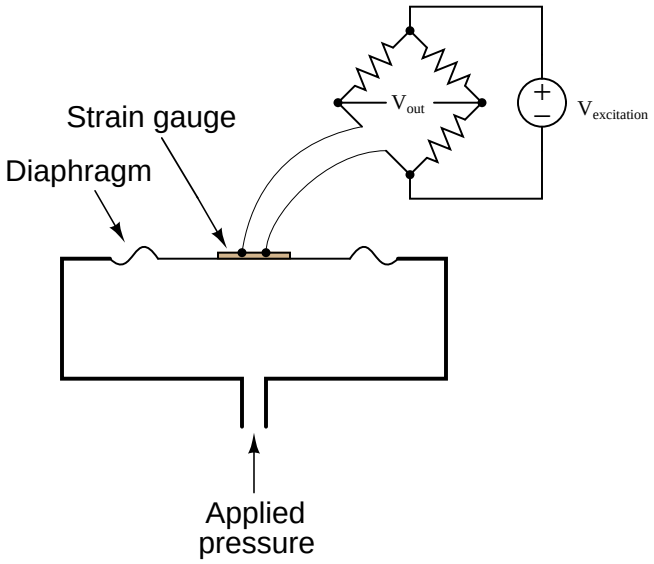
Attaching a strain gauge to a diaphragm results in a device that changes resistance with applied pressure. Pressure forces the diaphragm to deform, which in turn causes the strain gauge to change resistance. By measuring this change in resistance, we can infer the amount of pressure applied to the diaphragm.

The classic strain gauge system represented in the previous illustration is made of metal (both the test specimen and the strain gauge itself). Within its elastic limits, many metals exhibit good spring characteristics. Metals, however, are subject to *fatigue* over repeated cycles of strain (tension and compression), and they will begin to "flow" if strained beyond their elastic limit. This is a common source of error in metallic piezoresistive pressure instruments: if overpressured, they tend to lose accuracy due to damage of the spring and strain gauge elements.[3]

Modern manufacturing techniques have made possible the construction of strain gauges made of silicon instead of metal. Silicon exhibits very linear spring characteristics over its narrow range of motion, and a high resistance to fatigue. When a silicon strain gauge is over-stressed, it fails completely rather than "flows" as is the case with metal strain gauges. This is generally considered a better result, as it clearly indicates the need for sensor replacement (whereas a metallic strain sensor may give the false impression of continued function following an over-stress event).

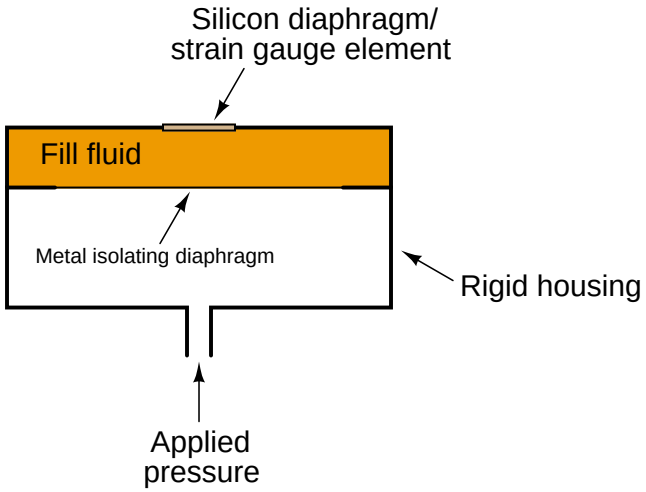
[3] For a simple demonstration of metal fatigue and metal "flow," simply take a metal paper clip and repeatedly bend it back and forth until you feel the metal wire weaken. Gentle force applied to the paper clip will cause it to deform in such a way that it returns to its original shape when the force is removed. Greater force, however, will exceed the paper clip's elastic limit, causing permanent deformation and also altering the spring characteristics of the clip.

Thus, most modern piezoresistive-based pressure instruments use silicon strain gauge elements to sense deformation of a diaphragm due to applied fluid pressure. A simplified illustration of a diaphragm / strain gauge pressure sensor is shown here:



As the diaphragm bows outward with applied fluid pressure, the strain gauge stretches to a greater length, causing its resistance to increase. This change in resistance imbalances the bridge circuit, causing a voltage (V_{out}) proportional to the amount of applied pressure. Thus, the strain gauge works to convert an applied pressure into a measurable voltage signal which may be amplified and converted into a 4-20 mA loop current signal (or into a digital "fieldbus" signal).

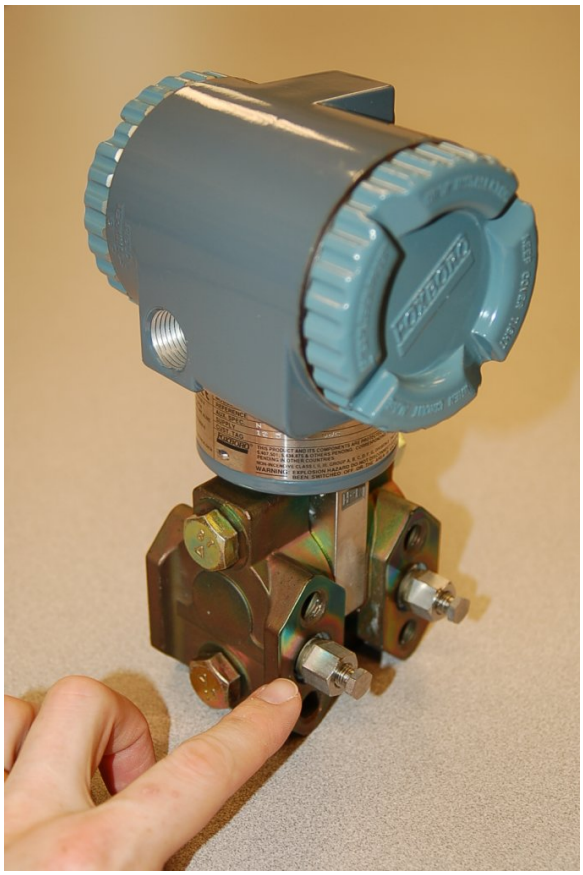
In some designs, a single silicon wafer serves as both the diaphragm and the strain gauge so as to fully exploit the excellent mechanical properties of silicon (high linearity and low fatigue). However, silicon is not chemically compatible with many process fluids, and so pressure must be transferred to the silicon diaphragm/sensor via a non-reactive *fill fluid* (commonly a silicone-based or fluorocarbon-based liquid). A metal *isolating diaphragm* transfers process fluid pressure to the fill fluid, which in turn transfers pressure to the silicon wafer. Another simplified illustration shows how this works:



The isolating diaphragm is designed to be much more flexible (less rigid) than the silicon diaphragm, because its purpose is to seamlessly transfer fluid pressure from the process fluid to the fill fluid, not to act as a spring element. In this way, the silicon sensor experiences the same pressure that it would if it were directly exposed to the process fluid, without having to contact the process fluid. The flexibility of the metal isolating diaphragm also means it experiences much less stress than the silicon sensing diaphragm, which avoiding the problems of metal fatigue experienced by transmitter designs using metal as the sensing (spring) element.

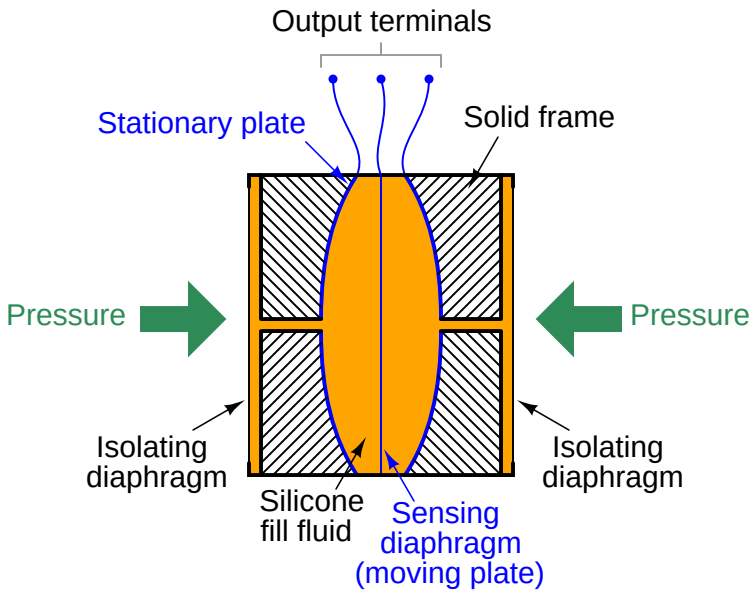
This use of a fill fluid to transfer pressure from an isolating diaphragm to a sensing diaphragm inside the transmitter is used in most if not all modern pressure transmitter designs, even those that are not piezoresistive.

An example of a pressure instrument utilizing a silicon strain gauge element is the Foxboro model IDP10 differential pressure transmitter, shown in the following photograph:



19.3.2 Differential capacitance sensors

Another common electrical pressure sensor design works on the principle of *differential capacitance*. In this design, the sensing element is a taut metal diaphragm located equidistant between two stationary metal surfaces[4], comprising three plates for a complementary pair of capacitors. An electrically insulating fill fluid (usually a liquid silicone compound) transfers motion from the isolating diaphragms to the sensing diaphragm, and also doubles as an effective dielectric for the two capacitors:



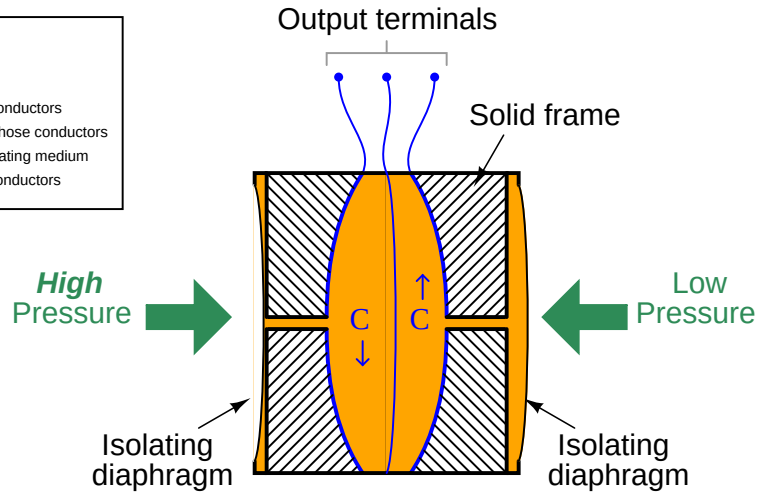
Any difference of pressure across the cell causes the diaphragm to flex in the direction of least pressure. The sensing diaphragm is a precision-manufactured spring element, meaning that its displacement is a predictable function of applied force. The applied force in this case can only be a function of differential pressure acting against the surface area of the diaphragm in accordance with the standard force-pressure-area equation $F = PA$. In this case, we have two forces caused by two fluid pressures working against each other, so our force-pressure-area equation may be re-written to describe *resultant* force as a function of differential pressure $(P_1 - P_2)$ and diaphragm area: $F = (P_1 - P_2)A$. Since diaphragm area is constant, and force is predictably related to diaphragm displacement, all we need now in order to infer differential pressure is to accurately measure displacement of the diaphragm.

[4] In the following diagram, both the sensing diaphragm and the stationary metal surfaces are shown colored blue, to distinguish these electrical elements from the other structural components of the device.

The diaphragm's secondary function as one plate of two capacitors provides a convenient method for measuring displacement. Since capacitance between conductors is inversely proportional to the distance separating them, capacitance on the low-pressure side will increase while capacitance on the high-pressure side will decrease:

$$C = \frac{A\varepsilon}{d}$$

C = Capacitance between two conductors
A = Overlapping area between those conductors
ε = Dielectric permittivity of insulating medium
d = Distance separating those conductors



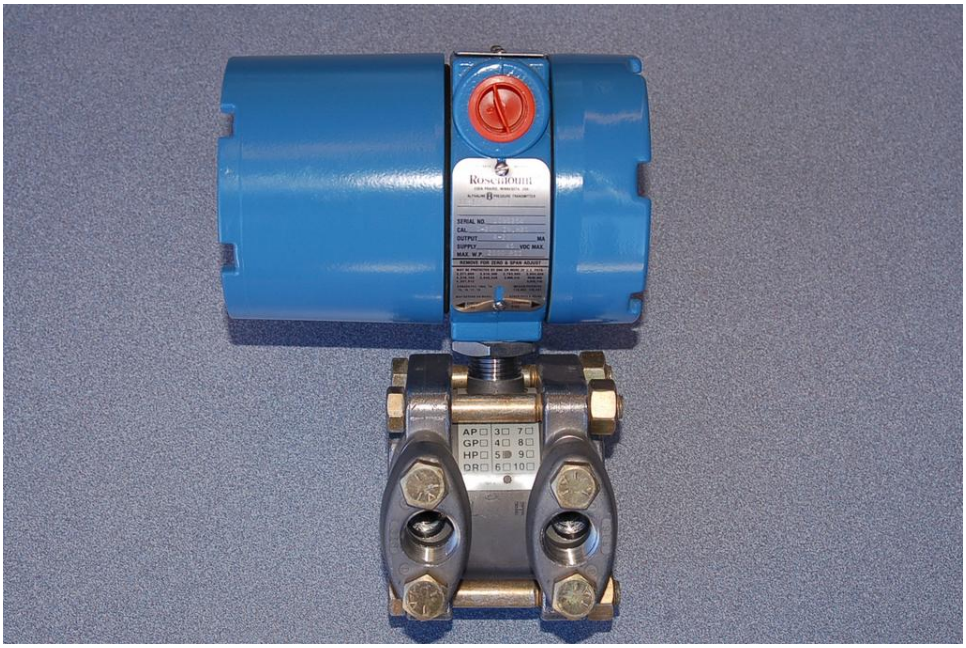
A capacitance detector circuit connected to this cell uses a high-frequency AC excitation signal to measure the different in capacitance between the two halves, translating that into a DC signal which ultimately becomes the signal output by the instrument representing pressure.

These pressure sensors are highly accurate, stable, and rugged. An interesting feature of this design – using two isolating diaphragms to transfer process fluid pressure to a single sensing diaphragm through an internal "fill fluid" – is that the solid frame bounds the motion of the two isolating diaphragms such that neither one is able to force the sensing diaphragm past its elastic limit. As the illustration shows, the higher-pressure isolating diaphragm gets pushed toward the metal frame, transferring its motion to the sensing diaphragm via the fill fluid. If too much pressure is applied to that side, the isolating diaphragm will merely "flatten" against the solid frame of the capsule and stop moving. This positively limits the isolating diaphragm's motion so that it cannot possibly exert any more force on the sensing diaphragm, even if additional process fluid pressure is applied. This use of isolating diaphragms and fill fluid to transfer motion to the sensing diaphragm, employed in other styles of differential pressure sensor as well, gives modern differential pressure instruments excellent resistance to overpressure damage.

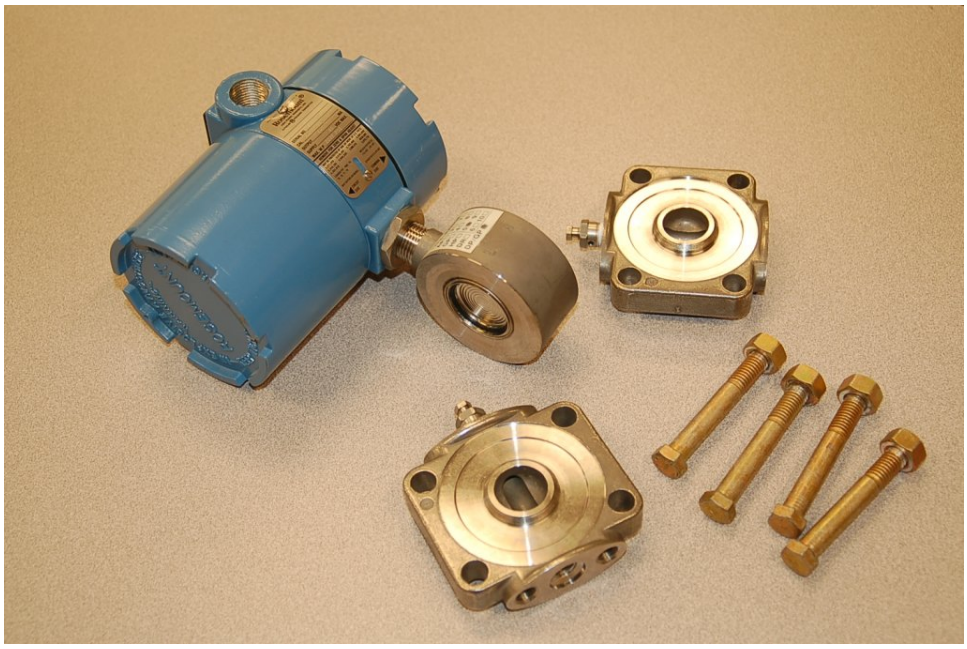
It should be noted that the use of a liquid fill fluid is key to this overpressure-resistant design. In order for the sensing diaphragm to accurately translate applied pressure into a proportional capacitance, it must not contact the conductive metal frame surrounding it. In order for any diaphragm to be protected against overpressure, however, it must contact a solid backstop to limit further travel. Thus, the need for non-contact (capacitance) and for contact (overpressure protection) are mutually exclusive, making it nearly impossible to perform both functions with a single sensing diaphragm. Using fill fluid to transfer pressure from isolating diaphragms to the sensing diaphragm allows us to separate the function of capacitive measurement (sensing diaphragm) from the function

of overpressure protection (isolation diaphragms) so that each diaphragm may be optimized for a separate purpose.

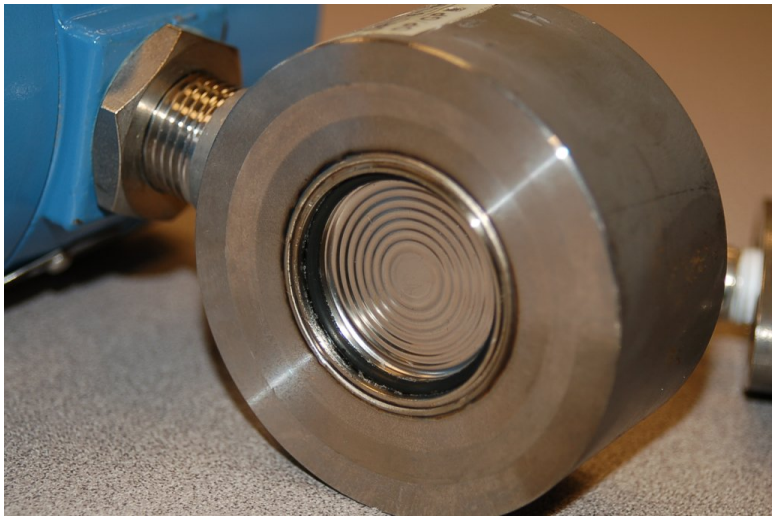
A classic example of a pressure instrument based on the differential capacitance sensor is the Rosemount model 1151 differential pressure transmitter, shown in assembled form in the following photograph:



By removing four bolts from the transmitter, we are able to remove two flanges from the pressure capsule, exposing the isolating diaphragms to plain view:



A close-up photograph shows the construction of one of the isolating diaphragms, which unlike the sensing diaphragm is designed to be very flexible. The concentric corrugations in the metal of the diaphragm allow it to easily flex with applied pressure, transmitting process fluid pressure through the silicone fill fluid to the taut sensing diaphragm inside the differential capacitance cell:



The interior of the same differential capacitance sensor (revealed by cutting a Rosemount model 1151 sensor in half with a chop saw[5]) shows the isolating diaphragms, the sensing diaphragm, and the ports connecting them together:



Here, the left-side isolating diaphragm is clearer to see than the right-side isolating diaphragm. A feature clearly evident in this photograph is the small clearance between the left-side isolating diaphragm and the internal metal frame, versus the spacious chamber in which the sensing diaphragm resides. Recall that these internal spaces are normally occupied by *fill fluid*, the purpose of which is to transfer pressure from the isolating diaphragms to the sensing diaphragm. As mentioned before, the solid metal frame limits the travel of each isolating diaphragm in such a way that the higher-pressure isolating diaphragm "bottoms out" on the metal frame before the sensing diaphragm can be pushed past its elastic limit. In this way, the sensing diaphragm is protected against damage from overpressure because the isolating diaphragms are simply not allowed to move any farther.

[5] A chop saw is admittedly not a tool of finesse, and it did a fair job of mangling this unfortunate differential capacitance cell. A bandsaw was tried at first, but made virtually no progress in cutting the hard stainless steel of the capsule assembly. The chop saw's abrasive wheel created a lot of heat, discoloring the metal and turning the silicone fill fluid into a crystalline mass which had to be carefully chipped out by hand using an ice pick so as to not damage the thin metal sensing diaphragm. Keep these labors in mind, dear reader, as you enjoy this textbook!

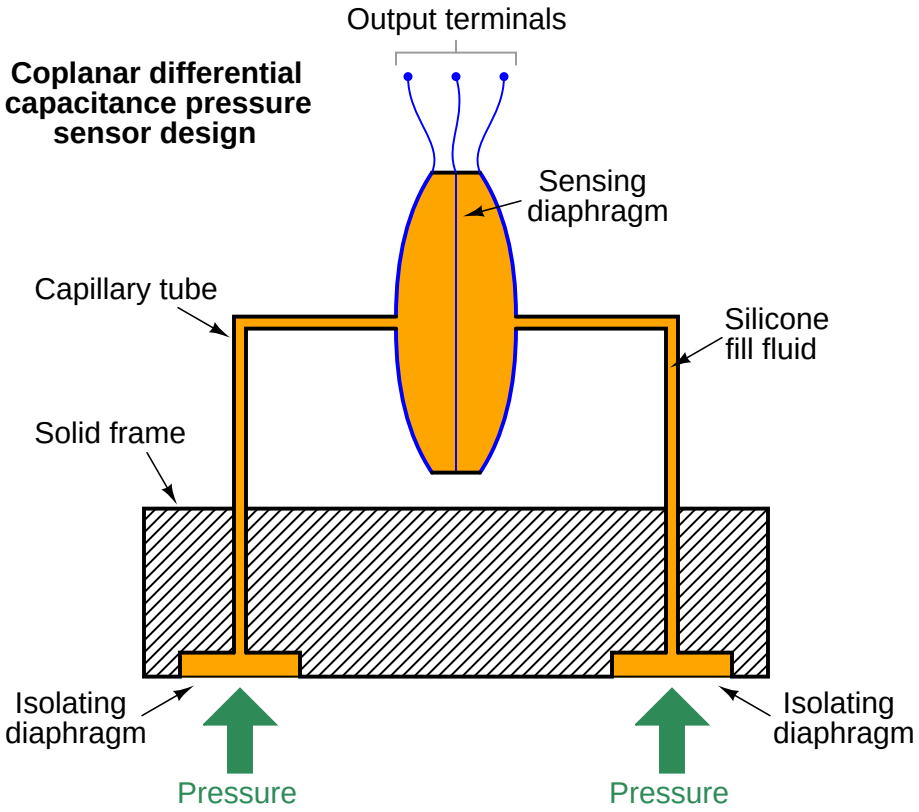
The differential capacitance sensor inherently measures *differences* in pressure applied between its two sides. In keeping with this functionality, this pressure instrument has two threaded ports into which fluid pressure may be applied. A later section in this chapter will elaborate on the utility of differential pressure transmitters (section 19.5 beginning on page 1345). All the electronic circuitry necessary for converting the sensor's differential capacitance into an electronic signal representing pressure is housed in the blue-colored structure above the capsule and flanges.

A more modern realization of the differential capacitance pressure-sensing principle is the Rosemount model 3051 differential pressure transmitter:



As is the case with all differential pressure devices, this instrument has *two* ports through which fluid pressure may be applied to the sensor. The sensor, in turn, responds only to the *difference* in pressure between the ports.

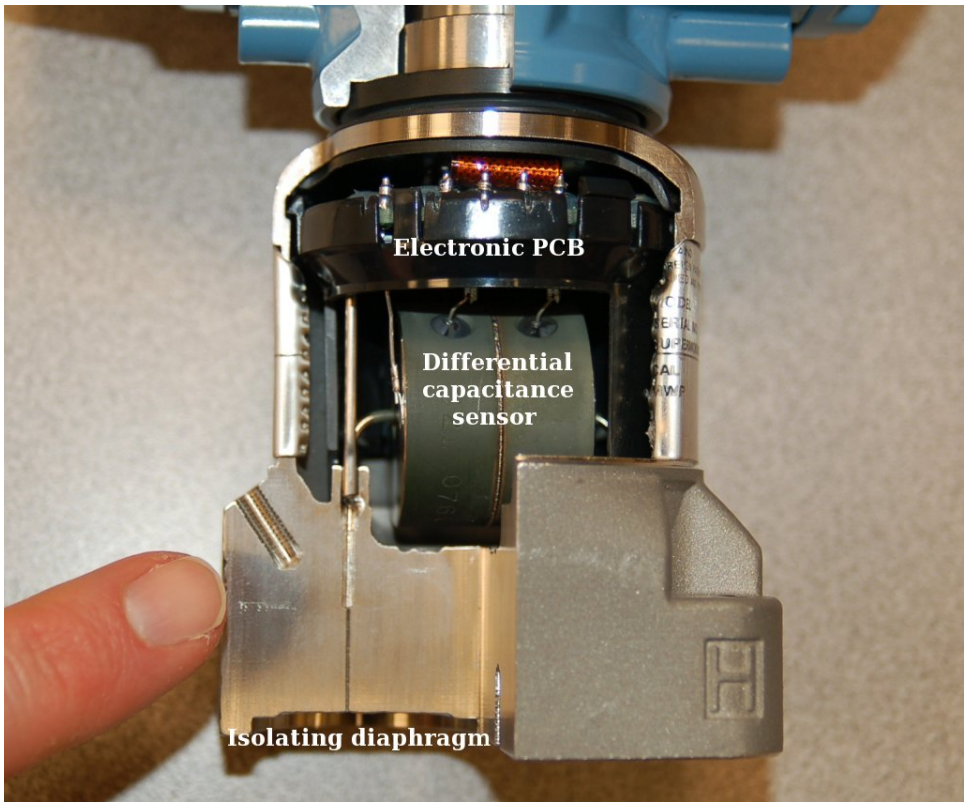
The differential capacitance sensor construction is more complex in this particular pressure instrument, with the plane of the sensing diaphragm perpendicular to the plane of the two isolating diaphragms. This "coplanar" design is more compact than the older style of sensor, and more importantly it isolates the sensing diaphragm from flange bolt stress – one of the main sources of error in the previous design[6].



Take particular note of how the sensor assembly is not embedded in the solid metal frame as was the case with the original Rosemount design. Instead, the sensor assembly is relatively isolated from the frame, connected only by two capillary tubes joining it to the isolating diaphragms. This way, stresses inside the metal frame imparted by flange bolts have virtually no effect on the sensor.

[6]Not only did applied torque of the four capsule bolts affect measurement accuracy in the older 1151 model design, but changes in temperature resulting in changing bolt tension also had a detrimental impact on accuracy. Most modern differential pressure transmitter designs strive to isolate the sensing diaphragm assembly from flange bolt stress for these reasons.

A cutaway model of a Rosemount model 3051S ("supermodule") DP transmitter shows how this all looks in real life:



Process fluid pressure applied to the isolating diaphragm(s) transfers to fill fluid inside the capillary tubes, conveying pressure to the taut diaphragm inside the differential capacitance sensor. Like the classic Rosemount model 1151 design, we see the fill fluid performing multiple functions:

- The fill fluid protects the delicate sensing diaphragm from contact with unclean or corrosive process fluids
- The fill fluid allows the isolating diaphragms to provide overpressure protection for the sensing diaphragm
- The fill fluid provides a medium of constant permittivity for the differential capacitance circuit to function

The "supermodule" series of Rosemount pressure transmitters shares the same coplanar design as the earlier 3051 models, but adds a new design feature: inclusion of the electronics within the stainless-steel module rather than the blue-painted upper housing. This feature allows the transmitter size to be significantly reduced if needed for applications with limited space.

19.3.3 Resonant element sensors

As any guitarist, violinist, or other stringed-instrument musician can tell you, the natural frequency of a tensed string increases with tension. This, in fact, is how stringed instruments are tuned: the tension on each string is precisely adjusted to achieve the desired resonant frequency.

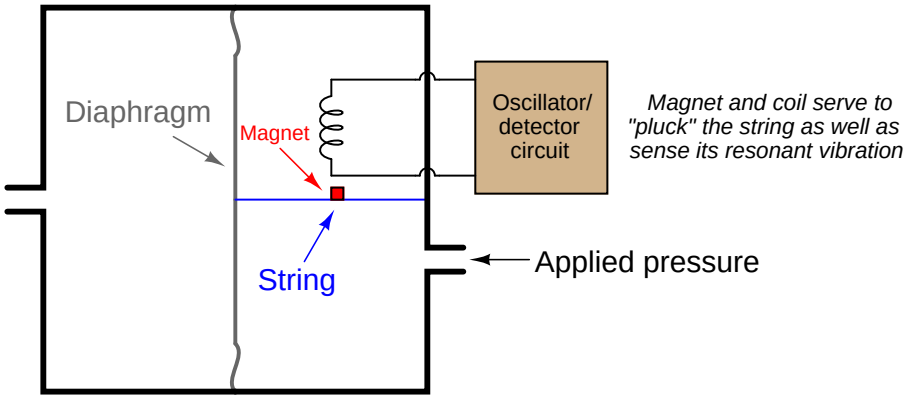
Mathematically, the resonant frequency of a string may be described by the following formula:

$$f = \frac{1}{2L}\sqrt{\frac{F_T}{\mu}}$$

Where,

f = Fundamental resonant frequency of string (Hertz)
L = String length (meters)
F_T = String tension (newtons)
μ = Unit mass of string (kilograms per meter)

It stands to reason, then, that a string may serve as a force sensor. All that is needed to complete the sensor is an oscillator circuit to keep the string vibrating at its resonant frequency, and that frequency becomes an indication of tension (force). If the force originates from pressure applied to some sensing element such as a bellows or diaphragm, the string's resonant frequency will indicate fluid pressure. A proof-of-concept device based on this principle might look like this:



It should be noted that this principle of force measurement is nonlinear[7], as indicated by the equation for resonant frequency (tension force F lies inside the radicand). This means the pressure transmitter must be designed with an electronic characterizing function to "linearize" the frequency measurement into a pressure measurement.

[7] For example, a doubling of force results in a frequency increase of 1.414 (precisely equal to $\sqrt{2}$). A *four*-fold increase in pressure would be necessary to *double* the string's resonant frequency. This particular form of nonlinearity, where diminishing returns are realized as the applied stimulus increases, yields excellent rangeability. In other words, the instrument is inherently more sensitive to changes in pressure at the low end of its sensing range, and "de-sensitizes" itself toward the high end of its sensing range.

The Foxboro company pioneered this concept in an early *resonant wire* design of pressure transmitter. Later, the Yokogawa corporation of Japan applied the concept using a pair of micro-machined[8] silicon resonator structures bonded to a single sensing diaphragm, which became the basis for their successful line of "DPharp" pressure transmitters.

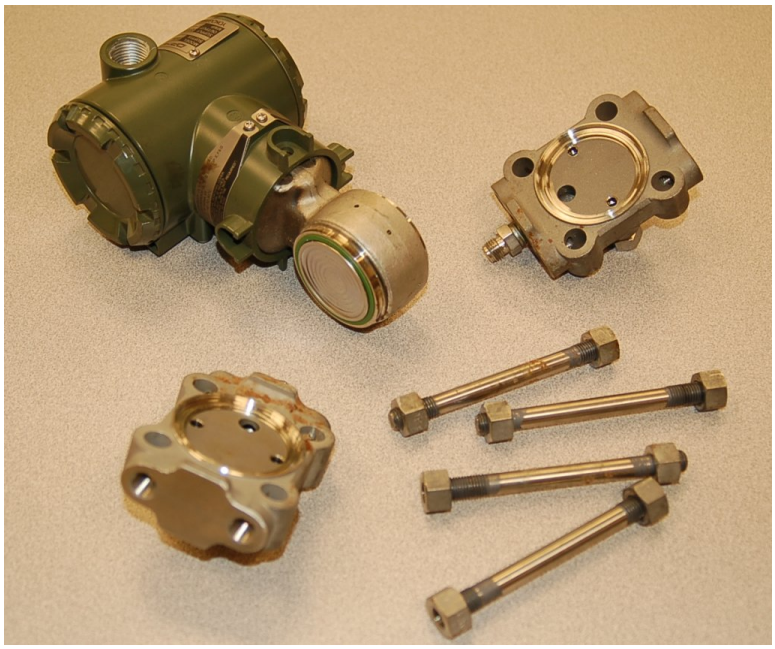
A photograph of a Yokogawa model EJA110 pressure transmitter with this technology is seen here:



Process pressure enters through ports in two flanges, presses against a pair of isolating diaphragms, transferring motion to a single sensing diaphragm via fill fluid where the resonant elements change frequency with diaphragm strain. Motion of the sensing diaphragm in either direction tenses one resonant element and compresses the other, causing their frequencies to deviate from each other. Electronic circuits within the upper housing measure the two resonant elements' frequencies and generate an output signal proportional to their frequency difference. This, of course, is a representation of applied differential pressure.

[8] This is an example of a micro-electro-mechanical system, or *MEMS*.

Even when disassembled, the transmitter does not look much different from the more common differential capacitance sensor design.



The important design differences are hidden from view, inside the sensing capsule. Functionally, though, this transmitter is much the same as its differential-capacitance and piezoresistive cousins. This design even uses fill fluid to protect the delicate silicon resonators from potentially destructive process fluids, just like differential capacitance sensors and most piezoresistive sensor designs.

An interesting advantage of the resonant element pressure sensor is that the sensor signal is easily digitized. The vibration of each resonant element is sensed by the electronics package as an AC frequency. This frequency signal is "counted" by a digital counter circuit over a given span of time and converted to a binary digital representation without any need for an analog-to-digital converter (ADC) circuit. Quartz crystal electronic oscillators are extremely precise, providing the stable frequency reference necessary for comparison in any frequency-based instrument.

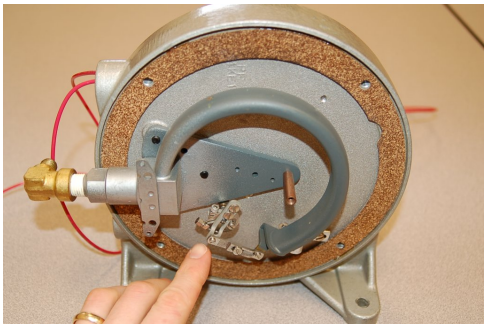
In the Yokogawa "DPharp" design, the two resonant elements oscillate at a nominal frequency of approximately 90 kHz. As the sensing diaphragm deforms with applied differential pressure, one resonator experiences tension while the other experiences compression, causing the frequency of the former to shift up and the latter to shift down (as much as $\pm$ 20 kHz). The signal conditioning electronics inside the transmitter measures this difference in resonator frequency to infer applied pressure.

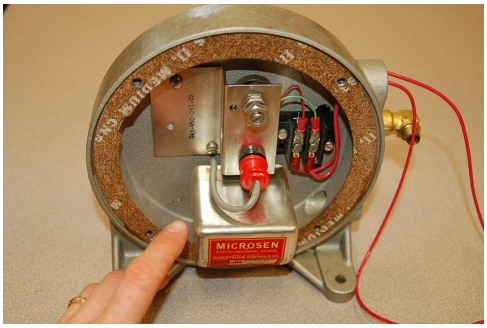
19.3.4 Mechanical adaptations

Most modern electronic pressure sensors convert very small diaphragm motions into electrical signals through the use of sensitive motion-sensing techniques (strain gauge sensors, differential capacitance cells, etc.). Diaphragms made from elastic materials behave as springs, but circular diaphragms exhibit very nonlinear behavior when significantly stretched unlike classic spring designs such as coil and leaf springs which exhibit linear behavior over a wide range of motion. Therefore, in order to yield a linear response to pressure, a diaphragm-based pressure sensor must be designed in such a way that the diaphragm stretches very little over the normal range of operation. Limiting the displacement of a diaphragm necessitates highly sensitive motion-detection techniques such as strain gauge sensors, differential capacitance cells, and mechanical resonance sensors to convert that diaphragm's very slight motion into an electronic signal.

An alternative approach to electronic pressure measurement is to use mechanical pressure-sensing elements with more linear pressure-displacement characteristics – such as bourdon tubes and spring-loaded bellows – and then detect the large-scale motion of the pressure element using a less-sophisticated electrical motion-sensing device such as a potentiometer, LVDT, or Hall Effect sensor. In other words, we take the sort of mechanism commonly found in a direct-reading pressure gauge and attach it to a potentiometer (or similar device) to derive an electrical signal from the pressure measurement.

The following photographs show front and rear views of an electronic pressure transmitter using a large C-shaped bourdon tube as the sensing element (seen in the left-hand photograph):



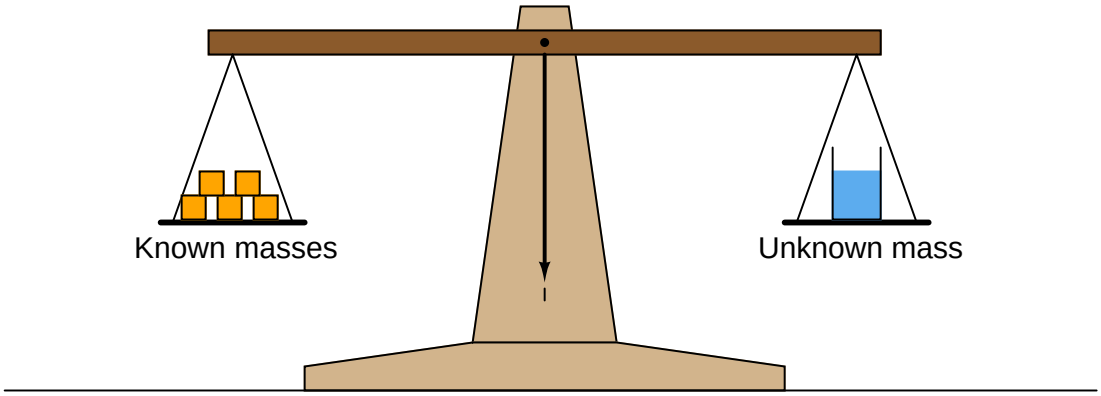


This alternative approach is undeniably simpler and less expensive to manufacture than the more sophisticated approaches used with diaphragm-based pressure instruments, but is prone to greater inaccuracies. Even bourdon tubes and bellows are not perfectly linear spring elements, and the substantial motions involved with using such pressure elements introduces the possibility of hysteresis errors (where the instrument does not respond accurately during reversals of pressure, where the mechanism changes direction of motion) due to mechanism friction, and deadband errors due to backlash (looseness) in mechanical connections.

You are likely to encounter this sort of pressure instrument design in direct-reading gauges equipped with electronic transmitting capability. An instrument manufacturer will take a proven product line of pressure gauge and add a motion-sensing device to it that generates an electric signal proportional to mechanical movement inside the gauge, resulting in an inexpensive pressure transmitter that happens to double as a direct-reading pressure gauge.

19.4 Force-balance pressure transmitters

An important legacy technology for all kinds of continuous measurement is the *self-balancing system*. A "self-balance" system continuously balances an adjustable quantity against a sensed quantity, the adjustable quantity becoming an indication of the sensed quantity once balance is achieved. A common manual-balance system is the type of scale used in laboratories to measure mass:

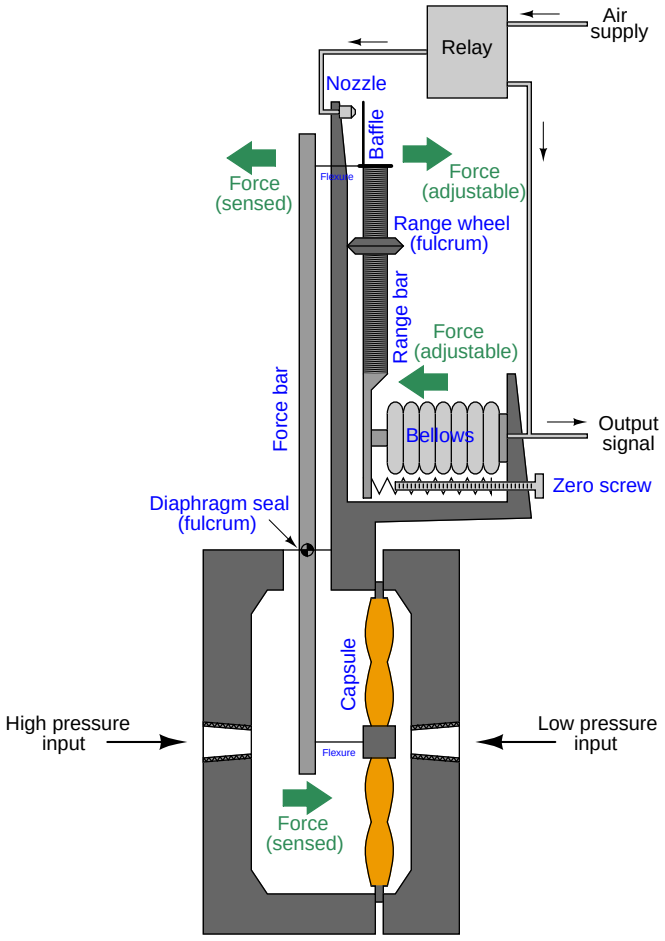


Here, the unknown mass is the sensed quantity, and the known masses are the adjustable quantity. A human lab technician applies as many masses to the left-hand side of the scale as needed to achieve balance, then counts up the sum total of those masses to determine the quantity of the unknown mass.

Such a system is perfectly linear, which is why these balance scales are popularly used for scientific work. The scale mechanism itself is the very model of simplicity, and the only thing the pointer needs to accurately sense is a condition of balance (equality between masses).

If the task of balancing is given to an automatic mechanism, the adjustable quantity will continuously change and adapt as needed to balance the sensed quantity, thereby becoming a representation of that sensed quantity. In the case of pressure instruments, pressure is easily converted into force by acting on the surface area of a sensing element such as a diaphragm or a bellows. A balancing force may be generated to exactly cancel the process pressure's force, making a *force-balance* pressure instrument. Like the laboratory balance scale, an industrial instrument built on the principle of balancing a sensed quantity with an adjustable quantity will be inherently linear, which is a tremendous advantage for measurement purposes.

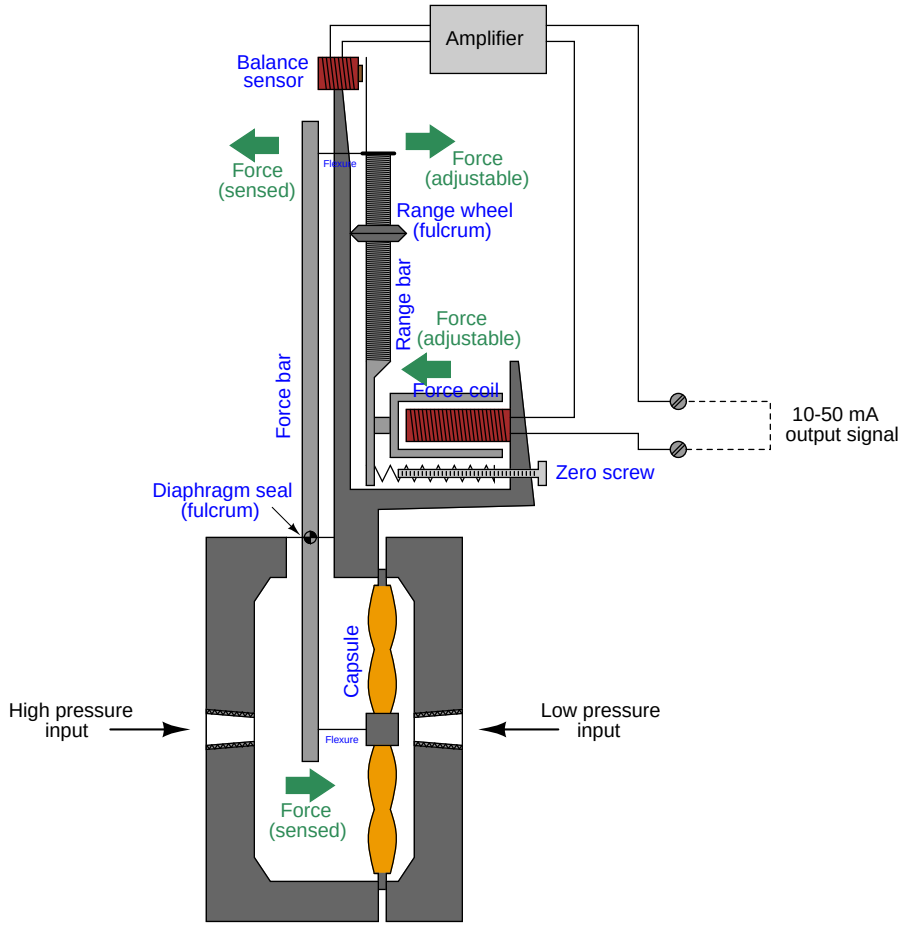
Here, we see a diagram of a force-balance pneumatic pressure transmitter[9], balancing a sensed differential pressure with an adjustable air pressure which becomes a pneumatic output signal:



Differential pressure is sensed by a liquid-filled diaphragm "capsule," which transmits force to a "force bar." If the force bar moves out of position due to this applied force, a highly sensitive "baffle" and "nozzle" mechanism senses it and causes a pneumatic amplifier (called a "relay") to send a different amount of air pressure to a bellows unit. The bellows presses against the "range bar" which pivots to counter-act the initial motion of the force bar. When the system returns to equilibrium, the air pressure inside the bellows will be a direct, linear representation of the process fluid pressure applied to the diaphragm capsule.

[9]Based on the design of Foxboro's popular model 13A pneumatic "DP cell" differential pressure transmitter.

With minor modifications to the design of this pressure transmitter[10], we may convert it from pneumatic to electronic force-balancing:



Differential pressure is sensed by the same type of liquid-filled diaphragm capsule, which transmits force to the force bar. If the force bar moves out of position due to this applied force, a highly sensitive electromagnetic sensor detects it and causes an electronic amplifier to send a different amount of electric current to a force coil. The force coil presses against the range bar which pivots to counteract the initial motion of the force bar. When the system returns to equilibrium, the milliampere current through the force coil will be a direct, linear representation of the process fluid pressure applied to the diaphragm capsule.

A distinct advantage of force-balance pressure instruments (besides their inherent linearity) is the constraining of sensing element motion. Unlike a modern diaphragm-based pressure transmitter which relies on the spring characteristics of the diaphragm to convert pressure into force and then

[10] Very loosely based on the design of Foxboro's now-obsolete E13 electronic "DP cell" differential pressure transmitter.

into motion (displacement) which is sensed and converted into an electronic signal, a force-balance transmitter works best when the diaphragm is slack and has no spring characteristics at all. Balance with the force of the process fluid pressure is achieved by the application of either an adjustable air pressure or an adjustable electric current, not by the natural tensing of a spring element. This makes a force-balance instrument far less susceptible to errors due to metal fatigue or any other degradation of spring characteristics.

Unfortunately, force-balance instruments have significant disadvantages as well. Force-balance mechanisms tend to be bulky[11], and they translate external vibration into inertial force which adds "noise" to the output signal. Also, the amount of electrical power necessary to provide adequate balancing force in an electronic force-balance transmitter is such that it is nearly impossible to limit below the level necessary to ensure intrinsic safety (protection against the accidental ignition of explosive atmospheres by limiting the amount of energy the instrument could possibly discharge into a spark).

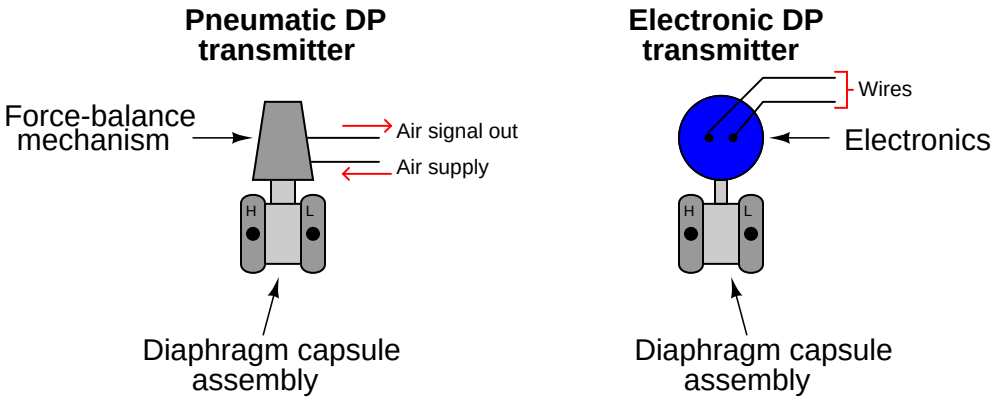
19.5 Differential pressure transmitters

One of the most common, and most useful, pressure measuring instruments in industry is the *differential pressure transmitter*. This device senses the difference in pressure between two ports and outputs a signal representing that pressure in relation to a calibrated range. Differential pressure transmitters may be based on any of the previously discussed pressure-sensing technologies, so this section focuses on application rather than theory.

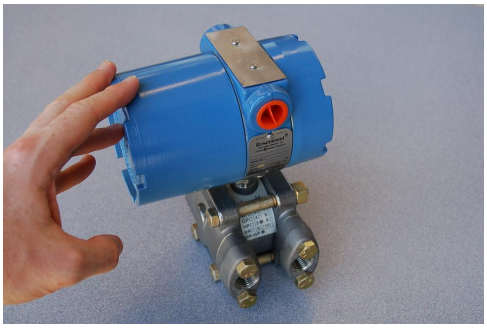
[11] One instrument technician I know referred to the Foxboro E13 differential pressure transmitter as "pig iron" after having to hoist it by hand to the top of a distillation column.

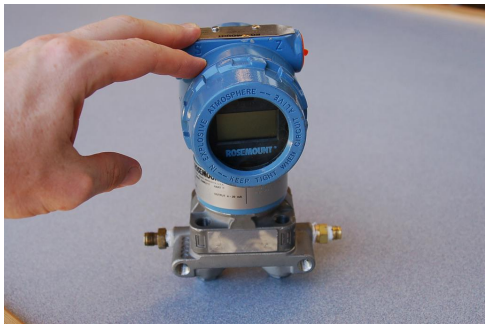
19.5.1 DP transmitter construction and behavior

Differential pressure transmitters constructed for industrial measurement applications typically consist of a strong (forged metal) body housing the sensing element(s), topped by a compartment housing the mechanical and/or electronic components necessary to translate the sensed pressure into a standard instrumentation signal (e.g. 3-15 PSI, 4-20 mA, digital fieldbus codes):

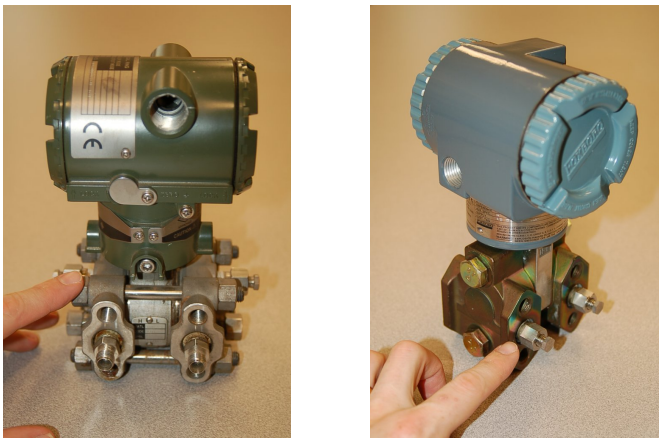


Two models of electronic differential pressure transmitter appear in the following photographs, the Rosemount model 1151 (left) and model 3051 (right):



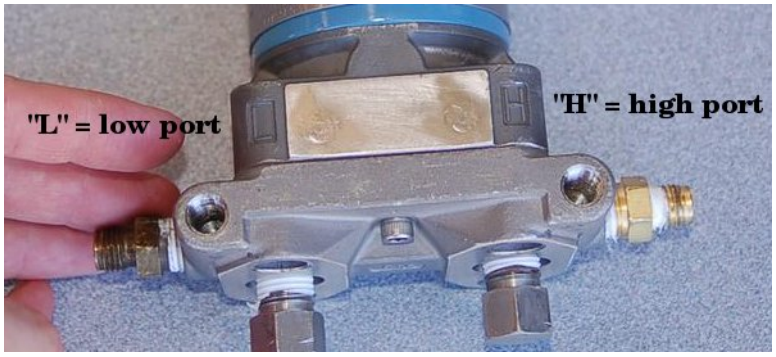


Two more models of electronic differential pressure transmitter are shown in the next photograph, the Yokogawa EJA110 (left) and the Foxboro IDP10 (right):



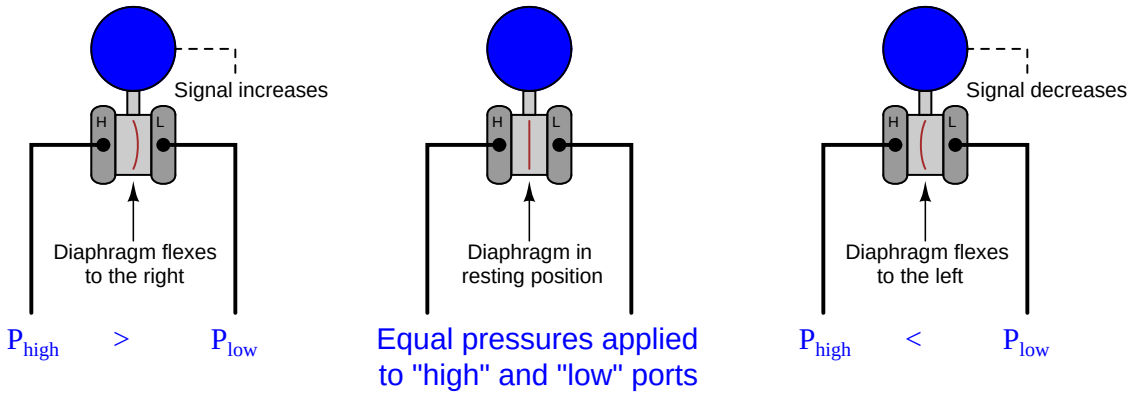
In each of these differential pressure transmitter examples, the pressure-sensing element is housed in the bottom half of the device (the forged-steel structure) while the electronics are housed in the top half (the colored, round, cast-aluminum structure).

Regardless of make or model, every differential pressure ("DP", "d/p", or ΔP)[12] transmitter has *two* pressure ports to sense different process fluid pressures. These ports typically have $\frac{1}{4}$ inch female NPT threads for convenient connection to the process. One of these ports is labeled "high" and the other is labeled "low". This labeling does not necessarily mean that the "high" port must always be at a greater pressure than the "low" port. What these labels represent is the effect any increasing fluid pressure applied to that port will have on the *direction* of the output signal's change.

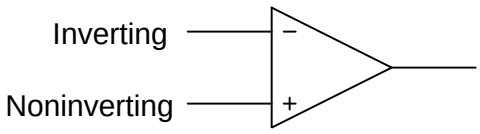


[12] As far as I have been able to determine, the labels "D/P" and "DP cell" were originally trademarks of the Foxboro Company. Those particular transmitter models became so popular that the term "DP cell" came to be applied to nearly *all* makes and models of differential pressure transmitter, much like the trademark "Vise-Grip" is often used to describe *any* self-locking pliers, or "Band-Aid" is often used to describe *any* form of self-adhesive bandage.

The most common sensing element used by modern DP transmitters is the diaphragm. One side of this diaphragm receives process fluid pressure from the "high" port, while the other receives process fluid pressure from the "low" port. Any difference of pressure between the two ports causes the diaphragm to flex from its normal resting (center) position. This flexing is then translated into an output signal by any number of different technologies, depending on the manufacturer and model of the transmitter:

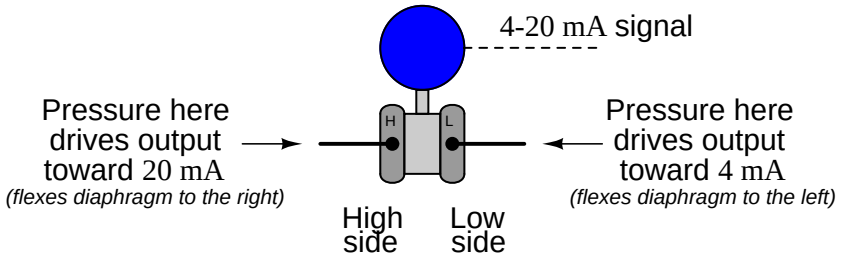


The concept of differential pressure instrument port labeling is very similar to the "inverting" and "noninverting" labels applied to operational amplifier input terminals:



The "+" and "−" symbols do not imply polarity of the input voltage(s); i.e. it is not as though the "+" input must be more positive than the "−" input. These symbols merely represent the different direction each input tends to drive the output signal. An increasing potential applied to the "+" input drives the opamp's output positive, while an increasing potential applied to the "−" input drives the opamp's output negative. Phrasing this in terms common to closed-loop control systems, we could say that the "+" input is *direct-acting* while the "−" input is *reverse-acting*.

Similarly, the "H" and "L" labels on a DP transmitter's ports do not imply magnitude of input pressures; i.e. it is not as though the "H" port's pressure must be greater than the "L" port's pressure. These symbols merely represent the different effects on the output signal resulting from pressure applied to each port. An increasing pressure applied to the "high" port of a DP transmitter will drive the output signal to a greater level (up), while an increasing pressure applied to the "low" port of a DP transmitter will drive the output signal to a lesser level (down)[13]:



The ability to arbitrarily connect a DP transmitter to a process in such a way that it is either direct-acting or reverse-acting is a great advantage, as we will later see.

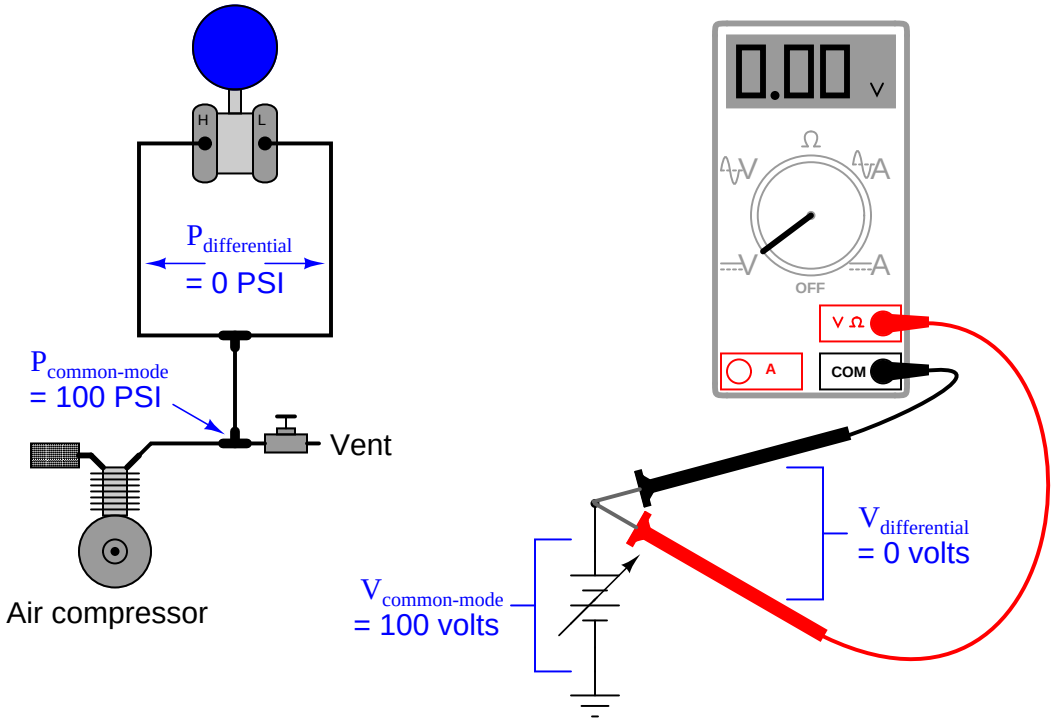
In the world of electronics, we refer to the ability of a differential voltage sensor (such as an operational amplifier) to sense small differences in voltage while ignoring large potentials measured with reference to ground by the phrase *common-mode rejection*. An ideal operational amplifier completely ignores the amount of voltage common to both input terminals, responding only to the *difference* in voltage *between* those terminals. This is precisely what a well-designed DP instrument does, except with fluid pressure instead of electrical voltage. A DP instrument ignores gauge pressure common to both ports, while responding only to *differences* in pressure *between* those two ports. Stated in other words, a differential pressure instrument (ideally[14]) responds only to differential pressure while ignoring common-mode pressure.

[13] One transmitter manufacturer I am aware of (ABB/Bailey) actually does use the "+" and "−" labels to denote high- and low-pressure ports rather than the more customary "H" and "L" labels found on other manufacturers' DP products.

[14] Perfect common-mode rejection is impossible for differential pressure instruments just as it is impossible for electronic voltage-measuring instruments, but in either case the effect is usually minimal. For differential pressure transmitters, the effect of common-mode pressure on the instrument's output signal is sometimes referred to as the *line pressure effect* or *static pressure effect*, typically stated as a percentage of the instrument's upper range limit per unit of common-mode pressure.

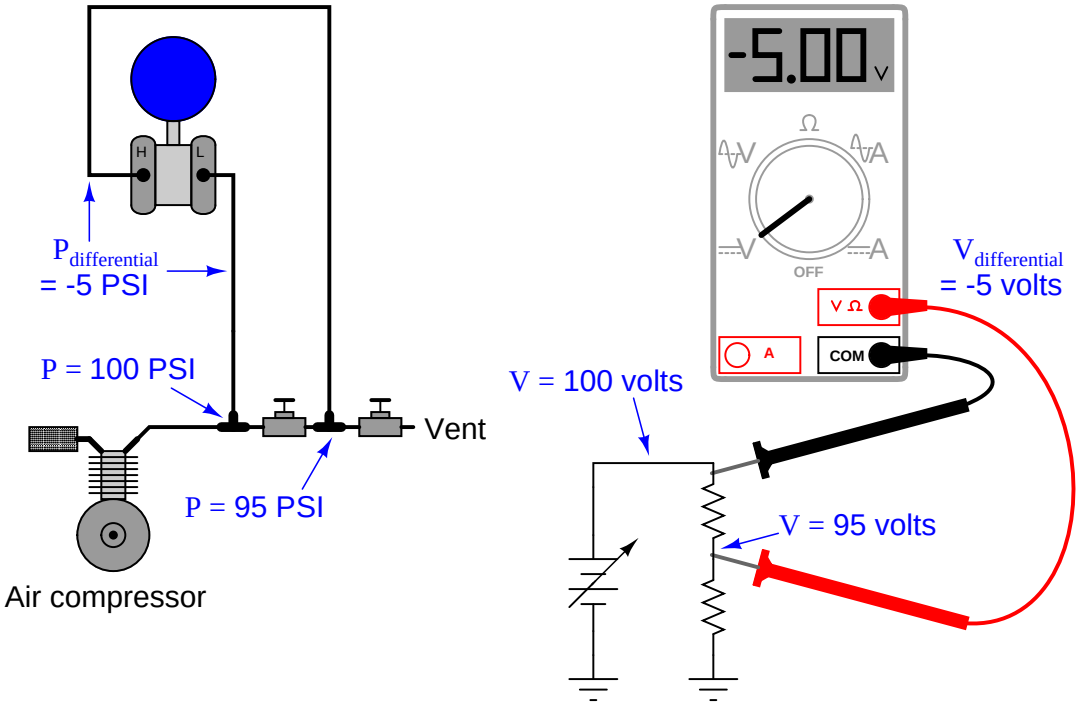
To illustrate, we may connect the "high" and "low" ports of a differential pressure transmitter together using pipe or tube, then expose both ports simultaneously to a source of fluid pressure such as pressurized air from an air compressor. If the transmitter is in good working order, it should continue to register zero differential pressure even as we vary the amount of static pressure applied to both ports. So long as the applied pressures to each port are equal, the transmitter's sensing diaphragm should experience zero net force pushing left or right. All force applied to the diaphragm from the "high" port's fluid pressure should be precisely countered (canceled) by force applied to the diaphragm from the "low" port's fluid pressure.

An electrical analogy to this would be connecting both red and black test leads of a voltmeter to a common point in an electrical circuit, then varying the amount of voltage between that point and earth ground. Since the voltmeter only registers *differences* of potential between its test leads, and those test leads are now electrically common to one another, the magnitude of common-mode voltage between that one point of the circuit and earth ground is irrelevant from the perspective of the voltmeter:



In each case the differential measurement device *rejects* the common-mode value, registering only the amount of difference (zero) between its sensing points.

The same common-mode rejection principle reveals itself in more complex fluid and electrical circuits. Consider the case of a DP transmitter and a voltmeter, both used to measure differential quantities in a "divider" circuit[15]:



In each case the differential measurement device responds only to the difference between the two measurement points, rejecting the common-mode value (97.5 PSI for the pressure transmitter, 97.5 volts for the voltmeter). Just to make things interesting in this example, the "high" side of each measuring instrument connects to the point of lesser value, such that the measured difference is a negative quantity. Like digital voltmeters, modern DP transmitters are equally capable of accurately measuring negative pressure differences as well as positive pressure differences.

[15]The electrical circuit shown on the right uses a pair of series-connected resistors to divide the source voltage into two parts, 5 volts and 95 volts. The pneumatic circuit shown on the left uses a pair of series-connected hand valves to divide the source pressure into two parts, 5 PSI and 95 PSI.

A vivid contrast between *differential* pressure and *common-mode* pressure for a DP instrument is seen in the pressure ratings shown on the nameplate of a Foxboro model 13A differential pressure transmitter:



This nameplate tells us that the transmitter has a calibrated differential pressure range of 50" H_2O (50 inches water column, which is only about 1.8 PSI). However, the nameplate also tells us that the transmitter has a *maximum working pressure* (MWP) of 1500 PSI. "Working pressure" refers to the amount of gauge pressure common to each port, not the differential pressure between ports. Taking these figures at face value means this transmitter will register zero (no differential pressure) even if the gauge pressure applied equally to both ports is a full 1500 PSI! In other words, this differential pressure transmitter will *reject* up to 1500 PSI of common-mode gauge pressure, and respond only to small differences in pressure between the ports (1.8 PSI differential being enough to stimulate the transmitter to full scale output).

19.5.2 DP transmitter applications

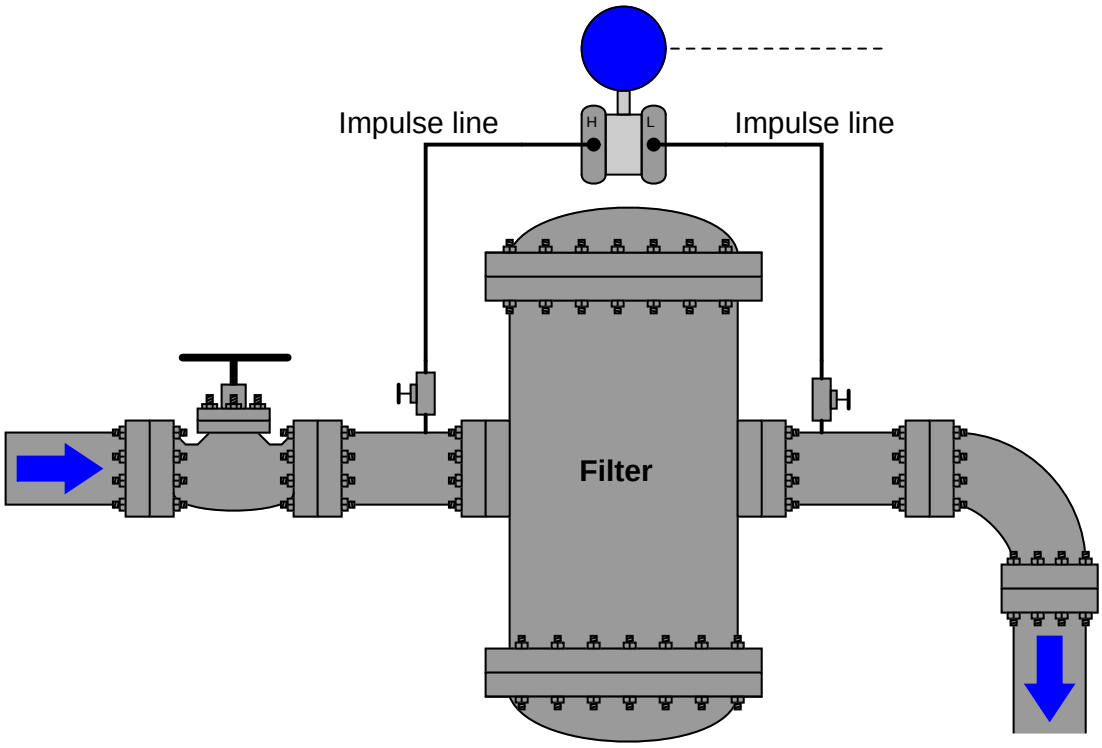
The combination of two differential pressure ports makes the DP transmitter very versatile as a pressure-measuring device. This one instrument may be used to measure pressure differences, positive (gauge) pressures, negative (vacuum) pressures, and even absolute pressures, just by connecting the "high" and "low" sensing ports differently.

In every DP transmitter application, there must be some means of connecting the transmitter's pressure-sensing ports to the points in a process. Metal or plastic tubes (or pipes) work well for this purpose, and are commonly called *impulse lines*, or *gauge lines*, or *sensing lines*[16]. This is equivalent to the test wires used to connect a voltmeter to points in a circuit for measuring voltage. Typically, these tubes are connected to the transmitter and to the process by means of *compression fittings* which allow for relatively easy disconnection and reconnection of tubes. For more information on instrument tube fittings, refer to section 8.2.1 beginning on page 580.

[16] Also called impulse *tubes*, gauge *tubes*, or sensing *tubes*.

Measuring process vessel clogging

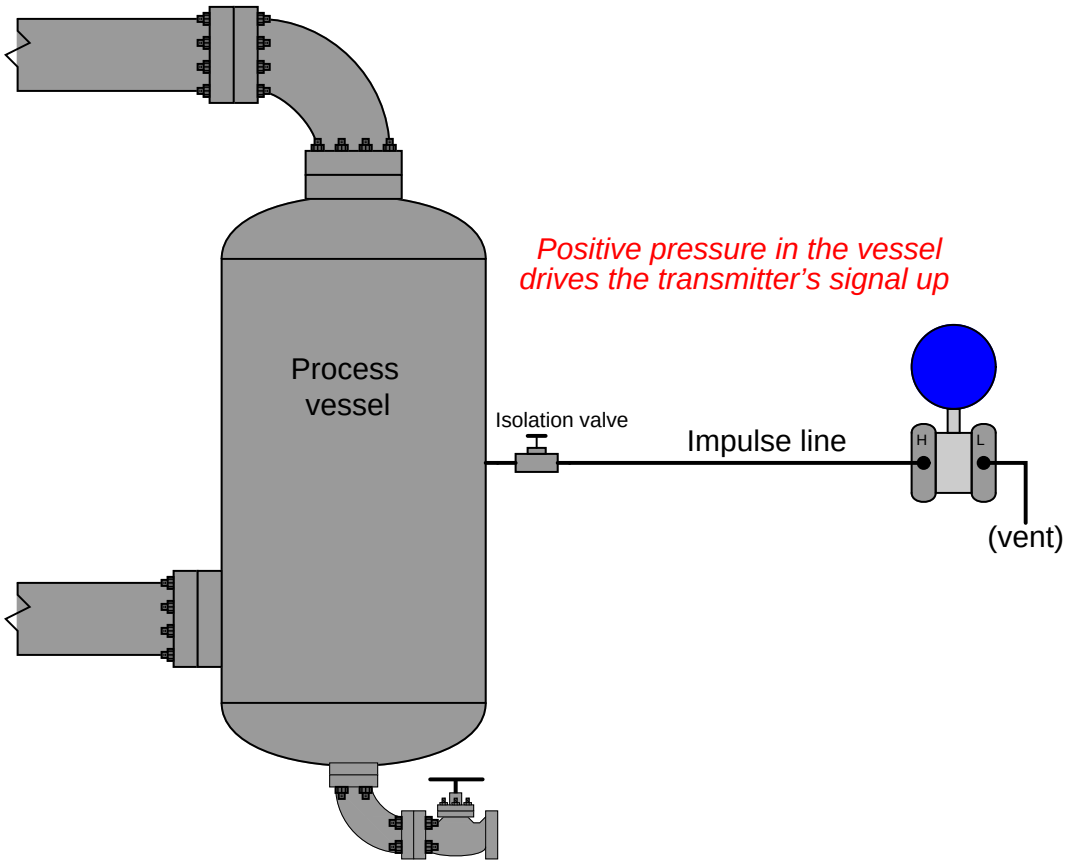
We may use the DP transmitter to measure an actual difference of pressure across a process vessel such as a filter, a heat exchanger, or a chemical reactor. The following illustration shows how a differential pressure transmitter may be used to measure clogging of a water filter:



Note how the high side of the DP transmitter connects to the upstream side of the filter, and the low side of the transmitter to the downstream side of the filter. This way, increased filter clogging will result in an increased transmitter output. Since the transmitter's internal pressure-sensing diaphragm only responds to *differences* in pressure between the "high" and "low" ports, the pressure in the filter and pipe relative to the atmosphere is completely irrelevant to the transmitter's output signal. The filter could be operating at a line pressure of 10 PSI or 10000 PSI – the only variable the DP transmitter measures is the pressure *drop* across the filter. If the upstream side is at 10 PSI and the downstream side is at 9 PSI, the differential pressure will be 1 PSI (sometimes labeled as PSID, "D" for *differential*). If the upstream pressure is 10000 PSI and the downstream pressure is 9999 PSI, the DP transmitter will still see a differential pressure of just 1 PSID. Likewise, the technician calibrating the DP transmitter on the workbench could use a precise air pressure of just 1 PSI (applied to the "high" port, with the "low" port vented to atmosphere) to simulate either of these real-world conditions. The DP transmitter simply cannot tell the difference between these three scenarios, nor should it be able to tell the difference if its purpose is to exclusively measure differential pressure.

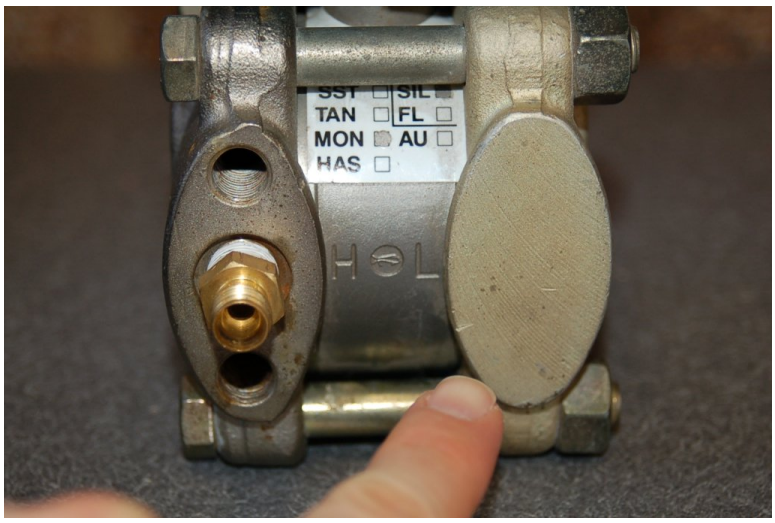
Measuring positive gauge pressure

DP instruments may also serve as simple *gauge pressure* instruments if needed, responding to pressures in excess of atmosphere. If we simply connect the "high" side of a DP instrument to a process vessel using an impulse tube, while leaving the "low" side vented to atmosphere, the instrument will interpret any positive pressure in the vessel as a positive *difference* between the vessel and atmosphere:

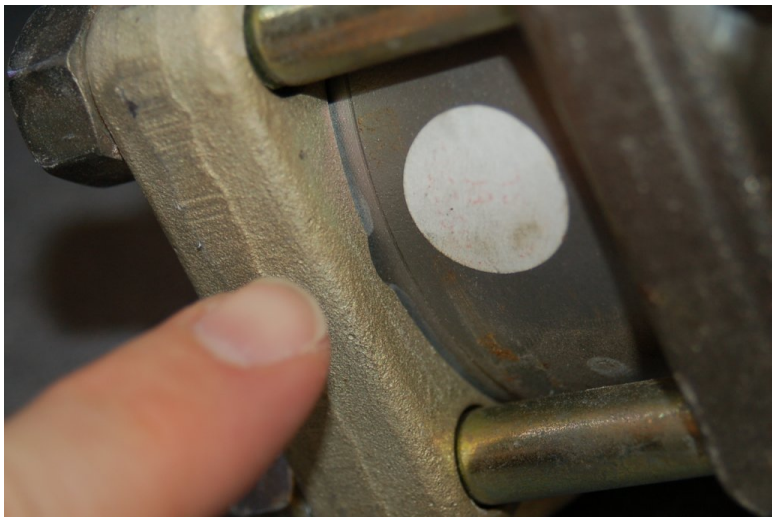


Although this may seem like a waste of the transmitter's abilities (why not just use a simpler gauge pressure transmitter with just one port?), it is actually a very common application for DP transmitters. This usage of a differential device may not actually be a "waste" if true-differential applications exist at the same facility for that pressure transmitter, which means only one spare transmitter need be stocked in the facility's warehouse instead of two spare transmitters (one of each type).

Most DP instrument manufacturers offer "gauge pressure" versions of their differential instruments, with the "high" side port open for connection to an impulse line and the "low" side of the sensing element capped off with a special vented flange, effectively performing the same function we see in the above example at a slightly lesser cost. A close-up photograph of a Rosemount model 1151GP gauge pressure transmitter shows the port-less flange on the "low" side of the pressure-sensing module. Only the "high" side of the sensor has a place for an impulse line to connect:

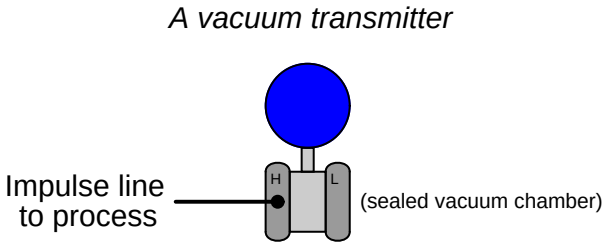


A closer look at this flange reveals a vent near the bottom, ensuring the "low" side of the pressure-sensing capsule always senses ambient (atmospheric) pressure:



Measuring absolute pressure

Absolute pressure is defined as the difference between a given fluid pressure and a perfect vacuum, as opposed to gauge pressure which is the difference between a fluid's pressure and the atmospheric air pressure. We may build an absolute pressure sensing instrument by taking a DP instrument and sealing the "low" side of its pressure-sensing element in connection to a vacuum chamber. This way, any pressure greater than a perfect vacuum will register as a positive difference:

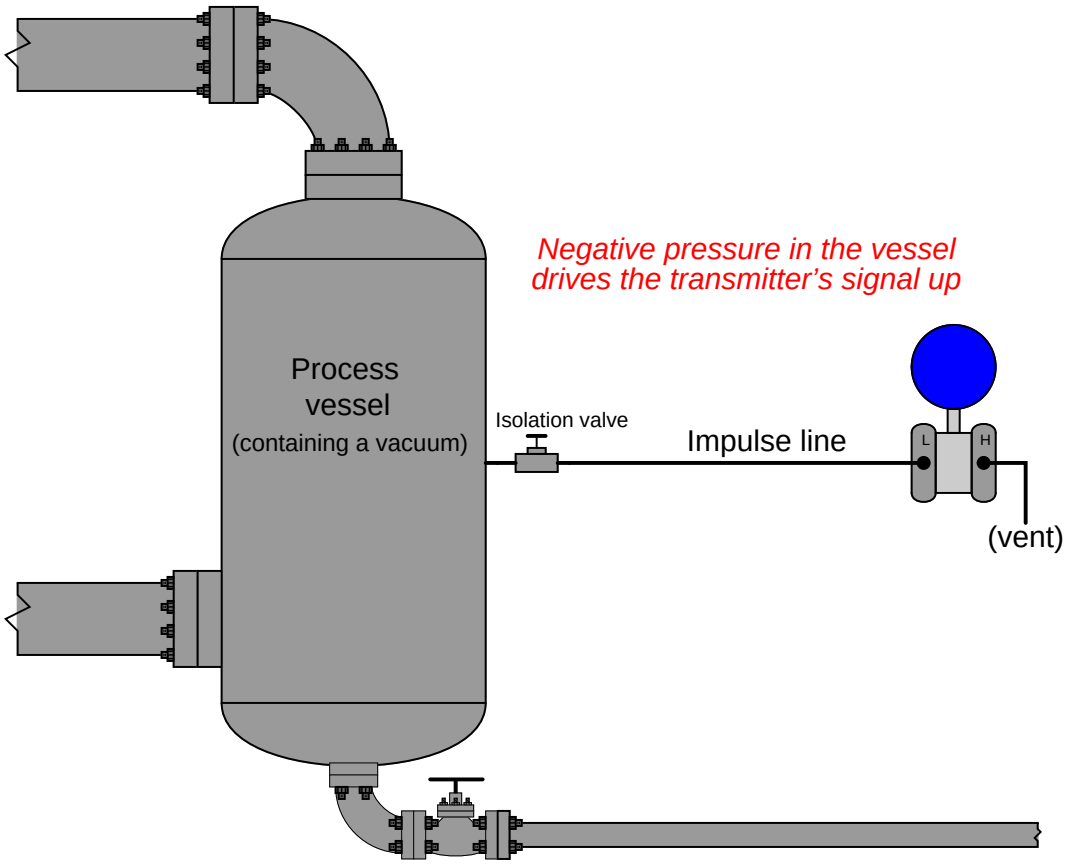


Most absolute pressure transmitters resemble "gauge pressure" adaptations of DP transmitters, with only one port available to connect an impulse line. Unlike gauge pressure transmitters, though, absolute pressure transmitters do *not* have vent holes on their "low" sides. The "low" side of an absolute pressure transmitter must be a sealed vacuum in order to accurately measure the "high" side fluid pressure in absolute terms.

Absolute pressure measurement is important for a variety of process applications, including boiling-point control and mass flow measurement of gases. The boiling temperature of any liquid is a function of the absolute pressure it experiences, and in applications where boiling temperature must be precisely controlled in order to achieve a certain outcome (e.g. vacuum distillation of crude oil, for example) the best type of pressure measurement to use absolute. When computing the mass flow rate of gases in a pipe, the relationship between volume and molecular count is a function of both temperature and pressure (both absolute), and so absolute pressure measurement is indispensable here as well.

Measuring vacuum

The same principle of connecting one port of a DP device to a process and venting the other works well as a means of measuring *vacuum* (pressures below that of atmosphere). All we need to do is connect the "low" side to the vacuum process and vent the "high" side to atmosphere:



Any pressure in the process vessel less than atmospheric will register to the DP transmitter as a *positive* difference (with P_{high} greater than P_{low}). Thus, the stronger the vacuum in the process vessel, the greater the signal output by the transmitter.

This last statement deserves some qualification. It used to be, the way analog pneumatic and electronic transmitters were designed many years ago, that the only way to obtain an increasing signal from a DP instrument was to ensure the "high" port pressure *rose* in relation to the "low" port pressure (or conversely stated, to ensure the "low" port pressure *dropped* in relation to the "high" side pressure). However, with the advent of digital electronic technology, it became rather easy to program a DP instrument with a *negative* range, for example 0 to −10 PSI. This way, a *decreasing* pressure as interpreted by the transmitter would yield an *increasing* output signal.

It is rare to find a pressure transmitter calibrated in such a way, but bear in mind that it is possible. This opens the possibility of using a regular "gauge" pressure transmitter (where the

"high" port connects to the process vessel and the "low" port is always vented to atmosphere by virtue of a special flange on the instrument) as a vacuum instrument. If a gauge pressure transmitter is given a negative calibration span, any decreasing pressure seen at the "high" port will yield an increasing output signal.

19.5.3 Inferential measurement applications

A very common technique in industrial instrumentation is to calculate the value of a process variable from the values of related variables which are easier to measure[17]. As it so happens, there are a host of variables which one may infer from readings of differential pressure. This makes DP transmitters very versatile devices, not just limited to measuring process variables of pressure and vacuum. This portion of the book will explore some of the more common inferred measurements possible with DP instruments.

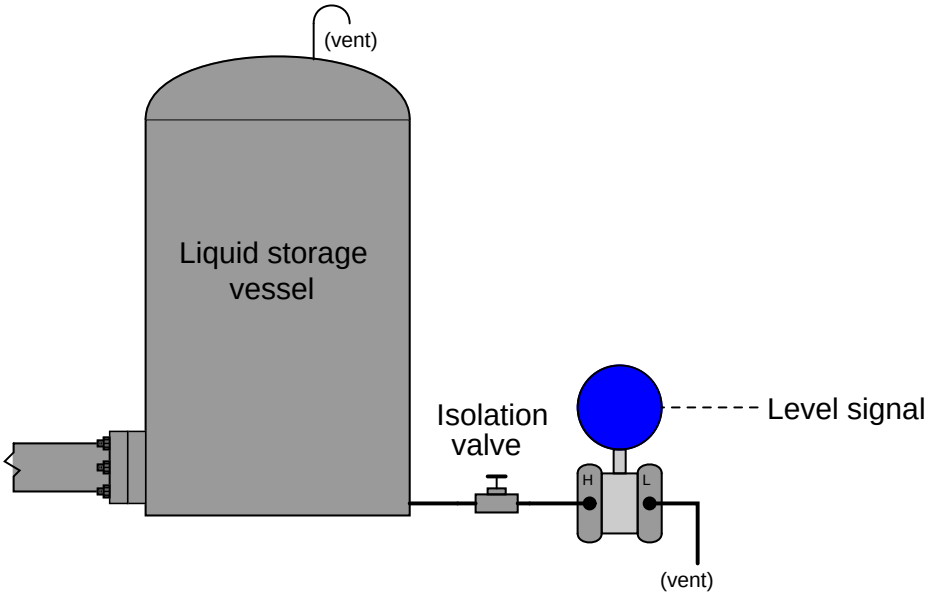
[17] Truth be told, *most* process variables are inferred rather than directly measured. Even pressure, which is being used here to infer measurements such as liquid level and fluid flow, is itself inferred from some other variable inside the DP instrument (e.g. capacitance, strain gauge resistance, resonant frequency)!

Inferring liquid level

Liquids generate pressure proportional to height (depth) due to their weight. The pressure generated by a vertical column of liquid is proportional to the column height (h), and liquid's mass density (ρ), and the acceleration of gravity (g):

$$P = \rho g h$$

Knowing this, we may use a DP transmitter as a liquid level-sensing device if we know the density of the liquid remains fairly constant[18]:

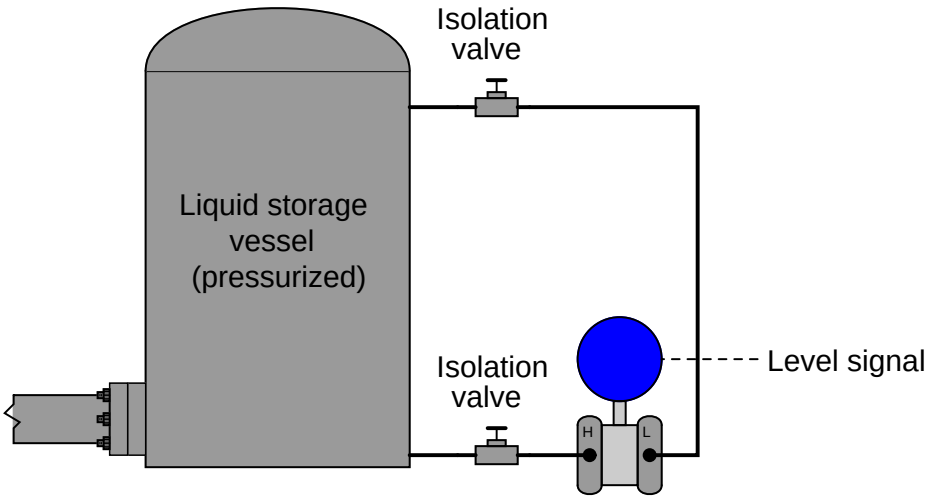


As liquid level in the vessel increases, the amount of hydrostatic pressure applied to the transmitter's "high" port increases in direct proportion. The width of the vessel is irrelevant to the amount of pressure produced – only the liquid *height* (h), *density* (ρ), and Earth's gravity (g) are significant. Thus, the transmitter's increasing signal represents the height of liquid inside the vessel, no matter the size or shape of the vessel:

$$h = \frac{P}{\rho g}$$

[18] We simply assume Earth's gravitational acceleration (g) to be constant as well.

This simple technique works even if the vessel is under pressure from a gas or a vapor (rather than being vented as was the case in the previous example). All we need to do to compensate for this other pressure is to connect the DP transmitter's "low" port to the top of the vessel so it senses nothing but the gas pressure:

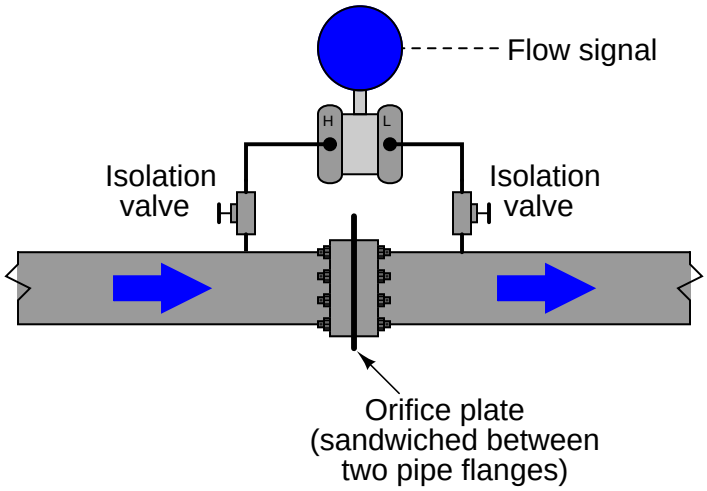


Since the transmitter responds only to differences of pressure between its two sensing ports, and the only cause for a difference of pressure in this application will be pressure generated by the height of a liquid column, the transmitter's signal becomes an exclusive representation of liquid level in the vessel, rejecting potential measurement errors caused by changes in gas pressure within the vessel. Any gas pressure within the vessel will be sensed equally by both ports on the transmitter as a "common-mode" pressure, thus canceling each other and having no effect on the differential pressure measurement. Only changes in liquid level within the vessel will cause the "high" port pressure to change independently of the "low" port pressure, changing the transmitter's output signal.

Inferring gas and liquid flow

Another common inferential measurement using DP transmitters is the measurement of fluid flow through a pipe. Pressure dropped across a constriction in the pipe varies in relation to flow rate (Q) and fluid density (ρ). So long as fluid density remains fairly constant, we may measure pressure drop across a piping constriction and use that measurement to infer flow rate.

The most common form of constriction used for this purpose is called an *orifice plate*, being nothing more than a metal plate with a precisely machined hole in the center. As fluid passes through this hole, its velocity changes, causing a pressure drop to form:



Once again, we see the common-mode rejection abilities of the pressure transmitter used for practical advantage. Since both ports of the transmitter connect to the same process line, static fluid pressure within that line has no effect on the measurement. Only *differences* of pressure between the upstream and downstream sides of the constriction (orifice plate) cause the transmitter to register flow.

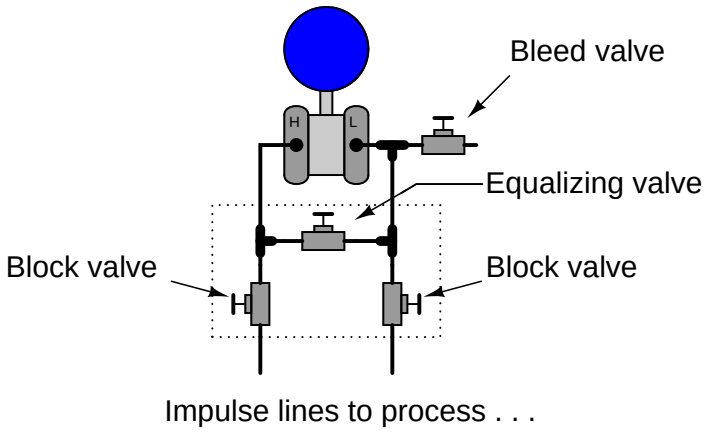
19.6 Pressure sensor accessories

Multiple accessories exist for pressure-sensing devices to function optimally in challenging process environments. Sometimes, we must use special accessories to protect the pressure instrument against hazards of certain process fluids. One such hazard is pressure *pulsation*, for example at the discharge of a piston-type (positive-displacement) high-pressure pump. Pulsating pressure can quickly damage mechanical sensors such as bourdon tubes, either by wear of the mechanism transferring pressure element motion to an indicating needle, and/or fatigue of the metal element itself.

19.6.1 Valve manifolds

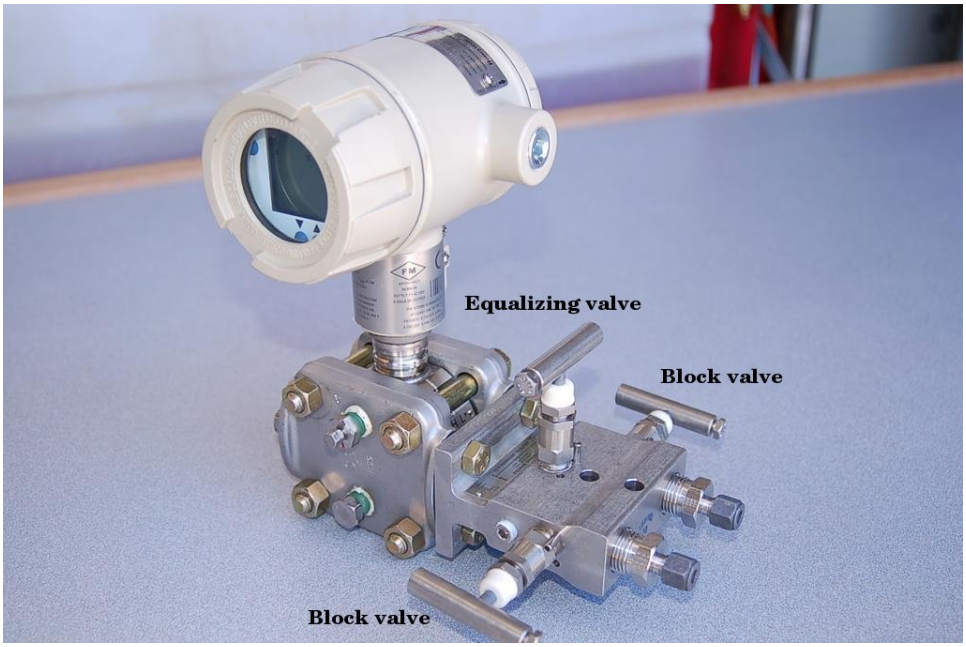
An important accessory to the DP transmitter is the *valve manifold*. This device incorporates manual valves to isolate and equalize pressure from the process to the transmitter, for maintenance and calibration purposes.

The following illustration shows the three valves comprising a three-valve manifold (within the dotted-line box), as well as a fourth valve called a "bleed" valve used to vent trapped fluid pressure to atmosphere:



While this illustration shows the three valves as separate devices, connected together and to the transmitter by tubing, three-valve manifolds are more commonly manufactured as monolithic devices: the three valves cast together into one block of metal, attaching to the pressure transmitter by way of a flanged face with O-ring seals. Bleed valves are most commonly found as separate devices threaded into one or more of the ports on the transmitter's diaphragm chambers.

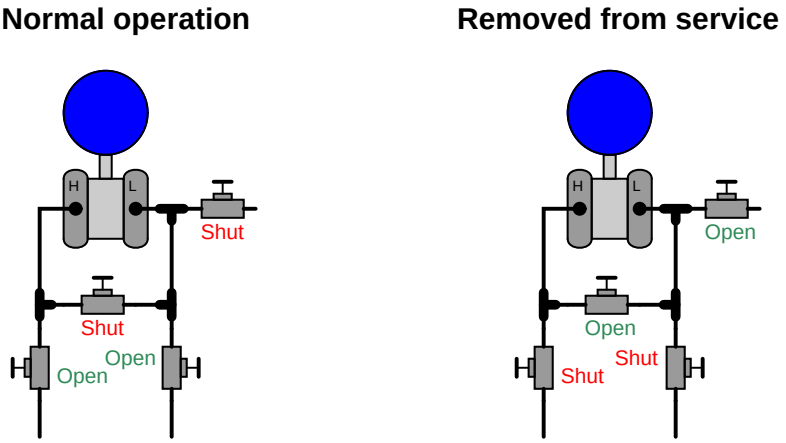
The following photograph shows a three-valve manifold bolted to a Honeywell model ST3000 differential pressure transmitter. A bleed valve fitting may be seen inserted into the upper port on the nearest diaphragm capsule flange:



In normal operation, the two block valves are left open to allow process fluid pressure to reach the transmitter. The equalizing valve is left tightly shut so no fluid can pass between the "high" and "low" pressure sides. To isolate the transmitter from the process for maintenance, one must close the block valves and open the equalizing valve. The best sequence to follow is to first close the high-pressure block valve, then open the equalizing valve, then close the low-pressure block valve. This sequence ensures the transmitter cannot be exposed to a high differential pressure during the isolation procedure, and that the trapped fluid pressure inside the transmitter will be as low as possible prior to "venting" to atmosphere. Finally, the "bleed" valve is opened at the very last step to relieve pent-up fluid pressure within the manifold and transmitter chambers[19]:

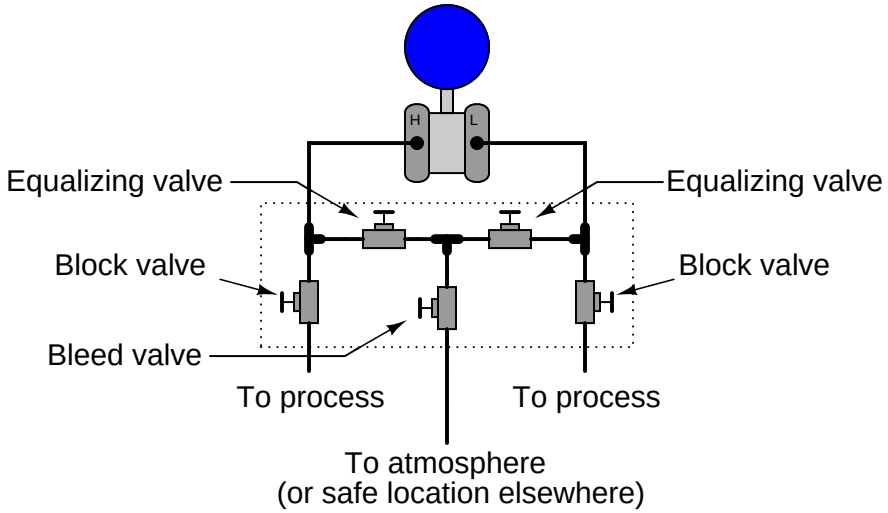
[19] To return the transmitter to live service, simply reverse these steps: close the bleed valve, open the low-pressure block valve, close the equalizing valve, and finally open the high-pressure block valve.

Final valve positions for both states are shown in the following illustrations:

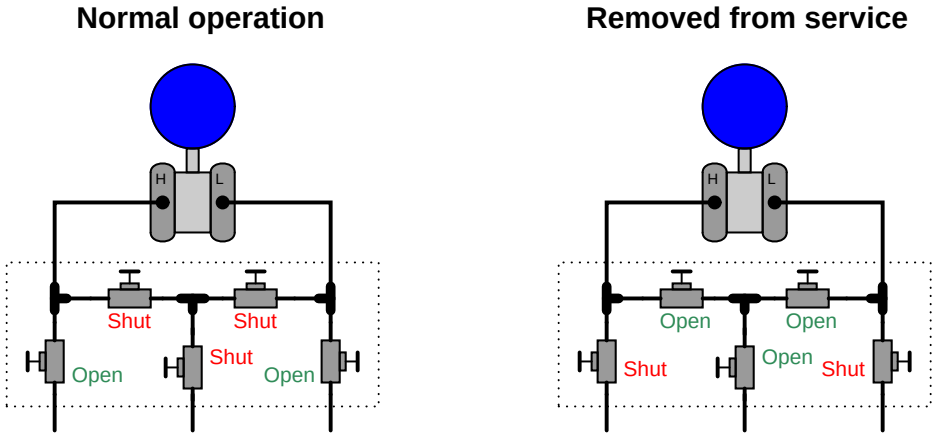


For added safety, shut block valves should be tagged (and possibly locked) so that no unauthorized people will open them up in a state when the transmitter is vented or removed from the manifold. In other words, the same safety procedure of *lock-out/tag-out* (LOTO) common to electrical maintenance work is applicable to isolation valves as well.

A variation on this theme is the *five-valve manifold*, shown in this illustration:



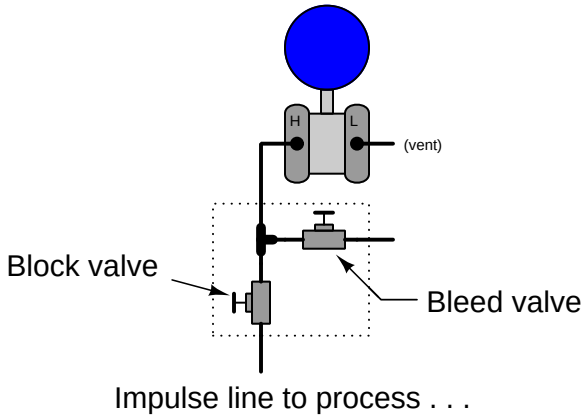
The presence of a built-in bleed valve in the five-valve manifold allows the technician to vent trapped pressure through a tube to some remote location, rather than directly venting at the transmitter. Valve positions for normal operation and maintenance on this manifold are as follows:



It is critically important that the equalizing valve(s) never be open on any transmitter manifold while both block valves are open! Doing so will allow process fluid to flow through the equalizing valve(s) from the high-pressure side of the process to the low-pressure side of the process. If the impulse tubes connecting the manifold to the process are intentionally filled with a *fill fluid* (such as glycerin, to displace process water from entering the impulse tubes; or water in a steam system), this fill fluid will be lost. Also, if the process fluid is dangerously hot or radioactive, a combination of open equalizing and block valves will let that dangerous fluid reach the transmitter and manifold, possibly causing damage or creating a personal hazard. Speaking from personal experience, I once

made this mistake on a DP transmitter connected to a steam system, causing hot steam to flow through the manifold and overheat the equalizing valve so that it seized open and could not be shut again! The only way I was able to stop the flow of hot steam through the manifold was to locate and shut a sliding-gate hand valve between the impulse tube and the process pipe. Fortunately, this cast steel valve was not damaged by the heat and was still able to shut off the flow.

Pressure transmitter valve manifolds also come in single block-and-bleed configurations, for gauge pressure applications. Here, the "low" pressure port of the transmitter is vented to atmosphere, with only the "high" pressure port connected to the impulse line:



The following photograph shows a bank of eight pressure transmitters, seven out of the eight being equipped with a single block-and-bleed manifold. The eighth transmitter (bottom row, second-from left) sports a 5-valve manifold:

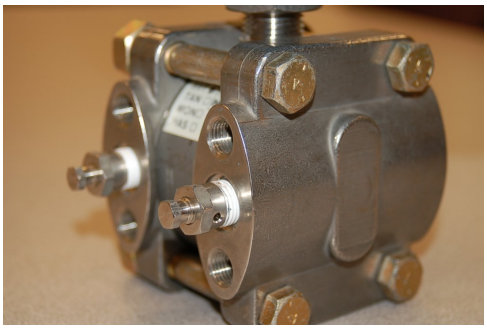


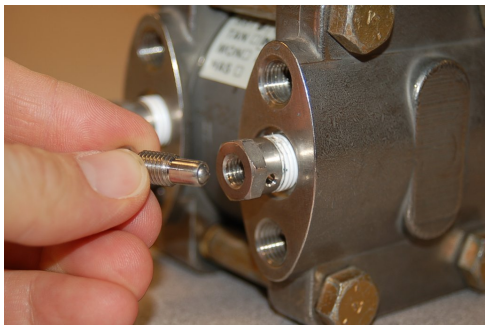
If you look closely at the photograph, you can see the bleed valve fittings installed on all the upper ports. Only the transmitter with the 5-valve manifold has *two* bleed valve fittings because it is the only DP transmitter of the group. The other seven transmitters are all *gauge pressure* units, and so only have one port to bleed.

A good habit to cultivate when operating valve handles on transmitter manifolds is to "back off" the open valves approximately one-quarter turn after opening. This discourages seizing in the full-open position, and also makes it possible for someone to more easily tell the states of the valves by feel: a closed valve will not easily turn (because it is tightened onto its seat) while an open valve is free to turn either direction a bit. Since there should be no flow going through the valves of a transmitter manifold, it is irrelevant whether an open manifold valve is 100% open or 90% open or 80% open, so there is no harm in "backing off" an open valve from the full-open position. It would of course be bad to do this with a closed valve, since any valve plug must be pressed tight into its seat in order to achieve positive shut-off.

19.6.2 Bleed (vent) fittings

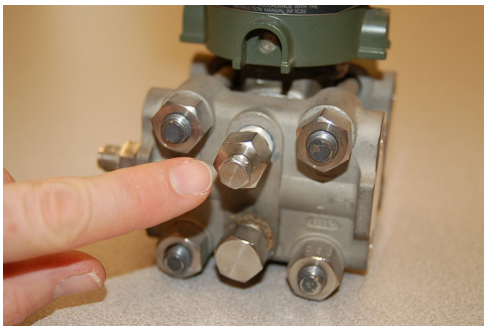
Before removing a pressure transmitter from live service, the technician must "bleed" or "vent" accumulated fluid pressure to atmosphere in order to achieve a *zero energy state* prior to disconnecting the transmitter from the impulse lines. Some valve manifolds provide a bleed valve for doing just this, but many do not[20]. An inexpensive and common accessory for pressure-sensing instruments (especially transmitters) is the *bleed valve fitting* or *vent valve fitting*, installed on the instrument as a discrete device. The most common bleed fitting is equipped with 1/4 inch male NPT pipe threads, for installation into one of the 1/4 inch female NPT pipe ports typically provided on pressure transmitter flanges. The bleed fitting is operated with a small wrench, loosening a ball-tipped plug off its seat to allow process fluid to escape through a small vent hole in the side of the fitting. The following photographs show close-up views of a bleed fitting both assembled (left) and with the plug fully extracted from the fitting (right). The bleed hole may be clearly seen in both photographs:

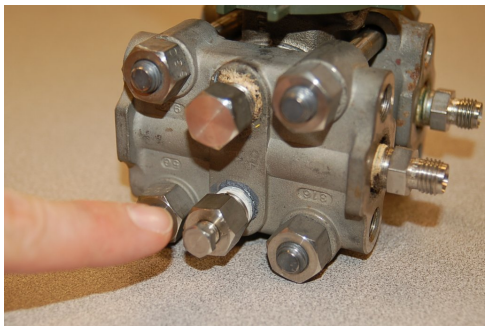




When installed directly on the flanges of a pressure instrument, these bleed valves may be used to bleed unwanted fluids from the pressure chambers, for example bleeding air bubbles from an instrument intended to sense water pressure, or bleeding condensed water out of an instrument intended to sense compressed air pressure.

The following photographs show bleed fittings installed two different ways on the side of a pressure transmitter flange, one way to bleed gas out of a liquid process (located on top) and the other way to bleed liquid out of a gas process (located on bottom):





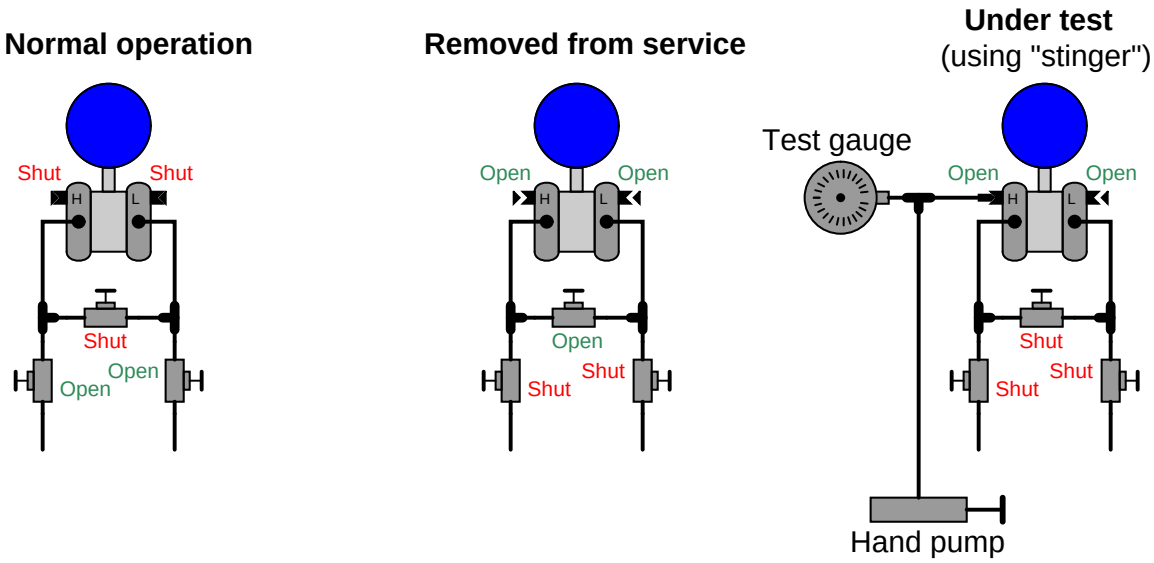
[20] The standard 3-valve manifold, for instance, does not provide a bleed valve – only block and equalizing valves.

With the bleed plug completely removed, the open bleed fitting provides a port through which one may apply air pressure for testing the response of the pressure transmitter. A special test fitting called a *bleed port adapter* or *DP transmitter calibration fitting* – colloquially known as a *stinger* – threads into the opened bleed fitting. A photograph of a bleed port adapter is shown here:



This special fitting allows a compression-style tube to be temporarily connected to the opened bleed port, which then allows the connection of an air pump and test pressure gauge to the transmitter. Thus, the bleed port adapter enables a technician to conveniently apply test pressures to the DP transmitter without having to loosen any of the instrument manifold bolts, tapered thread pipe connections, or impulse tube compression fittings.

When performing field checks of pressure transmitters, bleed port adapters substantially reduce the amount of time necessary to field-test pressure instruments. The following sequence of illustrations show how a bleed port adapter may be used in conjunction with a three-valve instrument manifold to isolate a DP transmitter from a process and then subject it to test pressures from a hand pump:

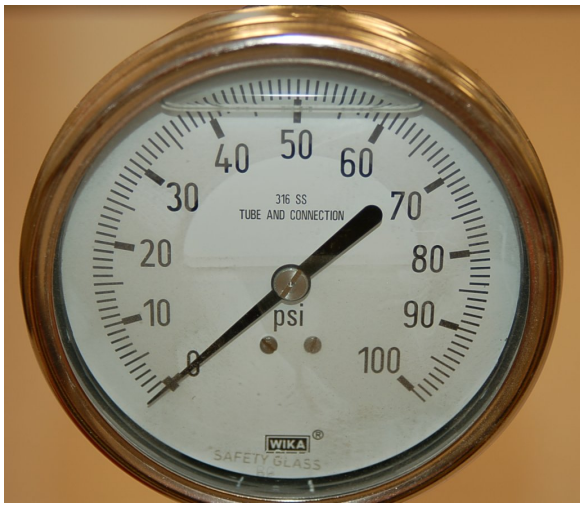


Note how both bleed vents must be opened, and the equalizing valve shut, in order to apply a test pressure to the DP transmitter. Although it is possible to safely bleed pressure from both sides of a DP instrument through just one bleed fitting (through the open equalizing valve), both bleeds must be open in order to perform a pressure test. If the "L" side bleed fitting is left in the shut position, some pressure may be trapped there as pressure is applied to the "H" side by the hand pump. If the equalizing valve is left open, no difference of pressure will be allowed to form across the DP instrument.

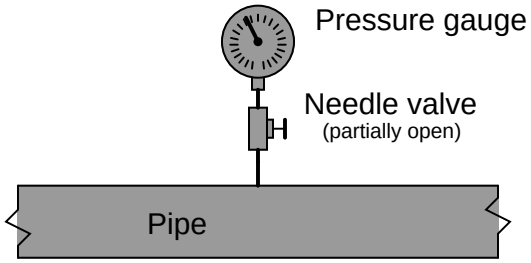
19.6.3 Pressure pulsation damping

A simple way to mitigate the effects of pulsation on a pressure gauge is to fill the inside of the gauge with a viscous liquid such as glycerin or oil. The inherent friction of this fill liquid has a "shock-absorber" quality which damps the gauge mechanism's oscillatory motion and helps protect against damage from pulsations or from external vibration. This method is ineffectual for high-amplitude pulsations, though.

An oil-filled pressure gauge may be seen in the following photograph. Note the air bubble near the top of the gauge face, which is the only visual indication of an oil filling:

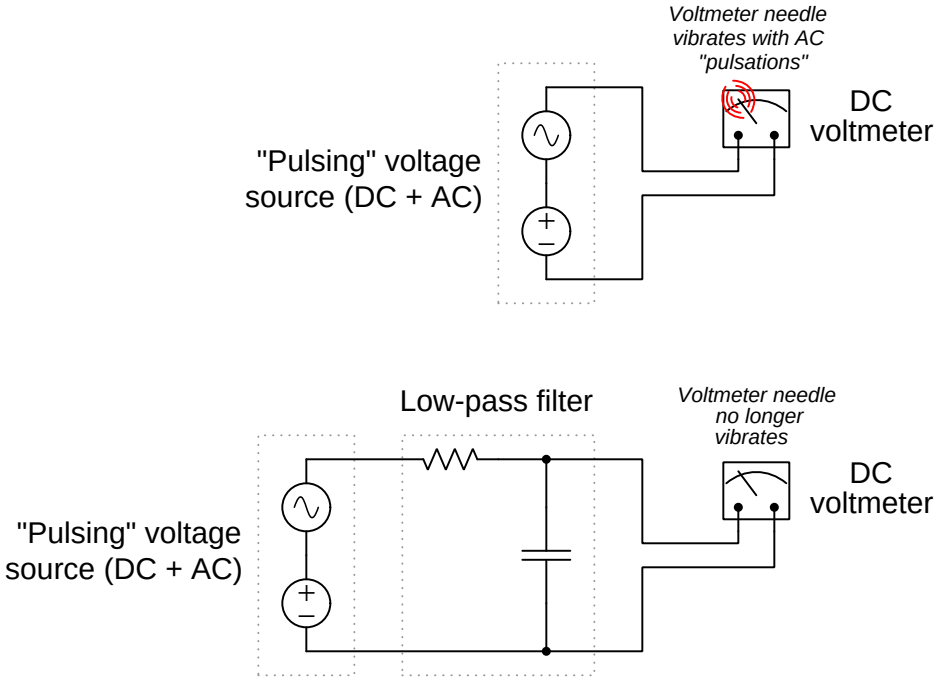


A more sophisticated method for damping pulsations seen by a pressure instrument is called a *snubber*, and it consists of a fluid restriction placed between with the pressure sensor and the process. The simplest example of a snubber is a simple *needle valve* (an adjustable valve designed for low flow rates) placed in a mid-open position, restricting fluid flow in and out of a pressure gauge:



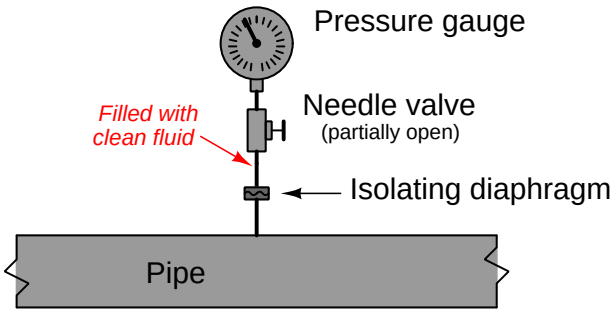
At first, the placement of a throttling valve between the process and a pressure-measuring instrument seems rather strange, because there should not be any continuous flow in or out of the gauge for such a valve to throttle! However, a *pulsing* pressure causes a small amount of *alternating* flow in and out of the pressure instrument, owing to the expansion and contraction of the mechanical pressure-sensing element (bellows, diaphragm, or bourdon tube). The needle valve provides a restriction for this flow which, when combined with the fluid capacitance of the pressure

instrument, combine to form a low-pass filter of sorts. By impeding the flow of fluid in and out of the pressure instrument, that instrument is prevented from "seeing" the high and low peaks of the pulsating pressure. Instead, the instrument registers a much steadier pressure over time. An electrical analogy for a pressure snubber is an RC low-pass filter circuit "damping" voltage pulsations from reaching a DC voltmeter:



One potential problem with the needle valve solution is that the small orifice inside the needle valve may plug up over time with debris from dirty process fluid. This, of course, would be bad because plugging will cause the pressure instrument to respond too slowly, or not at all if the plugging is complete.

A solution to this problem is to fill the pressure sensor mechanism with a clean liquid (called a *fill fluid*) and use that fill fluid to transfer pressure from the process fluid to the pressure-sensing element using a slack diaphragm or some other membrane separating the process fluid from the fill fluid:



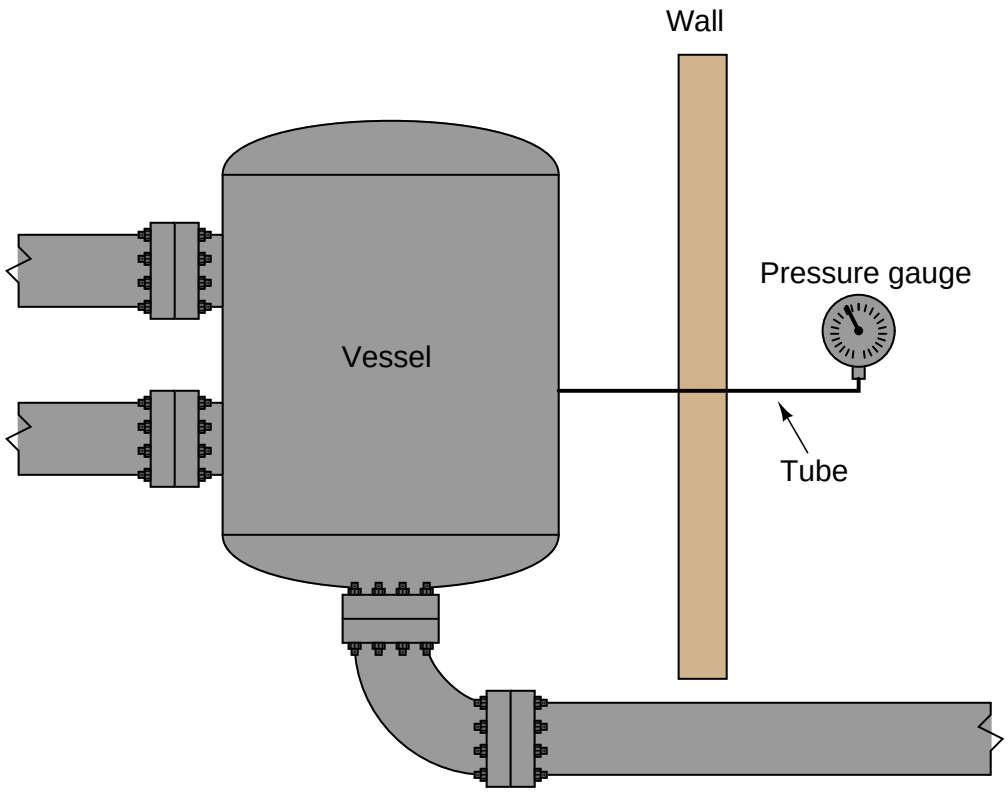
It should be noted that most pressure snubbers utilize a fixed-geometry orifice rather than an adjustable needle valve to dampen pressure pulsations seen at the pressure gauge.

In order for the fill fluid and isolating diaphragm to work effectively, there cannot be any gas bubbles in the fill fluid – it must be a "solid" hydraulic system from the diaphragm to the sensing element. Gas bubbles present in the filled system would make that volume compressible, which means the isolating diaphragm would have to move more than necessary to transfer pressure to the instrument's sensing element. This would mean motion at the isolating diaphragm caused by process pressure changes would be "lost" and not fully transferred to the instrument's sensing element, thereby introducing a pressure measurement error[21]. For this reason, isolating diaphragm systems for pressure instruments are usually "packed" with fill fluid at the point and time of manufacture, then sealed in such a way that they cannot be opened for any form of maintenance. Consequently, any fill fluid leak in such a system immediately ruins it.

[21] This concept will be immediately familiar to anyone who has ever had to "bleed" air bubbles out of an automobile brake system. With air bubbles in the system, the brake pedal has a "spongy" feel when depressed, and much pedal motion is required to achieve adequate braking force. After bleeding all air out of the brake fluid tubes, the pedal motion feels much more "solid" than before, with minimal motion required to achieve adequate braking force. Imagine the brake pedal being the isolating diaphragm, and the brake pads being the pressure sensing element inside the instrument. If enough gas bubbles exist in the tubes, the brake pedal might stop against the floor when fully pressed, preventing full force from ever reaching the brake pads. Likewise, if the isolating diaphragm hits a hard motion limit due to gas bubbles in the fill fluid, the sensing element will not experience full process pressure.

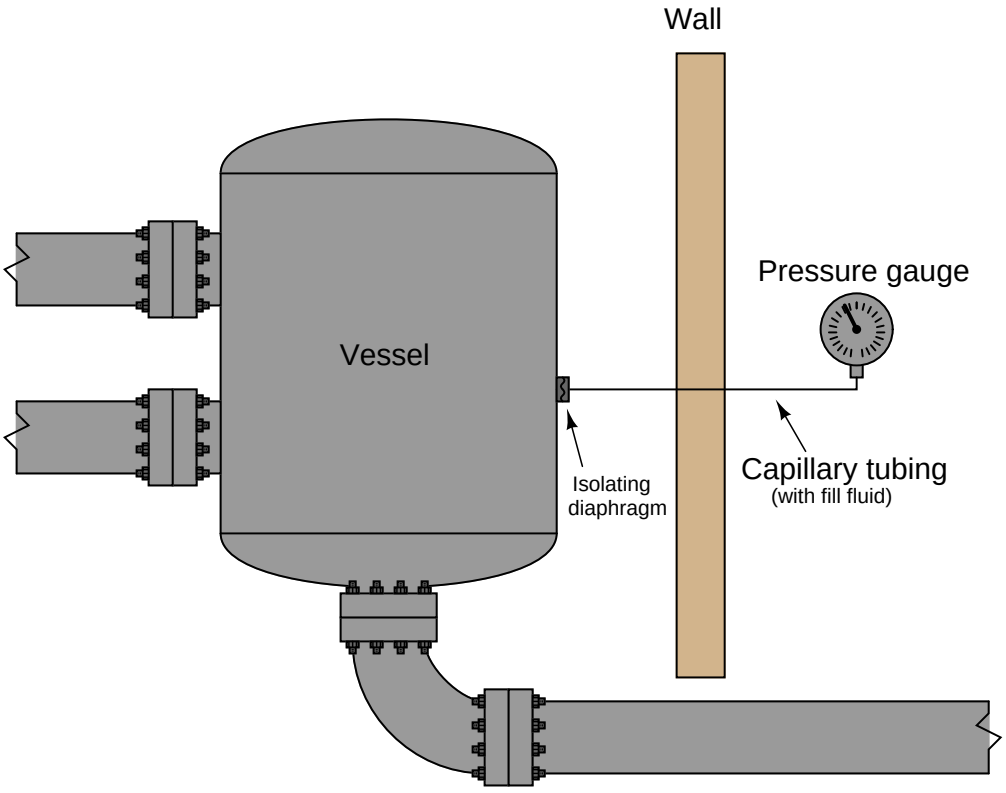
19.6.4 Remote and chemical seals

Isolating diaphragms have merit even in scenarios where pressure pulsations are not a problem. Consider the case of a food-processing system where we must remotely measure pressure inside a mixing vessel:



The presence of the tube connecting the vessel to the pressure gauge poses a hygiene problem. Stagnant process fluid (in this case, some liquid food product) inside the tube will encourage microbial growth, which will eventually contaminate the vessel no matter how well or how often the vessel is cleaned. Even automated *Clean-In-Place* and *Steam-In-Place* (*CIP* and *SIP*, respectively) protocols where the vessel is chemically purged between batches cannot prevent this problem because the cleaning agents never purge the entire length of the tubing (ultimately, to the bourdon tube or other sensing element inside the gauge).

A solution to this problem is to install an *isolating diaphragm* at the vessel, and a liquid-filled *capillary tube* to transfer sensed pressure to the instrument. Process pressure presses against this diaphragm, which in turn transfers[22] pressure to the "fill fluid" inside the capillary tube. This sealed fill fluid then presses against the instrument's sensing element (diaphragm, bourdon tube, bellows, etc.). Process (food) liquid cannot enter this sealed tube, and the isolating diaphragm will be cleaned with every CIP cycle. Thus, we completely eliminate the problem of microbial contamination:



Such systems are often referred to as *remote seals*, and they are available on a number of different pressure instruments including gauges, transmitters, and switches. If the purpose of an isolating diaphragm and fill fluid is to protect the sensitive instrument from corrosive or otherwise harsh chemicals, it is often referred to as a *chemical seal*.

[22] So long as the isolating diaphragm is "slack" (i.e. has no appreciable tautness or resistance to movement), the pressure of the fill fluid inside the capillary tube *will* be equal to the pressure of whatever fluid is within the process vessel. If any pressure imbalance were to develop between the process and fill fluids, the isolating diaphragm would immediately shift position away from the higher-pressure fluid and toward the lower-pressure fluid until equal pressures were re-established. In real practice, isolating diaphragms do indeed have some stiffness opposing motion, and therefore do not *perfectly* transfer pressure from the process fluid to the fill fluid. However, this pressure difference is usually negligible.

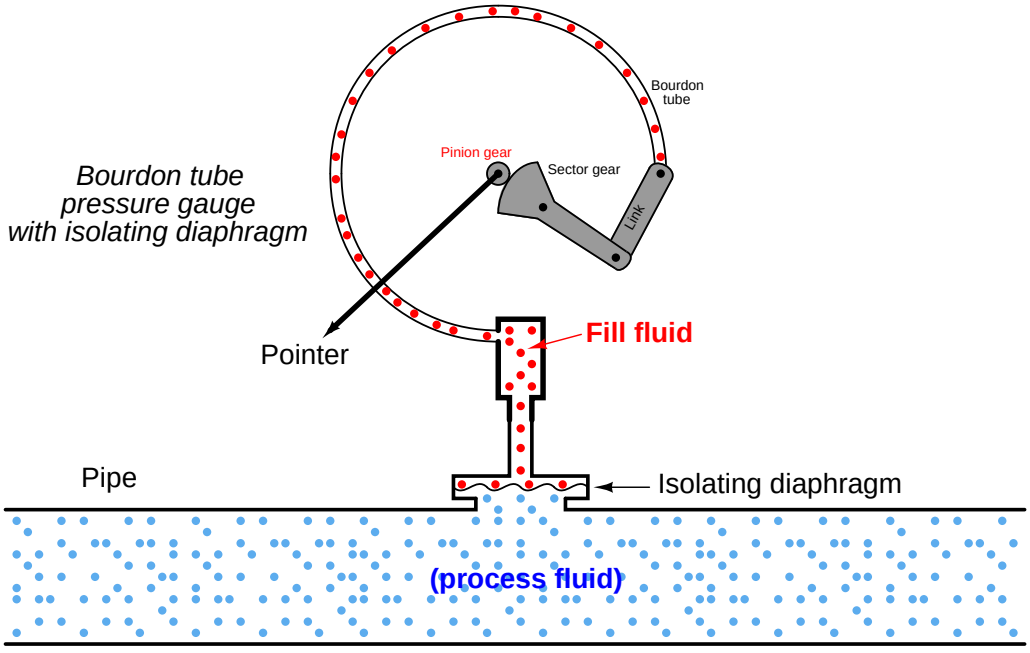
The following photograph shows a pressure gauge equipped with a chemical seal diaphragm. Note that the chemical seal on this particular gauge is close-coupled to the gauge, since the only goal here is protection of the gauge from harsh process fluids, not the ability to remotely mount the gauge:



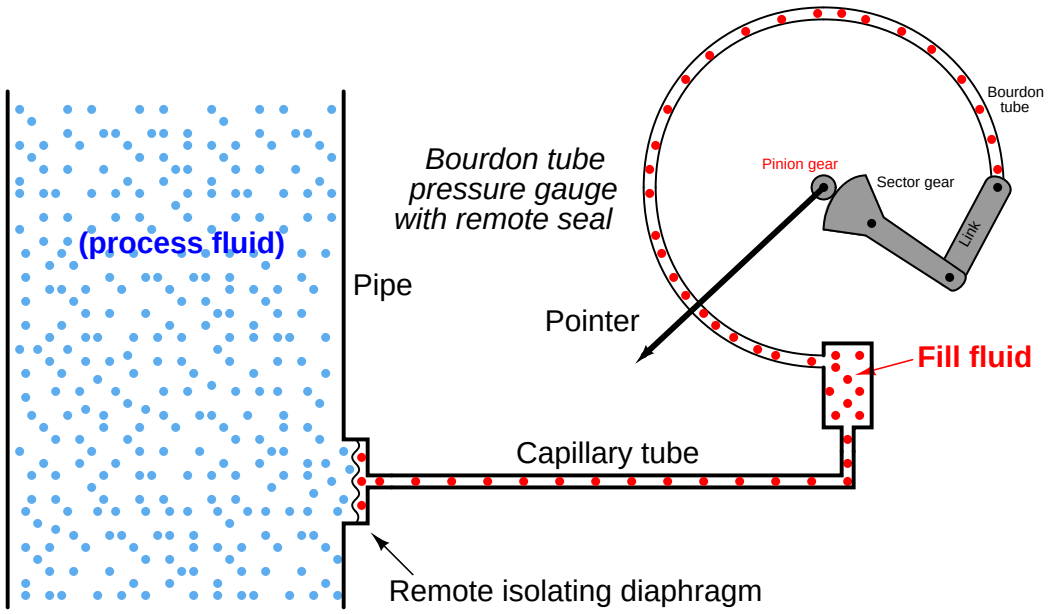
A view facing the bottom of the flange reveals the thin metal isolating diaphragm keeping process fluid from entering the gauge mechanism. Only inert fill fluid occupies the space between this diaphragm and the gauge's bourdon tube:



The following illustration shows how the fill fluid transfers process fluid pressure to the gauge's bourdon tube element while isolating that bourdon tube from the process fluid (shown here inside a pipe):



The only difference between this chemical-seal gauge and a remote-seal gauge is the small-diameter *capillary tubing* used to connect the gauge to a remote diaphragm. An illustration showing the internals of a remote seal system appears here:



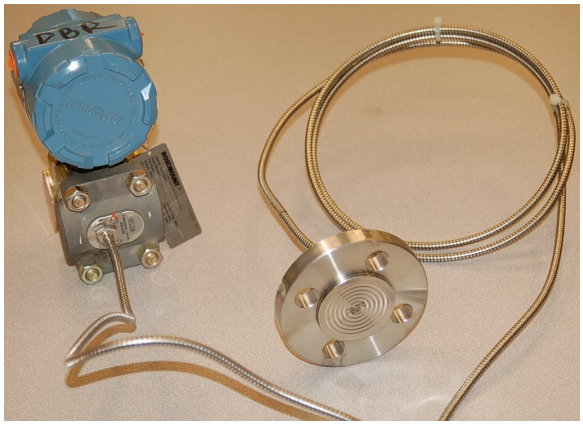
Direct-reading gauges are not the only type of pressure instruments benefiting from remote seals. Pressure switches and pressure transmitters may also employ remote seals for the same reasons: protection of the transmitter sensor from harsh process fluid, elimination of impulse tube clogging problems, and/or the prevention of "dead-end" tube lengths where organic process fluid would stagnate and harbor microbial growths. In this photograph, you see three pressure sensing devices (gauge, transmitter, and switch, from top to bottom), each one with its own remote seal to sense fluid pressure in a large pipe:



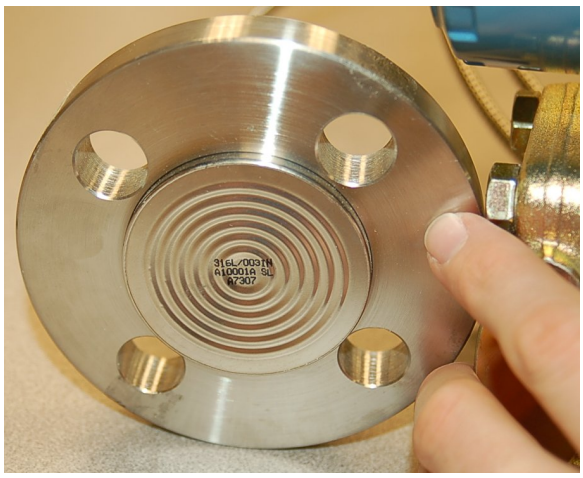
The pressure transmitter (a Yokogawa unit) is the only instrument shown here using a capillary tube. The other two instruments use short lengths of rigid pipe. The capillary is visible in this photograph as a coiled tube covered in black plastic. It is actually a very small-diameter metal tube enclosed in a spiral-metal protective sheath which is in turn covered by black plastic.

This particular application happens to be in wastewater treatment, where sludge has a tendency to clog instrument impulse lines connected to the main piping. With remote seals in place, that problem is completely eliminated.

The following photograph shows a Rosemount model 1151 electronic pressure transmitter equipped with a remote sealing diaphragm. Here we may see the coiled metal ("armor") sheath protecting the capillary tube from damage:



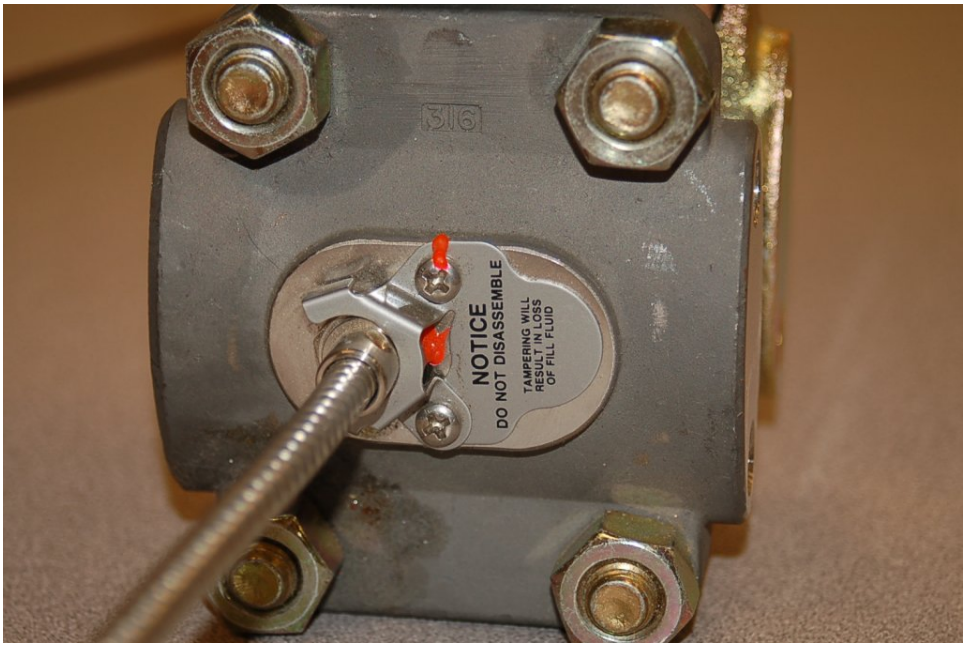
A close-up view of the sealing diaphragm shows its corrugated design, allowing the metal to easily flex and transfer pressure to the fill fluid within the capillary tubing[23]:



As with the isolating diaphragms of the pressure-sensing capsule, these remote diaphragms need only transfer process fluid pressure to the fill fluid and (ultimately) to the taut sensing element inside the instrument. Therefore, these diaphragms perform their function best if they are designed to easily flex. This allows the sensing diaphragm to provide the vast majority of the opposing force to the fluid pressure, as though it were the only spring element in the fluid system.

[23] Like all instrument diaphragms, this one is sensitive to damage from contact with sharp objects. If the diaphragm ever becomes nicked, dented, or creased, it will tend to exhibit hysteresis in its motion, causing calibration errors for the instrument. For this reason, isolating diaphragms are often protected from contact by a plastic plug when the instrument is shipped from the manufacturer. This plug must be removed from the instrument before placing it into service.

The connection point between the capillary tube and the transmitter's sensor capsule is labeled with a warning never to disassemble, since doing so would allow air to enter the filled system (or fill fluid to escape from the system) and thereby ruin its accuracy:

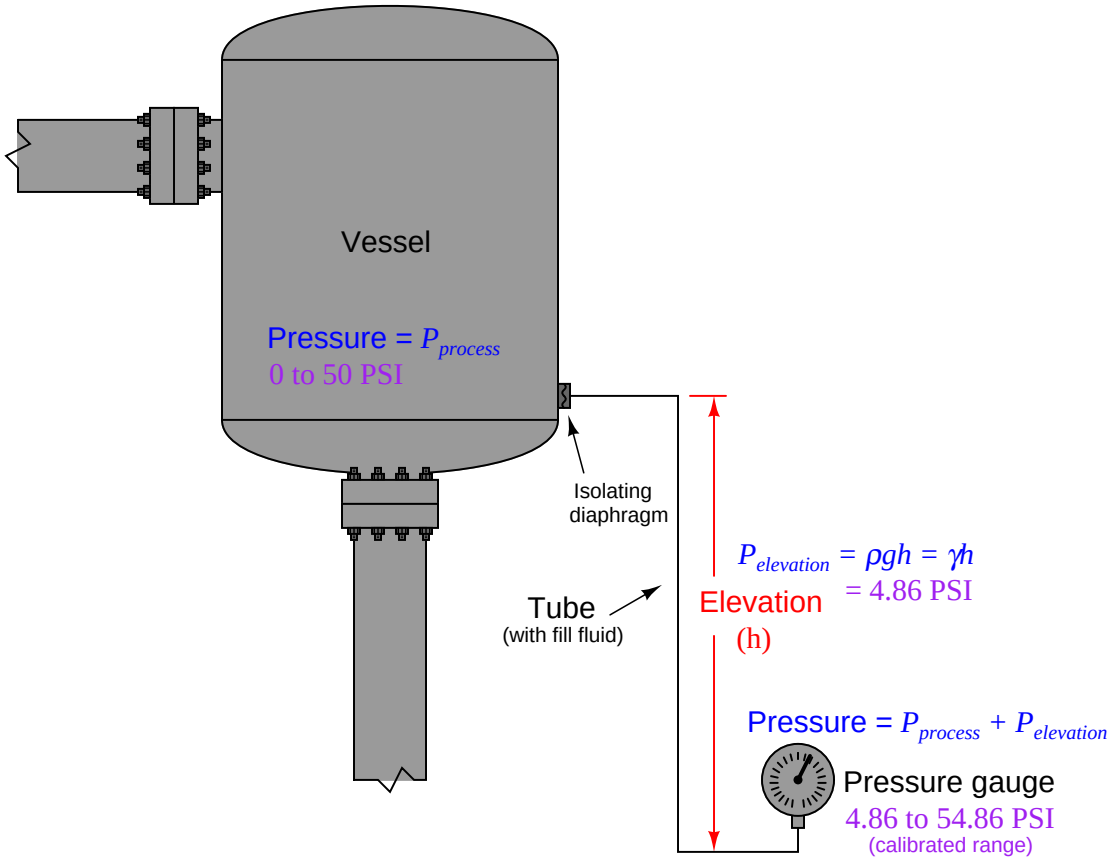


In order for a remote seal system to work, the hydraulic "connection" between the sealing diaphragm and the pressure-sensing element must be completely gas-free so there will be a "solid" transfer of motion from one end to the other[24]. The presence of gas bubbles in the filled system will cause some of the isolating diaphragms' motion to be "lost" rather than be fully transferred to the instrument's pressure-sensing element. For this reason, the capillary system must remain perfectly sealed at all times! Breaching this seal, even for just a brief moment, will ruin the system.

A protective measure visible in this photograph is the orange compound painted on the screw head and on the capillary tube connector. This is simply a visual indication that the factory seal is still intact, since any motion of the screw or of the tube connector would crack the brittle orange compound and betray the breach.

[24] Anyone familiar with "bleeding" air bubbles out of automotive hydraulic brake systems will understand this concept. In order for the pedal-operated hydraulic brakes in an automobile to function as designed, the hydraulic system must be gas-free. Incompressible liquid transfers pressure without loss of motion, whereas compressible gas bubbles will "give" in to pressure and result in lost brake pad motion for any given brake pedal motion. Thus, an hydraulic brake system with air bubbles in it will have a "spongy" feel at the brake pedal, and may not give full braking force when needed.

A potential problem with using remote diaphragms is the hydrostatic pressure generated by the fill fluid if the pressure instrument is located far away (vertically) from the process connection point. For example, a pressure gauge located far below the vessel it connects to will register a *greater* pressure than what is actually inside the vessel, because the vessel's pressure adds to the hydrostatic pressure caused by the liquid in the tubing:



This pressure may be calculated by the formula $P = \rho gh$ or $P = \gamma h$ where ρ is the mass density of the fill liquid or γ is the weight density of the fill liquid. For example, a 12 foot capillary tube height filled with a fill liquid having a weight density of 58.3 lb/ft^3 will generate an elevation pressure of almost 700 lb/ft^2, or 4.86 PSI. If the pressure instrument is located below the process connection point, this 4.86 PSI offset must be incorporated into the instrument's calibration range. If we desire this pressure instrument to accurately measure a process pressure range of 0 to 50 PSI, we would have to calibrate it for an actual range of 4.86 to 54.86 PSI.

The reverse problem exists where the pressure instrument is located *higher* than the process connection: here the instrument will register a *lower* pressure than what is actually inside the vessel, offset by the amount predicted by the hydrostatic pressure formulae ρgh or γh.

In all fairness, this problem is not limited to remote seal systems – even non-isolated systems

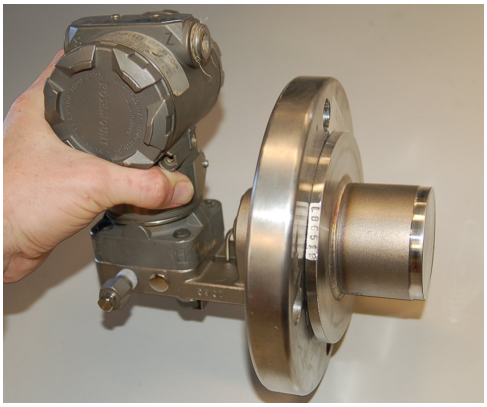
where the tubing is filled with process liquid will exhibit this offset error. However, in filled-capillary systems a vertical offset is *guaranteed* to produce a pressure offset because fill fluids are always liquid, and liquids generate pressure in direct proportion to the vertical height of the liquid column and its density. Many common remote seal fill fluids have *specific gravity* ratings greater than 1 (i.e. they are denser than water) which means the offset error resulting from hydrostatic pressure will be even greater than that of a water-filled tube.

A similar problem unique to isolated-fill pressure instruments is measurement error caused by temperature extremes. Suppose the liquid-filled capillary tube of a remote seal pressure instrument is placed too near a hot steam pipe, furnace, or some other source of high temperature. The expansion of the fill fluid may cause the isolation diaphragm to extend to the point where it begins to tense and add a pressure to the fill fluid above and beyond that of the process fluid. Cold temperatures may wreak havoc with filled capillary tubes as well, if the fill fluid congeals or even freezes such that it no longer flows as it should. Fill fluid expansion/contraction effects may be mitigated by keeping the volume of the fill fluid to a minimum, which is why capillary (small-diameter) tubes are used to connect remote seals with instruments.

Another problem with remote seal pressure instruments is a time delay caused by the viscosity of the fill fluid as it moves through the small-diameter capillary tubes. This makes the pressure instrument slow to respond to changes in process fluid pressure, as though the pressure instrument were equipped with an (undesired) pressure snubber. While minimal capillary tube diameter reduces the effects of temperature changes, it increases the effect of time delay.

Proper mounting of the instrument and proper selection of the fill fluid[25] will help to avoid such problems. All in all, the potential for trouble with remote- and chemical-seal pressure instruments is greatly offset by their benefits in the right applications.

Some diaphragm-sealed pressure transmitters are equipped with close-coupled seals rather than remote seals to minimize hydrostatic, temperature, and time delay effects caused by fill fluid inside a capillary tube. A Rosemount extended-diaphragm pressure transmitter appears in the left-hand photograph, while a Yokogawa transmitter of the same basic design is shown installed in a working process in the right-hand photograph:

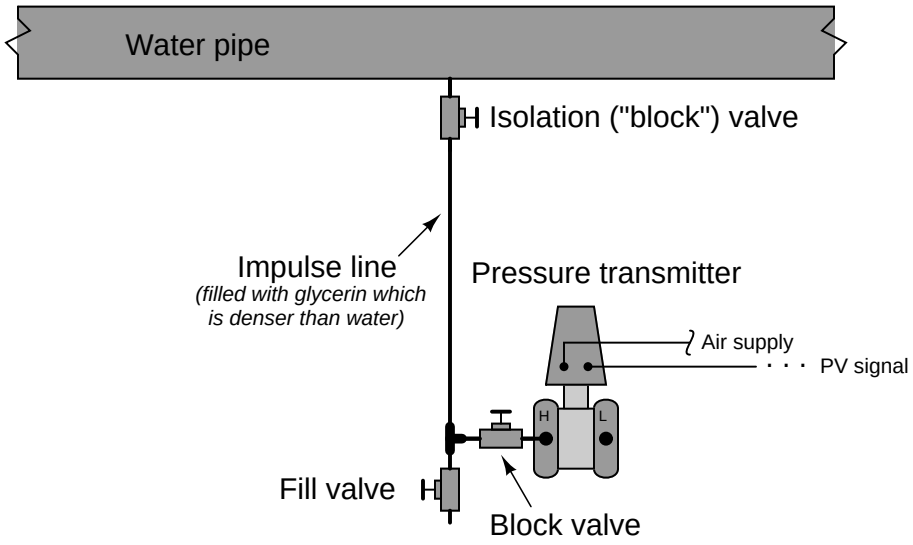




[25] Most pressure instrument manufacturers offer a range of fill fluids for different applications. Not only is temperature a consideration in the selection of the right fill fluid, but also potential contamination of or reaction with the process if the isolating diaphragm ever suffers a leak!

19.6.5 Filled impulse lines

An alternate method for isolating a pressure-sensing instrument from direct contact with process fluid is to *fill* the impulse lines with a harmless fluid, which in turn directly contacts the process fluid. Filling impulse tubes with a static fluid works when gravity is able to keep the fill fluid in place, such as in this example of a pressure transmitter connected to a water pipe by a glycerin-filled impulse line:



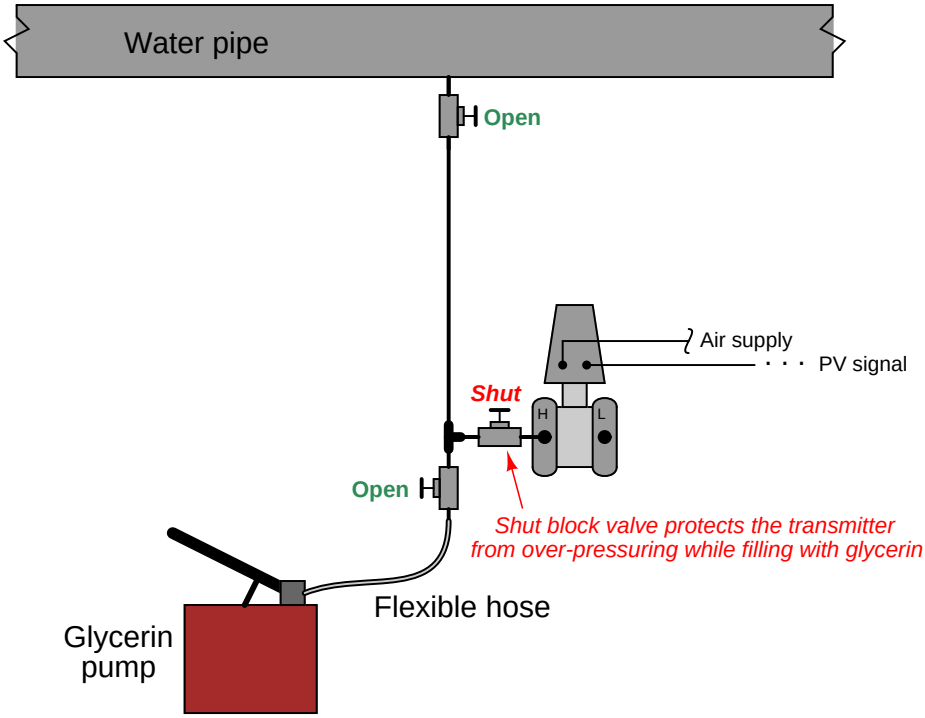
A reason someone might do this is for freeze protection, since glycerin freezes at a lower temperature than water. If the impulse line were filled with process water, it might freeze solid in cold weather conditions (the water in the pipe cannot freeze so long as it is forced to flow). The greater density of glycerin keeps it placed in the impulse line, below the process water line. A fill valve is provided near the transmitter so a technician may re-fill the impulse line with glycerin (using a hand pump) if ever needed.

As with a remote diaphragm, a filled impulse line will generate its own pressure proportional to the height difference between the point of process connection and the pressure-sensing element. If the height difference is substantial, the pressure offset resulting from this difference in elevation will require compensation by means of an intentional "zero shift" of the pressure instrument when it is calibrated.

With no isolating diaphragm to separate process fluid from the fill fluid, it is critical that the fill fluid be compatible[26] with the process fluid. Not only does this imply a total lack of chemical reactivity between the two fluids, but it also means the two fluids should not be readily *miscible* (capable of mixing in any proportion).

[26] Truth be told, this is a requirement for all pressure transmitter fill fluids even when isolating diaphragms are in place to prevent mixing of process and fill fluids, because no diaphragm is 100% guaranteed to seal forever. This means *every* pressure transmitter must be chosen for the application in mind, since modern DP transmitters all use fill fluid in their internal sensors, whether or not the impulse lines are also filled with a non-reactive fluid.

An important consideration in filled-line systems is how to refill the impulse line(s) without damaging the pressure instrument. Hand-operated pumps are commonly used to refill impulse lines, but such pumps are often capable of generating greater fluid pressures than the instrument can safely withstand. If we were to connect a glycerin pump to the filled system pictured previously, it would be advisable to shut the transmitter's block valve to ensure we did not accidently over-pressure the transmitter. This is especially true if the impulse line happens to become plugged with debris, and substantial glycerin pressure from the hand pump is required to dislodge the blockage:

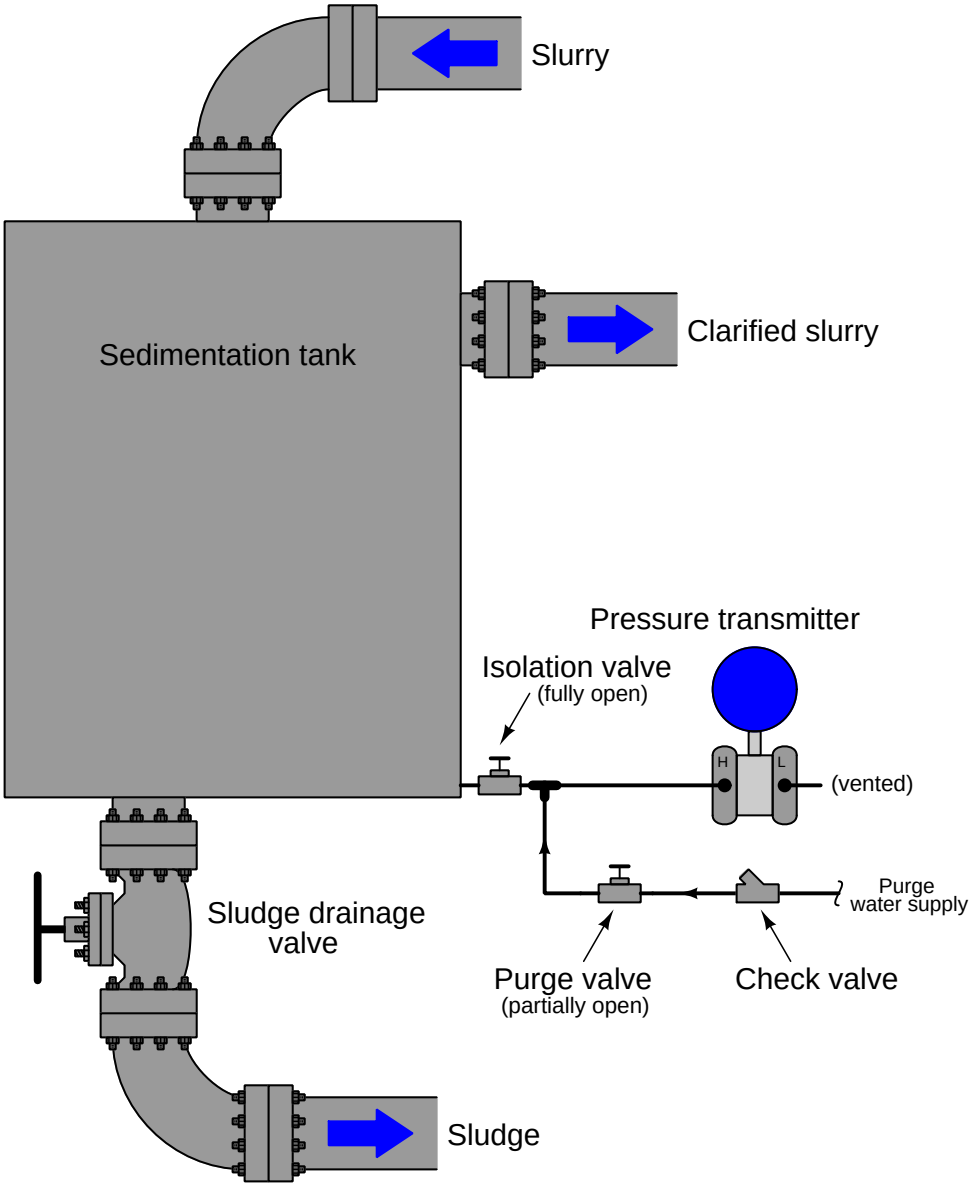


In fact, the issue of impulse tube plugging is another reason to consider filled-line connections between pressure instruments and process lines or vessels. If ever a plug develops in the line, re-pumping the lines with fresh fill fluid is a practical way to clear the plug without disassembling any part of the system.

For processes where impulse line plugging is a chronic problem, another solution exists called *purging* impulse lines, discussed in the next section.

19.6.6 Purged impulse lines

Another method for isolating a pressure instrument from direct contact with process fluid, particularly when the impulse line is prone to plugging, is *purging* the line with a continuous flow of clean fluid. Consider this example, where pressure is measured at the bottom of a sedimentation vessel:



In this system, a continuous flow of clean water enters through a pressure-dropping "purge valve" and flows through the impulse line, keeping it clear of sediment while still allowing the pressure instrument to sense pressure at the bottom of the vessel. A *check valve* guards against reverse flow through the purge line, in case process fluid pressure ever exceeds purge supply pressure. The continuous water purge maintains clean impulse tubing, and ensures the pressure transmitter never contacts process fluid directly.

A very important element of any purge system is a *restriction* between the purge supply and the connection with the process and pressure-sensing device. It is important that the pressure-sensing instrument senses the pressure of the process fluid and not the (higher) pressure of the purge fluid supply. In the example shown, the purge valve fulfills the role of this restriction, which is why it must be left in a partially-open condition, rather than fully-open.

If this purge restriction is not restrictive enough, the purge fluid flow rate will be too great, resulting in a dynamic pressure drop developed across the length of the impulse line. This pressure drop will add to the pressure of the process fluid inside the vessel, creating a positive pressure measurement error at the instrument (i.e. the instrument will register *more* pressure than there actually is in the vessel). The purge restriction should be set to allow just enough purge fluid flow to guard against plugging, and no more.

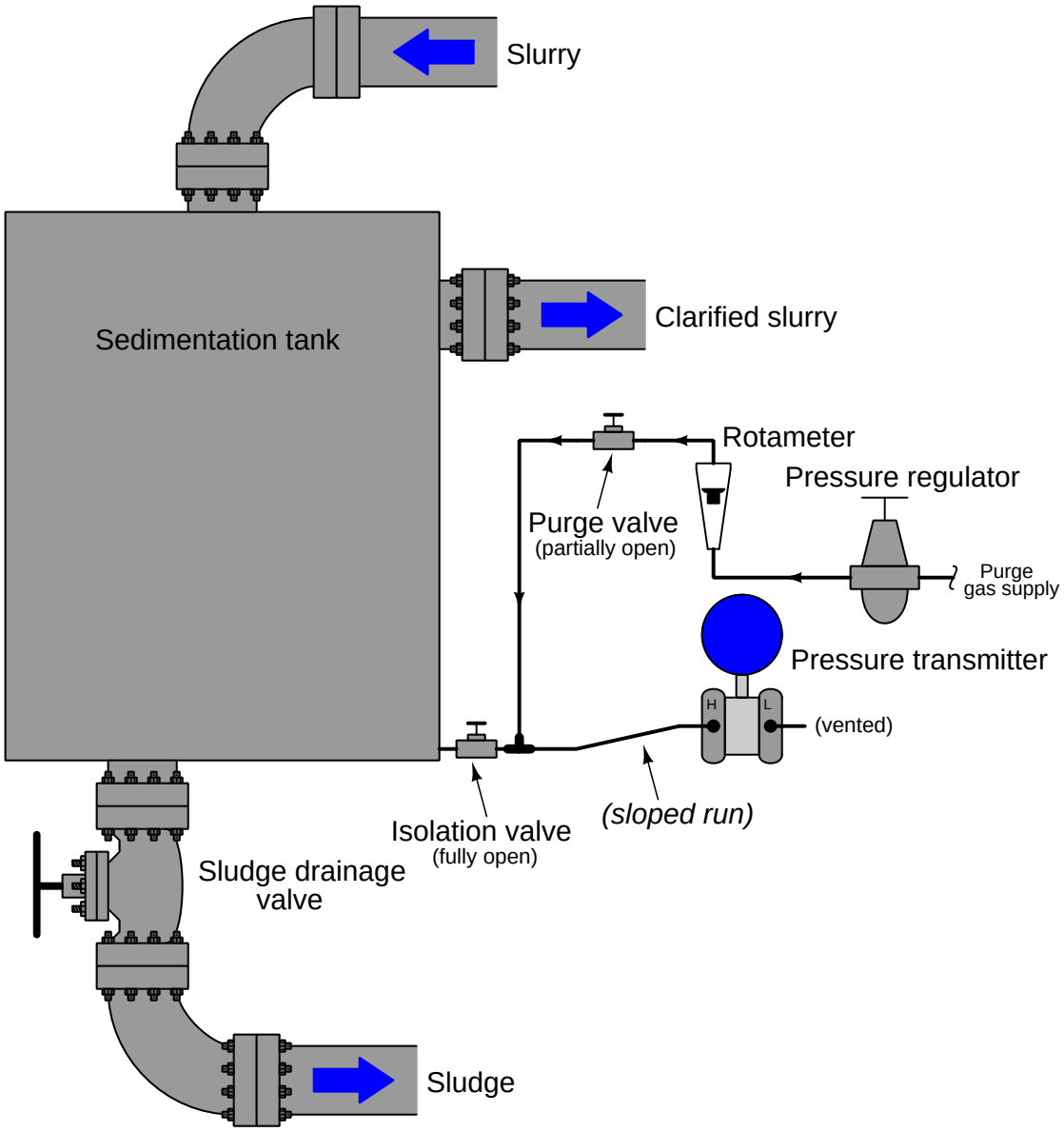
Purged systems are very useful, but a few requirements are necessary in order to ensure accurate and reliable operation:

- The purge fluid supply must be reliable: if the flow stops for any reason, the impulse line may plug!
- The purge fluid supply pressure must exceed the process pressure at all times, or else process fluid will flow backward *into* the impulse line!
- The purge fluid flow must be maintained at a low rate, to avoid pressure measurement errors.
- The purge fluid should be introduced into the impulse line as close to the process connection as possible, to minimize errors due to the purge flow rate through long lengths of tubing.
- The purge fluid must not adversely react with the process.
- The purge fluid must not contaminate the process.
- The purge fluid must be reasonable in cost, since it will be continuously consumed over time.

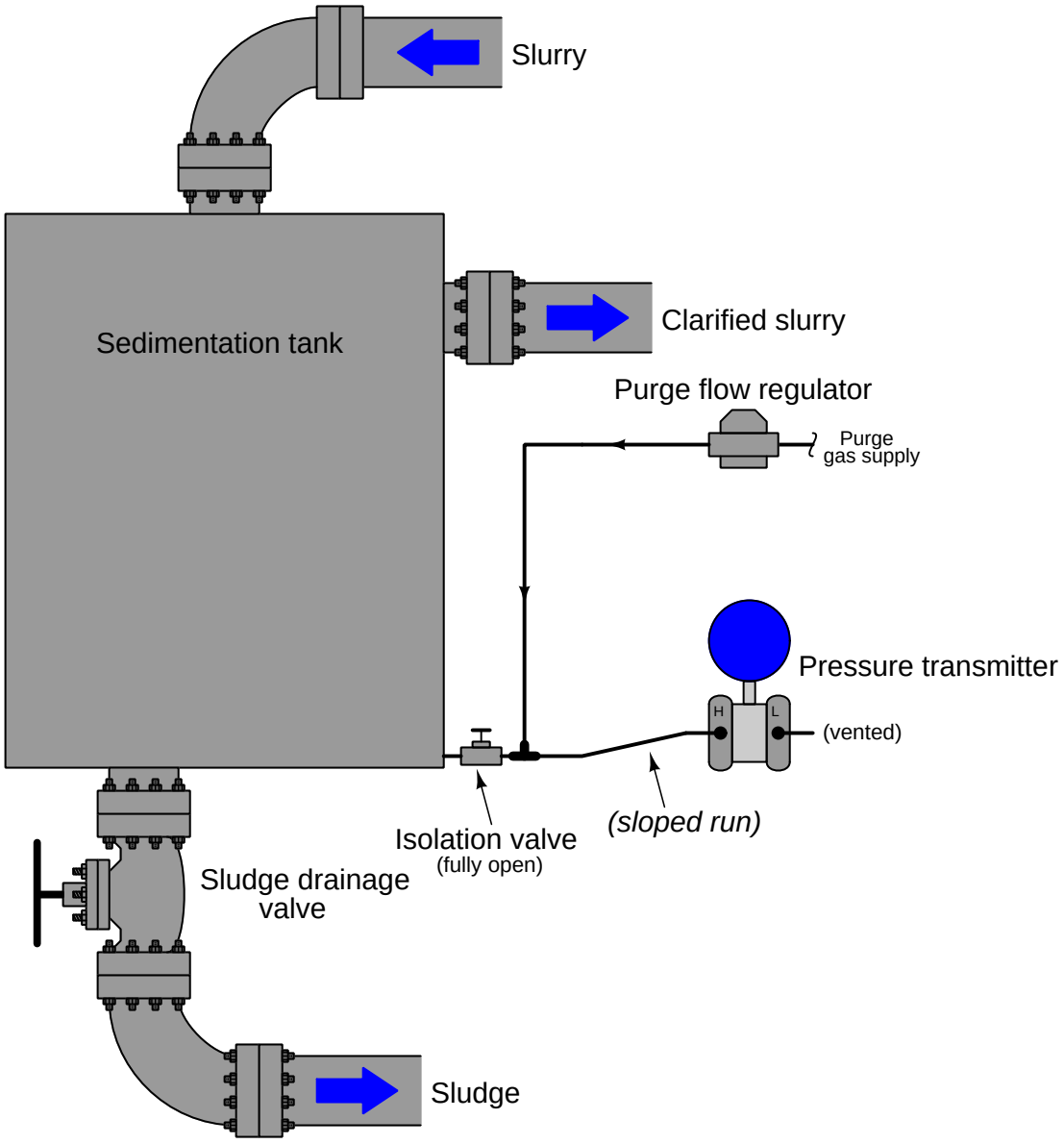
A useful accessory to include in any purge system is a visual flow indicator such as a *rotameter*. Such an indicator is useful for manual adjustment of purge flow rate, and also as a troubleshooting aid, to indicate if anything happens to halt the purge flow.

In the previous example, the purge fluid was clean water. Many options exist for purge fluids other than water, though. Gases such as air, nitrogen, or carbon dioxide are often used in purged systems, for both gas and liquid process applications.

An illustration of a gas-purged pressure measurement system is shown here:



An alternative to the pressure regulator, rotameter, and purge valve is a self-contained unit called a *purge flow regulator* which automatically adjusts to maintain a constant flow rate of purge gas into the purged impulse line:

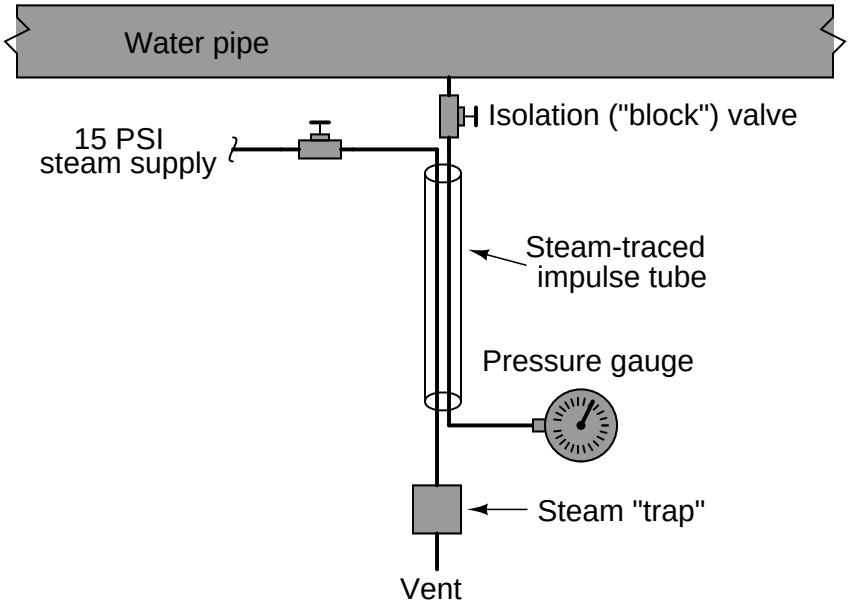


It should be noted that liquid-purged impulse lines – just like filled lines and diaphragm-isolated lines – will generate hydrostatic pressure with vertical height. This is not a problem with gas-purged lines.

19.6.7 Heat-traced impulse lines

If impulse lines are filled with liquid, there may exist a possibility for that liquid to freeze in cold-weather conditions. This possibility depends, of course, on the type of liquid filling the impulse lines and how cold the weather gets in that geographic location.

One safeguard against impulse line freezing is to *trace* the impulse lines with some form of active heating medium, steam and electrical being the most common. "Steam tracing" consists of a copper tube carrying low-pressure steam, bundled alongside one or more impulse tubes, enclosed in a thermally insulating jacket.



Steam flows through the shut-off valve, through the tube in the insulated bundle, transferring heat to the impulse tube as it flows past. Cooled steam condenses into water and collects in the *steam trap* device located at the lowest elevation on the steam trace line. When the water level builds up to a certain level inside the trap, a float-operated valve opens to vent the water. This allows more steam to flow into the tracing tube, keeping the impulse line continually heated.

The steam trap naturally acts as a sort of thermostat as well, even though it only senses condensed water level and not temperature. The rate at which steam condenses into water depends on how cold the impulse tube is. The colder the impulse tube (caused by colder ambient conditions), the more heat energy drawn from the steam, and consequently the faster condensation rate of steam into water. This means water will accumulate faster in the steam trap, which means it will "blow down" more often. More frequent blow-down events means a greater flow rate of steam into the tracing tube, which adds more heat to the tubing bundle and raises its temperature. Thus, the system is naturally regulating, with its own negative feedback loop to maintain bundle temperature at a relatively stable point[27].

[27] In fact, after you become accustomed to the regular "popping" and "hissing" sounds of steam traps blowing down, you can interpret the blow-down frequency as a crude ambient temperature thermometer! Steam traps seldom

The following photograph shows a picture of a steam trap:



Steam traps are not infallible, being susceptible to freezing (in *very* cold weather) and sticking open (wasting steam by venting it directly to atmosphere). However, they are generally reliable devices, capable of adding tremendous amounts of heat to impulse tubing for protection against freezing.

blow down during warm weather, but their "popping" is much more regular (one every minute or less) when ambient temperatures drop well below the freezing point of water.

Electrically traced impulse lines are an alternative solution for cold-weather problems. The "tracing" used is a twin-wire cable (sometimes called *heat tape*) that acts as a resistive heater. When power is applied, the cable heats up, thus imparting thermal energy to the impulse tubing it is bundled with. This next photograph shows the end of a section of electrical heat tape, rated at 33 watts per meter (10 watts per foot) at 10 degrees Celsius (50 degrees Fahrenheit):



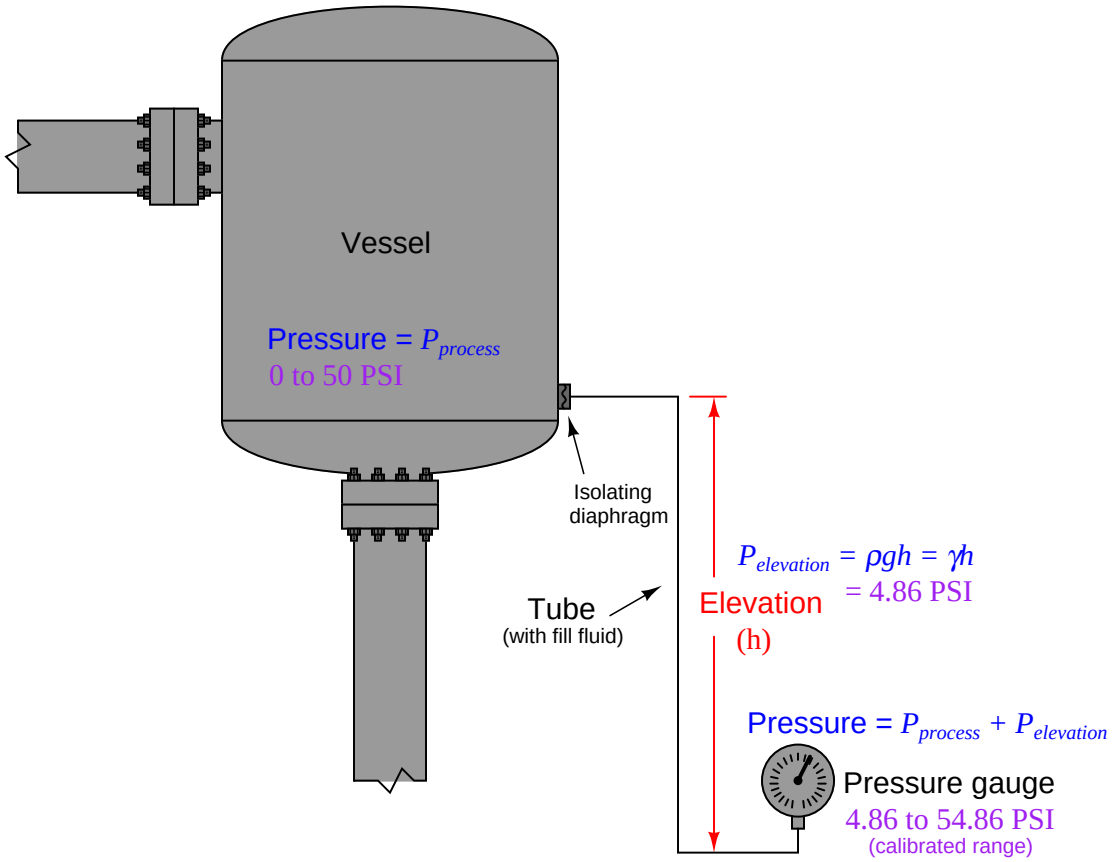
This particular heat tape also has a maximum current rating of 20 amps (at 120 volts). Since heat tape is really just a continuous parallel circuit, longer lengths of it draw greater current. This maximum total current rating therefore places a limit on the usable length of the tape.

Heat tape may be self-regulating, or controlled with an external thermostat. Self-regulating heat tape exhibits an electrical resistance that varies with temperature, automatically self-regulating its own temperature without the need for external controls.

Both steam and electrical heat tracing are used to protect instruments themselves from cold weather freezing, not just the impulse lines. In these applications it is important to remember that only the liquid-filled portions of the instrument require freeze protection, not the electronic portions!

19.6.8 Self-purged impulse lines

One of the advantages of purging a pressure instrument impulse line with gas rather than with liquid is the elimination of measurement error due to the pressure generated by a vertical column of liquid. We investigated an example of this phenomenon in the "Remote and Chemical Seals" subsection where a liquid-filled vertical run of capillary tube 12 feet high created an additional hydrostatic pressure of 4.86 PSI sensed by the pressure gauge. This hydrostatic pressure caused the gauge to read falsely high, so that instead of registering 0 to 50 PSI as the process vessel pressure ranges from 0 to 50 PSI, the gauge instead reads 4.86 PSI *extra* at all points: reading 4.86 to 54.86 PSI as the process vessel pressure goes from 0 to 50 PSI:



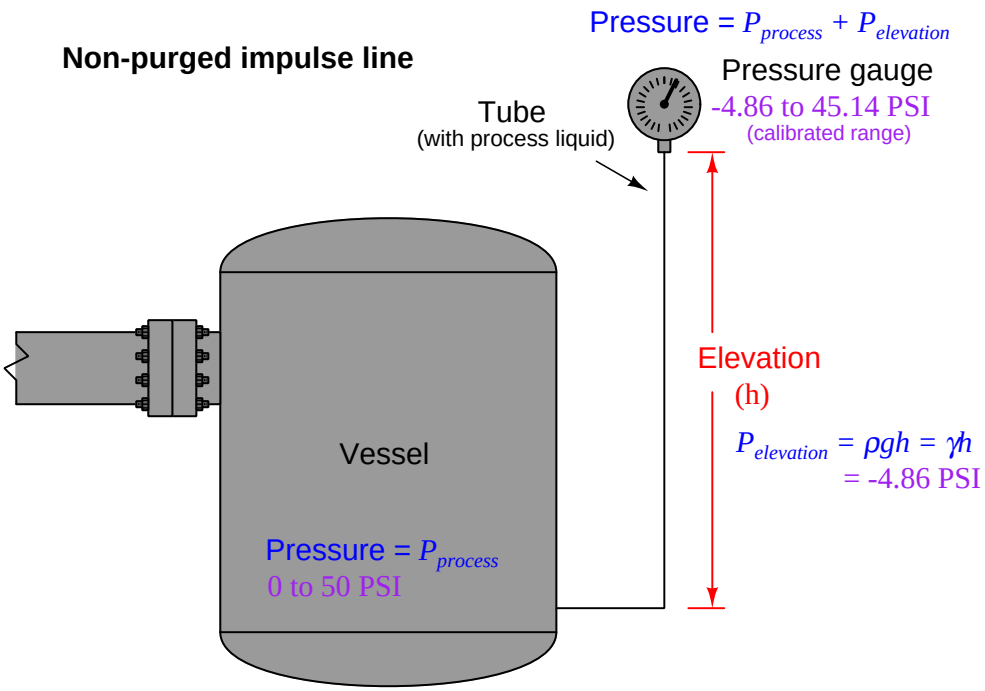
This measurement error may be compensated by shifting the "zero" calibration of the pressure gauge by 4.86 PSI, forcing it to register 4.86 PSI *less* than the pressure it senses at its input port. Only by custom-calibrating the pressure gauge in this manner can we solve the problem created by that 4.86 PSI hydrostatic pressure ($P_{elevation}$).

We also learned in the "Filled Impulse Lines" subsection that this hydrostatic effect is not limited to remote-seal capillary systems but is endemic to *any* significant vertical length of tube filled with

any liquid. A gas-purged impulse line, by contrast, generates negligible pressure due to elevation differences simply because the density of most gases is negligibly small.

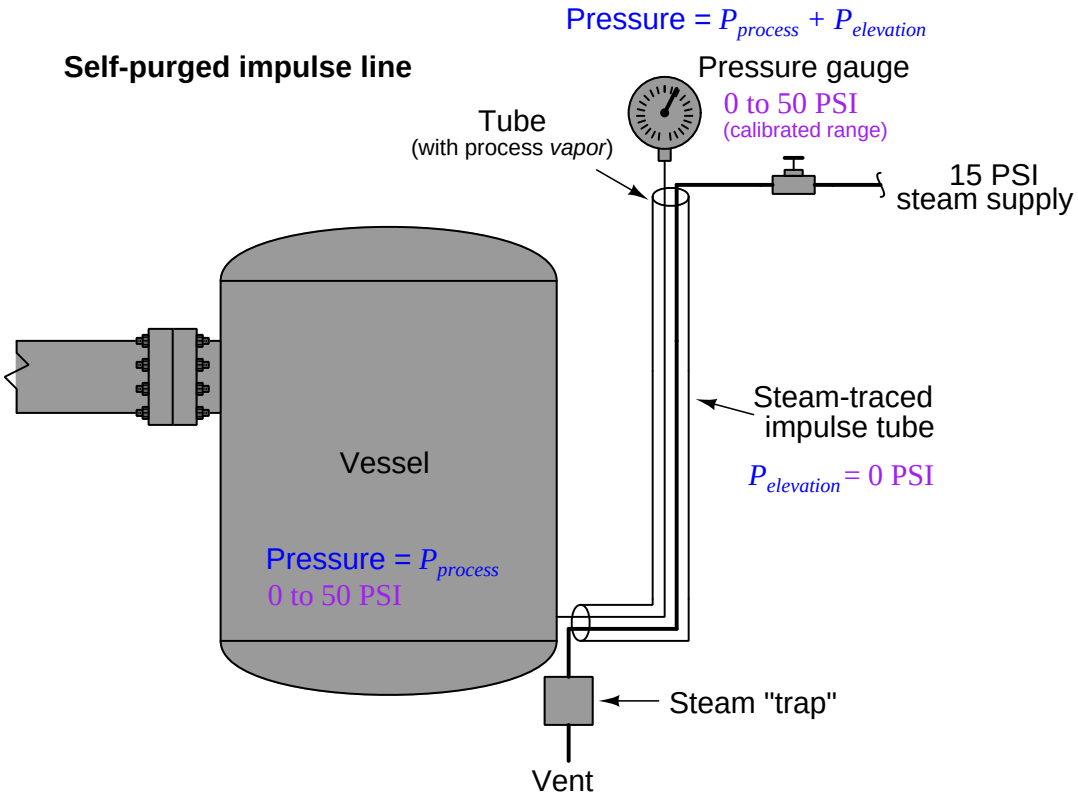
An interesting variation on the theme of gas-purging for instrument impulse lines is the use of an external heat source on those impulse lines to cause the process liquid to *boil* and vaporize within the lines. This technique, of course, only works for process liquids that are easily vaporized with modest applications of heat, but in many processes this is practical. Examples of process liquids amenable to this treatment are propane, butane, and any cryogenic[28] liquid. If we use heat-traced impulse lines, the thermal energy added to the lines maintains their interiors in a gaseous state rather than a liquid state, eliminating any vertical liquid columns inside the lines and therefore eliminating any pressure measurement error resulting from elevation between the pressure instrument and the process connection point.

Self-purging works best in installations where the pressure-sensing instrument is mounted *above* the process liquid level, and where the presence of liquid inside the vertical run of impulse line extending down from the pressure instrument to the vessel would otherwise create a *negative* pressure offset. Shown here is an example of an installation where a vertical impulse line creates a negative pressure measurement error:



[28] "Cryogenic" simply refers to a condition of extremely low temperature required to condense a gas into liquid. Such liquids will flash into vapor if raised to room temperature, and so it is quite easy to make impulse lines self-purging in such cases.

Heating this impulse line causes any liquid inside of it to vaporize, forcing remaining liquid to flow out the bottom of the line and back into the process vessel, "self-purging" the line with vapor and thereby ensuring the pressure instrument senses actual process vessel pressure:



The ideal impulse line heat-trace temperature is greater than the *critical temperature* of the process fluid, so that no amount of process pressure can make it liquefy. This will ensure a liquid-free state inside the heated impulse lines even if process pressure happens to increase.

If any *other* liquids exist inside the vessel that will not vaporize at the same temperature, it becomes necessary to install a liquid *trap* at the bottom of the impulse line where this other liquid can accumulate without filling up the impulse line. Periodically draining this trap of accumulated liquid then becomes a regular maintenance task.

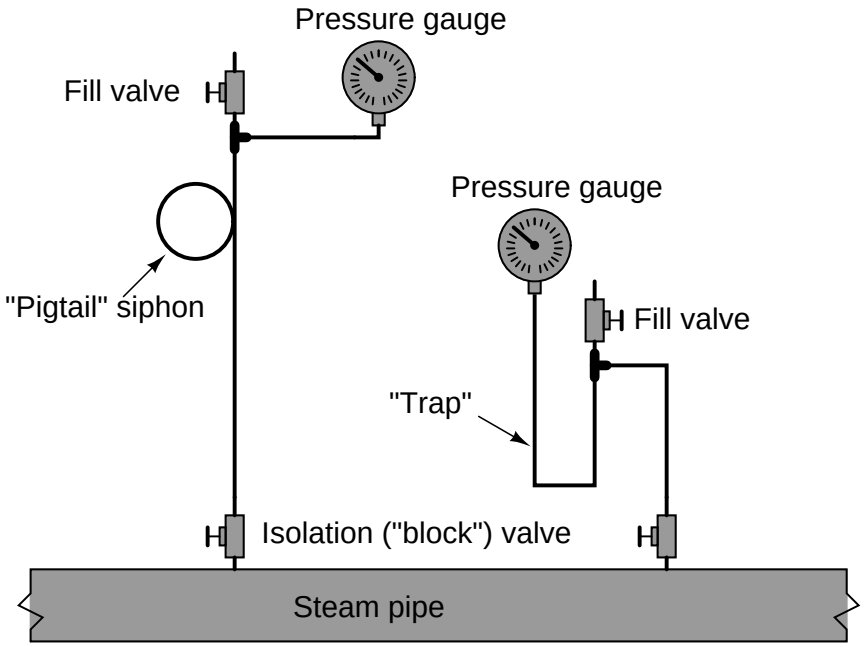
Self-purging does not work as well in installations where the process vessel is located above of the pressure-sensing instrument because liquid will continuously find its way into impulse line by gravity, where the sudden expansion from liquid into vapor will create pressure surges inside of the line. This will cause the pressure instrument to register intermittent "surges" of pressure, which is a worse problem[29] than having an hydrostatic offset.

[29] At least in the case of a liquid-filled impulse line generating its own hydrostatic pressure, that pressure is constant

19.6.9 Water traps and pigtail siphons

Many industrial processes utilize high-pressure steam for direct heating, performing mechanical work, combustion control, and as a chemical reactant. Measuring the pressure of steam is important both for its end-point use and its generation (in a boiler). One problem with doing this is the relatively high temperature of steam at the pressures common in industry, which can cause damage to the sensing element of a pressure instrument if directly connected.

A simple yet effective solution to this problem is to intentionally create a "low" spot in the impulse line where condensed steam (water) will accumulate and act as a liquid barrier to prevent hot steam from reaching the pressure instrument. The principle is much the same as a plumber's trap used underneath sinks, creating a liquid seal to prevent noxious gases from entering a home from the sewer system. A loop of tube or pipe called a *pigtail siphon* achieves the same purpose:



and may be compensated by "zero-shifting" the range of the pressure instrument. An impulse line that generates random *surges* of pressure cannot be compensated at all!

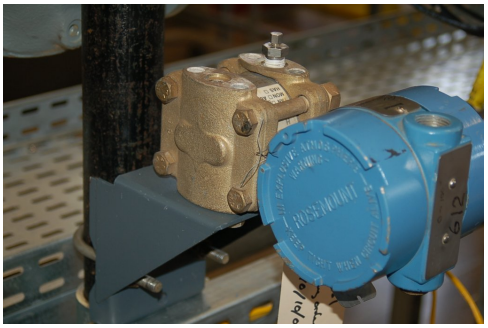
The following photograph shows a pigtail siphon connected to a pressure gauge sensing pressure on a steam line:

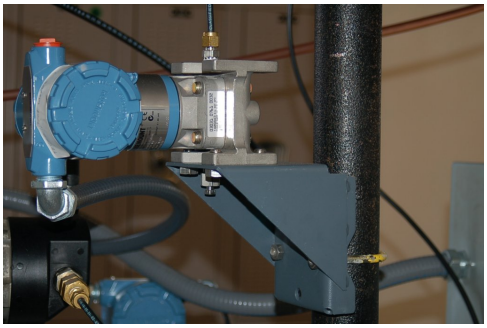


19.6.10 Mounting brackets

An accessory specifically designed for a variety of field-mounted instruments including DP transmitters is the *2 inch pipe mounting bracket*. Such a bracket is manufactured from heavy-gauge sheet metal and equipped with a U-bolt designed to clamp around any 2 inch black iron pipe. Holes stamped in the bracket match mounting bolts on the capsule flanges of most common DP transmitters, providing a mechanically stable means of attaching a DP transmitter to a framework in a process area.

The following photographs show several different instruments mounted to pipe sections using these brackets:











19.6.11 Heated enclosures

In installations where the ambient temperature may become very cold, a protective measure against fluid freezing inside a pressure transmitter is to house the transmitter in an insulated, heated enclosure. The next photograph shows just such an enclosure with the cover removed:



Not surprisingly, this installation works well to protect all kinds of temperature-sensitive instruments from extreme cold. Here, we see an explosive gas sensor mounted inside a slightly different style of insulated enclosure, with the lid opened up for inspection:



19.7 Process/instrument suitability

On a fundamental level, pressure is universal. Regardless of the fluid in question; liquid or gas, hot or cold, corrosive or inert, pressure is nothing more than the amount of force exerted by that fluid over a unit area:

$$P = \frac{F}{A}$$

It should come as no surprise, then, that the common mechanical sensing elements for measuring pressure (bellows, diaphragm, bourdon tube, etc.) are equally applicable to all fluid pressure measurement applications, at least in principle. It is normally a matter of proper material selection and element strength (material thickness) to make a pressure instrument suitable for any range of process fluids.

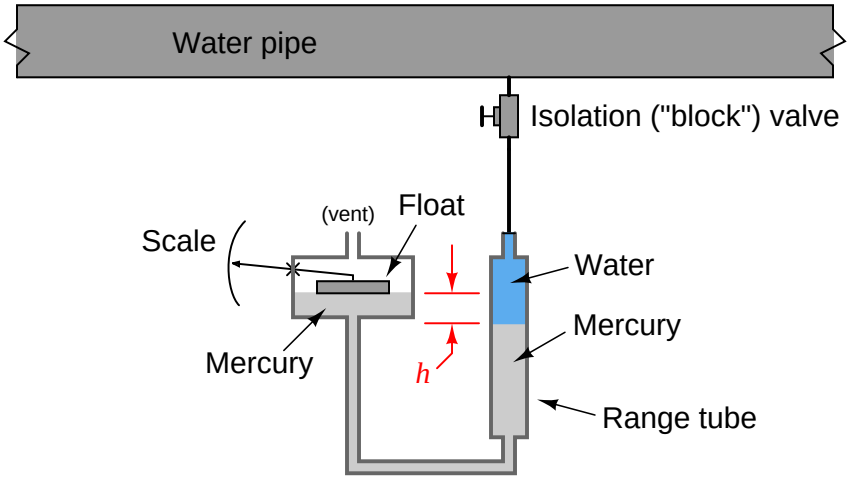
Fill fluids used in pressure instruments – whether it be the dielectric liquid inside a differential capacitance sensor, the fill liquid of a remote or chemical seal system, or liquid used to fill a vertical section of impulse tubing – must be chosen so as to not adversely react with or contaminate the process.

Pure oxygen processes require that no system component have traces of hydrocarbon fluids present. While oxygen itself is not explosive, it greatly accelerates the combustion and explosive potential of any flammable substance. Therefore, a pressure gauge calibrated using oil as the working fluid in a deadweight tester would definitely *not* be suitable for pure oxygen service! The same may be said for a DP transmitter with a hydrocarbon-based fill inside its pressure-sensing capsule[30].

Pharmaceutical, medical, and food manufacturing processes require strict purity and the ability to disinfect all elements in the process system at will. Stagnant lines are not allowed in such processes, as microbe cultures may flourish in such "dead end" piping. Remote seals are very helpful in overcoming this problem, but the fill fluids used in remote systems must be chosen such that a leak in the isolating diaphragm will not contaminate the process.

[30] Although this fluid would not *normally* contact pure oxygen in the process, it could if the isolating diaphragm inside the transmitter were to ever leak.

Manometers, of course, are rather limited in their application, as their operation depends on direct contact between process fluid and manometer liquid. In the early days of industrial instrumentation, liquid mercury was a very common medium for process manometers, and it was not unusual to see a mercury manometer used in direct contact with a process fluid such as oil or water to provide pressure indication:



Thankfully, those days are gone. Mercury (chemical symbol "Hg") is a toxic metal and therefore hazardous to work with. Calibration of these manometers was also challenging due to the column height of the process liquid in the impulse line and the range tube. When the process fluid is a gas, the difference in mercury column height directly translates to sensed pressure by the hydrostatic pressure formula $P = \rho g h$ or $P = \gamma h$. When the process fluid is a liquid, though, the shifting of mercury columns also creates a change in height of the process liquid column, which means the indicated pressure is a function of the height difference (h) *and* the difference in density between the process liquid and mercury. Consequently, the indications provided by mercury manometers in liquid pressure applications were subject to correction according to process liquid density.

19.8 Review of fundamental principles

Shown here is a partial listing of principles applied in the subject matter of this chapter, given for the purpose of expanding the reader's view of this chapter's concepts and of their general interrelationships with concepts elsewhere in the book. Your abilities as a problem-solver and as a life-long learner will be greatly enhanced by mastering the applications of these principles to a wide variety of topics, the more varied the better.

- **Definition of pressure**: $P = \frac{F}{A}$ (pressure is the amount of force applied over a specified area by a fluid.
- **Pascal's principle**: changes in fluid pressure are transmitted evenly throughout an enclosed fluid volume. Relevant to pressure measurement, as fluid pressure in all parts of an enclosed system will experience the same changes in pressure.
- **Hydrostatic pressure**: fluids having substantial weight generate pressure proportional to their density and to their vertical height ($P = \gamma h$ and $P = \rho g h$). Relevant to pressure offsets generated in vertical spans of impulse or capillary tubing, causing a pressure instrument to register more or less pressure than that at the process vessel connection.
- **Self-balancing pneumatic mechanisms**: all self-balancing pneumatic instruments work on the principle of negative feedback maintaining a nearly constant baffle-nozzle gap. Force-balance mechanisms maintain this constant gap by balancing force against force with negligible motion, like a tug-of-war. Motion-balance mechanisms maintain this constant gap by balancing one motion with another motion, like two dancers moving in unison.

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Chapter 20

Continuous level measurement

Many industrial processes require the accurate measurement of fluid or solid (powder, granule, etc.) height within a vessel. Some process vessels hold a stratified combination of fluids, naturally separated into different layers by virtue of differing densities, where the height of the *interface* point between liquid layers is of interest.

A wide variety of technologies exist to measure the level of substances in a vessel, each exploiting a different principle of physics. This chapter explores the major level-measurement technologies in current use.

20.1 Level gauges (sightglasses)

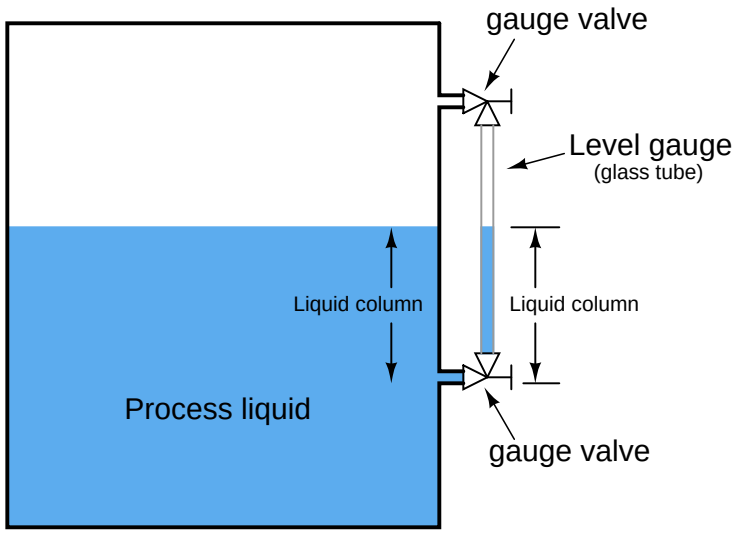
Level gauges are perhaps the simplest indicating instrument for liquid level in a vessel. They are often found in industrial level-measurement applications, even when another level-measuring instrument is present, to serve as a direct indicator for an operator to monitor in case there is doubt about the accuracy of the other instrument.

20.1.1 Basic concepts of sightglasses

The *level gauge*, or *sightglass* is to liquid level measurement as manometers are to pressure measurement: a very simple and effective technology for direct visual indication of process level. In its simplest form, a level gauge is nothing more than a clear tube through which process liquid may be seen. The following photograph shows a simple example of a sightglass:



A functional diagram of a sightglass shows how it visually represents the level of liquid inside a vessel such as a storage tank:

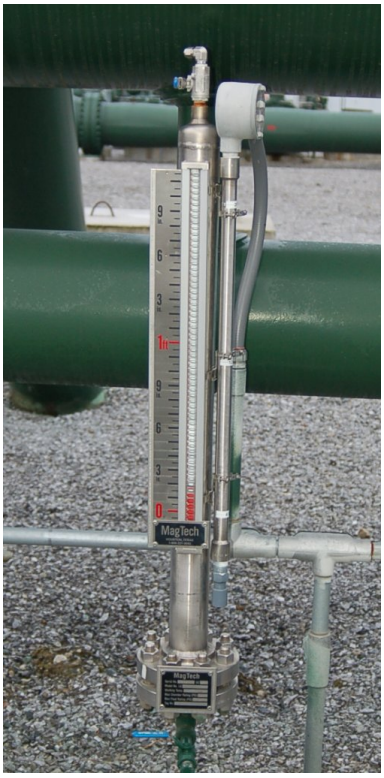


A level gauge is not unlike a U-tube manometer, with equal pressures applied to both liquid columns (one column being the liquid in the gauge sightglass, the other column being the liquid in the vessel).

Level gauge valves exist to allow replacement of the glass tube without emptying or depressurizing the process vessel. These valves are usually equipped with flow-limiting devices in the event of a tube rupture, so too much process fluid does not escape even when the valves are fully open.

Some level gauges called *reflex gauges* are equipped with special optics to facilitate the viewing of clear liquids, which is problematic for simple glass-tube sightglasses.

A weakness of glass-tube level gauges is the glass tube itself. The tube must be kept in a clean condition in order for the liquid level to be clearly visible, which may be a problem in a dirty-liquid service. Also, glass tubes may rupture if subjected to thermal or mechanical shock. One solution to this problem is to eliminate the glass tube entirely, replacing it with a non-magnetic metal tube (e.g. stainless steel) containing a magnetized float, with magnet-sensing indicator flags outside of this tube to visually indicate level. Here is one example of such a level gauge, manufactured by MagTech:

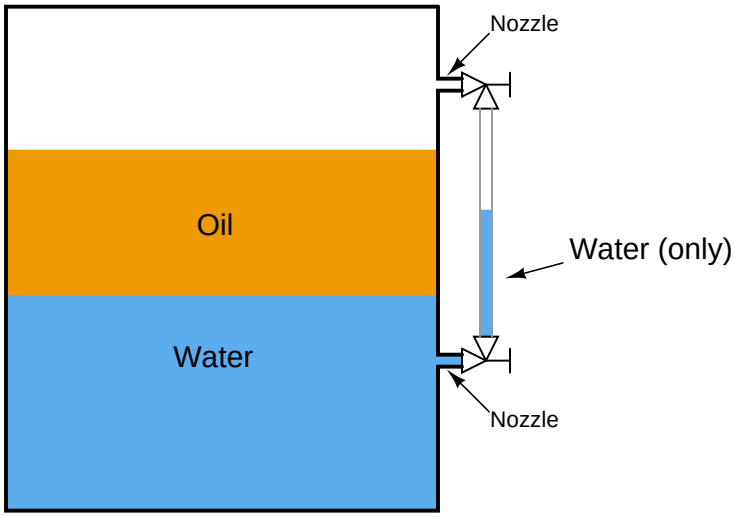


In this instrument, you can see red-colored flags toward the bottom of the scale which have been "flipped" by the motion of the magnetic float inside the stainless-steel tube. The height of the red zone (i.e. how many flags have been flipped to show their red sides) indicates the height of the liquid inside the tube.

Some magnetic level gauges even have high- and low-level magnetic switches located at strategic points along the tube's height, providing discrete sensing capability for alarms and/or shutdown controls, if the liquid level ever goes outside of safe operating limits. These switches will open and close as the magnetic float passes by, remotely signaling liquid level at that height.

20.1.2 Interface problems

As simple and apparently trouble-free as level gauges may seem, there are special circumstances where they will register incorrectly. One such circumstance is in the presence of a lighter liquid layer existing between the connection ports of the gauge. If a lighter (less dense) liquid exists above a heavier (denser) liquid in the process vessel, the level gauge may not show the proper interface, if at all:

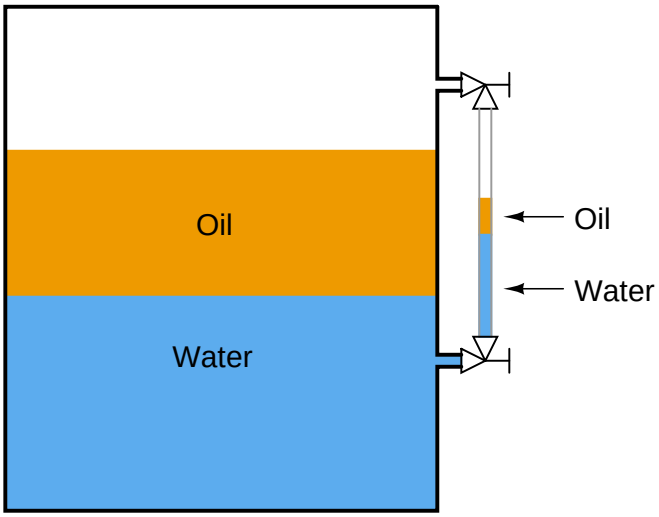


A practical application of level measurement where two liquids form an interface is where water exists in the presence of petroleum substances, such as diesel fuel. Water is a denser liquid than most oils, and these two liquids are immiscible[1], which means the denser water forms a separate layer beneath the lighter oil. Another application of interface measurement is in the oil and gas extraction industry, where water must be separated from petroleum fluids coming out of wells drilled deep into the earth. Fluid from an oil well enters a special vessel called a "separator" where gravity causes the water to separate from the petroleum liquids, and petroleum vapors to separate from all liquids. These water/oil/gas separator vessels are critical components of any petroleum well system, with the liquid-liquid interface between water and oil being an important process variable to measure and control.

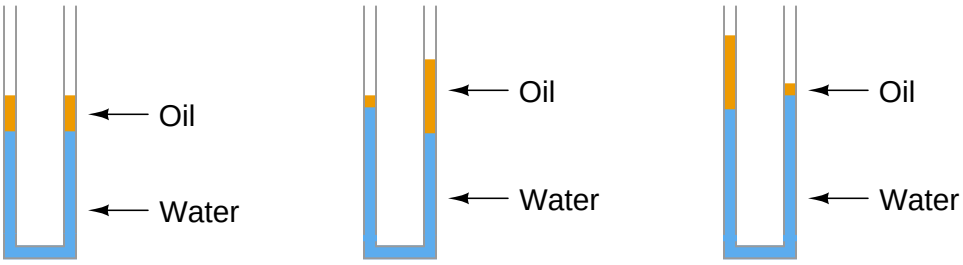
In the above illustration we see how a column of water in the sightglass shows less (total) level than the combination of water and oil inside a process vessel. Since the oil lies between the two level gauge ports into the vessel (sometimes called *nozzles*), it cannot enter the sightglass tube, and therefore the level gauge will continue to show just water.

[1] Liquids are considered "miscible" if they may be mixed in any proportion to each other to form a solution. Immiscible liquids refuse to mix thoroughly, and therefore tend to separate.

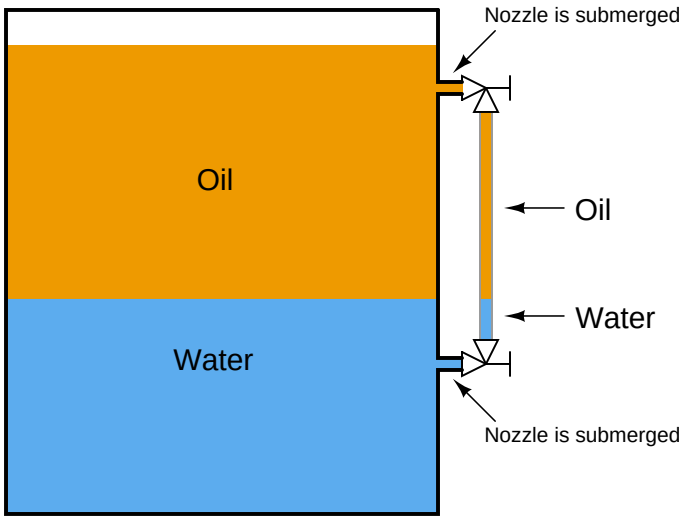
If by chance some oil does find its way into the sightglass tube – either by the interface level dropping below the lower nozzle or the total level rising above the upper nozzle – the oil/water interface shown inside the level gauge may not continue to reflect the true interface inside the vessel once the interface and total levels return to their previous positions:



Recall that the level gauge and vessel together form a U-tube manometer. So long as the pressures from each liquid column are the same, the columns balance each other. The problem is, many different liquid-liquid interface columns can have the same hydrostatic pressure without being identical to one another:

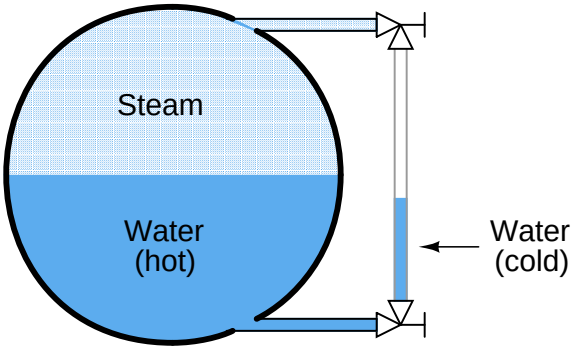


The only way to ensure proper two-part liquid interface level indication in a sightglass is to keep both ports (nozzles) submerged:



20.1.3 Temperature problems

Another troublesome scenario for level gauges is when the liquid inside the vessel is substantially hotter than the liquid in the gauge, causing the densities to be different. This is commonly seen on boiler level gauges, where the water inside the sightglass cools off substantially from its former temperature inside the boiler drum:

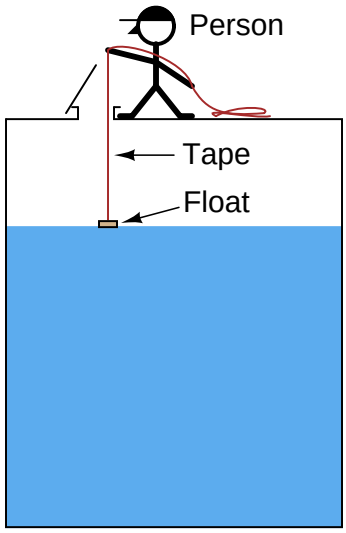


Looking at the sightglass as a U-tube manometer again, we see that unequal-height liquid columns may indeed balance each other's hydrostatic pressures if the two columns are comprised of liquids with different densities. The weight density of water is 62.4 lb/ft^3 at standard temperature, but may be as low as only 36 lb/ft^3 at temperatures common for power generation boilers.

20.2 Float

Perhaps the simplest form of solid or liquid level measurement is with a *float*: a device that rides on the surface of the fluid or solid within the storage vessel. The float itself must be of substantially lesser density than the substance of interest, and it must not corrode or otherwise react with the substance.

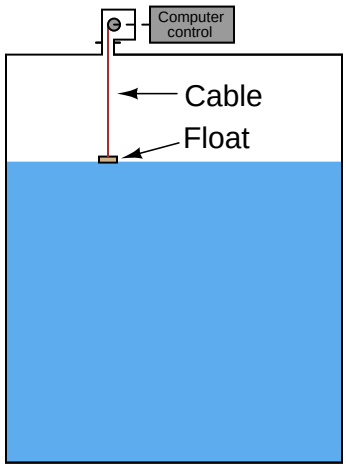
Floats may be used for manual "gauging" of level, as illustrated here:



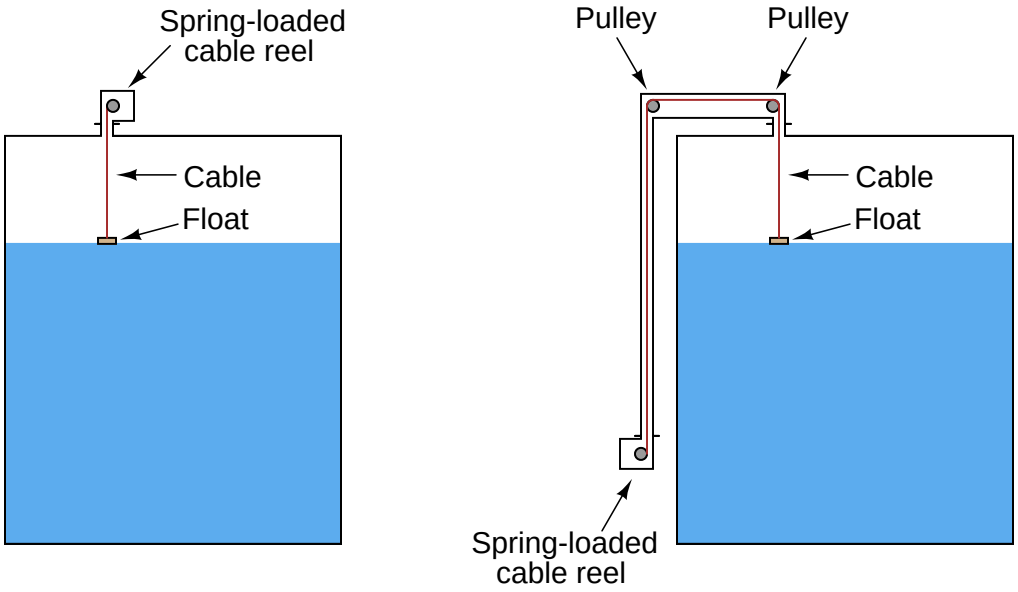
A person lowers a float down into a storage vessel using a flexible measuring tape, until the tape goes slack due to the float coming to rest on the material surface. At that point, the person notes the length indicated on the tape (reading off the lip of the vessel access hole). This distance is called the *ullage*, being the distance from the top of the vessel to the surface of the process material. *Fillage* of the vessel may be determined by subtracting this "ullage" measurement from the known height of the vessel.

Obviously, this method of level measurement is tedious and may pose risk to the person conducting the measurement. If the vessel is pressurized, this method is simply not applicable.

If we automate the person's function using a small winch controlled by a computer – having the computer automatically lower the float down to the material surface and measure the amount of cable played out at each measurement cycle – we may achieve better results without human intervention. Such a level gauge may be enclosed in such a way to allow pressurization of the vessel, too:

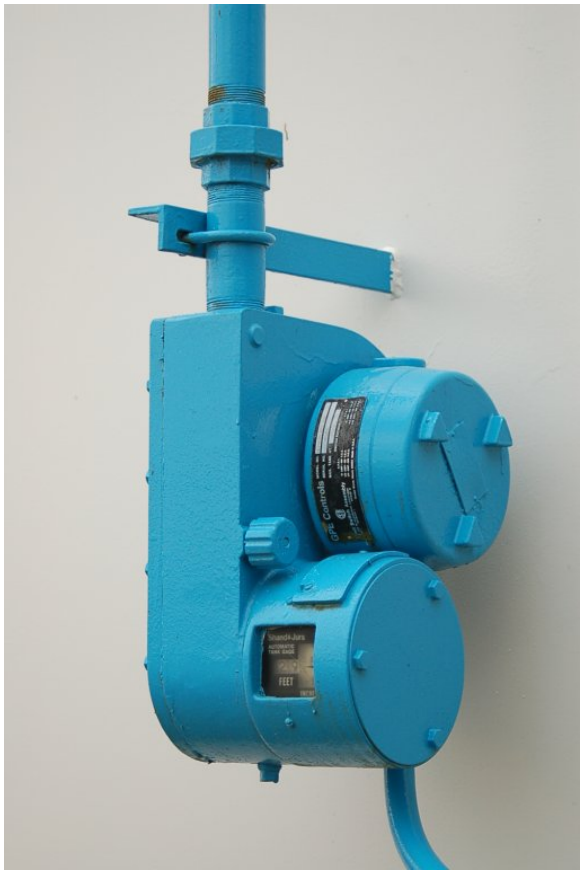


A simpler version of this technique uses a spring-reel to constantly tension the cable holding the float, such that the float continuously rides on the surface of the liquid in the vessel[2]:



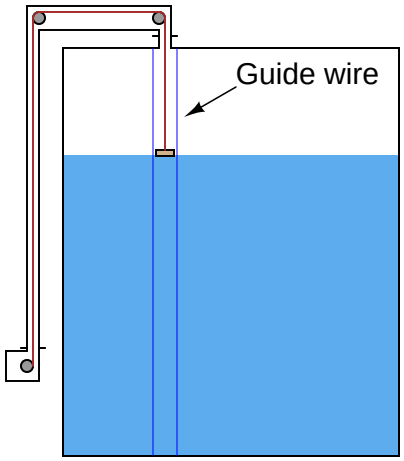
[2] A spring-loaded cable float only works with liquid level measurement, while a retracting float will measure liquids and solids with equal ease. The reason for this limitation is simple: a float that always contacts the material surface is likely to become buried if the material in question is a solid (powder or granules), which must be fed into the vessel from above.

The following photograph shows the "measurement head" of a spring-reel tape-and-float liquid level transmitter, with the vertical pipe housing the tape on its way to the top of the storage tank where it will turn 180 degrees via two pulleys and attach to the float inside the tank:



The spring reel's angular position may be measured by a multi-turn potentiometer or a rotary encoder (located inside the "head" unit), then converted to an electronic signal for transmission to a remote display, control, and/or recording system. Such systems are used extensively for measurement of water and fuel in storage tanks.

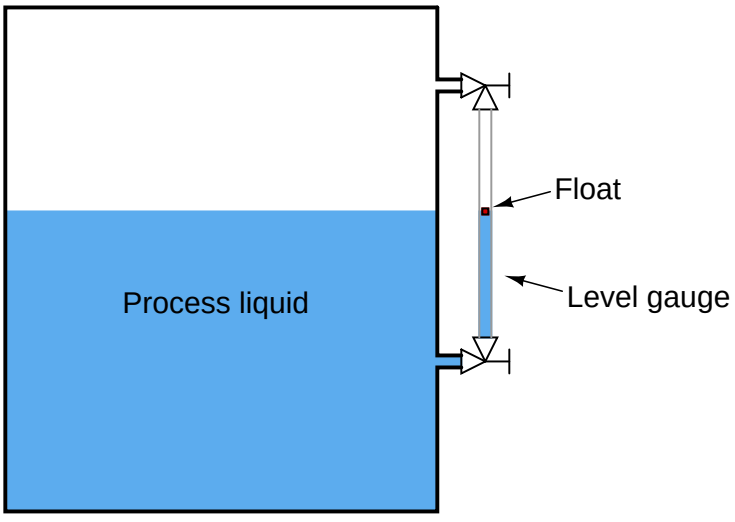
If the liquid inside the vessel is subject to turbulence, *guide wires* may be necessary to keep the float cable in a vertical orientation:



The guide wires are anchored to the floor and roof of the vessel, passing through ring lugs on the float to keep it from straying laterally.

One of the potential disadvantages of tape-and-float level measurement systems is fouling of the tape (and guide wires) if the substance is sticky or unclean.

A variation on the theme of float level measurement is to place a small float inside the tube of a sightglass-style level gauge:



The float's position inside the tube may be readily detected by ultrasonic waves, magnetic sensors or any other applicable means. Locating the float inside a tube eliminates the need for guide wires or a sophisticated tape retraction or tensioning system. If no visual indication is necessary, the

level gauge tube may be constructed out of metal instead of glass, greatly reducing the risk of tube breakage. All the problems inherent to sightglasses, however, still apply to this form of float instrument.

The following photograph shows just such a float-style level indicator, manufactured by K-Tek. This particular level indicator is removed from the process, sitting on a concrete floor for the photograph. A magnetic float rides on the surface of a liquid column inside a non-magnetic metal tube, while a brightly-colored indicating flag tracks the magnetic float's position inside a transparent plastic tube for convenient viewing:



Two major advantages of a magnetically-coupled float is increased pressure rating and safety (since the float tube need not be constructed of clear material such as plastic or glass), and increased readability (since the viewing tube will never get dirty with process fluid residue, and the float may be brightly colored).

This particular level indicator also has a special pneumatic valve mounted to the side of the non-magnetic metal tube. This valve is actuated by the magnetic field of the float, turning a pneumatic "circuit" on and off based on the float's position:



This is just one example of auxiliary functions possible with magnetic float level indicators. Such a pneumatic valve may be used to control a larger process valve to redirect liquid to or from the vessel based on level, to trip an operator alarm, or any number of other automatic functions.

Another variation on the theme of auxiliary float functions is a principle called *magnetostriction* to detect the position of the float along a metal guide rod called a *waveguide*. This instrument design is discussed in significant detail later in this chapter (see 20.5.4 beginning on page 1490).

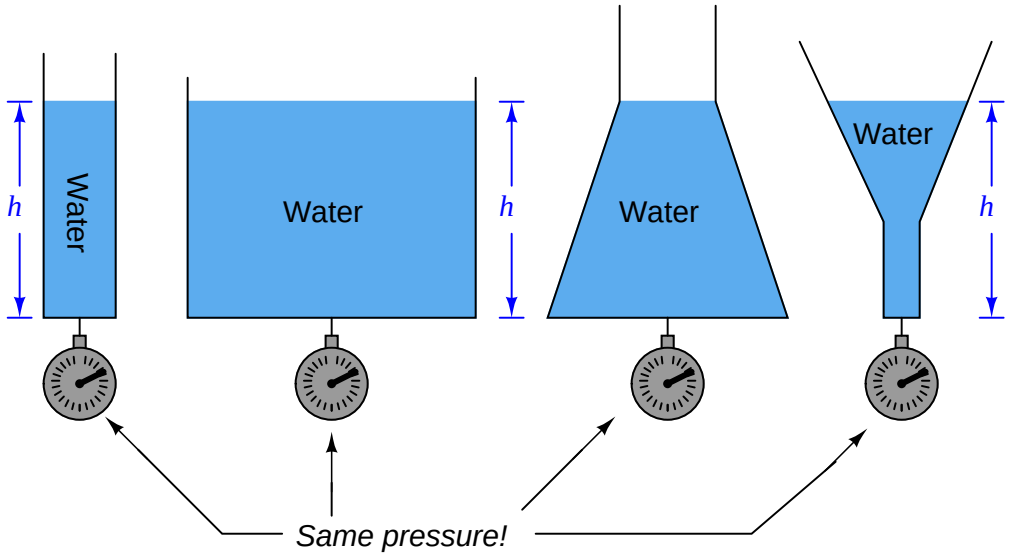
20.3 Hydrostatic pressure

A vertical column of fluid generates a pressure at the bottom of the column owing to the action of gravity on that fluid. The greater the vertical height of the fluid, the greater the pressure, all other factors being equal. This principle allows us to infer the level (height) of liquid in a vessel by pressure measurement.

20.3.1 Pressure of a fluid column

A vertical column of fluid exerts a pressure due to the column's weight. The relationship between column height and fluid pressure at the bottom of the column is constant for any particular fluid (density) regardless of vessel width or shape.

This principle makes it possible to infer the height of liquid in a vessel by measuring the pressure generated at the bottom:



The mathematical relationship between liquid column height and pressure is as follows:

$$P = \rho g h \qquad\qquad P = \gamma h$$

Where,

P = Hydrostatic pressure

ρ = Mass density of fluid in kilograms per cubic meter (metric) or slugs per cubic foot (British)

g = Acceleration of gravity

γ = Weight density of fluid in newtons per cubic meter (metric) or pounds per cubic foot (British)

h = Height of vertical fluid column above point of pressure measurement

For example, the pressure generated by a column of oil 12 feet high (h) having a weight density of 40 pounds per cubic foot (γ) is:

$$P = \gamma h$$

$$P_{oil} = \left(\frac{40 \text{ lb}}{\text{ft}^3}\right)\left(\frac{12 \text{ ft}}{1}\right) = \frac{480 \text{ lb}}{\text{ft}^2}$$

Note the cancellation of units, resulting in a pressure value of 480 pounds per square foot (PSF). To convert into the more common pressure unit of pounds per square inch, we may multiply by the proportion of square feet to square inches, eliminating the unit of square feet by cancellation and leaving square inches in the denominator:

$$P_{oil} = \left(\frac{480 \text{ lb}}{\text{ft}^2}\right)\left(\frac{1^2 \text{ ft}^2}{12^2 \text{ in}^2}\right)$$

$$P_{oil} = \left(\frac{480 \text{ lb}}{\text{ft}^2}\right)\left(\frac{1 \text{ ft}^2}{144 \text{ in}^2}\right)$$

$$P_{oil} = \frac{3.33 \text{ lb}}{\text{in}^2} = 3.33 \text{ PSI}$$

Thus, a pressure gauge attached to the bottom of the vessel holding a 12 foot column of this oil would register 3.33 PSI. It is possible to customize the scale on the gauge to read directly in feet of oil (height) instead of PSI, for convenience of the operator who must periodically read the gauge. Since the mathematical relationship between oil height and pressure is both linear and direct, the gauge's indication will always be proportional to height.

An alternative method for calculating pressure generated by a liquid column is to relate it to the pressure generated by an equivalent column of water, resulting in a pressure expressed in units of water column (e.g. inches W.C.) which may then be converted into PSI or any other unit desired.

For our hypothetical 12-foot column of oil, we would begin this way by calculating the *specific gravity* (i.e. how dense the oil is compared to water). With a stated weight density of 40 pounds per cubic foot, the specific gravity calculation looks like this:

$$\text{Specific Gravity of oil} = \frac{\gamma_{oil}}{\gamma_{water}}$$

$$\text{Specific Gravity of oil} = \frac{40 \text{ lb/ft}^3}{62.4 \text{ lb/ft}^3}$$

$$\text{Specific Gravity of oil} = 0.641$$

The hydrostatic pressure generated by a column of water 12 feet high, of course, would be 144 inches of water column (144 "W.C.). Since we are dealing with an oil having a specific gravity of 0.641 instead of water, the pressure generated by the 12 foot column of oil will be only 0.641 times (64.1%) that of a 12 foot column of water, or:

$$P_{oil} = (P_{water})(\text{Specific Gravity})$$

$$P_{oil} = (144 \text{ "W.C.})(0.641)$$

$$P_{oil} = 92.3 \text{ "W.C.}$$

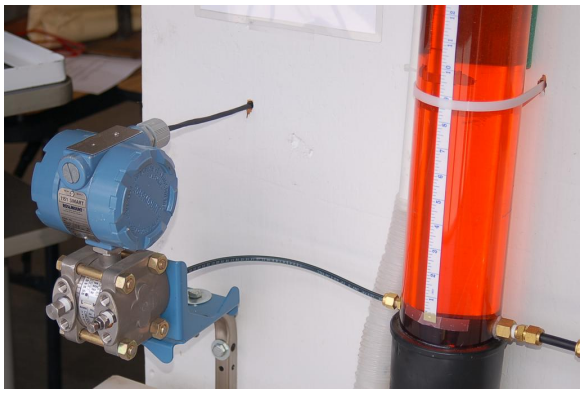
We may convert this pressure value into units of PSI simply by dividing by 27.68, since we know 27.68 inches of water column is equivalent to 1 PSI:

$$P_{oil} = \left(\frac{92.3 \text{ "W.C.}}{1}\right)\left(\frac{1 \text{ PSI}}{27.68 \text{ "W.C.}}\right)$$

$$P_{oil} = 3.33 \text{ PSI}$$

As you can see, we arrive at the same result as when we applied the $P = \gamma h$ formula. Any difference in value between the two methods is due to imprecision of the conversion factors used (e.g. 27.68 "W.C., 62.4 lb/ft^3 density for water).

Any type of pressure-sensing instrument may be used as a liquid level transmitter by means of this principle. In the following photograph, you see a Rosemount model 1151 pressure transmitter being used to measure the height of colored water inside a clear plastic tube:

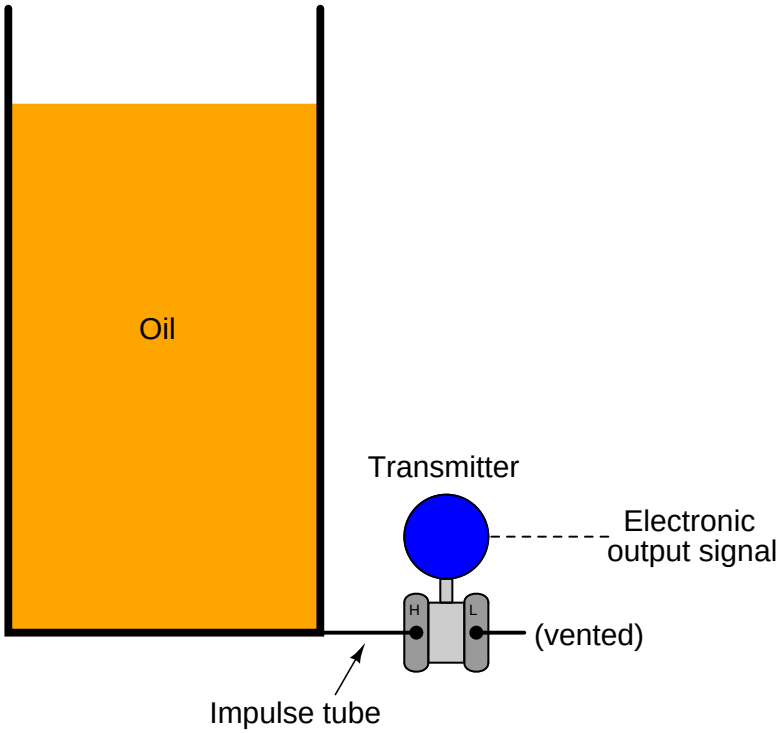


In most level-measurement applications, we are concerned with knowing the *volume* of the liquid contained within a vessel, and we infer this volume by using instruments to sense the *height* of the fluid column. So long as the vessel's cross-sectional area is constant throughout its height, liquid height will be directly proportional to stored liquid volume. Pressure measured at the bottom of a vessel can give us a proportional indication of liquid height if and only if the density of that liquid is known and constant. This means liquid density is a critically important factor for volumetric measurement when using hydrostatic pressure-sensing instruments. If liquid density is subject to random change, the accuracy of any hydrostatic pressure-based level or volume instrument will be correspondingly unreliable.

It should be noted, though, that changes in liquid density will have absolutely no effect on hydrostatic measurement of liquid *mass*, so long as the vessel has a constant cross-sectional area throughout its entire height. A simple thought experiment proves this: imagine a vessel partially full of liquid, with a pressure transmitter attached to the bottom to measure hydrostatic pressure. Now imagine the temperature of that liquid increasing, such that its volume expands and has a lower density than before. Assuming no addition or loss of liquid to or from the vessel, any increase in liquid level will be strictly due to volume expansion (density decrease). Liquid level inside this vessel will rise, but the transmitter will sense the exact same hydrostatic pressure as before, since the rise in level is precisely countered by the decrease in density (if h increases by the same factor

that γ decreases, then $P = \gamma h$ must remain the same!). In other words, hydrostatic pressure is seen to be directly proportional to the amount of liquid *mass* contained within the vessel, regardless of changes in liquid density. This is useful to know in applications where true mass measurement of a liquid (rather than volume measurement) is either preferable or necessary[3].

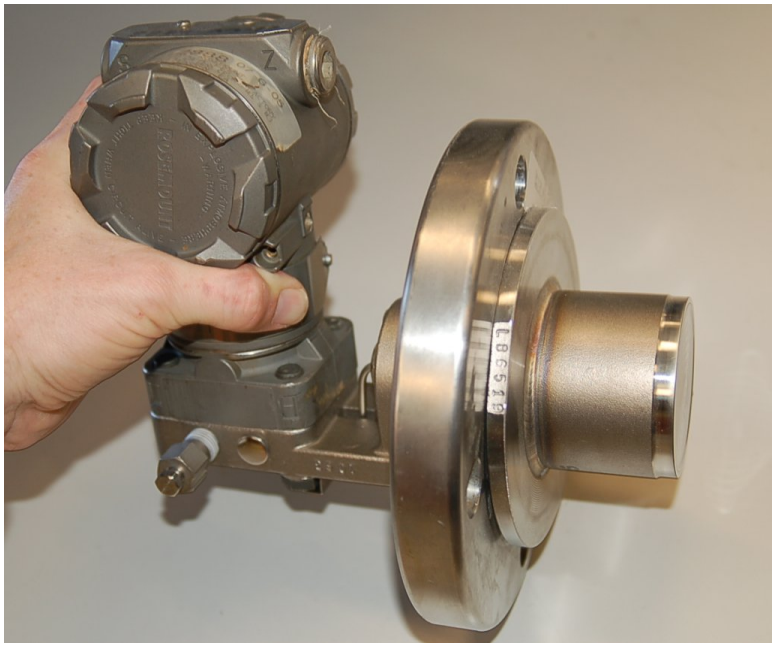
Differential pressure transmitters are the most common pressure-sensing device used in this capacity to infer liquid level within a vessel. In the hypothetical case of the oil vessel just considered, the transmitter would connect to the vessel in this manner (with the high side toward the process and the low side vented to atmosphere):



Connected as such, the differential pressure transmitter functions as a gauge pressure transmitter, responding to hydrostatic pressure exceeding ambient (atmospheric) pressure. As liquid level increases, the hydrostatic pressure applied to the "high" side of the differential pressure transmitter also increases, driving the transmitter's output signal higher.

[3] We may prove this mathematically by algebraic substitution. Given that the total mass (m) of any liquid sample is equal to the product of that liquid's mass density and its sample volume ($m = \rho V$), that volume (V) for any vessel of constant cross-sectional area (A) is given by the expression $V = Ah$, and that hydrostatic pressure is equal to $P = \rho g h$, we may combine these three equations to arrive at $m = \frac{AP}{g}$. This final equation demonstrates how the total mass of liquid stored in a vessel (m) of constant cross-sectional area (A) is directly proportional to pressure (P), and independent of density (ρ).

Some pressure-sensing instruments are built specifically for hydrostatic measurement of liquid level in vessels, eliminating with impulse tubing altogether in favor of a special kind of sealing diaphragm extending slightly into the vessel through a flanged pipe entry (commonly called a *nozzle*). A Rosemount hydrostatic level transmitter with an extended diaphragm is shown here:



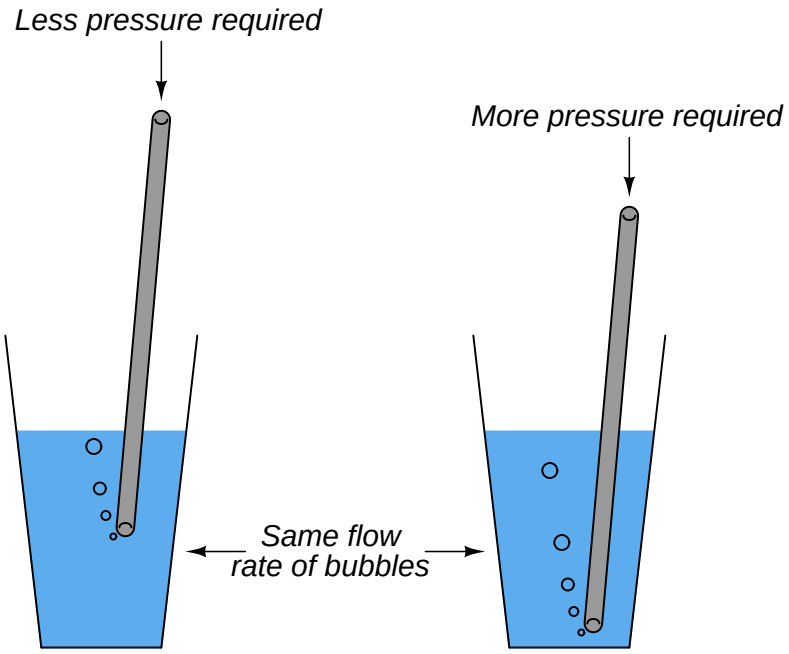
The calibration table for a transmitter close-coupled to the bottom of an oil storage tank would be as follows, assuming a zero to twelve foot measurement range for oil height, an oil density of 40 pounds per cubic foot, and a 4-20 mA transmitter output signal range:

Oil level	**Percent of range**	**Hydrostatic pressure**	**Transmitter output**
0 ft	0 %	0 PSI	4 mA
3 ft	25 %	0.833 PSI	8 mA
6 ft	50 %	1.67 PSI	12 mA
9 ft	75 %	2.50 PSI	16 mA
12 ft	100 %	3.33 PSI	20 mA

20.3.2 Bubbler systems

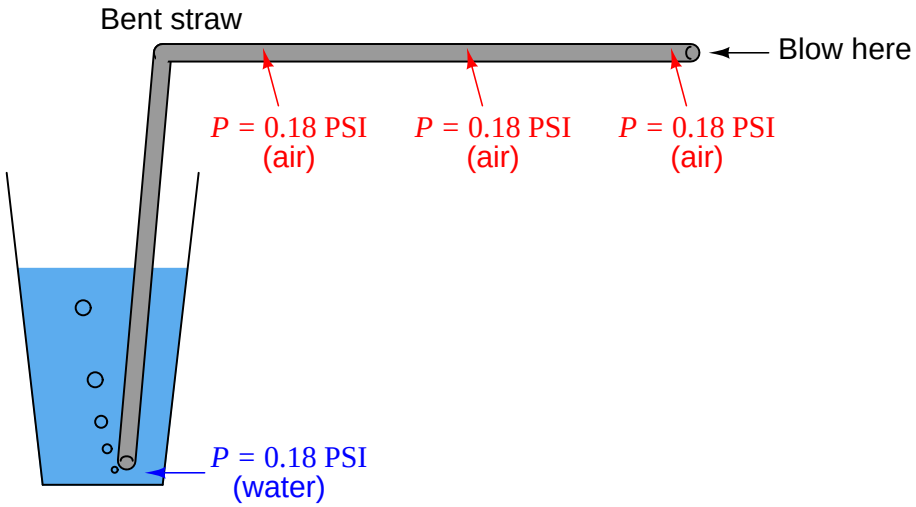
An interesting variation on this theme of direct hydrostatic pressure measurement is the use of a purge gas to measure hydrostatic pressure in a liquid-containing vessel. This eliminates the need for direct contact of the process liquid against the pressure-sensing element, which can be advantageous if the process liquid is corrosive.

Such systems are often called *bubble tube* or *dip tube* systems, the former name being appropriately descriptive for the way purge gas bubbles out the end of the tube as it is submerged in process liquid. A very simple bubbler system may be simulated by gently blowing air through a straw into a glass of water, maintaining a steady rate of bubbles exiting the straw while changing the depth of the straw's end in the water:



The deeper you submerge the straw, the harder it becomes to blow bubbles out the end with your breath. The hydrostatic pressure of the water at the straw's tip becomes translated into air pressure in your mouth as you blow, since the air pressure must just exceed the water's pressure in order to escape out the end of the straw. So long as the flow rate of air is modest (no more than a few bubbles per second), the air pressure will be very nearly equal to the water pressure, allowing measurement of water pressure (and therefore water depth) at any point along the length of the air tube.

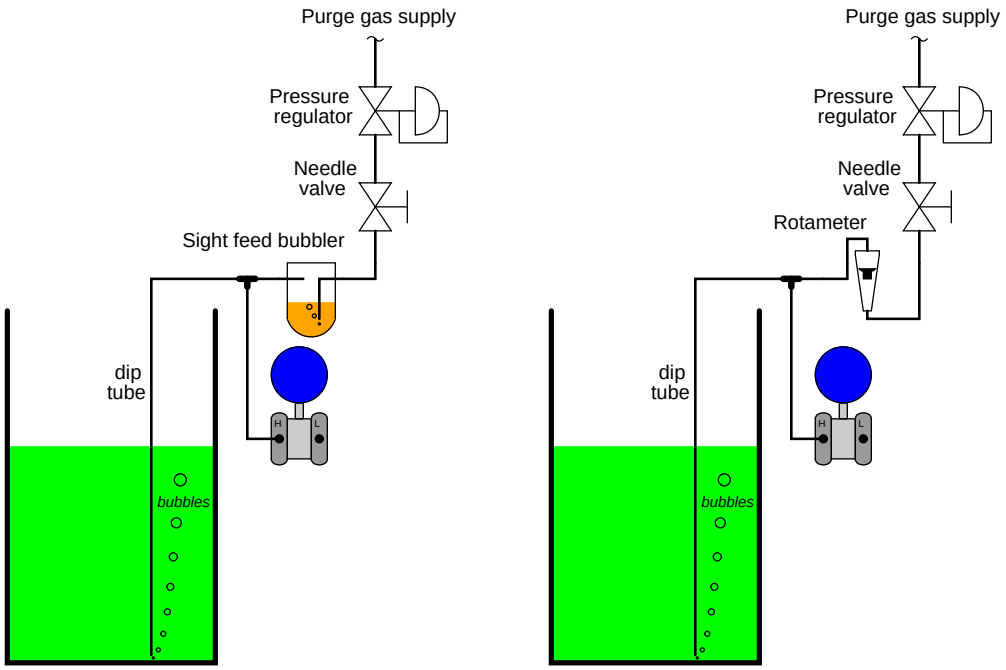
If we lengthen the straw and measure pressure at all points throughout its length, it will be the same as the pressure at the submerged tip of the straw (assuming negligible friction between the moving air molecules and the straw's interior walls):



This is how industrial "bubbler" level measurement systems work: a *purge gas* is slowly introduced into a "dip tube" submerged in the process liquid, so that no more than a few bubbles per second of gas emerge from the tube's end. Gas pressure inside all points of the tubing system will (very nearly) equal the hydrostatic pressure of the liquid at the tube's submerged end. Any pressure-measuring device tapped anywhere along the length of this tubing system will sense this pressure and be able to infer the depth of the liquid in the process vessel without having to directly contact the process liquid.

Bubbler-style liquid level measurement systems are especially useful when the process liquid in question is highly corrosive, prone to plugging sample ports, or in any other way objectionable to have in direct contact with a pressure sensor. Unlike pressure sensors which must use diaphragms or other flexible (usually metallic) sensing elements and therefore may only be constructed from a limited range of materials, a dip tube need not be flexible and therefore may be constructed of *any* material capable of withstanding the process liquid. A process liquid so corrosive that non-metallic vessels are required to hold it would preclude direct contact with a metal pressure gauge or pressure transmitter, but would be easily measured with a bubbler system provided the dip tube were made out of plastic, ceramic, or some other material immune to corrosion. A process liquid so laden with solids that it plugs up any non-flowing port would preclude pressure measurement via a sample port and impulse line, but would be easily measured by a bubbler system where the dip tube is continuously purged with clean gas. Level measurement applications where direct contact with the pressure sensor would render access to that sensor inconvenient or even impossible are made much more practical by using a bubbler system, where the pressure sensor may be located *anywhere* along the dip tube's length and therefore easily located where maintenance personnel can access it.

Excessive purge gas flow through the tube will result in additional pressure caused by frictional pressure drop along the tube's length, causing the pressure-sensing instrument to falsely register high. A key detail of any practical bubble tube system, therefore, is some means to monitor and control gas flow through the tube. A common construction uses either a *rotameter* or a *sightfeed bubbler* to monitor purge gas flow rate, with a needle valve to restrict that flow:



A more sophisticated solution to the problem of purge gas flow rate is to install a *flow-regulator* in lieu of a pressure regulator and needle valve, a mechanism designed to automatically monitor and throttle gas flow to maintain a constant purge rate. Flow regulators compensate for changes in dip tube pressure and in gas supply pressure, eliminating the need for a human operator to periodically adjust a needle valve.

Limiting the flow of purge gas is also important if that purge gas is expensive to obtain. For bottled gases such as nitrogen (necessary in processes requiring a non-reactive purge), the cost of purchasing tanks of compressed gas is obvious. For air-purged systems the cost is still present, but not so obvious: the cost of running an air compressor to maintain continuous purge air pressure. Either way, limiting the flow rate of purge gas in a bubbler system yields economic benefits aside from increased measurement accuracy.

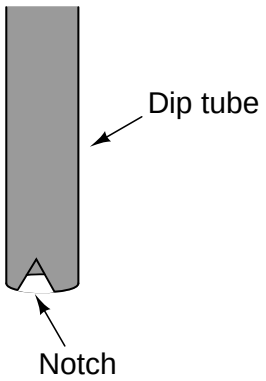
As with all purged systems, certain criteria must be met for successful operation. Listed here are some of them:

- The purge gas supply must be reliable: if the flow stops for any reason, the level measurement will cease to be accurate, and the dip tube may even plug with debris!
- The purge gas supply pressure must exceed the hydrostatic pressure at all times, or else the level measurement range will fall below the actual liquid level.
- The purge gas flow must be maintained at a low rate, to avoid pressure drop errors (i.e. excess pressure measured due to friction of the purge gas through the tube).
- The purge gas must not adversely react with the process.
- The purge gas must not contaminate the process.
- The purge gas must be reasonable in cost, since it will be continuously consumed over time.

One measurement artifact of a bubble tube system is a slight variation in pressure each time a new bubble breaks away from the end of the tube. The amount of pressure variation is approximately equal to the hydrostatic pressure of process fluid at a height equal to the diameter of the bubble, which in turn will be approximately equal to the diameter of the bubble tube. For example, a $\frac{1}{4}$ inch diameter dip tube will experience pressure oscillations with a peak-to-peak amplitude of approximately $\frac{1}{4}$ inch elevation of process liquid. The frequency of this pressure oscillation, of course, will be equal to the rate at which individual bubbles escape out the end of the dip tube.

Usually, this is a small variation when considered in the context of the measured liquid height in the vessel. A pressure oscillation of approximately $\frac{1}{4}$ inch compared to a measurement range of 0 to 10 feet, for example, is only about 0.2% of span. Modern pressure transmitters have the ability to "filter" or "damp" pressure variations over time, which is a useful feature for minimizing the effect such a pressure variation will have on system performance.

A way to help minimize this effect is to place small V-shaped notches at the end of the dip tube, to help bubbles escape at sizes smaller than the tube's diameter:

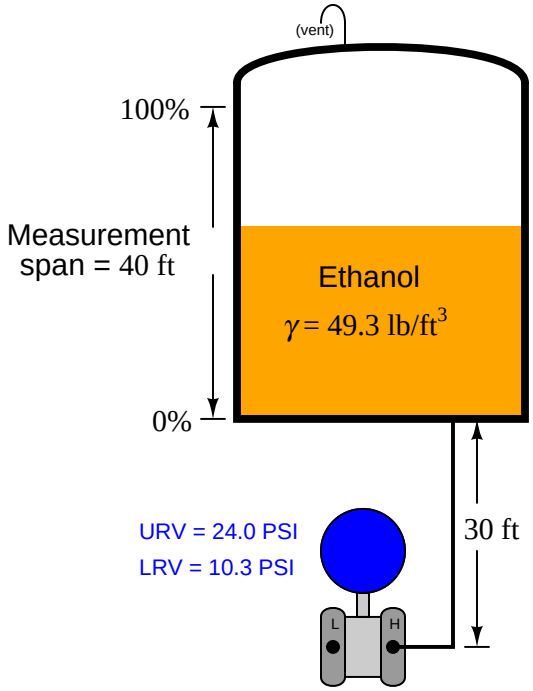


20.3.3 Transmitter suppression and elevation

A very common scenario for liquid level measurement is where the pressure-sensing instrument is not located at the same level as the 0% measurement point. The following photograph shows an example of this, where a Rosemount model 3051 differential pressure transmitter is being used to sense hydrostatic pressure of colored water inside a (clear) vertical plastic tube:



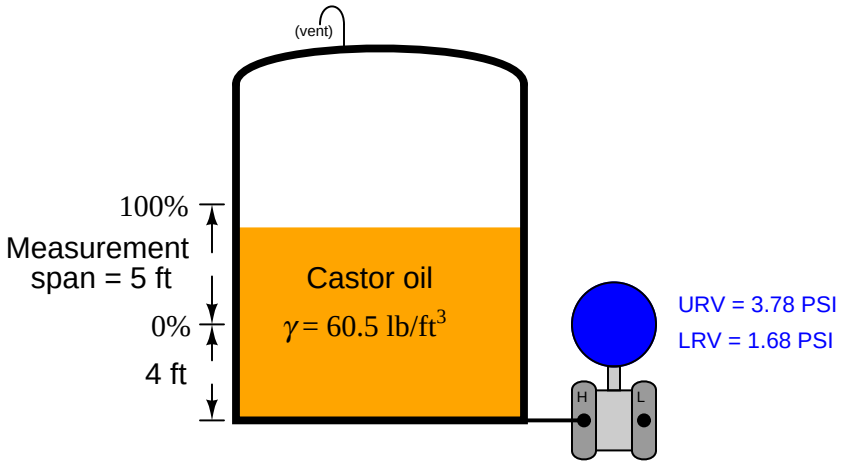
Consider the example of a pressure sensor measuring the level of liquid ethanol in a storage tank. The measurement range for liquid height in this ethanol storage tank is 0 to 40 feet, but the transmitter is located 30 feet below the tank:



This means the transmitter's impulse line contains a 30-foot elevation head of ethanol, so the transmitter "sees" 30 feet of ethanol when the tank is empty and 70 feet of ethanol when the tank is full. A 3-point calibration table for this instrument would look like this, assuming a 4 to 20 mA DC output signal range:

Ethanol level in tank	**Percent of range**	**Pressure** (inches of water)	**Pressure** (PSI)	**Output** (mA)
0 ft	0 %	284 "W.C.	10.3 PSI	4 mA
20 ft	50 %	474 "W.C.	17.1 PSI	12 mA
40 ft	100 %	663 "W.C.	24.0 PSI	20 mA

Another common scenario is where the transmitter is mounted at or near the vessel's bottom, but the desired level measurement range does not extend to the vessel bottom:



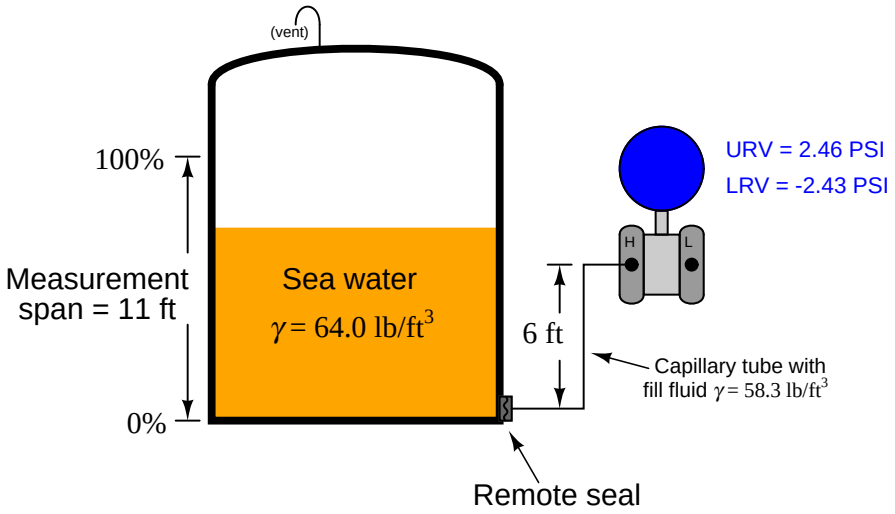
In this example, the transmitter is mounted exactly at the same level as the vessel bottom, but the level measurement range begins at 4 feet up from the vessel bottom. At the level of castor oil deemed 0%, the transmitter "sees" a hydrostatic pressure of 1.68 PSI (46.5 inches of water column) and at the 100% castor oil level the transmitter "sees" a pressure of 3.78 PSI (105 inches water column). Thus, these two pressure values would define the transmitter's lower and upper range values (LRV and URV), respectively.

The term for describing either of the previous scenarios, where the lower range value (LRV) of the transmitter's calibration is a positive number, is called *zero suppression*[4]. If the zero offset is reversed (e.g. the transmitter mounted at a location *higher* than the 0% process level), it is referred to as *zero elevation*[5].

[4] Or alternatively, zero *depression*.

[5] There is some disagreement among instrumentation professionals as to the definitions of these two terms. According to Béla G. Lipták's *Instrument Engineers' Handbook, Process Measurement and Analysis* (Fourth Edition, page 67), "suppressed zero range" refers to the transmitter being located below the 0% level (the LRV being a positive pressure value), while "suppression," "suppressed range," and "suppressed span" mean exactly the opposite (LRV is a negative value). The Yokogawa Corporation defines "suppression" as a condition where the LRV is a positive pressure ("Autolevel" Application Note), as does the Michael MacBeth in his CANDU Instrumentation & Control course (lesson 1, module 4, page 12), Foxboro's technical notes on bubble tube installations (pages 4 through 7), and Rosemount's product manual for their 1151 Alphaline pressure transmitter (page 3-7). Interestingly, the Rosemount document defines "zero range suppression" as synonymous with "suppression," which disagrees with Lipták's distinction. My advice: draw a picture if you want the other person to clearly understand what you mean!

If the transmitter is elevated above the process connection point, it will most likely "see" a negative pressure (vacuum) with an empty vessel owing to the pull of liquid in the line leading down from the instrument to the vessel. It is vitally important in elevated transmitter installations to use a *remote seal* rather than an open impulse line, so liquid cannot dribble out of this line and into the vessel[6]:



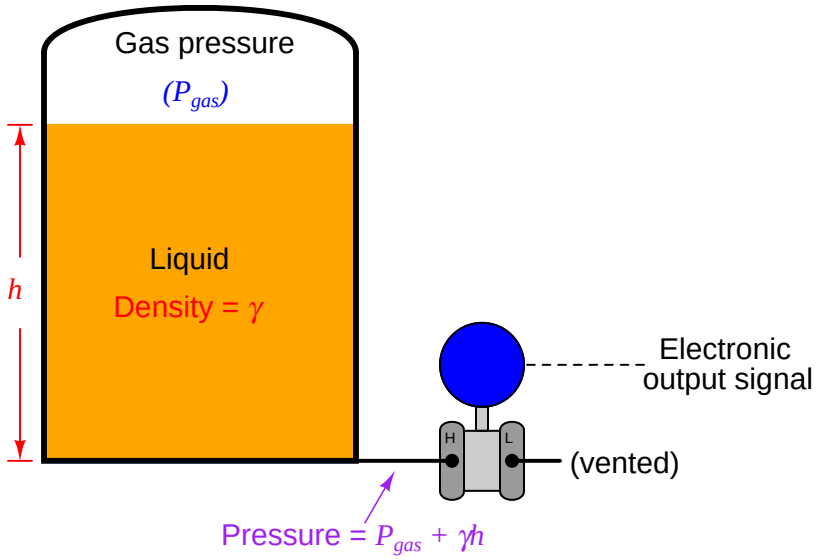
In this example, we see a remote seal system with a fill fluid having a density of 58.3 lb/ft^3, and a process level measurement range of 0 to 11 feet of sea water (density = 64 lb/ft^3). The transmitter elevation is 6 feet, which means it will "see" a vacuum of −2.43 PSI (−67.2 inches of water column) when the vessel is completely empty. This, of course, will be the transmitter's calibrated lower range value (LRV). The upper range value (URV) will be the pressure "seen" with 11 feet of sea water in the vessel. This much sea water will contribute an additional 4.89 PSI of hydrostatic pressure at the level of the remote seal diaphragm, causing the transmitter to experience a pressure of +2.46 PSI[7].

[6] As you are about to see, the calibration of an elevated transmitter depends on us knowing how much hydrostatic pressure (or vacuum, in this case) is generated within the tube connecting the transmitter to the process vessel. If liquid were to ever escape from this tube, the hydrostatic pressure would be unpredictable, and so would be the accuracy of our transmitter as a level-measuring instrument. A remote seal diaphragm guarantees no fill fluid will be lost if and when the process vessel goes empty.

[7] The sea water's positive pressure at the remote seal diaphragm adds to the negative pressure already generated by the downward length of the capillary tube's fill fluid (−2.43 PSI), which explains why the transmitter only "sees" 2.46 PSI of pressure at the 100% full mark.

20.3.4 Compensated leg systems

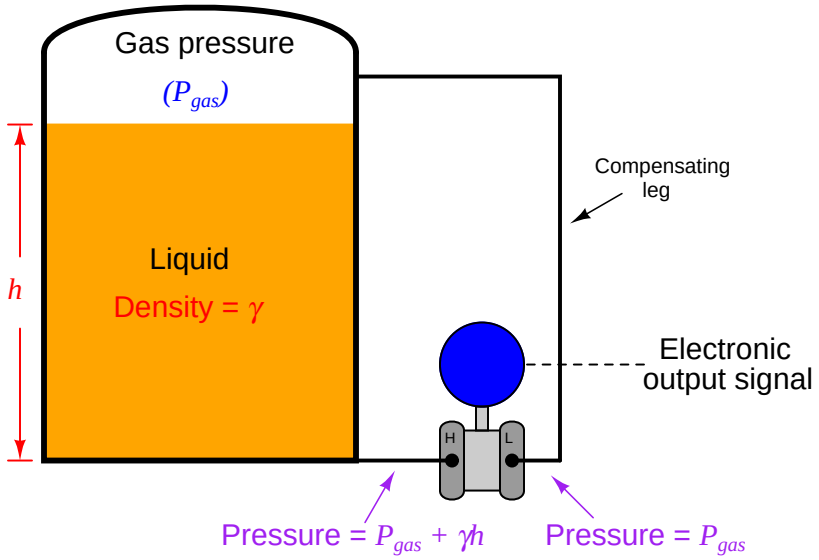
The simple and direct relationship between liquid height in a vessel and pressure at the bottom of that vessel is ruined if another source of pressure exists inside the vessel other than hydrostatic (elevation head). This is virtually guaranteed to be the case if the vessel in question is unvented. Any gas or vapor pressure accumulation in an enclosed vessel will add to the hydrostatic pressure at the bottom, causing any pressure-sensing instrument to falsely register a high level:



A pressure transmitter has no way of "knowing" how much of the sensed pressure is due to liquid elevation and how much of it is due to pressure existing in the vapor space above the liquid. Unless a way can be found to compensate for any non-hydrostatic pressure in the vessel, this extra pressure will be interpreted by the transmitter as additional liquid level.

Moreover, this error will change as gas pressure inside the vessel changes, so it cannot simply be "calibrated away" by a static zero shift within the instrument. The only way to hydrostatically measure liquid level inside an enclosed (non-vented) vessel is to continuously compensate for gas pressure.

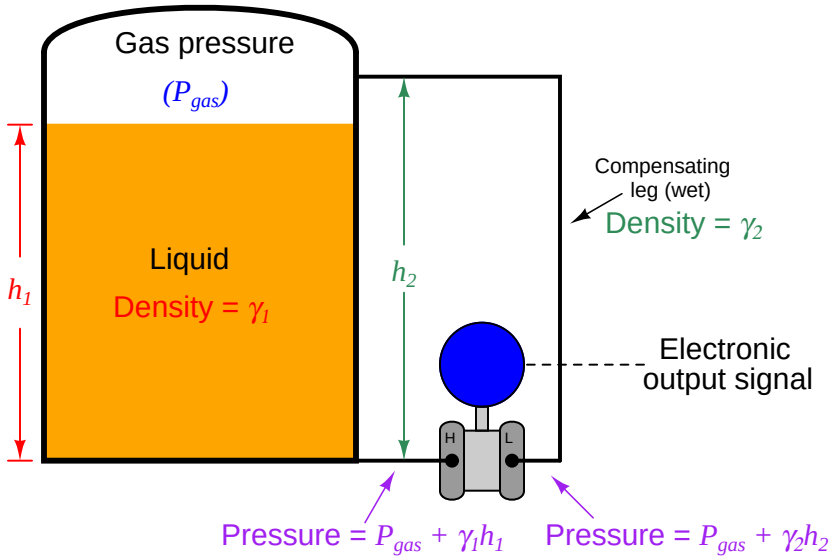
Fortunately, the capabilities of a *differential* pressure transmitter make this a simple task. All we need to do is connect a second impulse line (called a *compensating leg*), from the "Low" port of the transmitter to the top of the vessel, so the "Low" side of the transmitter experiences nothing but the gas pressure enclosed by the vessel, while the "High" side experiences the *sum* of gas and hydrostatic pressures. Since a differential pressure transmitter responds only to *differences* in pressure between "High" and "Low" sides, it will naturally subtract the gas pressure (P_{gas}) to yield a measurement based solely on hydrostatic pressure (γh):



$$(P_{gas} + \gamma h) - P_{gas} = \gamma h$$

The amount of gas pressure inside the vessel now becomes completely irrelevant to the transmitter's indication, because its effect is canceled at the differential pressure instrument's sensing element. If gas pressure inside the vessel were to increase while liquid level remained constant, the pressure sensed at *both* ports of the differential pressure transmitter would increase by the exact same amount, with the pressure *difference* between the "high" and "low" ports remaining absolutely constant with the constant liquid level. This means the instrument's output signal is a representation of hydrostatic pressure only, which represents liquid height (assuming a known liquid density γ).

Unfortunately, it is common for enclosed vessels to hold condensible vapors, which may over time fill a compensating leg full of liquid. If the tube connecting the "Low" side of a differential pressure transmitter fills completely with a liquid, this will add a hydrostatic pressure to that side of the transmitter, causing another calibration shift. This *wet leg* condition makes level measurement more complicated than a *dry leg* condition where the only pressure sensed by the transmitter's "Low" side is gas pressure (P_{gas}):

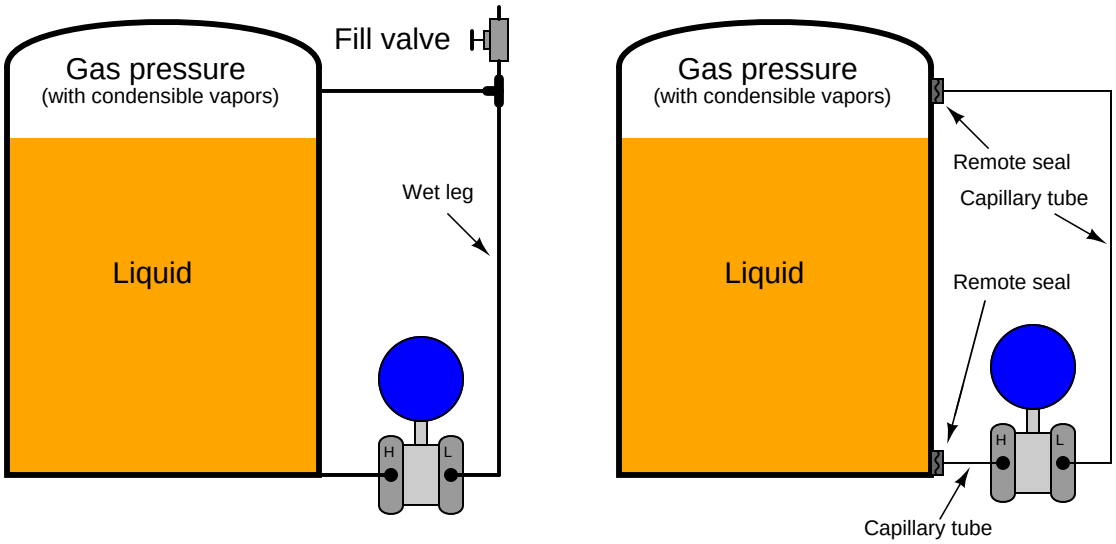


$$(P_{gas} + \gamma_1 h_1) - (P_{gas} + \gamma_2 h_2) = \gamma_1 h_1 - \gamma_2 h_2$$

Gas pressure still cancels due to the differential nature of the pressure transmitter, but now the transmitter's output indicates a difference of hydrostatic pressures between the vessel and the wet leg, rather than just the hydrostatic pressure of the vessel's liquid level. Fortunately, the hydrostatic pressure generated by the wet leg will be constant, so long as the density of the condensed vapors filling that leg (γ_2) is constant. If the wet leg's hydrostatic pressure is constant, we can compensate for it by calibrating the transmitter with an intentional zero shift, so it indicates as though it were measuring hydrostatic pressure on a vented vessel.

$$\text{Differential pressure} = \gamma_1 h_1 - \text{Constant}$$

We may ensure a constant density of wet leg liquid by intentionally filling that leg with a liquid known to be denser than the densest condensed vapor inside the vessel and non-miscible with the process fluid. We could also use a differential pressure transmitter with remote seals and capillary tubes filled with liquid of known density:



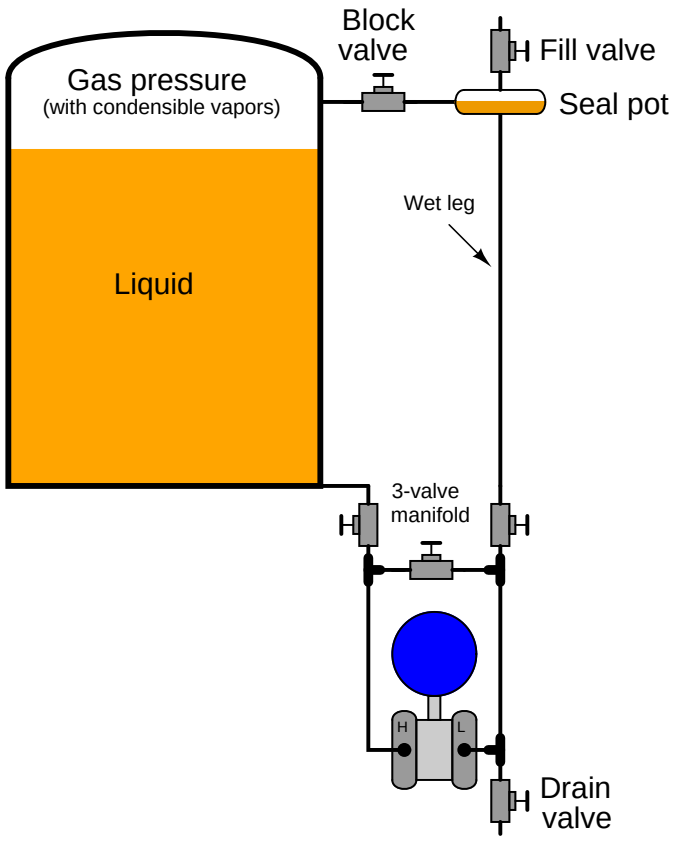
Remote seals are very useful in applications such as this, as the wet leg never requires re-filling.

An actual level transmitter installation using two remote seals (in this case, a Foxboro IDP10 differential pressure transmitter) appears in this photograph:



The vessel itself is insulated, and covered in sheet aluminum to protect the thermal insulation from impact and weather-related damage. White-painted flanged "nozzles" protrude from the vessel through the insulation to provide places for the level-sensing instrument to connect. You can see the two flanged remote seals (painted blue) where the armored (stainless steel) capillary tubes terminate. Note how the long capillary tube on the "wet" leg is neatly coiled to reduce the possibility of damage by snagging on any moving object.

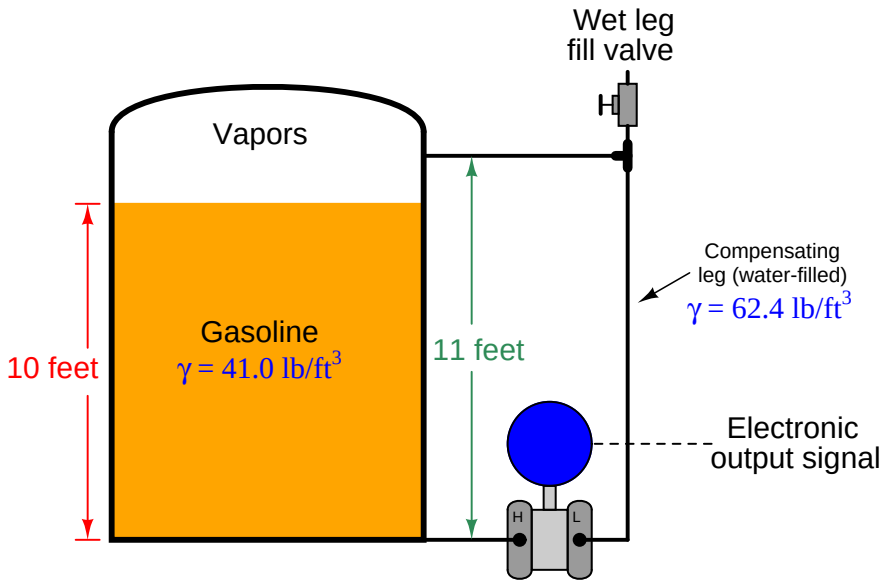
An accessory commonly used with non-sealed (non-capillary) "wet leg" systems is a *seal pot*. This is a chamber at the top of the wet leg joining the wet leg line with the impulse line to the upper connection point on the process vessel. This "seal pot" maintains a small volume of liquid in it to allow for occasional liquid loss during transmitter maintenance procedures without greatly affecting the height of the liquid column in the wet leg:



Regular operation of the transmitter's three-valve manifold (and drain valve) during routine instrument maintenance inevitably releases some liquid volume from the wet leg. Without a seal pot, even a small loss of liquid in the wet leg may create a substantial loss in liquid column height within that tube, given the tube's small diameter. With a seal pot, the (comparatively) large liquid volume held by the pot allows for some liquid loss through the transmitter's manifold without substantially affecting the height of the liquid column within the wet leg.

Seal pots are standard on level measurement systems for boiler steam drums, where steam readily condenses in the upper impulse tube to naturally form a wet leg. Although steam will condense over time to refill the wet leg following a loss of water in that leg, the level measurements taken during that re-fill time will be in error. The presence of a seal pot practically eliminates this error as the steam condenses to replenish the water lost from the pot, since the amount of height change inside the pot due to a small volume loss is trivial compared to the height change in a wet leg lacking a seal pot.

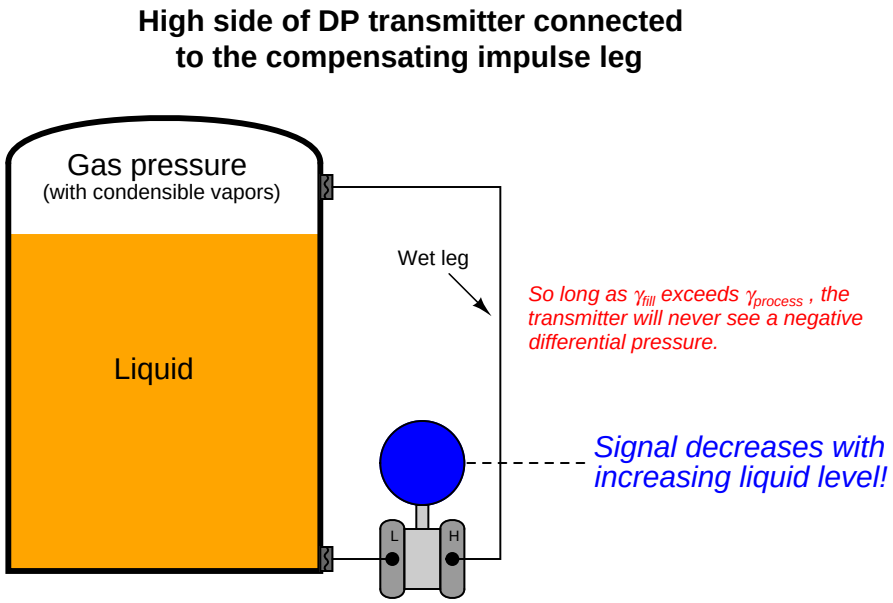
The following example shows the calibration table for a compensated-leg (wet) hydrostatic level measurement system, for a gasoline storage vessel and water as the wet leg fill fluid. Here, I am assuming a density of 41.0 lb/ft^3 for gasoline and 62.4 lb/ft^3 for water, with a 0 to 10 foot measurement range and an 11 foot wet leg height:



Gasoline level	Percent of range	Differential pressure at transmitter	Transmitter output
0 ft	0 %	−4.77 PSI	4 mA
2.5 ft	25 %	−4.05 PSI	8 mA
5 ft	50 %	−3.34 PSI	12 mA
7.5 ft	75 %	−2.63 PSI	16 mA
10 ft	100 %	−1.92 PSI	20 mA

Note that due to the superior density and height of the wet (water) leg, the transmitter *always* sees a negative pressure (pressure on the "Low" side exceeds pressure on the "High" side).

With some older differential pressure transmitter designs, this negative pressure was a problem. Consequently, it is common to see "wet leg" hydrostatic transmitters installed with the "Low" port connected to the bottom of the vessel and the "High" port connected to the compensating leg. In fact, it is *still* common to see modern differential pressure transmitters installed in this manner[8], although modern transmitters may be ranged for negative pressures just as easily as for positive pressures. It is vitally important to recognize that any differential pressure transmitter connected as such (for any reason) will respond in reverse fashion to increases in liquid level. That is to say, as the liquid level in the vessel rises, the transmitter's output signal will *decrease* instead of increase:

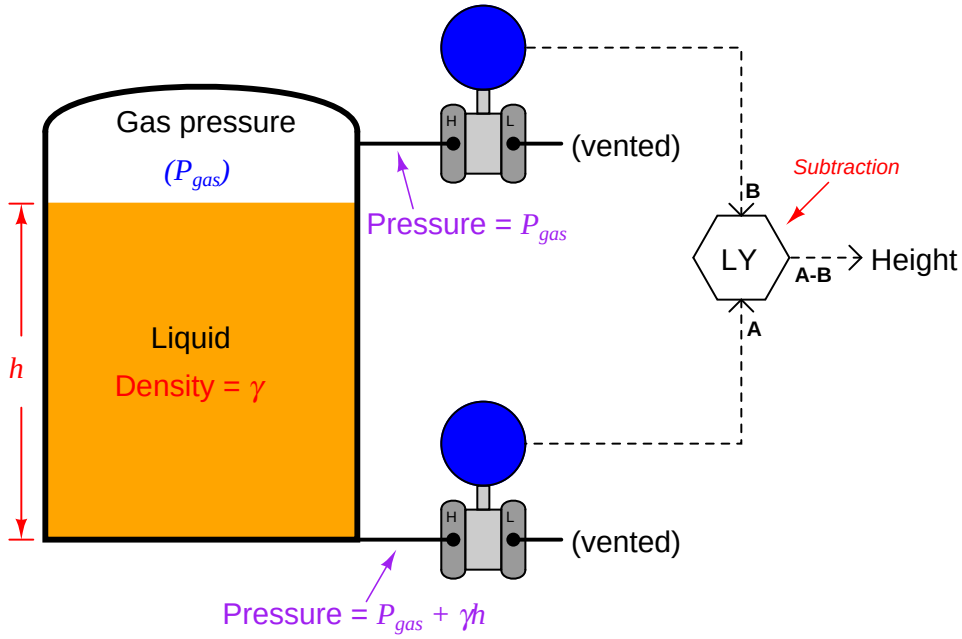


Either way of connecting the transmitter to the vessel will suffice for measuring liquid level, so long as the instrumentation receiving the transmitter's signal is properly configured to interpret the signal. The choice of which way to connect the transmitter to the vessel should be driven by fail-safe system design, which means to design the measurement system such that the most probable system failures – including broken signal wires – result in the control system "seeing" the most dangerous process condition and therefore taking the safest action.

[8] Sometimes this is done out of habit, other times because instrument technicians do not know the capabilities of new technology.

20.3.5 Tank expert systems

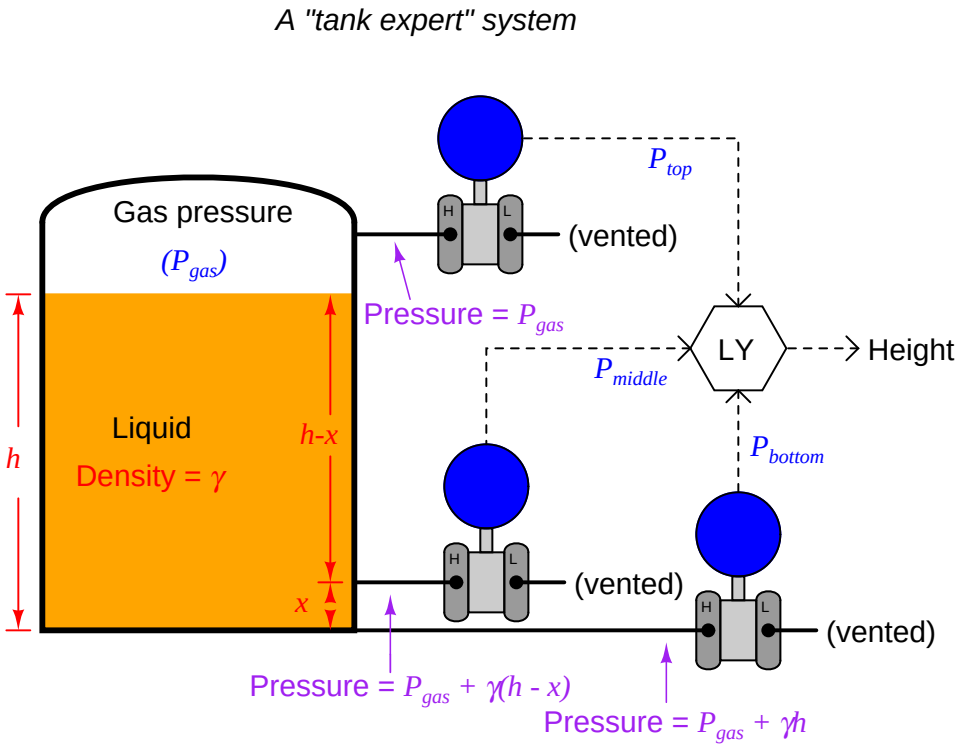
An alternative to using a compensating leg to subtract gas pressure inside an enclosed vessel is to simply use a second pressure transmitter and electronically subtract the two pressures in a computing device:



This approach enjoys the distinct advantage of avoiding a potentially wet compensating leg, but suffers the disadvantages of extra cost and greater error due to the potential calibration drift of *two* transmitters rather than just one. Such a system is also impractical in applications where the gas pressure is substantial compared to the hydrostatic (elevation head) pressure[9].

[9] This is due to limited transmitter resolution. Imagine an application where the elevation head was 10 PSI (maximum) yet the vapor space pressure was 200 PSI. The majority of each transmitter's working range would be "consumed" measuring gas pressure, with hydrostatic head being a mere 5% of the measurement range. This would make precise measurement of liquid level very difficult, akin to trying to measure the sound intensity of a whisper in a noisy room.

If we add a third pressure transmitter to this system, located a known distance (x) above the bottom transmitter, we have all the pieces necessary for what is called a *tank expert system*. These systems are used on large storage tanks operating at or near atmospheric pressure, and have the ability to measure infer liquid height, liquid density, total liquid volume, and total liquid mass stored in the tank:



The pressure difference between the bottom and middle transmitters will change only if the liquid density changes[10], since the two transmitters are separated by a known and fixed height difference.

[10] Assuming the liquid level is equal to or greater than x. Otherwise, the pressure difference between P_{bottom} and P_{middle} will depend on liquid density *and* liquid height. However, this condition is easy to check: the level computer simply checks to see if P_{middle} and P_{top} are unequal. If so, then the computer knows the liquid level exceeds x and it is safe to calculate density. If not, and P_{middle} registers the same as P_{top}, the computer knows those two transmitters are both registering gas pressure only, and it knows to stop calculating density.

Algebraic manipulation shows us how the measured pressures may be used by the level computer (LY) to continuously calculate liquid density (γ):

$$P_{bottom} - P_{middle} = (P_{gas} + \gamma h) - [P_{gas} + \gamma(h - x)]$$

$$P_{bottom} - P_{middle} = P_{gas} + \gamma h - P_{gas} - \gamma(h - x)$$

$$P_{bottom} - P_{middle} = P_{gas} + \gamma h - P_{gas} - \gamma h + \gamma x$$

$$P_{bottom} - P_{middle} = \gamma x$$

$$\frac{P_{bottom} - P_{middle}}{x} = \gamma$$

Once the computer knows the value of γ, it may calculate the height of liquid in the tank with great accuracy based on the pressure measurements taken by the bottom and top transmitters:

$$P_{bottom} - P_{top} = (P_{gas} + \gamma h) - P_{gas}$$

$$P_{bottom} - P_{top} = \gamma h$$

$$\frac{P_{bottom} - P_{top}}{\gamma} = h$$

With all the computing power available in the LY, it is possible to characterize the tank such that this height measurement converts to a precise volume measurement[11] (V), which may then be converted into a total mass (m) measurement based on the mass density of the liquid (ρ) and the acceleration of gravity (g). First, the computer calculates mass density based on the proportionality between mass and weight (shown here starting with the equivalence between the two forms of the hydrostatic pressure formula):

$$\rho g h = \gamma h$$

$$\rho g = \gamma$$

$$\rho = \frac{\gamma}{g}$$

[11] The details of this math depend entirely on the shape of the tank. For vertical cylinders – the most common shape for vented storage tanks – volume and height are related by the simple formula $V = \pi r^2 h$ where r is the radius of the tank's circular base. Other tank shapes and orientations may require much more sophisticated formulae to calculate stored volume from height. See section 26.3 beginning on page 2069, for more details on this subject.

Armed with the mass density of the liquid inside the tank, the computer may now calculate total liquid mass stored inside the tank:

$$m = \rho V$$

Dimensional analysis shows how units of mass density and volume cancel to yield only units of mass in this last equation:

$$[\text{kg}] = \left[\frac{\text{kg}}{\text{m}^3}\right] \left[\text{m}^3\right]$$

Here we see a vivid example of how several measurements may be inferred from just a few actual process (in this case, pressure) measurements. Three pressure measurements on this tank allow us to compute four inferred variables: liquid density, liquid height, liquid volume, and liquid mass.

The accurate measurement of liquids in storage tanks is not just useful for process operations, but also for conducting business. Whether the liquid represents raw material purchased from a supplier, or a processed product ready to be pumped out to a customer, both parties have a vested interest in knowing the exact quantity of liquid bought or sold. Measurement applications such as this are known as *custody transfer*, because they represent the transfer of custody (ownership) of a substance exchanged in a business agreement. In some instances, both buyer and seller operate and maintain their own custody transfer instrumentation, while in other instances there is only one instrument, the calibration of which validated by a neutral party.

A photograph showing the two lower pressure transmitters of a tank expert system on a refrigerated butane storage vessel appears here:

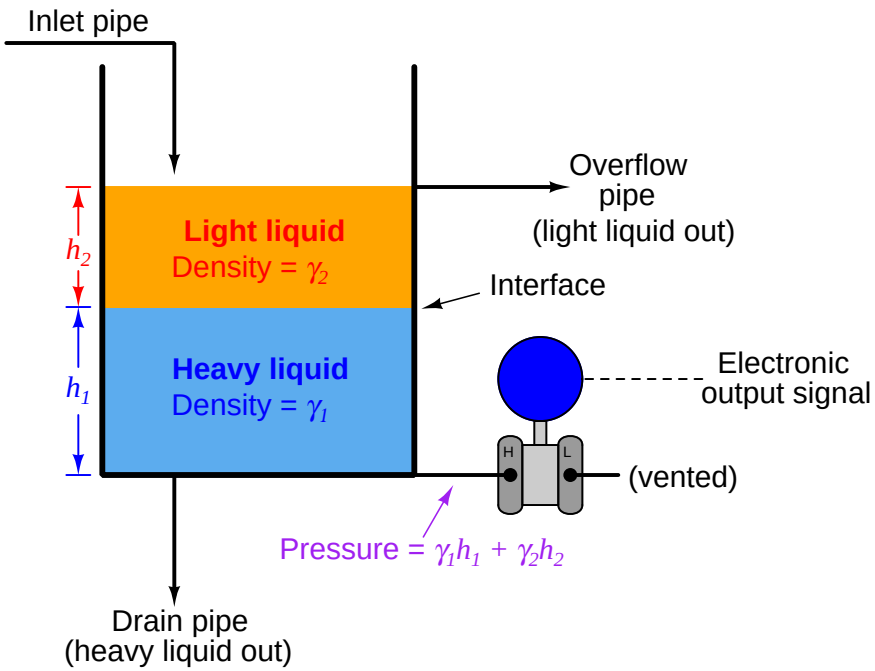


The upper and lower instruments are pressure transmitters, while the middle instrument is a temperature sensor used to report the temperature of the refrigerated butane to the control system.

20.3.6 Hydrostatic interface level measurement

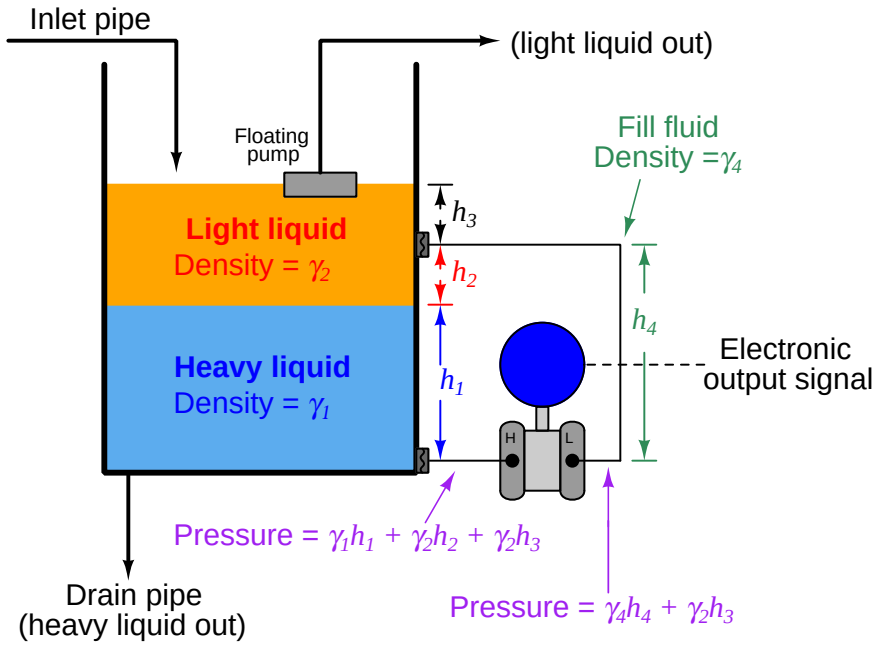
Hydrostatic pressure sensors may be used to detect the level of a liquid-liquid interface, if and only if the total height of liquid never drops down into the sensing range of the interface level instrument. This is critically important because any single hydrostatic-based level instrument cannot discriminate between a changing interface level and a changing total level within the same range, and therefore the latter must be fixed in order to measure the former.

One way to ensure this condition is to fix the total liquid height to some constant value by using an overflow pipe, and ensuring drain flow is always less than incoming flow (forcing some flow to always go through the overflow pipe). This strategy naturally lends itself to separation processes, where a mixture of light and heavy liquids are separated by their differing densities:



Here we see a practical application for liquid-liquid interface level measurement. If the goal is to separate two liquids of differing densities from one another, we need only the light liquid to exit out the overflow pipe and only the heavy liquid to exit out the drain pipe. This means we must control the interface level to stay between those two piping points on the vessel. If the interface drifts too far up, heavy liquid will be carried out the overflow pipe; and if we let the interface drift too far down, light liquid will flow out of the drain pipe. The first step in controlling any process variable is to measure that variable, and so here we are faced with the necessity of measuring the interface point between the light and heavy liquids.

Another way to address the problem of varying total liquid height is to use a compensating leg located at a point on the vessel below the total liquid height. In this example, a transmitter with remote seals is used:



Since both sides of the differential pressure transmitter "see" the hydrostatic pressure generated by the liquid column above the top connection point ($\gamma_2 h_3$), this term naturally cancels. The result of this is that total liquid height becomes irrelevant, so long as it remains above the upper connection point:

$$(\gamma_1 h_1 + \gamma_2 h_2 + \gamma_2 h_3) - (\gamma_4 h_4 + \gamma_2 h_3)$$

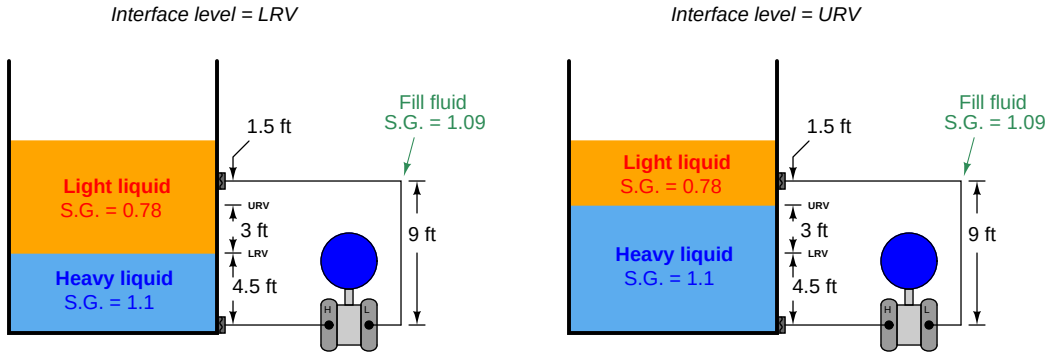
$$\gamma_1 h_1 + \gamma_2 h_2 + \gamma_2 h_3 - \gamma_4 h_4 - \gamma_2 h_3$$

$$\gamma_1 h_1 + \gamma_2 h_2 - \gamma_4 h_4$$

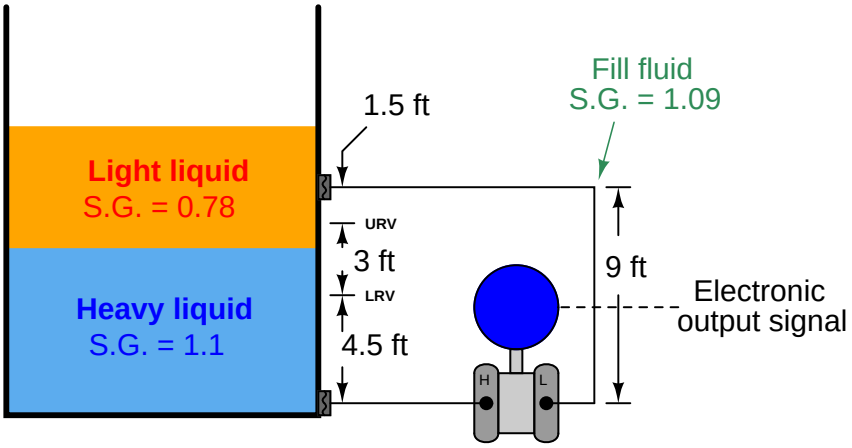
The hydrostatic pressure in the compensating leg is constant ($\gamma_4 h_4 = \text{Constant}$), since the fill fluid never changes density and the compensating leg's height never changes. This means the transmitter's sensed pressure will differ from that of an uncompensated transmitter merely by a constant offset, which may be "calibrated out" so as to have no impact on the measurement:

$$\gamma_1 h_1 + \gamma_2 h_2 - \text{Constant}$$

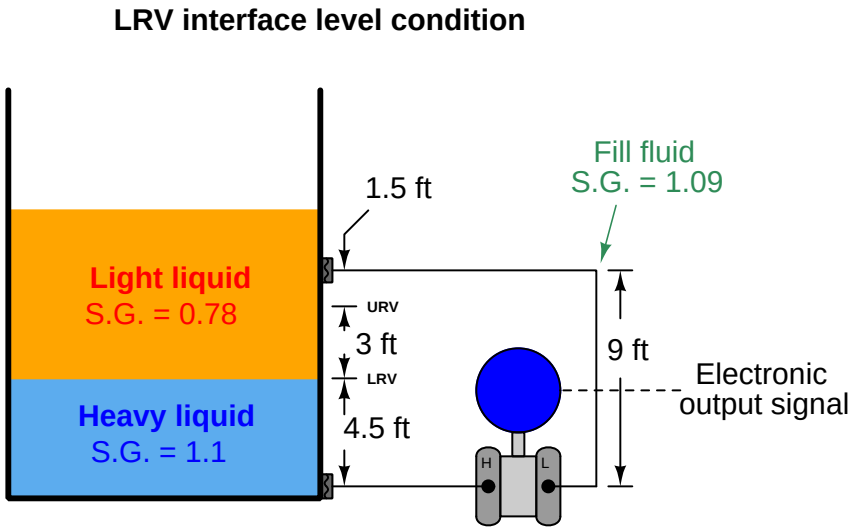
At first, it may seem as though determining the calibration points (lower- and upper-range values: LRV and URV) for a hydrostatic interface level transmitter is impossibly daunting given all the different pressures involved. A recommended problem-solving technique to apply here is that of a *thought experiment*, where we imagine what the process will "look like" at lower-range value condition and at the upper-range value condition, drawing two separate illustrations:



For example, suppose we must calibrate a differential pressure transmitter to measure the interface level between two liquids having specific gravities of 1.1 and 0.78, respectively, over a span of 3 feet. The transmitter is equipped with remote seals, each containing a halocarbon fill fluid with a specific gravity of 1.09. The physical layout of the system is as follows:



As the first step in our "thought experiment," we imagine what the process would look like with the interface at the LRV level, calculating hydrostatic pressures seen at each side of the transmitter. For this I recommend actually *sketching* a simple diagram showing what the fluid levels would look like with the interface at the LRV point, so the height dimensions of each liquid layer become obvious:



We know from our previous exploration of this setup that any hydrostatic pressure resulting from liquid level *above* the top remote seal location is irrelevant to the transmitter, since it is "seen" on both sides of the transmitter and thus cancels out. All we must do, then, is calculate hydrostatic pressures as though the total liquid level stopped at that upper diaphragm connection point.

First, calculating the hydrostatic pressure "seen" at the high port of the transmitter[12]:

$$P_{high} = 4.5 \text{ feet of heavy liquid} + 4.5 \text{ feet of light liquid}$$

$$P_{high} = 54 \text{ inches of heavy liquid} + 54 \text{ inches of light liquid}$$

$$P_{high} \text{ "W.C.} = (54 \text{ inches of heavy liquid})(1.1) + (54 \text{ inches of light liquid})(0.78)$$

$$P_{high} \text{ "W.C.} = 59.4 \text{ "W.C.} + 42.12 \text{ "W.C.}$$

$$P_{high} = 101.52 \text{ "W.C.}$$

Next, calculating the hydrostatic pressure "seen" at the low port of the transmitter:

$$P_{low} = 9 \text{ feet of fill fluid}$$

$$P_{low} = 108 \text{ inches of fill fluid}$$

$$P_{low} \text{ "W.C.} = (108 \text{ inches of fill fluid})(1.09)$$

$$P_{low} = 117.72 \text{ "W.C.}$$

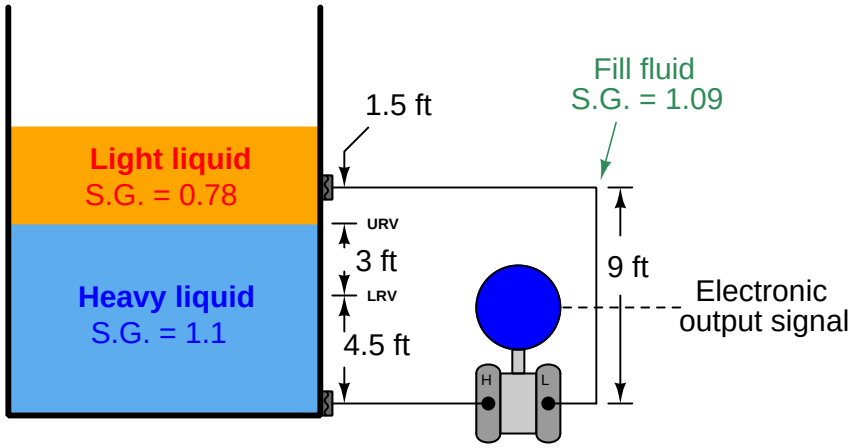
The differential pressure applied to the transmitter in this condition is the difference between the high and low port pressures, which becomes the lower range value (LRV) for calibration:

$$P_{LRV} = 101.52 \text{ "W.C.} - 117.72 \text{ "W.C.} = -16.2 \text{ "W.C.}$$

[12]Here I will calculate all hydrostatic pressures in units of inches water column. This is relatively easy because we have been given the specific gravities of each liquid, which make it easy to translate actual liquid column height into column heights of pure water.

As the second step in our "thought experiment," we imagine what the process would look like with the interface at the URV level, calculating hydrostatic pressures seen at each side of the transmitter. Again, sketching a simple diagram of what the liquid layers would look like in this condition helps us visualize the scenario:

URV interface level condition



$$P_{high} = 7.5 \text{ feet of heavy liquid} + 1.5 \text{ feet of light liquid}$$

$$P_{high} = 90 \text{ inches of heavy liquid} + 18 \text{ inches of light liquid}$$

$$P_{high} \text{ "W.C.} = (90 \text{ inches of heavy liquid})(1.1) + (18 \text{ inches of light liquid})(0.78)$$

$$P_{high} \text{ "W.C.} = 99 \text{ "W.C.} + 14.04 \text{ "W.C.}$$

$$P_{high} = 113.04 \text{ "W.C.}$$

The hydrostatic pressure of the compensating leg is exactly the same as it was before: 9 feet of fill fluid having a specific gravity of 1.09, which means there is no need to calculate it again. It will still be 117.72 inches of water column. Thus, the differential pressure at the URV point is:

$$P_{URV} = 113.04 \text{ "W.C.} - 117.72 \text{ "W.C.} = -4.68 \text{ "W.C.}$$

Using these two pressure values and some interpolation, we may generate a 5-point calibration table (assuming a 4-20 mA transmitter output signal range) for this interface level measurement system:

Interface level	Percent of range	Differential pressure at transmitter	Transmitter output
4.5 ft	0 %	−16.2 "W.C.	4 mA
5.25 ft	25 %	−13.32 "W.C.	8 mA
6 ft	50 %	−10.44 "W.C.	12 mA
6.75 ft	75 %	−7.56 "W.C.	16 mA
7.5 ft	100 %	−4.68 "W.C.	20 mA

When the time comes to bench-calibrate this instrument in the shop, the easiest way to do so will be to set the two remote diaphragms on the workbench (at the same level), then apply 16.2 to 4.68 inches of water column pressure to the *low* remote seal diaphragm with the other diaphragm at atmospheric pressure to simulate the desired range of negative differential pressures[13].

The more mathematically inclined reader will notice that the span of this instrument (Span = URV − LRV) is equal to the span of the interface level (3 feet, or 36 inches) multiplied by the difference in specific gravities (1.1 − 0.78):

$$\text{Span in "W.C.} = (36 \text{ inches})(1.1 - 0.78)$$

$$\text{Span} = 11.52 \text{ "W.C.}$$

Looking at our two "thought experiment" illustrations, we see that the only difference between the two scenarios is the type of liquid filling that 3-foot region between the LRV and URV marks. Therefore, the only difference between the transmitter's pressures in those two conditions will be the difference in height multiplied by the difference in density. Not only is this an easy way for us to quickly calculate the necessary transmitter span, but it also is a way for us to check our previous work: we see that the difference between the LRV and URV pressures is indeed a difference of 11.52 inches water column just as this method predicts.

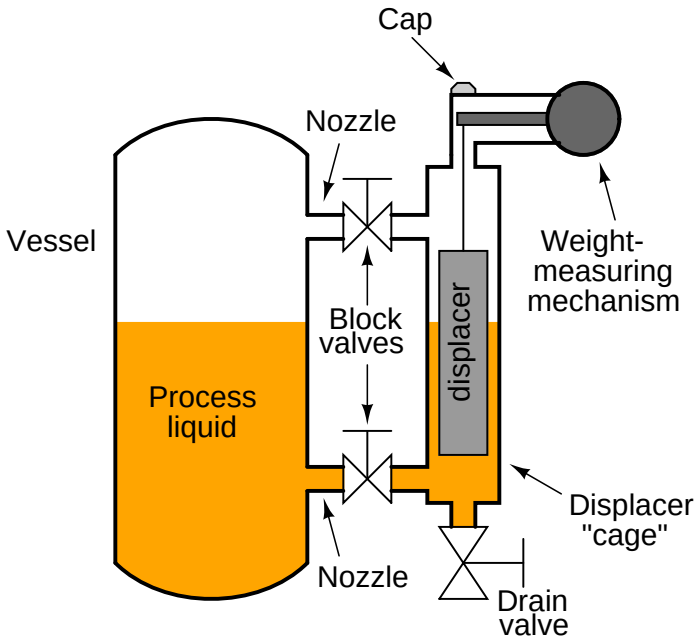
20.4 Displacement

Displacer level instruments exploit *Archimedes' Principle* to detect liquid level by continuously measuring the weight of an object (called the *displacer*) immersed in the process liquid. As liquid level increases, the displacer experiences a greater buoyant force, making it appear lighter to the sensing instrument, which interprets the loss of weight as an increase in level and transmits a proportional output signal.

[13] Remember that a differential pressure instrument cannot "tell the difference" between a positive pressure applied to the low side, an equal vacuum applied to the high side, or an equivalent difference of two positive pressures with the low side's pressure exceeding the high side's pressure. Simulating the exact process pressures experienced in the field to a transmitter on a workbench would be exceedingly complicated, so we "cheat" by simplifying the calibration setup and applying the equivalent difference of pressure only to the "low" side.

20.4.1 Buoyant-force instruments

In practice a displacer level instrument usually takes the following form. Process piping in and out of the vessel has been omitted for simplicity – only the vessel and its displacer level instrument are shown:



The displacer itself is usually a sealed metal tube, weighted sufficiently so it cannot float in the process liquid. It hangs within a pipe called a "cage" connected to the process vessel through two block valves and nozzles. These two pipe connections ensure the liquid level inside the cage matches the liquid level inside the process vessel, much like a sightglass.

If liquid level inside the process vessel rises, the liquid level inside the cage rises to match. This will submerge more of the displacer's volume, causing a buoyant force to be exerted upward on the displacer. Remember that the displacer is too heavy to float, so it does not "bob" on the surface of the liquid nor does it rise the same amount as the liquid's level – rather, it hangs in place inside the cage, becoming "lighter[14]" as the buoyant force increases. The weight-sensing mechanism detects this buoyant force when it perceives the displacer becoming lighter, interpreting the decreased (apparent) weight as an increase in liquid level. The displacer's apparent weight reaches a minimum when it is fully submerged, when the process liquid has reached the 100% point inside the cage.

It should be noted that static pressure inside the vessel will have negligible effect on a displacer instrument's accuracy. The only factor that matters is the density of the process fluid, since buoyant force is directly proportional to fluid density ($F = \gamma V$).

[14] This is not unlike the experience of feeling lighter when you are standing in a pool of water just deep enough to submerge most of your body with your feet touching the bottom. This reduction of apparent weight is due to the buoyant force of the water upward on your body, equal to the weight of water that your body displaces.

The following photograph shows a Fisher "Level-Trol" model pneumatic transmitter measuring condensate level in a *knockout drum*[15] for natural gas service. The instrument itself appears on the right-hand side of the photo, topped by a grey-colored "head" with two pneumatic pressure gauges visible. The displacer "cage" is the vertical pipe immediately behind and below the head unit. Note that a sightglass level gauge appears on the left-hand side of the knockout chamber (or *condensate boot*) for visual indication of condensate level inside the process vessel:



The purpose of this particular displacer instrument is to measure the amount of condensate liquid collected inside the "boot." This model of Fisher Level-Trol comes complete with a pneumatic controller mechanism sending an air pressure signal to a drain valve to automatically drain the condensate out of the boot.

[15] So-called for its ability to "knock out" (separate and collect) condensible vapors from the gas stream. This particular photograph was taken at a natural gas compression facility, where it is very important the gas to be compressed is dry (since liquids are essentially incompressible). Sending even relatively small amounts of liquid into a compressor may cause the compressor to catastrophically fail!

Two photos of a disassembled Level-Trol displacer instrument appear here, showing how the displacer fits inside the cage pipe:

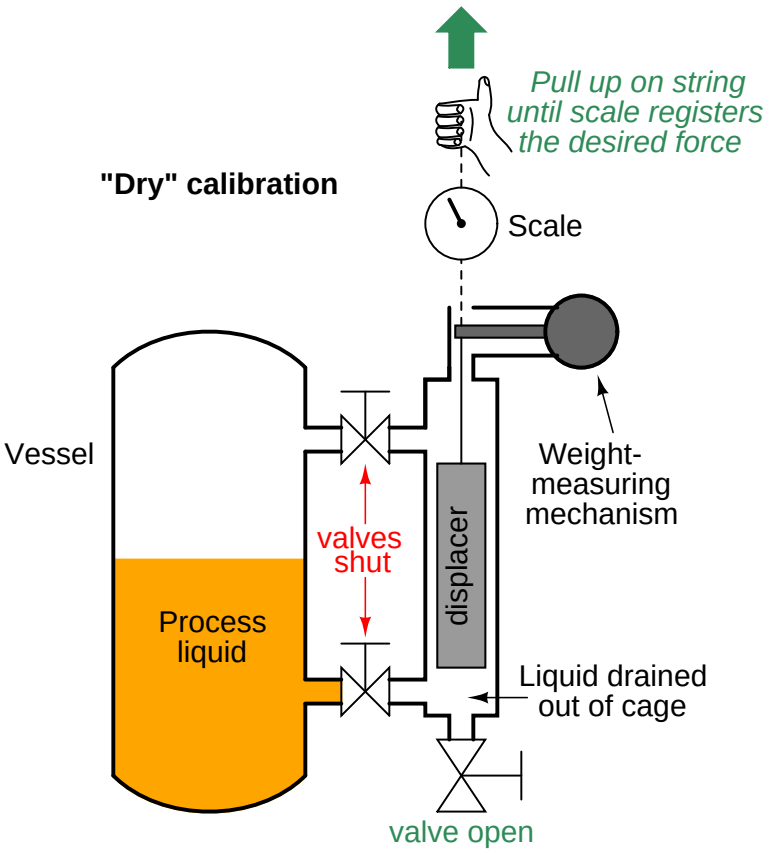




The cage pipe is coupled to the process vessel through two block valves, allowing isolation from the process. A drain valve allows the cage to be emptied of process liquid for instrument service and zero calibration.

Some displacer-type level sensors do not use a cage, but rather hang the displacer element directly in the process vessel. These are called "cageless" sensors. Cageless instruments are of course simpler than cage-style instruments, but they cannot be serviced without de-pressurizing (and perhaps even emptying) the process vessel in which they reside. They are also susceptible to measurement errors and "noise" if the liquid inside the vessel is agitated, either by high flow velocities in and out of the vessel, or by the action of motor-turned impellers installed in the vessel to provide thorough mixing of the process liquid(s).

Full-range calibration may be performed by flooding the cage with process liquid (a *wet* calibration), or by suspending the displacer with a string and precise scale (a *dry* calibration), pulling upward on the displacer at just the right amount to simulate buoyancy at 100% liquid level:



Calculation of this buoyant force is a simple matter. According to Archimedes' Principle, buoyant force is always equal to the weight of the fluid volume displaced. In the case of a displacer-based level instrument at full range, this usually means the entire volume of the displacer element is submerged in the liquid. Simply calculate the volume of the displacer (if it is a cylinder, $V = \pi r^2 l$, where r is the cylinder radius and l is the cylinder length) and multiply that volume by the weight density (γ):

$$F_{buoyant} = \gamma V$$

$$F_{buoyant} = \gamma \pi r^2 l$$

For example, if the weight density of the process fluid is 57.3 pounds per cubic foot and the displacer is a cylinder measuring 3 inches in diameter and 24 inches in length, the necessary force to simulate a condition of buoyancy at full level may be calculated as follows:

$$\gamma = \left(\frac{57.3 \text{ lb}}{\text{ft}^3}\right)\left(\frac{1 \text{ ft}^3}{12^3 \text{ in}^3}\right) = 0.0332\frac{\text{lb}}{\text{in}^3}$$

$$V = \pi r^2 l = \pi(1.5 \text{ in})^2(24 \text{ in}) = 169.6 \text{ in}^3$$

$$F_{buoyant} = \gamma V = \left(0.0332\frac{\text{lb}}{\text{in}^3}\right)\left(169.6 \text{ in}^3\right) = 5.63 \text{ lb}$$

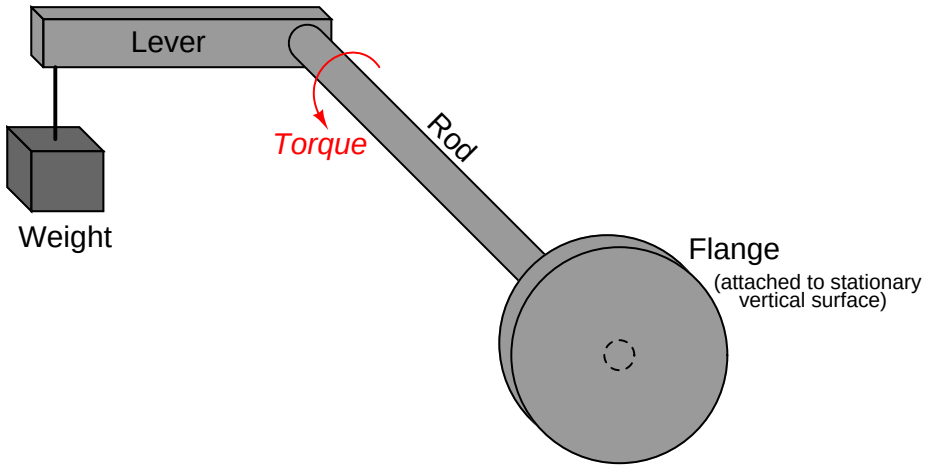
Note how important it is to maintain consistency of units! The liquid density was given in units of pounds per cubic *foot* and the displacer dimensions in *inches*, which would have caused serious problems without a conversion between feet and inches. In my example work, I opted to convert density into units of pounds per cubic inch, but I could have just as easily converted the displacer dimensions into feet to arrive at a displacer volume in units of cubic feet.

In a "wet" calibration, the 5.63 pound buoyant force will be created by the liquid itself, the technician ensuring there is enough liquid inside the cage to simulate a 100% level condition. In a "dry" calibration, the buoyant force will be simulated by tension applied upward on the displacer with a hand scale and string, the technician pulling with an upward force of 5.63 pounds to make the instrument "think" it is sensing 100% liquid level when in fact the displacer is completely dry, hanging in air.

20.4.2 Torque tubes

An interesting design problem for displacement-type level transmitters is how to transfer the sensed weight of the displacer to the transmitter mechanism while positively sealing process vapor pressure from that same mechanism. The most common solution to this problem is an ingenious mechanism called a *torque tube*. Unfortunately, torque tubes can be rather difficult to understand unless you have direct hands-on access to one, and so this section will explore the concept in more detail than is customarily available in reference manuals.

Imagine a solid, horizontal, metal rod with a flange at one end and a perpendicular lever at the other end. The flange is mounted to a stationary surface, and a weight suspended from the end of the lever. A dashed-line circle shows where the rod is welded to the center of the flange:

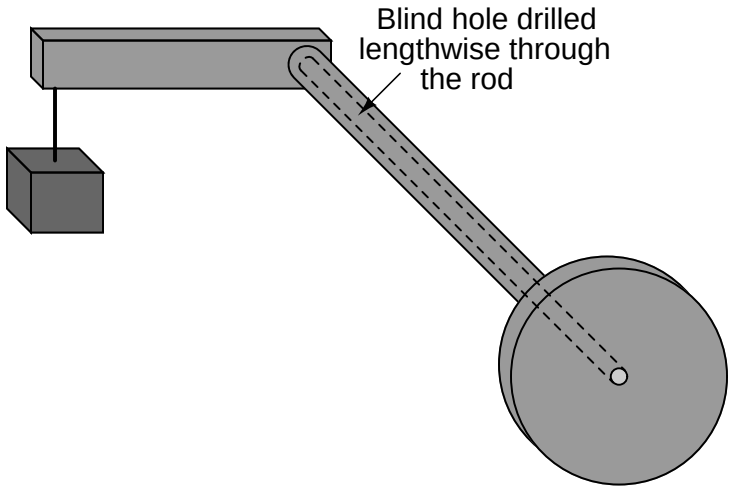


The downward force of the weight acting on the lever imparts a twisting force (torque) to the rod, causing it to slightly twist along its length. The more weight hung at the end of the lever, the more the rod will twist[16]. So long as the torque applied by the weight and lever never exceeds the elastic limit of the rod, the rod will continue to act as a spring. If we know the "spring constant" of the rod, and measured its torsional deflection, we can in fact use this slight motion to measure the magnitude of the weight hung at the end of the lever.

Applied to a displacer-type level instrument, a displacer takes the place of the weight at the lever's end, the torsional deflection of this rod serving to indicate buoyant force. As liquid rises, buoyant force on the displacer increases, making the displacer seem lighter from the rod's perspective. The rod's slight motion resulting from this apparent weight change, then, indicates liquid level.

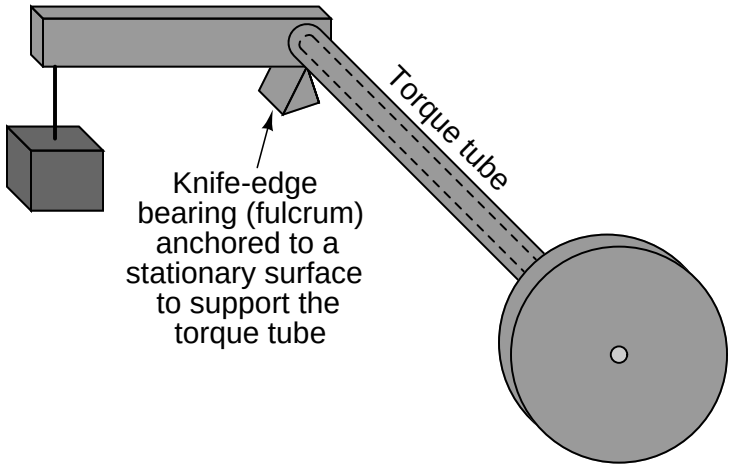
[16] To anyone familiar with the front suspension of a 1960's vintage Chevrolet truck, or the suspension of the original Volkswagen "Beetle" car, the concept of a *torsion bar* should be familiar. These vehicles used straight, spring-steel rods to provide suspension force instead of the more customary coil springs used in modern vehicles. However, even the familiar coil spring is an example of torsional forces at work: a coil spring is nothing more than a torsion bar bent in a coil shape. As a coil spring is stretched or compressed, torsional forces develop along the circumferential length of the spring coil, which is what makes the spring "try" to maintain a fixed height.

Now imagine drilling a long hole through the rod, lengthwise, that almost reaches the end where the lever attaches. In other words, imagine a *blind hole* through the center of the rod, starting at the flange and ending just shy of the lever:

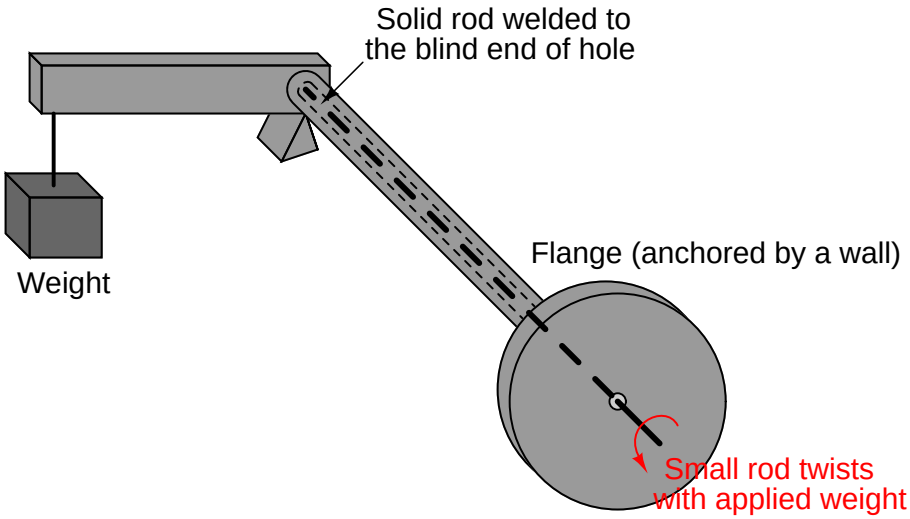


The presence of this long hole does not change much about the behavior of the assembly, except perhaps to alter the rod's spring constant. With less solid metal, the rod will be a weaker spring, and will twist to a greater degree with applied weight at the end of the lever. More importantly for the purpose of this discussion, though, the long hole transforms the rod into a *tube* with a sealed end. Instead of a being a "torsion bar," the rod is now more properly called a *torque tube*, twisting ever so slightly with applied weight at the end of the lever.

In order to give the torque tube vertical support so it does not sag downward with the applied weight, a supporting *knife-edge bearing* is often placed underneath the end of the lever where it attaches to the torque tube. The purpose of this fulcrum is to provide vertical support for the weight while forming a virtually frictionless pivot point, ensuring the only stress applied to the torque tube is *torque* from the lever:



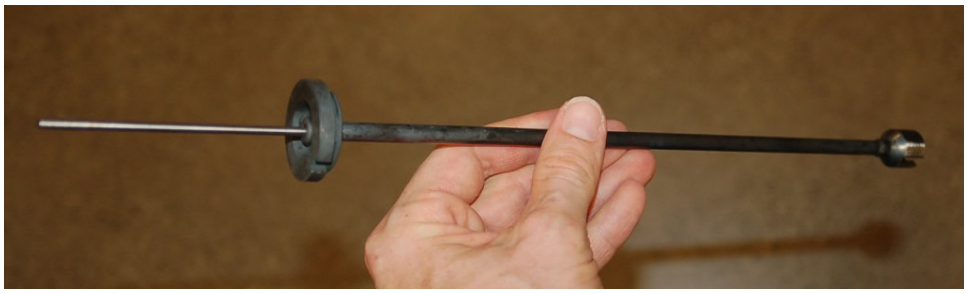
Finally, imagine another solid metal rod (slightly smaller diameter than the hole) spot-welded to the far end of the blind hole, extending out beyond the end of the flange:



The purpose of this smaller-diameter rod is to transfer the twisting motion of the torque tube's far end to a point past the flange where it may be sensed. Imagine the flange anchored to a vertical wall, while a variable weight tugs downward at the end of the lever. The torque tube will flex in a twisting motion with the variable force, but now we are able to see just how much it twists by watching the rotation of the smaller rod on the near side of the wall. The weight and lever may be completely hidden from our view by this wall, but the small rod's twisting motion nevertheless reveals how much the torque tube yields to the force of the weight.

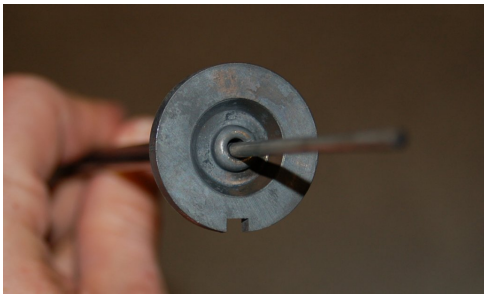
We may apply this torque tube mechanism to the task of measuring liquid level in a pressurized vessel by replacing the weight with a displacer, attaching the flange to a nozzle welded to the vessel, and aligning a motion-sensing device with the small rod end to measure its rotation. As liquid level rises and falls, the apparent weight of the displacer varies, causing the torque tube to slightly twist. This slight twisting motion is then sensed at the end of the small rod, in an environment isolated from the process fluid pressure.

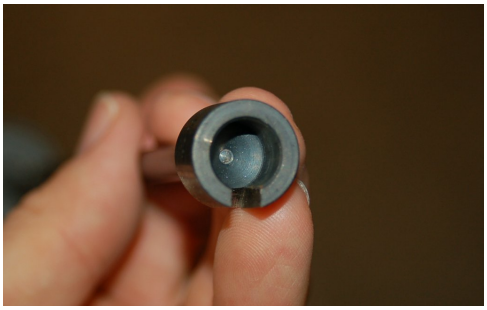
A photograph taken of a real torque tube from a Fisher "Level-Trol" level transmitter shows its external appearance:



The dark-colored metal is the elastic steel used to suspend the weight by acting as a torsional spring, while the shiny portion is the inner rod used to transfer motion. As you can see, the torque tube itself is not very wide in diameter. If it were, it would be far too stiff of a spring to be of practical use in a displacer-type level instrument, since the displacer is not typically very heavy, and the lever is not long.

Looking closer at each end of the torque tube reveals the open end where the small-diameter rod protrudes (left) and the "blind" end of the tube where it attaches to the lever (right):

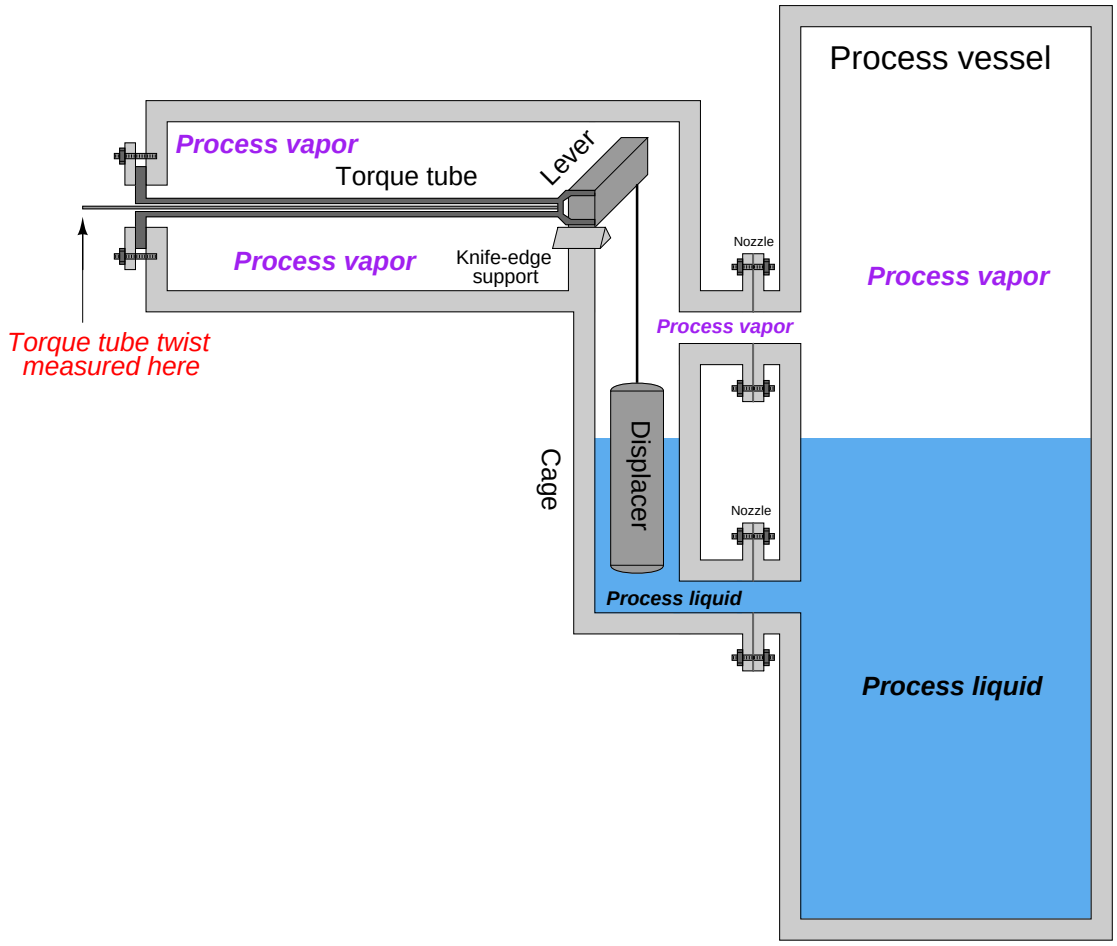




If we were to slice the torque tube assembly in half, lengthwise, its cross-section would look something like this:



This next illustration shows the torque tube as part of a whole displacement-style level transmitter[17]:



As you can see from this illustration, the torque tube serves three distinct purposes when applied to a displacer-type level measurement application: (1) to serve as a torsional spring suspending the weight of the displacer, (2) to seal off process fluid pressure from the position-sensing mechanism, and (3) to transfer motion from the far end of the torque tube into the sensing mechanism.

[17] This illustration is simplified, omitting such details as access holes into the cage, block valves between the cage and process vessel, and any other pipes or instruments attached to the process vessel. Also, the position-sensing mechanism normally located at the far left of the assembly is absent from this drawing.

In pneumatic level transmitters, the sensing mechanism used to convert the torque tube's twisting motion into a pneumatic (air pressure) signal is typically of the *motion-balance* design. The Fisher Level-Trol mechanism, for example, uses a C-shaped bourdon tube with a nozzle at the end to follow a baffle attached to the small rod. The center of the bourdon tube is aligned with the center of the torque tube. As the rod rotates, the baffle advances toward the nozzle at the bourdon tube tip, causing backpressure to rise, which in turn causes the bourdon tube to flex. This flexing draws the nozzle away from the advancing baffle until a balanced condition exists. Rod motion is therefore balanced by bourdon tube motion, making this a motion-balance pneumatic system:



20.4.3 Displacement interface level measurement

Displacer level instruments may be used to measure liquid-liquid interfaces just the same as hydrostatic pressure instruments. One important requirement is that the displacer always be fully submerged ("flooded"). If this rule is violated, the instrument will not be able to discriminate between a low (total) liquid level and a low interface level. This criterion is analogous to the use of compensated-leg differential pressure instruments to measure liquid-liquid interface levels: in order for the instrument to solely respond to changes in interface level and not be "fooled" by changes in total liquid level, both process connection points must be submerged.

If the displacer instrument has its own "cage," it is important that both pipes connecting the cage to the process vessel (sometimes called "nozzles") be submerged. This ensures the liquid interface inside the cage matches the interface inside the vessel. If the upper nozzle ever goes dry, the same problem can happen with a caged displacer instrument as with a "sightglass" level gauge (see section 20.1.2 beginning on page 1412 for a detailed explanation of this problem.).

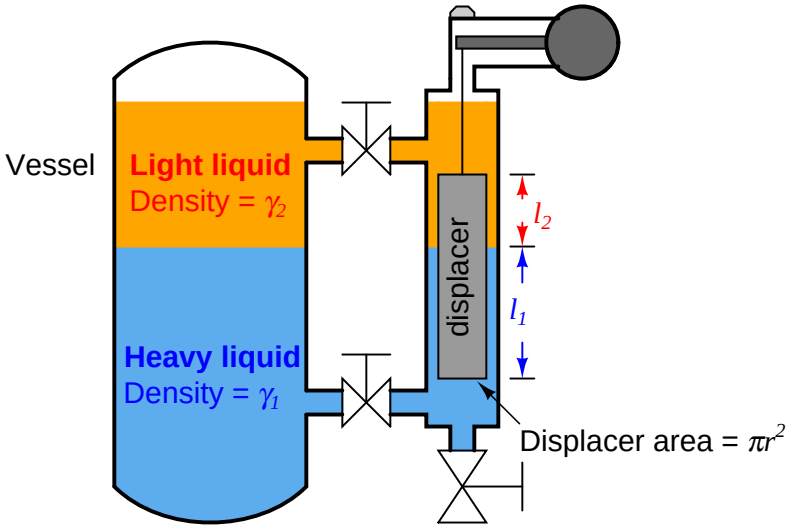
Calculating buoyant force on a displacer element due to a combination of two liquids is not as difficult as it may sound. Archimedes' Principle still holds: that buoyant force is equal to the weight of the fluid(s) displaced. All we need to do is calculate the combined weights and volumes of the displaced liquids to calculate buoyant force. For a single liquid, buoyant force is equal to the weight density of that liquid (γ) multiplied by the volume displaced (V):

$$F_{buoyant} = \gamma V$$

For a two-liquid interface, the buoyant force is equal to the sum of the two liquid weights displaced, each liquid weight term being equal to the weight density of that liquid multiplied by the displaced volume of that liquid:

$$F_{buoyant} = \gamma_1 V_1 + \gamma_2 V_2$$

Assuming a displacer of constant cross-sectional area throughout its length, the volume for each liquid's displacement is simply equal to the same area (πr^2) multiplied by the length of the displacer submerged in that liquid:



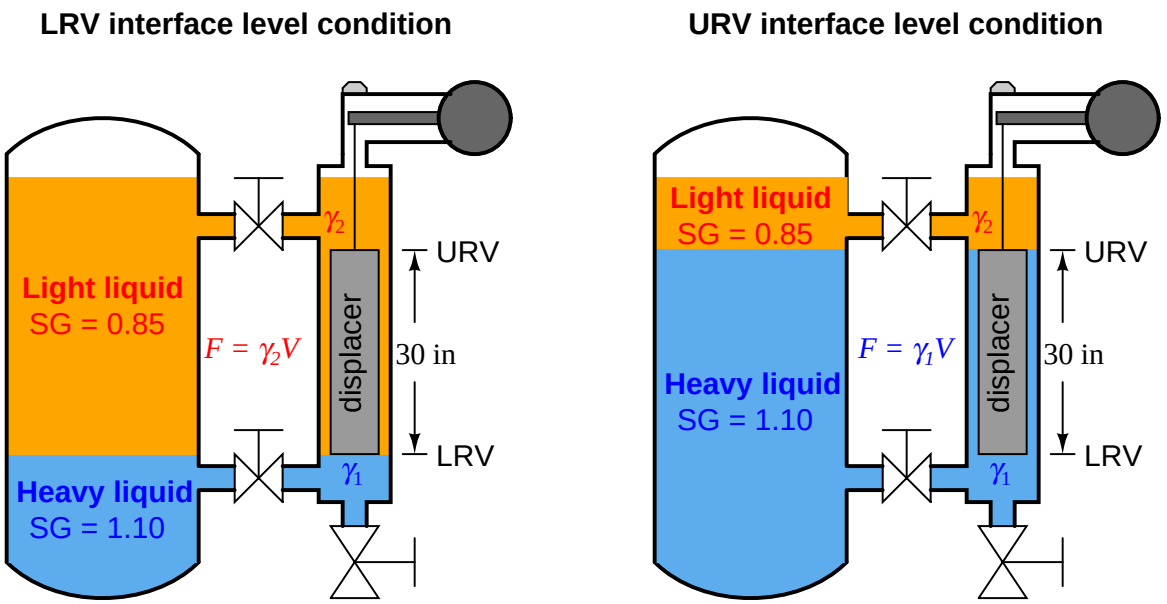
$$F_{buoyant} = \gamma_1 \pi r^2 l_1 + \gamma_2 \pi r^2 l_2$$

Since the area (πr^2) is common to both buoyancy terms in this equation, we may factor it out for simplicity's sake:

$$F_{buoyant} = \pi r^2 (\gamma_1 l_1 + \gamma_2 l_2)$$

Determining the calibration points of a displacer-type level instrument for interface applications is relatively easy if the LRV and URV conditions are examined as a pair of "thought experiments" just as we did with hydrostatic interface level measurement. First, we imagine what the displacer's condition would "look like" with the interface at the lower range value, then we imagine a different scenario with the interface at the upper range value. Sketching illustrations of each scenario is recommended for clarity.

Suppose we have a displacer instrument measuring the interface level between two liquids having specific gravities of 0.850 and 1.10, with a displacer length of 30 inches and a displacer diameter of 2.75 inches (radius = 1.375 inches). Let us further suppose the LRV in this case is where the interface is at the displacer's bottom and the URV is where the interface is at the displacer's top. The placement of the LRV and URV interface levels at the extreme ends of the displacer's length simplifies our LRV and URV calculations, as the LRV "thought experiment" will simply be the displacer completely submerged in light liquid and the URV "thought experiment" will simply be the displacer completely submerged in heavy liquid.



Calculating the LRV buoyant force:

$$F_{buoyant}\ (\text{LRV}) = \gamma_2 V = \gamma_2 \pi r^2 l$$

Calculating the URV buoyant force:

$$F_{buoyant}\ (\text{URV}) = \gamma_1 V = \gamma_1 \pi r^2 l$$

Showing the actual calculations for this hypothetical example:

$$\gamma_1 = \left(62.4\ \frac{\text{lb}}{\text{ft}^3}\right)(1.10) = 68.6\ \frac{\text{lb}}{\text{ft}^3} = 0.0397\ \frac{\text{lb}}{\text{in}^3}$$

$$\gamma_2 = \left(62.4\ \frac{\text{lb}}{\text{ft}^3}\right)(0.85) = 53.0\ \frac{\text{lb}}{\text{ft}^3} = 0.0307\ \frac{\text{lb}}{\text{in}^3}$$

$$F_{buoyant}\ (\text{LRV}) = \left(0.0307\ \frac{\text{lb}}{\text{in}^3}\right)\pi(1.375\ \text{in})^2(30\ \text{in}) = 5.47\ \text{lb}$$

$$F_{buoyant}\ (\text{URV}) = \left(0.0397\ \frac{\text{lb}}{\text{in}^3}\right)\pi(1.375\ \text{in})^2(30\ \text{in}) = 7.08\ \text{lb}$$

The buoyancy for any measurement percentage between the LRV (0%) and URV (100%) may be calculated by interpolation:

Interface level (inches)	**Buoyant force** (pounds)
0	5.47
7.5	5.87
15	6.27
22.5	6.68
30	7.08

20.5 Echo

A completely different way of measuring liquid level in vessels is to bounce a traveling wave off the surface of the liquid – typically from a location at the top of the vessel – using the time-of-flight for the waves as an indicator of distance[18], and therefore an indicator of liquid height inside the vessel. Echo-based level instruments enjoy the distinct advantage of immunity to changes in liquid density, a factor crucial to the accurate calibration of hydrostatic and displacement level instruments. In this regard, they are quite comparable with float-based level measurement systems.

From a historical perspective, hydrostatic and displacement level instruments have a richer pedigree. These instruments are simpler in nature than echo-based instruments, and were practical long before the advent of modern electronic technology. Echo-based instruments require precision timing and wave-shaping circuitry, plus sensitive (and rugged!) transceiver elements, demanding a much higher level of technology. However, modern electronic design and instrument manufacturing practices are making echo-based level instruments more and more practical for industrial applications. At the time of this writing (2008), it is common practice in some industries to replace old displacer level instruments with guided-wave radar instruments, even in demanding applications operating at high pressures[19].

Liquid-liquid interfaces may also be measured with some types of echo-based level instruments, most commonly guided-wave radar.

The single most important factor to the accuracy of any echo-based level instrument is the speed at which the wave travels en route to the liquid surface and back. This wave propagation speed is as fundamental to the accuracy of an echo instrument as liquid density is to the accuracy of a hydrostatic or displacer instrument. So long as this velocity is known and stable, good level measurement accuracy is possible. Although it is true that the calibration of an echo-based level instrument does not depend on process fluid density for the reason it does in hydrostatic- or displacement-based level instruments, this does not necessarily mean the calibration of an echo-based level instrument remains fixed as process fluid density changes. The propagation velocity of the wave used in an echo-based level instrument may indeed be subject to change as the process fluids change temperature or composition. For ultrasonic (sound) echo instruments, the speed of sound is a function of medium density. Thus, an ultrasonic level transmitter measuring time-of-flight through a vapor above the liquid may drift out of calibration if the speed of sound through that vapor changes substantially, which may happen if the vapor's temperature or pressure happens to change. If the sound wave time-of-flight is measured while the waves pass through liquid, the calibration may drift if the speed of sound in that liquid changes substantially, which may happen if the liquid's temperature changes. For radar (radio wave) echo instruments, the speed of radio wave propagation varies according to the dielectric permittivity of the medium. Permittivity is also affected by changes in density for the fluid medium, and so even radar level instruments may suffer calibration drift with process fluid density changes.

To summarize these effects, the speed of sound through any medium is a function of density and bulk modulus (the "compressibility" of the medium), with density generally being the more variable of the two. For gases and vapors, this means the speed of sound is strongly affected by changes in

[18] The general term for this form of measurement is *time domain reflectometry*.

[19] My own experience with this trend is within the oil refining industry, where legacy displacer instruments (typically Fisher brand "Level-Trol" units) are being replaced with new guided-wave radar transmitters, both for vapor-liquid and liquid-liquid interface applications.

gas pressure and/or gas temperature. For liquids, this means the speed of sound is strongly affected by temperature. For solids, this means the speed of sound is weakly affected by temperature. The degree to which the speed of sound will be affected by temperature changes is directly related to the degree the medium's density changes with temperature: solid materials generally expand and contract less than liquids over the same temperature range, thus the strong temperature effect for liquids and the weak temperature effect for solids.

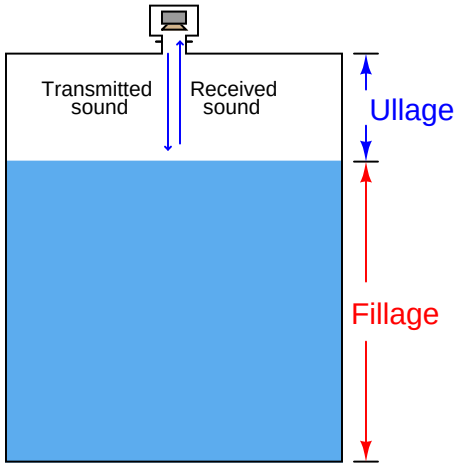
Radio wave velocity is a function of dielectric permittivity, which is also a function of density. However, the degree of change in dielectric permittivity resulting from changes in pressure and/or temperature are generally much less than the degree of change in speed of sound for the same media and the same changes in pressure and/or temperature. This means that – all other factors being equal – an echo-based level instrument using radio waves will suffer far less calibration error than an echo-based level instrument using sound waves when process fluid pressure and/or temperature change. However, it should be noted that process fluid *composition* (i.e. its chemical make-up) may have a strong effect on radio wave propagation, not just on its time-of-flight but also on its ability to produce an adequate echo at the interface between two fluids.

Echo-based level instruments may also be "fooled" by layers of foam resting on top of the liquid, and the liquid-to-liquid interface detection models may have difficulty detecting non-distinct interfaces (such as emulsions). Irregular structures residing within the vapor space of a vessel (such as access portals, mixer paddles and shafts, ladders, etc.) may wreak havoc with echo-based level instruments by casting false echoes back to the instrument, although this problem may be mitigated by installing guide tubes for the waves to travel in, or using wave probes as in the cases of guided-wave radar instruments. Liquid streams pouring in to the vessel through the vapor space may similarly cause problems for an echo instrument. Additionally, all echo-based instruments have *dead zones* where liquid level is too close to the transceiver to be accurately measured or even detected (the echo time-of-flight being too short for the receiving electronics to distinguish from the incident pulse).

As you can see, echo-based level instruments have strengths and weaknesses just like any other type of level instrument. There is no "perfect" level instrument, but rather a wide array of choices from which the end-user must judiciously select for the particular application in mind. Beware of sales pitches urging you to buy the "perfect" level meter! The wise approach is to first research the underlying physics of the instrument, then determine how strongly its accuracy will be affected by realistic changes in process conditions (e.g. pressure, temperature, composition).

20.5.1 Ultrasonic level measurement

Ultrasonic level instruments measure the distance from the transmitter (located at some high point) to the surface of a process material located farther below using reflected sound waves. The frequency of these waves extend beyond the range of human hearing, which is why they are called *ultra*sonic. The time-of-flight for a sound pulse indicates this distance, and is interpreted by the transmitter electronics as process level. These transmitters may output a signal corresponding either to the fullness of the vessel (*fillage*) or the amount of empty space remaining at the top of a vessel (*ullage*).



Ullage is the "natural" mode of measurement for this sort of level instrument, because the sound wave's time-of-flight is a direct function of how much empty space exists between the liquid surface and the top of the vessel. Total tank height will always be the sum of fillage and ullage, though. If the ultrasonic level transmitter is programmed with the vessel's total height, it may calculate fillage via simple subtraction:

$$\text{Fillage} = \text{Total height} - \text{Ullage}$$

If a sound wave encounters a sudden change in the material's speed of sound, some of that wave's energy will be reflected in the form of another wave in the opposite direction. In other words, the sound wave will "echo" when it encounters a material having a different sonic velocity[20]. This is the basis of all ultrasonic ranging devices. Thus, in order for an ultrasonic level transmitter to function reliably, the difference in sonic velocities at the interface between liquid and gas must be large. Distinct interfaces of liquid and gas almost always exhibit huge differences in their speeds of sound, and so are relatively easy to detect using ultrasonic waves. Liquids with a heavy layer of foam floating on top are more difficult, since the foam is less dense than the liquid, but considerably

[20] The speed of sound through any substance is a function of both the substance's density and its bulk modulus (i.e. the compressibility of a substance). Mathematically, $c = \sqrt{\frac{B}{\rho}}$ where c is the sonic velocity, B is the bulk modulus, and ρ is the mass density. Water and air provide an excellent illustration of this principle: the speed of sound through water happens to be much faster than the speed of sound through air despite the vastly greater mass density of water, only because of the even greater disparity in bulk modulus between water and air.

denser than the gas above. A weak echo will be generated at the interface of foam and gas, and another generated at the interface of liquid and foam, with the foam acting to scatter and dissipate much of the second echo's energy.

The instrument itself consists of an electronics module containing all the power, computation, and signal processing circuits; plus an ultrasonic transducer[21] to send and receive the sound waves. This transducer is typically piezoelectric in nature, being the equivalent of a very high-frequency audio speaker. The following photographs show a typical electronics module (left) and sonic transducer (right):

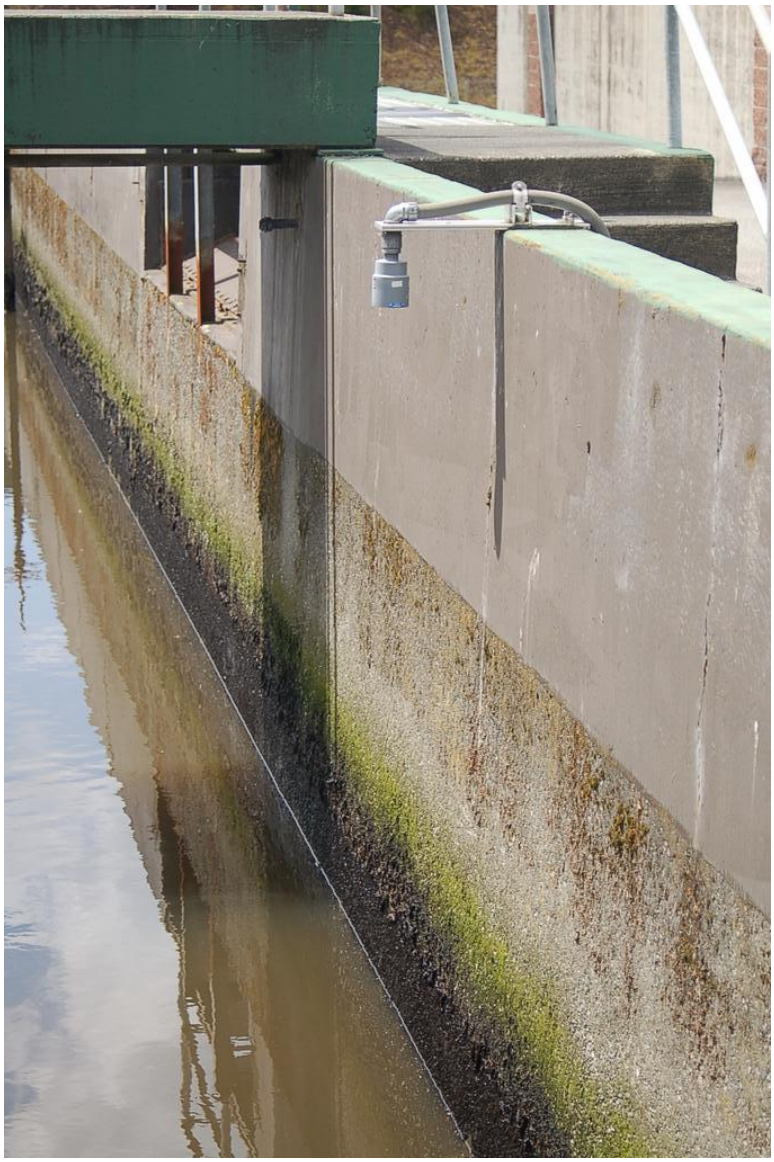




The ISA-standard designations for each component would be "LT" (level transmitter) for the electronics module and "LE" (level element) for the transducer, respectively. Even though we call the device responsible for transmitting and receiving the sound waves a *transducer* (in the scientific sense of the word), its function as a process instrument is to be the *primary sensing element* for the level measurement system, and therefore it is more properly designated a "level element" (LE).

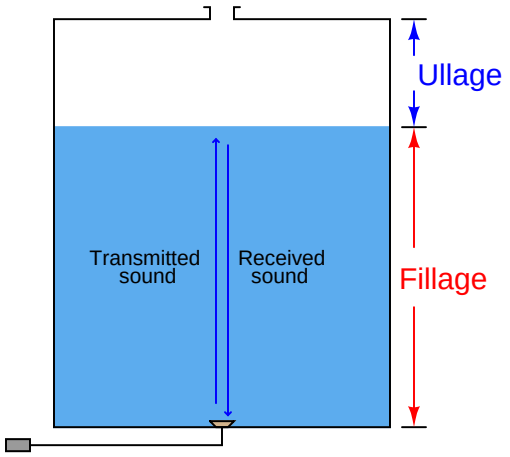
[21] In the industrial instrumentation world, the word "transducer" usually has a very specific meaning: a device used to process or convert standardized instrumentation signals, such as 4-20 mA converted into 3-15 PSI, etc. In the general scientific world, however, the word "transducer" describes any device converting one form of energy into another. It is this latter definition of the word that I am using when I describe an ultrasonic "transducer" – a device used to convert electrical energy into ultrasonic sound waves, and vice-versa.

This photograph shows a typical installation for an ultrasonic level-sensing element (LE), here sensing the level of wastewater in an open channel:



Electrical conduit serves to protect the signal cable from exposure to the elements as it routes back to an indoor location where the level transmitter (LT) is located.

If the ultrasonic transducer is rugged enough, and the process vessel sufficiently free of sludge and other sound-damping materials accumulating at the vessel bottom, the transducer may be mounted at the bottom of the vessel, bouncing sound waves off the liquid surface through the liquid itself rather than through the vapor space. As stated previously, any significant difference in sonic velocity between the two materials is sufficient to reflect a sound wave. This being the case, it shouldn't matter which material the incident sound wave propagates through *first*:

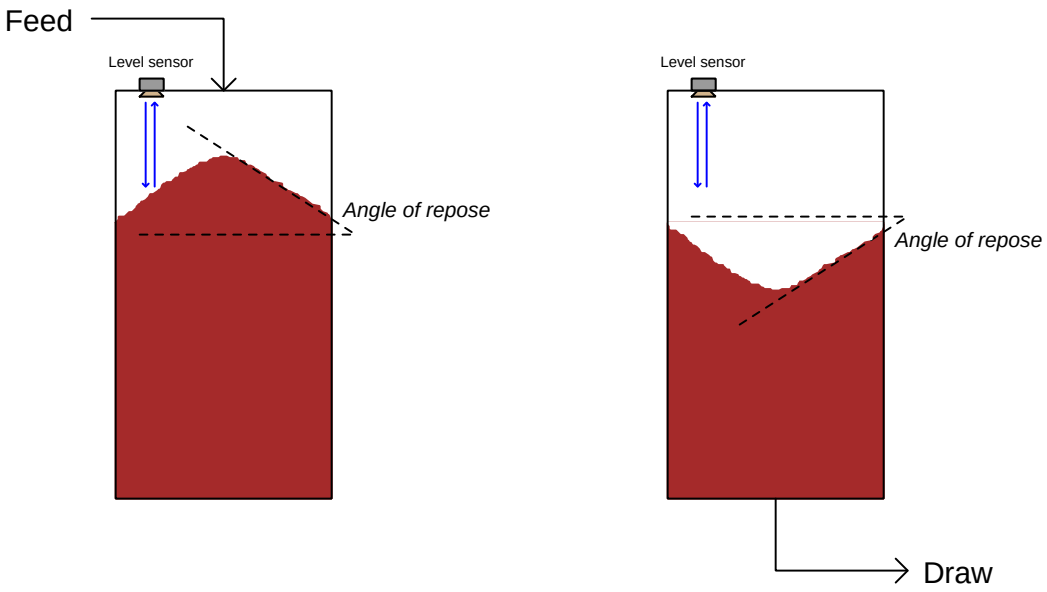


This arrangement makes fillage the natural measurement, and ullage a derived measurement (calculated by subtraction from total vessel height).

$$\text{Ullage} = \text{Total height} - \text{Fillage}$$

As mentioned previously, the calibration of an ultrasonic level transmitter depends on the speed of sound through the medium between the transducer and the interface. For top-mounted transducers, this is the speed of sound through the air (or vapor) over the liquid, since this is the medium through which the incident and reflected wave travel time is measured. For bottom-mounted transducers, this is the speed of sound through the liquid. In either case, to ensure good accuracy, one must make sure the speed of sound through the "timed" travel path remains reasonably constant (or else compensate for changes in the speed of sound through that medium by use of temperature or pressure measurements and a compensating algorithm).

Ultrasonic level instruments enjoy the advantage of being able to measure the height of solid materials such as powders and grains stored in vessels, not just liquids. Again, the fundamental criterion for detecting a level of material is that the speeds of sound through the upper and lower materials must differ (the greater the difference, the stronger the echo). A unique challenge to solids measurement is the distinct possibility of uneven material profiles. A classic problem encountered when measuring the level of a powdered or granular material in a vessel is the *angle of repose* formed by the material as a result of being fed into the vessel at one point:



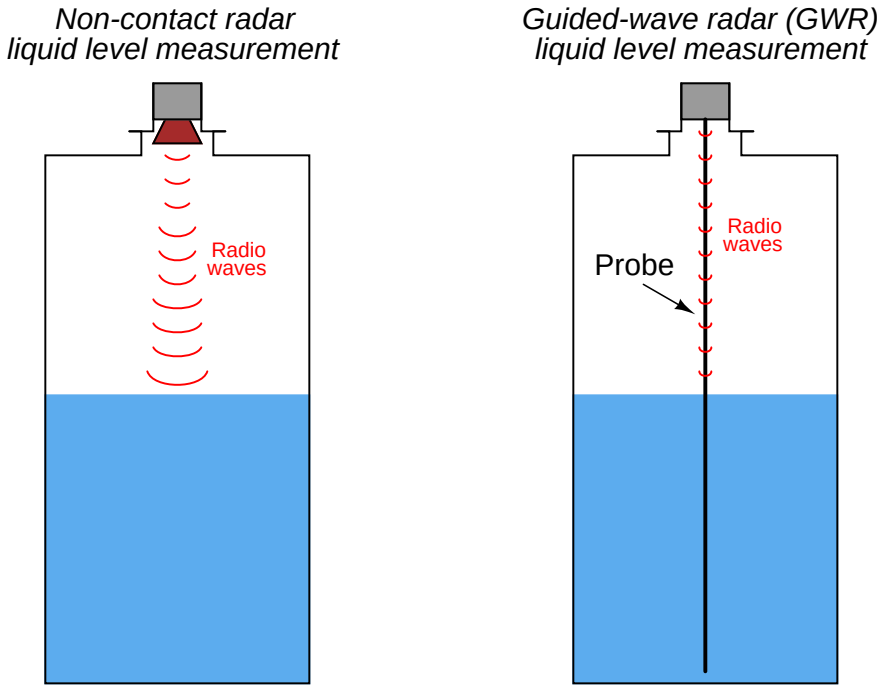
This angled surface is difficult for an ultrasonic device to detect because it tends to scatter the sound waves laterally instead of reflecting them strongly back toward the instrument. However, even if the scattering problem is not significant, there still remains the problem of interpretation: what is the instrument actually measuring? The detected level near the vessel wall will certainly register less than at the center, but the level detected mid-way between the vessel wall and vessel center may not be an accurate average of those two heights. Moreover, this angle may decrease over time if mechanical vibrations cause the material to "flow" and tumble from center to edge.

For this reason, solids storage measurement applications demanding high accuracy generally use other techniques, such as weight-based measurement (see section 20.6 for more information) or three-dimensional scanning (see section 26.3 for more information).

20.5.2 Radar level measurement

Radar[22] level instruments measure the distance from the transmitter (located at some high point) to the surface of a process material located farther below in much the same way as ultrasonic transmitters – by measuring the time-of-flight of a traveling wave. The fundamental difference between a radar instrument and an ultrasonic instrument is the type of wave used: radio waves instead of sound waves. Radio waves are electromagnetic in nature (comprised of alternating electric and magnetic fields), and very high frequency (in the microwave frequency range – GHz). Sound waves are *mechanical* vibrations (transmitted from molecule to molecule in a fluid or solid substance) and of much lower frequency (tens or hundreds of kilohertz – still too high for a human being to detect as a tone) than radio waves. In any case, a wave will refect off of an interface of two different substances if those two substances possess different wave-propagation velocities.

Some radar level instruments use waveguide "probes" to guide the electromagnetic waves to and from the process liquid while others send electromagnetic waves out through open space to reflect off the process material. The instruments using waveguides are called *guided-wave radar* instruments, whereas the radar instruments relying on open space for signal propagation are called *non-contact radar*. The differences between these two varieties of radar instruments is shown in the following illustration:



[22] "Radar" is an acronym: RAdio Detection And Ranging. First used as a method for detecting enemy ships and aircraft at long distances over the ocean in World War II, this technology is used for detecting the presence, distance, and/or speed of objects in a wide variety of applications.

Photographs of non-contact (left) and guided-wave (right) radar level transmitters are shown below. The non-contact transmitter is placed on a table for inspection while the guided-wave transmitter is installed in a "cage"[23] similar to that of a displacement-style level transmitter attached to the vessel by two pipes:

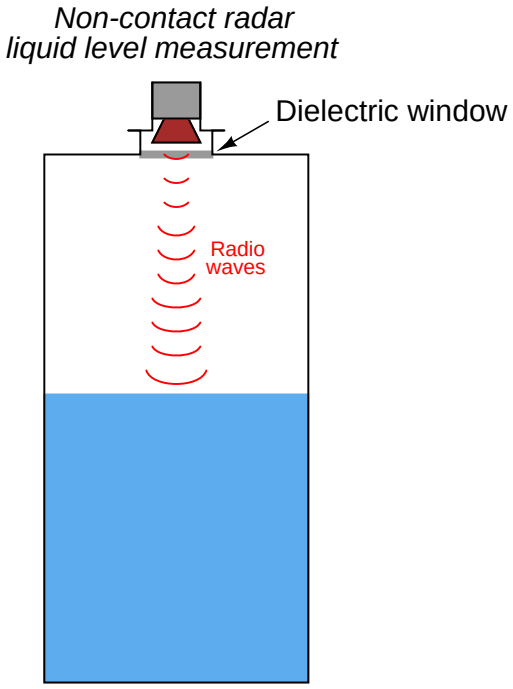




Non-contact radar devices suffer much more signal loss than guided-wave radar devices, due to the natural tendency of electromagnetic radiation to disperse over space. Waveguides combat this signal loss by channeling the radio energy along a straight-line path. Probes used in guided-wave radar instruments may be single metal rods, parallel pairs of metal rods, or a coaxial metal rod-and-tube structure. Single-rod probes suffer the greatest energy losses, while coaxial probes excel at guiding the microwave energy to the liquid interface and back. However, single-rod probes are much more tolerant of process fouling than two-rod or (especially) coaxial probes, where sticky masses of viscous liquid and/or solid matter cling to the probe. Such fouling deposits, if severe enough, will cause electromagnetic wave reflections that "look" to the transmitter like the reflection from an actual liquid level or interface.

[23] In fact, it is a common retrofit practice to install a guided-wave radar level transmitter in the exact same cage that once housed a displacement-style level transmitter.

Non-contact radar instruments rely on antennas to direct microwave energy into the vessel, and to receive the echo (return) energy. These antennas must be kept clean and dry, which may be a problem if the liquid being measured emits condensible vapors. For this reason, non-contact radar instruments are often separated from the vessel interior by means of a *dielectric window* (made of some substance such as plastic that is relatively "transparent" to electromagnetic waves yet acts as an effective vapor barrier):



Electromagnetic waves travel at the speed of light[24], 2.9979×10^8 meters per second in a perfect vacuum. The velocity of an electromagnetic wave through space depends on the dielectric permittivity (symbolized by the Greek letter "epsilon," ϵ) of that space. A formula relating wave velocity (v) to relative permittivity (the ratio of a substance's permittivity to that of a perfect vacuum, symbolized as ϵ_r and sometimes called the *dielectric constant* of the substance) and the speed of light in a perfect vacuum (c) is shown here[25]:

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

[24] In actuality, both radio waves and light waves are electromagnetic in nature. The only difference between the two is frequency: while the radio waves used in radar systems are classified as "microwaves" with frequencies in the gigahertz (GHz) region, visible light waves range in the hundred of terahertz (THz)!

[25] This formula assumes lossless conditions: that none of the wave's energy is converted to heat while traveling through the dielectric. For many situations, this is true enough to assume.

As mentioned previously, the calibration of any echo-based level transmitter depends on knowing the speed of wave propagation through the medium separating the instrument from the process fluid interface. For radar transmitters sensing a single liquid below a gas or vapor, this speed is the speed of light through that gas or vapor space, which we know to be a function of electrical permittivity.

The relative permittivity of air at standard pressure and temperature is very nearly unity (1). This means the speed of light in air under atmospheric pressure and ambient temperature will very nearly be the same as it is for a perfect vacuum (2.9979×10^8 meters per second). If, however, the vapor space above the liquid is not ambient air, and is subject to large changes in temperature and/or pressure[26] which cause the vapor's density to change, the permittivity of that vapor may substantially change and consequently skew the speed of light, and therefore the calibration of the level instrument. This calibration shift is sometimes referred to as the *gas phase effect*.

A formula useful for calculating the permittivity of any gas or vapor based on both pressure and temperature is shown here[27]:

$$\epsilon_r = 1 + (\epsilon_{ref} - 1)\frac{PT_{ref}}{P_{ref}T}$$

Where,

ϵ_r = Relative permittivity of a gas at a given pressure (P) and temperature (T)
ϵ_{ref} = Relative permittivity of the same gas at standard pressure (P_{ref}) and temperature (T_{ref})
P = Absolute pressure of gas (bars)
P_{ref} = Absolute pressure of gas under standard conditions ($\approx$ 1 bar)
T = Absolute temperature of gas (Kelvin)
T_{ref} = Absolute temperature of gas under standard conditions ($\approx$ 273 K)

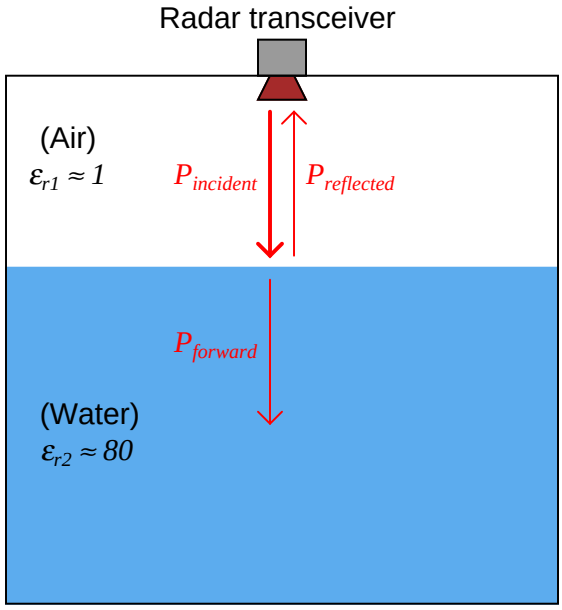
This formula is based on the principle that bulk permittivity is a function of density. We may see why this is by running a "thought experiment" in which a sample of gas becomes denser. As gas density increases, more gas molecules will become packed into the same volume of space. If each gas molecule's permittivity is greater than the permittivity of empty space, then having more of those gas molecules present will mean the permittivity of that volume increases. Greater permittivity, of course, decreases the velocity of light through the gas, and thereby affects the calibration of the radar instrument.

Relating this concept to pressure and temperature variations in the gas, we can see that the permittivity of a gas increases with increasing pressure (by increasing gas density), and decreases with increasing temperature (by decreasing gas density). This means the speed of light through a gas decreases with increasing pressure, and increases with increasing temperature. For radar level instruments operating in gas environments subject to significant pressure and temperature (i.e. density) variations, the consequent variations in the speed of light through that gas will compromise the instrument's accuracy.

[26] Or if the chemical composition of the gas or vapor changes dramatically.

[27] The pressure and temperature factors in this formula come from the Ideal Gas Law ($PV = nRT$), manipulating that equation to express molecular gas density in terms of pressure and temperature ($\rho = \frac{n}{V} = \frac{P}{RT}$). The fraction $\frac{PT_{ref}}{P_{ref}T}$ expresses a ratio of molecular densities: $\frac{\rho}{\rho_{ref}}$.

For any echo-based level instruments, the necessary condition for an echo to occur is that the wave encounters a sudden change in propagation velocity. With ultrasonic level instruments, the velocity of propagation for the sound wave depends on both the densities and the bulk moduli (incompressibilities) of the substances, so that a sudden change in either parameter from one substance to another will cause the sound wave to reflect. With radar level instruments, the necessary condition for wave reflection is a sudden change in dielectric permittivity[28] (ϵ). When an electromagnetic wave encounters a sudden change in dielectric permittivity, some of that wave's energy will be reflected in the form of another wave traveling the opposite direction, while the balance of the wave's energy continues forward to propagate into the new material. The strength of the reflected signal depends on how greatly the two materials' permittivities differ:



This same principle explains reflected signals in copper transmission lines as well. Any discontinuities (sudden changes in characteristic impedance) along the length of a transmission line will reflect a portion of the electrical signal's power back to the source. In a transmission line, continuities may be formed by pinches, breaks, or short-circuits. In a radar level measurement system, any sudden change in electrical permittivity is a discontinuity that reflects some of the incident wave energy back to the source. Thus, radar level instruments function best when there is a large difference in permittivity between the two substances at the interface. As shown in the previous illustration, air and water meet this criterion, having an 80:1 permittivity ratio.

[28] Dielectric permittivity is one of the factors determining the speed of any electromagnetic wave through a substance, but not the only one. The material's *magnetic permeability* is another factor, but it is far more common to encounter interfaces of gas-liquid or liquid-liquid where differences in permittivity rather than differences in permeability constitute the major reason for differences in radio wave velocity.

The ratio of reflected power to incident (transmitted) power at any interface of materials is called the *power reflection factor* (R). This may be expressed as a unitless ratio, or more often as a decibel figure. The relationship between dielectric permittivity and reflection factor is as follows:

$$R = \frac{\left(\sqrt{\epsilon_{r2}} - \sqrt{\epsilon_{r1}}\right)^2}{\left(\sqrt{\epsilon_{r2}} + \sqrt{\epsilon_{r1}}\right)^2}$$

Where,

R = Power reflection factor at interface, as a unitless ratio

ϵ_{r1} = Relative permittivity (dielectric constant) of the first medium

ϵ_{r2} = Relative permittivity (dielectric constant) of the second medium

The fraction of incident power transmitted through the interface ($\frac{P_{forward}}{P_{incident}}$) is, of course, the mathematical complement of the power reflection factor: $1 - R$.

For situations where the first medium is air or some other low-permittivity gas, the formula simplifies to the following form (with ϵ_r being the relative permittivity of the reflecting substance):

$$R = \frac{\left(\sqrt{\epsilon_r} - 1\right)^2}{\left(\sqrt{\epsilon_r} + 1\right)^2}$$

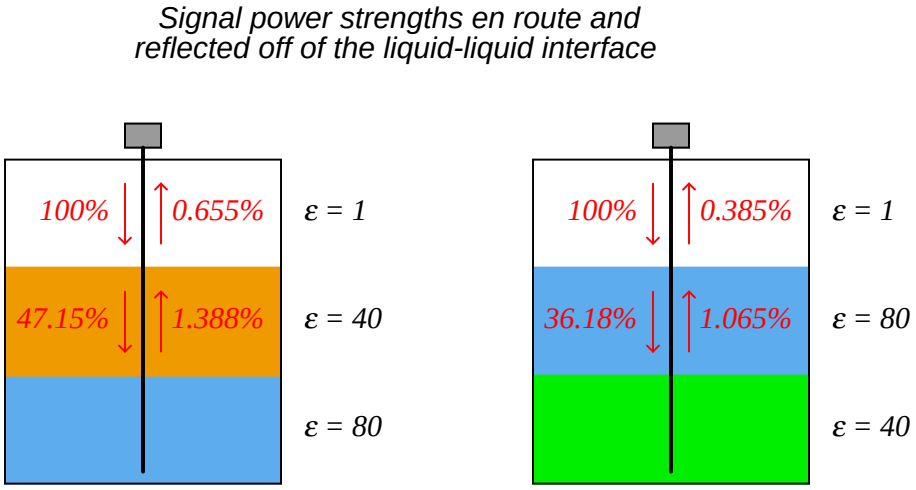
In the previous illustration, the two media were air ($\epsilon_r \approx 1$) and water ($\epsilon_r \approx 80$) – a nearly ideal scenario for strong signal reflection. Given these relative permittivity values, the power reflection factor has a value of 0.638 (63.8%), or −1.95 dB. This means well over half the incident power reflects off the air/water interface to form a strong echo signal, with the remaining 0.362 (36.2%) of the wave's power traveling through the air-water interface and propagating into water. If the liquid in question is gasoline rather than water (having a rather low relative permittivity value of approximately 2), the power reflection ratio will only be 0.0294 (2.94%) or −15.3 dB, with the vast majority of the wave's power successfully penetrating the air-gasoline interface.

The longer version of the power reflection factor formula suggests liquid-liquid interfaces should be detectable using radar, and indeed they are. All that is needed is a sufficiently large difference in permittivity between the two liquids to create a strong enough echo to reliably detect. Liquid-liquid interface level measurement with radar works best when the upper liquid has a substantially lesser permittivity value than the lower liquid[29]. A layer of hydrocarbon oil on top of water (or any aqueous solution such as an acid or a caustic) is a good candidate for guided-wave radar level measurement. An example of a liquid-liquid interface that would be very difficult for a radar instrument to detect is water ($\epsilon_r \approx 80$) above glycerin ($\epsilon_r \approx 42$).

If the radar instrument uses a digital network protocol to communicate information with a host system (such as HART or any number of "fieldbus" standards), it may perform as a multi-variable transmitter, transmitting *both* the interface level measurement and the total liquid level measurement simultaneously. This capability is rather unique to guided-wave radar transmitters, and is very useful in some processes because it eliminates the need for multiple instruments measuring multiple levels.

[29] Rosemount's "Replacing Displacers with Guided Wave Radar" technical note states that the difference in dielectric constant between the upper and lower liquids *must* be at least 10.

One reason why a lesser-ϵ fluid above a greater-ϵ fluid is easier to detect than the inverse is due to the necessity of the signal having to travel through a gas-liquid interface above the liquid-liquid interface. With gases and vapors having such small ϵ values, the signal would have to pass through the gas-liquid interface first in order to reach the liquid-liquid interface. This gas-liquid interface, having the greatest difference in ϵ values of any interface within the vessel, will be *most* reflective to electromagnetic energy *in both directions*. Thus, only a small portion of the incident wave will ever reach the liquid-liquid interface, and a similarly small portion of the wave reflected off the liquid-liquid interface (which itself is a fraction of the forward wave power that made it through the gas-liquid interface on its way down) will ever make it through the gas-liquid interface on its way *back up* to the instrument. The situation is much improved if the ϵ values of the two liquid layers are inverted, as shown in this hypothetical comparison (all calculations[30] assume no power dissipation along the way, only reflection at the interfaces):



As you can see in the illustration, the difference in power received back at the instrument is nearly two to one, just from the upper liquid having the lesser of two identical ϵ values. Of course, in real life you do not have the luxury of *choosing* which liquid will go on top of the other (this being determined by fluid density), but you do have the luxury of choosing the appropriate liquid-liquid interface level measurement technology, and as you can see here certain orientations of ϵ values are less detectable with radar than others.

Another factor working against radar as a liquid-liquid interface measurement technology for interfaces where the upper liquid has a greater dielectric constant is that fact that many high-ϵ liquids are aqueous in nature, and water readily dissipates microwave energy. This fact is exploited in microwave ovens, where microwave radiation excites water molecules in the food, dissipating energy in the form of heat. For a radar-based level measurement system consisting of gas/vapor over water over some other (heavier) liquid, the echo signal will be extremely weak because the signal must pass through the "lossy" water layer *twice* before it returns to the radar instrument.

[30] $R = 0.5285$ for the 1/40 interface; $R = 0.02944$ for the 40/80 interface; and $R = 0.6382$ for the 1/80 interface, all based on the formula $R = \frac{(\sqrt{\epsilon_r}-1)^2}{(\sqrt{\epsilon_r}+1)^2}$ using the pair of permittivity values at each interface.

Electromagnetic energy losses are important to consider in radar level instrumentation, even when the detected interface is simply gas (or vapor) over liquid. The power reflection factor formula only predicts the ratio of reflected power to incident power *at an interface of substances*. Just because an air-water interface reflects 63.8% of the incident power does not mean 63.8% of the incident power will actually return to the transceiver antenna! Any dissipative losses between the transceiver and the interface(s) of concern will weaken the signal, to the point where it may become difficult to distinguish from noise.

Another important factor in maximizing reflected power is the degree to which the microwaves disperse on their way to the liquid interface(s) and back to the transceiver. Guided-wave radar instruments receive a far greater percentage of their transmitted power than non-contact radar instruments because the metal probe used to guide the microwave signal pulses help prevent the waves from spreading (and therefore weakening) throughout the liquids as they propagate. In other words, the probe functions as a transmission line to direct and focus the microwave energy, ensuring a straight path from the instrument into the liquid, and a straight echo return path from the liquid back to the instrument. This is why guided-wave radar is the only practical radar technology for measuring liquid-liquid interfaces.

A critically important factor in accurate level measurement using radar instruments is that the dielectric permittivity of every substance lying between the radar instrument and the interface of interest be accurately known. The reason for this is rooted in the dependence of electromagnetic wave propagation velocity to relative permittivity. Recalling the wave velocity formula shown earlier:

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

Where,

v = Velocity of electromagnetic wave through a particular substance
c = Speed of light in a perfect vacuum ($\approx 3 \times 10^8$ meters per second)
ϵ_r = Relative permittivity (dielectric constant) of the substance

In the case of a single-liquid application where nothing but gas or vapor exists above the liquid, the permittivity of that gas or vapor must be precisely known. In the case of a two-liquid interface with gas or vapor above, the relative permittivities of *both* gas and upper liquids must be accurately known in order to accurately measure the liquid-liquid interface. Changes in dielectric constant value of the medium or media through which the microwaves must travel and echo will cause the microwave radiation to propagate at different velocities. Since all radar measurement is based on time-of-flight through the media separating the radar transceiver from the echo interface, changes in wave velocity through this media will affect the amount of time required for the wave to travel from the transceiver to the echo interface, and reflect back to the transceiver. Therefore, changes in dielectric constant are relevant to the accuracy of any radar level measurement[31].

Factors influencing the dielectric constant of gases include pressure and temperature, which means the accuracy of a radar level instrument will vary as gas pressure and/or gas temperature vary! This is often referred to as the *gas phase effect*. Whether or not this variation is substantial

[31] It should be noted that the dielectric constant of the lowest medium (the liquid in a simple, non-interface, level measurement application) is irrelevant for calibration purposes. All we are concerned with is the propagation time of the signal to and from the level of interest, nothing below it.

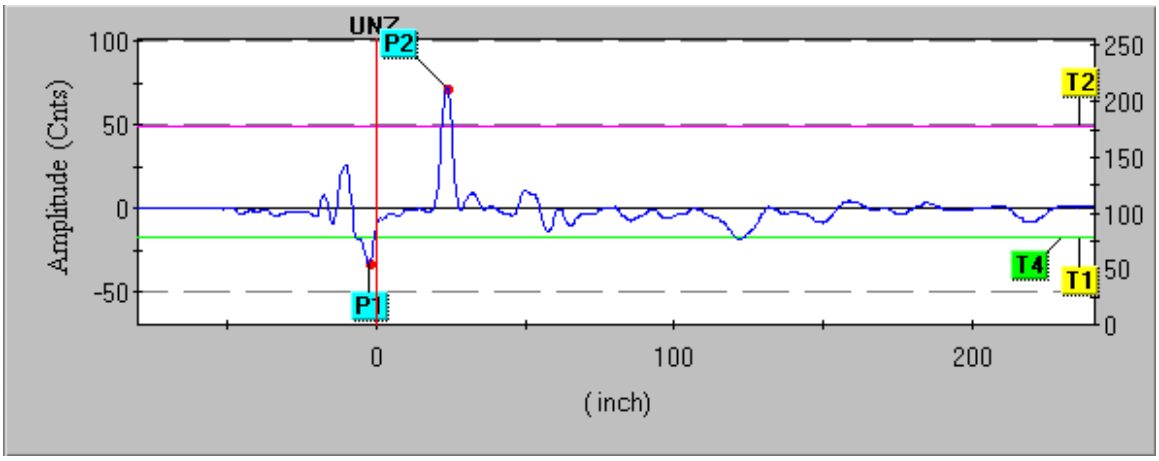
enough to consider for any application depends on the desired measurement accuracy and the degree of permittivity change from one pressure/temperature extreme to another. In no case should a radar instrument be considered for any level measurement application unless the dielectric constant value(s) of the upper media are precisely known. This is analogous to the dependence on liquid density that hydrostatic level instruments face. It is futile to attempt level measurement based on hydrostatic pressure if liquid density is unknown or widely varying, and it is just as futile to attempt level measurement based on radar if the dielectric constants are unknown[32] or varies widely.

One way to compensate for the gas phase effect in radar level instruments is to equip the instrument with a *reference probe* of fixed length oriented in such a way that its entire length is always above the liquid level (i.e. it only senses gas). If the permittivity of the gas is constant, the echo time along this reference probe will remain the same. If, however, the gas permittivity changes, the reference probe's echo time will correspondingly change, allowing the instrument's microprocessor to measure gas permittivity and consequently adjust calculations for liquid level based on this known change. This concept is analogous to the *compensating probe* sometimes used in capacitive level sensors, designed to measure fluid permittivity so as to compensate for any changes in this critical parameter.

As with ultrasonic level instruments, radar level instruments can sense the level of solid substances in vessels (e.g. powders and granules) and not just liquids. The same caveat of repose angle applicable to ultrasonic level measurement (see section 20.5.1 beginning on page 1476), however, is a factor for radar measurement as well. Also, low particulate solid density (i.e. significant amounts of air between the solid particles) tends to reduce the material's dielectric constant and thereby weaken the radar echo.

[32] For vented-tank level measurement applications where air is the only substance above the point of interest, the relative permittivity is so close to a value of 1 that there is little need for further consideration on this point. Where the permittivity of fluids becomes a problem for radar is in high-pressure (non-air) gas applications and liquid-liquid interface applications, especially where the upper substance composition is subject to change.

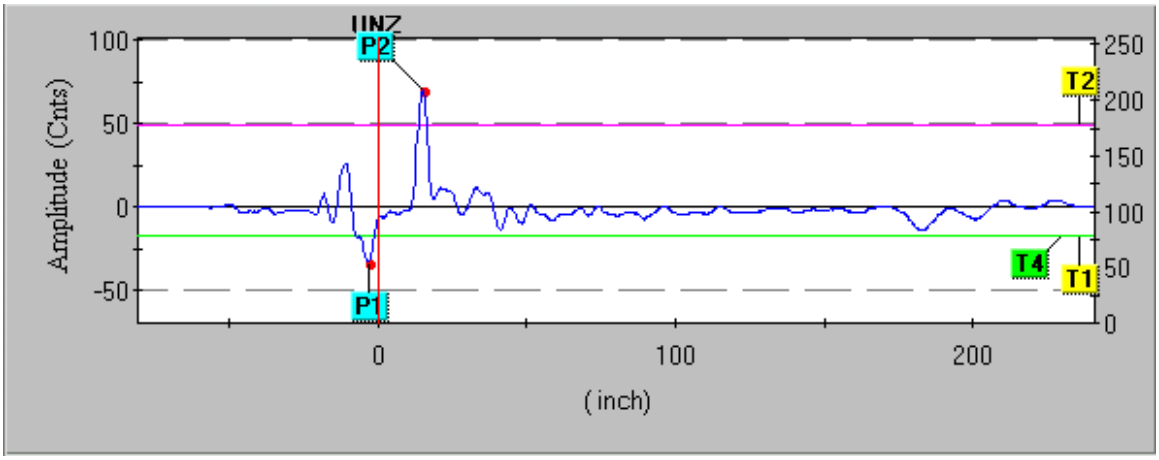
Modern radar level instruments provide a wealth of diagnostic information to aid in troubleshooting. One of the most informative is the *echo curve*, showing each reflected signal received by the instrument along the incident signal's path of travel. The following image is a screen capture of a computer display, from software used to configure a Rosemount model 3301 guided-wave radar level transmitter with a coaxial probe:



To view a flip-book animation showing how a guided-wave radar (GWR) instrument detects both liquid surface level and liquid-liquid interface level, turn to Appendix A.5 beginning on page 3059.

Pulse `P1` is the *reference* or *fiducial* pulse, resulting from the change in dielectric permittivity between the extended "neck" of the probe (connecting the transmitter to the probe tube) and the coaxial probe itself. This pulse marks the top of the probe, thereby establishing a point of reference for ullage measurement.

This next screen capture shows the same level transmitter measuring a water level that is 8 inches higher than before. Note how pulse P2 is further to the left (indicating an echo received sooner in time), indicating a lesser ullage (greater level) measurement:



Several *threshold* settings determine how the transmitter categorizes each received pulse. Threshold T1 for this particular radar instrument defines which pulse is the reference (fiducial). Thus, the first echo in time to exceed the value of threshold T1 is interpreted by the instrument to be the reference point. Threshold T2 defines the upper product level, so the first echo in time to exceed this threshold value is interpreted as the vapor/liquid interface point. Threshold T3 for this particular transmitter is used to define the echo generated by a liquid-liquid interface. However, threshold T3 does not appear in this echo plot because the interface measurement option was disabled during this experiment. The last threshold, T4, defines the end-of-probe detection. Set at a negative value (just like the reference threshold T1), threshold T4 looks for the first pulse in time to exceed that value and interprets that pulse as the one resulting from the signal reaching the probe's end.

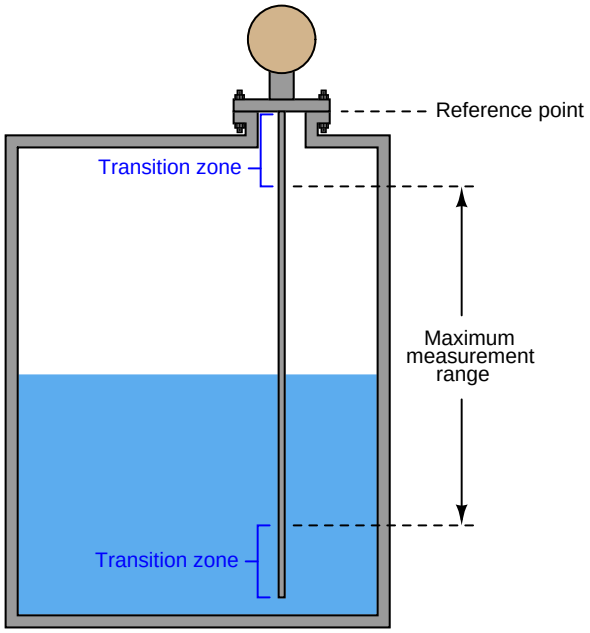
All along the echo curve you can see weak echo signals showing up as bumps. These echoes may be caused by discontinuities along the probe (solid deposits, vent holes, centering spacers, etc.), discontinuities in the process liquid (suspended solids, emulsions, etc.), or even discontinuities in the surrounding process vessel (for non-coaxial probes which exhibit varying degrees of sensitivity to surrounding objects). A challenge in configuring a radar level transmitter is to set the threshold values such that "false" echoes are not interpreted as real liquid or interface levels.

A simple way to eliminate false echoes near the reference point is to set a *null zone* where any echoes are ignored. The upper null zone (UNZ) setting on the Rosemount 3301 radar level transmitter whose screen capture image was shown previously was set to zero, meaning it would be sensitive to any and all echoes near the reference point. If a false echo from a tank nozzle or some other discontinuity near the probe's entry point into the process vessel created a measurement problem, the upper null zone (UNZ) value could be set just beyond that point so the false echo would not be interpreted as a liquid level echo, regardless of the threshold value for T2. A "null zone" is sometimes referred to as a *hold-off distance*.

Some radar level instruments allow thresholds to be set as curves themselves rather than straight lines. Thus, thresholds may be set high during certain periods along the horizontal (time/distance)

axis to ignore false echoes, and set low during other periods to capture legitimate echo signals.

Regardless of how null zones and thresholds are set for any guided-wave radar level transmitter, the technician must be aware of *transition zones* near each end of the probe. Measurements of liquid level or interface level within these zones may not be accurate or even linearly responsive. Thus, it is strongly advised to range the instrument in such a way that the lower- and upper-range values (LRV and URV) lie between the transition zones:



The size of these transition zones depends on both the process substances and the probe type[33]. The instrument manufacturer will provide you with appropriate data for determining transition zone dimensions.

[33] Probe mounting style will also influence the lower transition zone, in the case of flexible probes anchored to the bottom of the process vessel.

20.5.3 Laser level measurement

The least-common form of echo-based level measurement is *laser*, which uses pulses of laser light reflected off the surface of a liquid to detect the liquid level. Perhaps the most limiting factor with laser measurement is the necessity of having a sufficiently reflective surface for the laser light to "echo" off of. Many liquids are not reflective enough for this to be a practical measurement technique, and the presence of dust or thick vapors in the space between the laser and the liquid will disperse the light, weakening the light signal and making the level more difficult to detect.

However, lasers have been applied with great success in measuring distances between objects. Applications of this technology include motion control on large machines, where a laser points at a moving reflector, the laser's electronics calculating distance to the reflector based on the amount of time it takes for the laser "echo" to return. The advent of mass-produced, precision electronics has made this technology practical and affordable for many applications. At the time of this writing (2008), it is even possible for the average American consumer to purchase laser "tape measures" for use in building construction.

20.5.4 Magnetostrictive level measurement

A variation on the theme of echo-based level instruments, where the level of some process material in a vessel is measured by timing the travel of a wave between the instrument and the material interface, is one applied to float-type instruments: *magnetostriction*.

In a magnetostrictive level instrument, liquid level is sensed by a lightweight, donut-shaped float containing a magnet. This float is centered around a long metal rod called a *waveguide*, hung vertically in the process vessel (or hung vertically in a protective cage like the type used for displacement-style level instruments) so that the float may rise and fall with process liquid level. The magnetic field from the float's magnet at that point, combined with the magnetic field produced by an electric current pulse periodically sent through the rod, generates a torsional stress pulse[34] at the precise location of the float. This torsional (twisting) stress travels at the speed of sound through the rod toward either end. At the bottom end is a dampener device designed to absorb the mechanical wave[35].

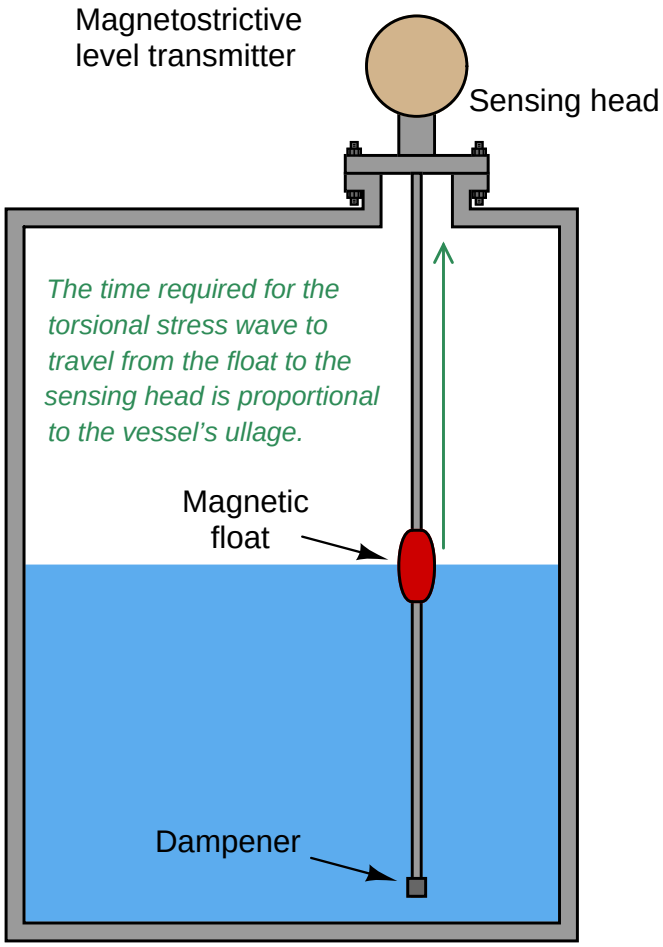
One might argue that a magnetostrictive instrument is not an "echo" technology in the strictest sense of the word. Unlike ultrasonic, radar, and laser instruments, we are not reflecting a wave off a discontinuous interface between materials. Instead, a mechanical wave (pulse) is *generated* at the location of a magnetic float in response to an electrical pulse. However, the principle of measuring distance by the wave's travel time is the same. At the top end of the rod (above the process liquid level) is a sensor and electronics package designed to detect the arrival of the mechanical wave. A precision electronic timing circuit measures the time elapsed between the electric current pulse (called the *interrogation pulse*) and the received mechanical pulse. So long as the speed of sound through the metal waveguide rod remains fixed, the time delay is strictly a function of distance between the float and the sensor, which we already know is called *ullage*.

[34] An approximate analogy for understanding the nature of this pulse may be performed using a length of rope. Laying a long piece of rope in a straight line on the ground, pick up one end and quickly move it in a tight circle using a "flip" motion of your wrist. You should be able to see the torsional pulse travel down the length of the rope until it either dies out from dissipation or it reaches the rope's end. As with the torsional pulse in a magnetostrictive waveguide, this pulse in the rope is mechanical in nature: a movement of the rod's (rope's) molecules. As a mechanical wave, it may be properly understood as a form of *sound*.

[35] This "dampener" is the mechanical equivalent of a *termination resistor* in an electrical transmission line: it makes the traveling wave "think" the waveguide is infinitely long, preventing any reflected pulses. For more information on electrical transmission lines and termination resistors, see section 5.10 beginning on page 475.

The following photograph (left) and illustration (right) show a magnetostrictive level transmitter[36] propped up against a classroom wall and the same style of transmitter installed in a liquid-holding vessel, respectively:



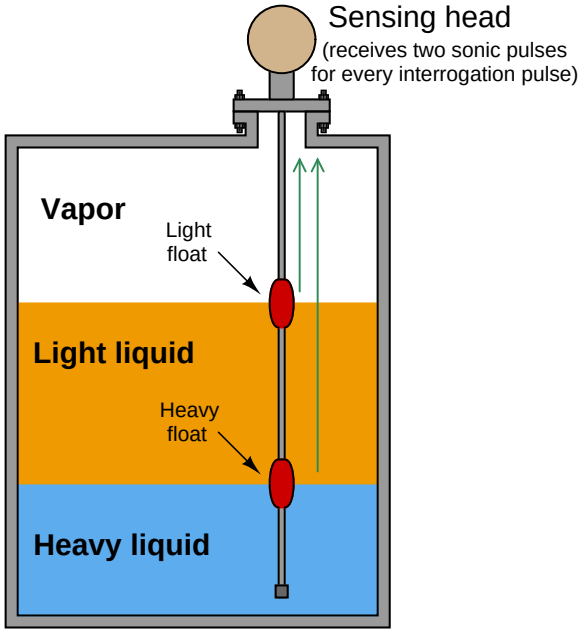


The design of this instrument is reminiscent of a guided-wave radar transmitter, where a metal *waveguide* hangs vertically into the process liquid, guiding a pulse to the sensor head where the sensitive electronic components are located. The major difference here is that the pulse is a sonic vibration traveling through the metal of the waveguide rod, not an electromagnetic pulse as is the case with radar. Like all sound waves, the torsional pulse in a magnetostriction-based level transmitter is much slower-traveling[37] than electromagnetic waves.

[36] This particular transmitter happens to be one of the "M-Series" models manufactured by MTS.

[37] One reference gives the speed of sound in a magnetostrictive level instrument as 2850 meters per second. Rounding this up to 3×10^3 m/s, we find that the speed of sound in the magnetostrictive waveguide is at least *five orders of magnitude slower* than the speed of light in a vacuum (approximately 3×10^8 m/s). This relative slowness of

It is even possible to measure liquid-liquid interfaces with magnetostrictive instruments. If the waveguide is equipped with a float of such density that it floats on the interface between the two liquids (i.e. the float is denser than the light liquid and less dense than the heavy liquid), the sonic pulse generated in the waveguide by that float's position will represent interface level. Magnetostrictive instruments may even be equipped with two floats: one to sense a liquid-liquid interface, and the other to sense the liquid-vapor interface, so that it may measure both the interface and total levels simultaneously just like a guided-wave radar transmitter:



With such an instrument, each electrical "interrogation" pulse returns *two* sonic pulses to the sensor head: the first pulse representing the total liquid level (upper, light float) and the second pulse representing the interface level (lower, heavy float). If the instrument has digital communication capability (e.g. HART, FOUNDATION Fieldbus, Profibus, etc.), both levels may be reported to the control system over the same wire pair, making it a "multivariable" instrument.

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wave propagation is a good thing for our purposes here, as it gives more time for the electronic timing circuit to count, yielding a more precise measurement of distance traveled by the wave. This fact grants superior resolution of measurement to magnetostrictive level sensors over radar-based and laser-based level sensors. Open-air ultrasonic level instruments deal with propagation speeds even slower than this (principally because the bulk moduli of gases and vapors is far less than that of a solid metal rod) which at first might seem to give these level sensors the advantage in precision. However, open-air level sensors experience far greater propagation velocity variations caused by changes in pressure and temperature than magnetostrictive sensors. Unlike the speed of sound in gases or liquids, the speed of sound in a solid metal rod is very stable over a large range of process temperatures, and practically constant for a large range of process pressures. Another factor adding to the calibration stability of magnetostrictive instruments is that the composition of the medium never changes. With instruments measuring time-of-flight through process fluids, the chemical composition of those fluids often affects the wave velocity. In a magnetostrictive instrument, the waves are always traveling through the same material – the metal of the waveguide bar – and thus are not subject to variation with process changes.

Perhaps the greatest limitation of magnetostrictive level instruments is mechanical interference between the float and the rod. In order for the magnetostrictive effect to be strong, the magnet inside the float must be in close proximity to the rod. This means the inside diameter of the donut-shaped float must fit closely to the outside diameter of the waveguide. Any fouling of the waveguide's or float's surfaces by suspended solids, sludge, or other semi-solid materials may cause the float to bind and therefore not respond to changes in liquid level.

20.6 Weight

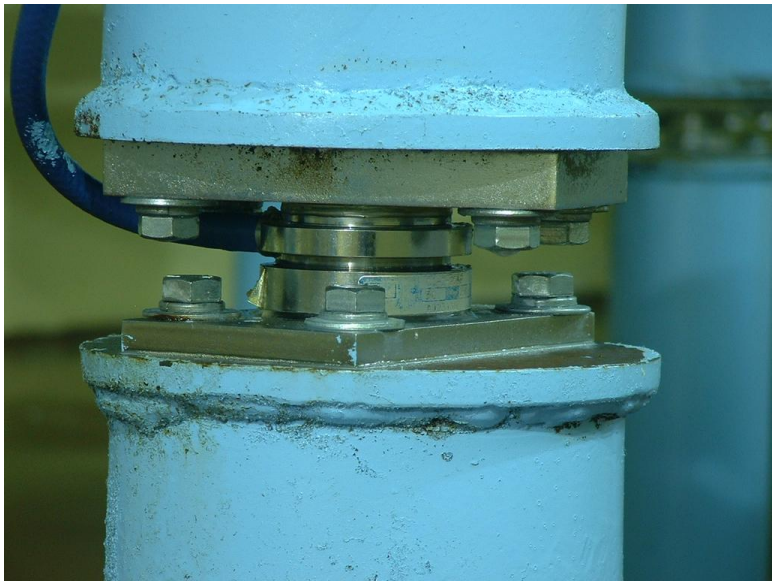
Weight-based level instruments sense process level in a vessel by directly measuring the weight of the vessel. If the vessel's empty weight (*tare weight*) is known, process weight becomes a simple calculation of total weight minus tare weight. Obviously, weight-based level sensors can measure both liquid and solid materials, and they have the benefit of providing inherently linear mass storage measurement[38]. *Load cells* (strain gauges bonded to a steel element of precisely known modulus) are typically the primary sensing element of choice for detecting vessel weight. As the vessel's weight changes, the load cells compress or relax on a microscopic scale, causing the strain gauges inside to change resistance. These small changes in electrical resistance become a direct indication of vessel weight.

The following photograph shows three bins used to store powdered milk, each one supported by pillars equipped with load cells near their bases:



[38] Regardless of the vessel's shape or internal structure, the measurement provided by a weight-sensing system is based on the true mass of the stored material. Unlike height-based level measurement technologies (float, ultrasonic, radar, etc.), no characterization will ever be necessary to convert a measurement of height into a measurement of mass.

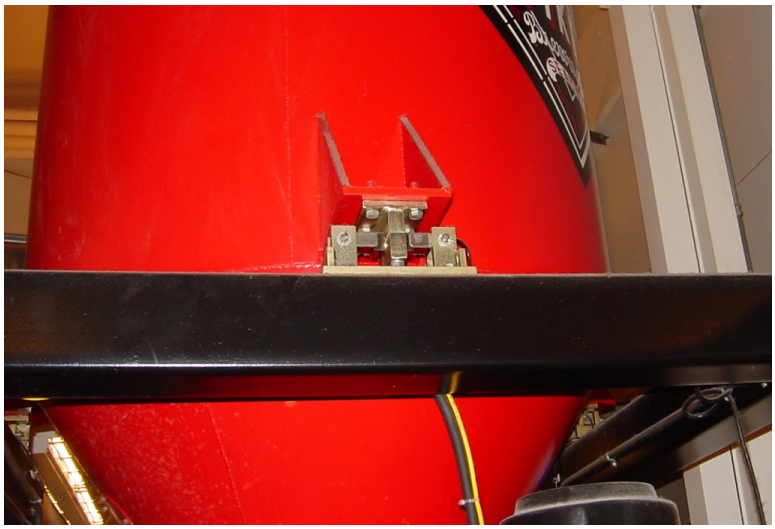
A close-up photograph shows one of the load cell units in detail, near the base of a pillar:



When multiple load cells are used to measure the weight of a storage vessel, the signals from all load cell units must be added together ("summed") to produce a signal representative of the vessel's *total* weight. Simply measuring the weight at one suspension point is insufficient[39], because one can never be sure the vessel's weight is distributed equally amongst all the supports.

[39] If we happened to know, somehow, that the vessel's weight *was* in fact equally shared by all supports, it would be sufficient to simply measure stress at one support to infer total vessel weight. In such an installation, assuming three supports, the total vessel weight would be the stress at any one support multiplied by three.

This next photograph shows a smaller-scale load cell installation used to measure the quantity of material fed into a beer-brewing process[40]:

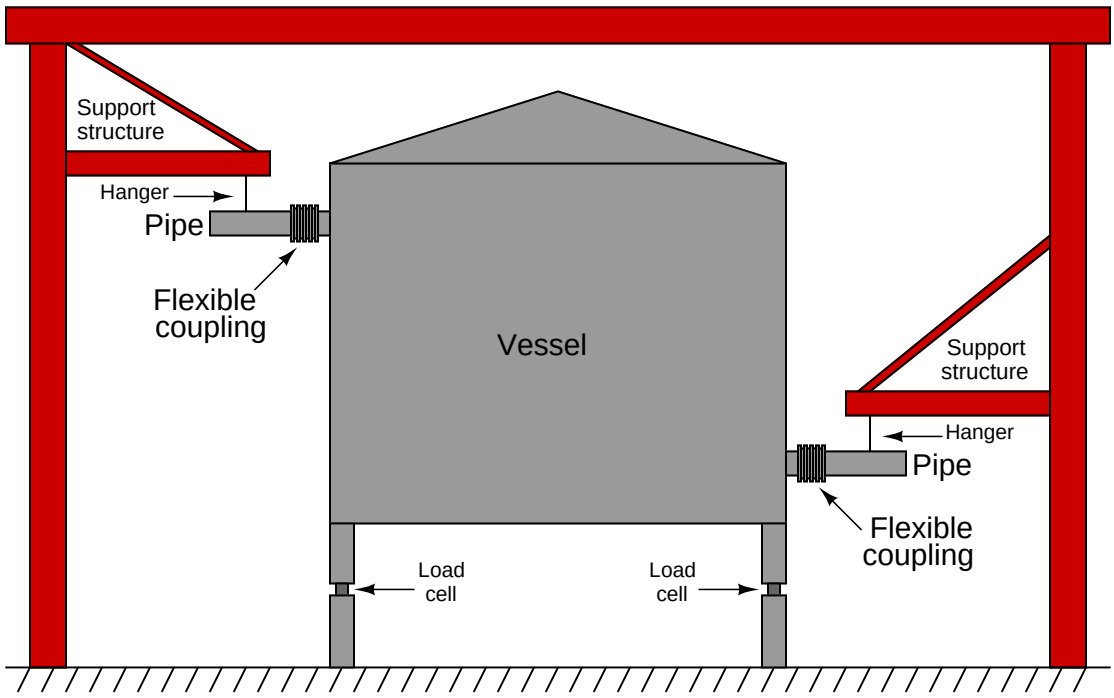


Weight-based measurements are often employed where the true mass of a quantity must be ascertained, rather than the level. So long as the material's density is a known constant, the relationship between weight and level for a vessel of constant cross-sectional area will be linear and predictable. Constant density is not always the case, especially for solid materials, and so weight-based inference of vessel level may be problematic.

In applications where batch mass is more important than height (level), weight-based measurement is often the preferred method for portioning batches. You will find weight-based portion measurements used frequently in the food processing industries (e.g. consistently filling bags and boxes with product), and also for custody transfer of certain materials (e.g. coal and metal ore).

[40]The particular "micro-brewery" process shown here is at the Pike's Place Market in downtown Seattle, Washington. Three load cells measure the weight of a hopper filled with ingredients prior to brewing in the "mash tun" vessel.

One very important caveat for weight-based level instruments is to isolate the vessel from any external mechanical stresses generated by pipes or machinery. The following illustration shows a typical installation for a weight-based measurement system, where all pipes attaching to the vessel do so through flexible couplings, and the weight of the pipes themselves is borne by outside structures through *pipe hangers*:



Stress relief is very important because any forces acting upon the storage vessel will be interpreted by the load cells as more or less material stored in the vessel. The only way to ensure that the load cell's measurement is a direct indication of material held inside the vessel is to ensure that no other forces act upon the vessel except the gravitational weight of the material.

A similar concern for weight-based batch measurement is *vibration* produced by machinery surrounding (or on) the vessel. Vibration is nothing more than oscillatory *acceleration*, and the acceleration of any mass produces a reaction force ($F = ma$). Any vessel suspended by weight-sensing elements such as load cells will induce oscillating forces on those load cells if shaken by vibration. This concern in particular makes it quite difficult to install and operate *agitators* or other rotating machinery on a weighed vessel[41].

An interesting problem associated with load cell measurement of vessel weight arises if there are ever electric currents traveling through the load cell(s). This is not a normal state of affairs,

[41] One practical solution to this problem is to shut down the source of vibration (e.g. agitator motor, pump, etc.) for a long enough time to take a sample weight measurement, then run the machine again between measurements. So long as intermittent weight measurement is adequate for the needs of the process, the interference of machine vibration may be dealt with in this manner.

but it can happen if maintenance workers incorrectly attach arc welding equipment to the support structure of the vessel, or if certain electrical equipment mounted on the vessel such as lights or motors develop ground faults. The electronic amplifier circuits interpreting a load cell's resistance will detect voltage drops created by such currents, interpreting them as changes in load cell resistance and therefore as changes in material level. Sufficiently large currents may even cause permanent damage to load cells, as is often the case when the currents in question are generated by arc welding equipment.

A variation on this theme is the so-called *hydraulic load cell* which is a piston-and-cylinder mechanism designed to translate vessel weight directly into hydraulic (liquid) pressure. A normal pressure transmitter then measures the pressure developed by the load cell and reports it as material weight stored in the vessel. Hydraulic load cells completely bypass the electrical problems associated with resistive load cells, but are more difficult to network for the calculation of total weight (using multiple cells to measure the weight of a large vessel).

20.7 Capacitive

Capacitive level instruments measure electrical capacitance of a conductive rod inserted vertically into a process vessel. As process level increases, capacitance increases between the rod and the vessel walls, causing the instrument to output a greater signal.

The basic principle behind capacitive level instruments is the capacitance equation:

$$C = \frac{\epsilon A}{d}$$

Where,

C = Capacitance

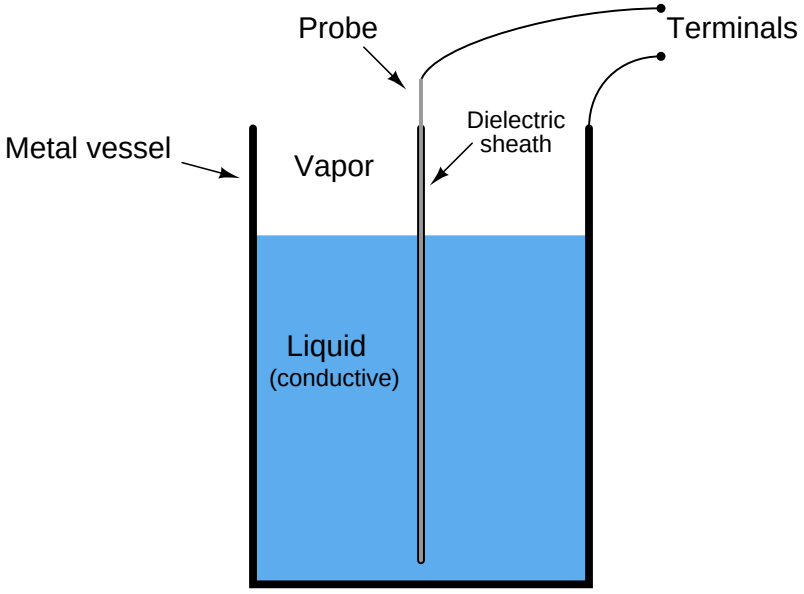
ϵ = Permittivity of dielectric (insulating) material between plates

A = Overlapping area of plates

d = Distance separating plates

The amount of capacitance exhibited between a metal rod inserted into the vessel and the metal walls of that vessel will vary only with changes in permittivity (ϵ), area (A), or distance (d). Since A is constant (the interior surface area of the vessel is fixed, as is the area of the rod once installed), only changes in ϵ or d can affect the probe's capacitance.

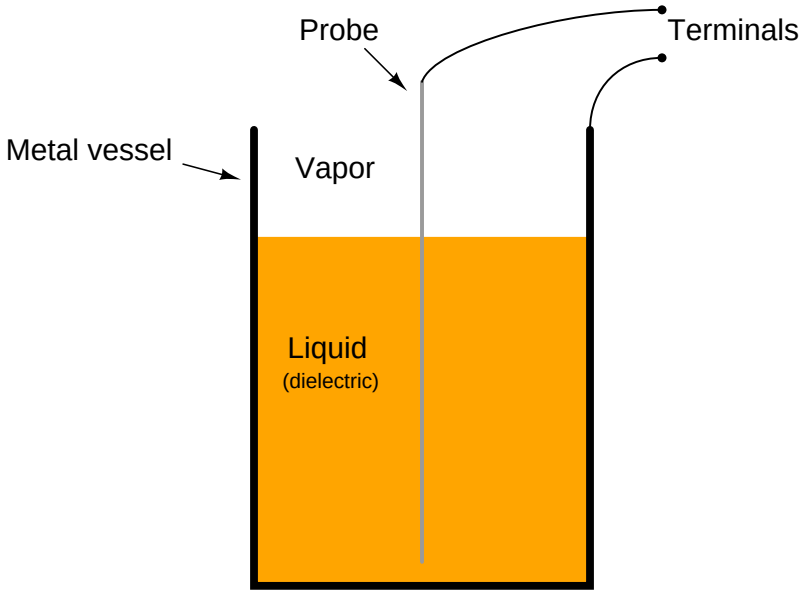
Capacitive level probes come in two basic varieties: one for conductive liquids and one for non-conductive liquids. If the liquid in the vessel is conductive, it cannot be used as the dielectric (insulating) medium of a capacitor. Consequently, capacitive level probes designed for conductive liquids are coated with plastic or some other dielectric substance, so the metal probe forms one plate of the capacitor and the conductive liquid forms the other:



In this style of capacitive level probe, the variables are permittivity (ϵ) and distance (d), since a rising liquid level displaces low-permittivity gas and essentially acts to bring the vessel wall

electrically closer to the probe. This means total capacitance will be greatest when the vessel is full (ϵ is greatest and effective distance d is at a minimum), and least when the vessel is empty (ϵ of the gas is in effect, and over a much greater distance).

If the liquid is non-conductive, it may be used as the dielectric itself, with the metal wall of the storage vessel forming the second capacitor plate. The probe is just a bare metal cable or rod:



In this style of capacitive level probe, the only variable affecting probe capacitance is permittivity (ϵ), provided the liquid has a substantially greater permittivity than the vapor space above the liquid. This means total capacitance will be greatest when the vessel is full (average permittivity ϵ is at a maximum), and least when the vessel is empty. Distance (d) is constant with a non-conducting process liquid, being the radius of the vessel (assuming the probe is mounted in the center).

Permittivity of the process substance is a critical variable in the non-conductive style of capacitance level probe, and so good accuracy may be obtained with this kind of instrument only if the process material permittivity is accurately known. A clever way to ensure good level measurement accuracy when the material's permittivity is not stable over time is to equip the instrument with a special *compensating probe* (sometimes called a *composition probe*) below the LRV point in the vessel that will always be submerged. Since this compensating probe is always immersed, and always experiences the same A and d dimensions, its capacitance is purely a function of the substance's permittivity (ϵ). This gives the instrument a way to continuously measure material permittivity, which it then uses to calculate the level of that material in the vessel based on the capacitance of the main probe. The inclusion of a compensating probe to measure and compensate for changes in permittivity is analogous to the inclusion of a third pressure transmitter in a hydrostatic *tank expert* system to continuously measure and compensate for density. It is a way to correct for changes in the one remaining system variable that is not related to changes in level.

Capacitive level instruments may be used to measure the level of solids (powders and granules) in

addition to liquids. In these applications, the material in question is almost always non-conductive, and therefore the permittivity of the substance becomes a factor in measurement accuracy. This can be problematic, as moisture content variations in the solid may greatly affect permittivity, as can variations in granule size. Compensating probes may not be very useful, either, because their location (at the bottom of the vessel) may not expose them to the same degree of material granularity and moisture content experienced by the main probe.

Capacitive level instruments are generally found in applications where precision is not important. These instruments tend to suffer from errors arising from changes in process substance permittivity, changes in process vapor-space permittivity, and errors caused by stray capacitance in probe cables.

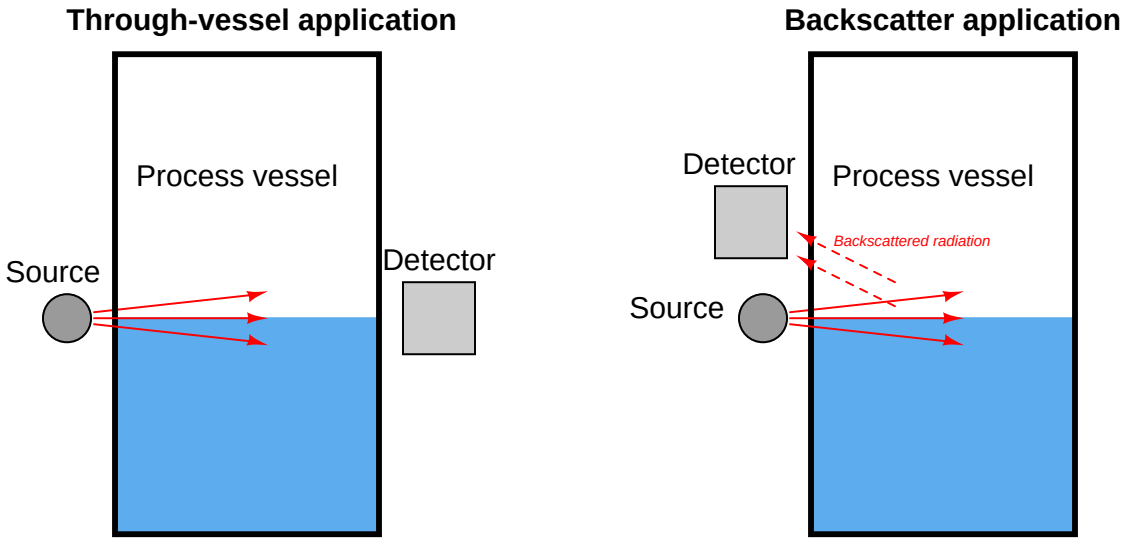
20.8 Radiation

Certain types of nuclear radiation easily penetrates the walls of industrial vessels, but is attenuated by traveling through the bulk of material stored within those vessels. By placing a radioactive source on one side of the vessel and measuring the radiation reaching the other side of the vessel, an approximate indication of level within that vessel may be obtained. Other types of radiation are *scattered* by process material in vessels, which means the level of process material may be sensed by sending radiation into the vessel through one wall and measuring *back-scattered* radiation returning through the same wall.

The four most common forms of nuclear radiation are *alpha particles* (α), *beta particles* (β), *gamma rays* (γ), and *neutrons* (n). Alpha particles are helium nuclei (2 protons bound together with 2 neutrons) ejected at high velocity from the nuclei of certain decaying atoms. They are easy to detect, but have very little penetrating power and so are not used for industrial level measurement. Beta particles are electrons[42] ejected at high velocity from the nuclei of certain decaying atoms. Like alpha particles, though, they have little penetrating power and so are not used for industrial level measurement. Gamma rays are electromagnetic in nature (like X-rays and light waves) and have great penetrating power. Neutron radiation also penetrates metal very effectively, but is attenuated and scattered by any substance containing hydrogen (e.g. water, hydrocarbons, and many other industrial fluids), which makes it almost ideal for detecting the presence of a great many process fluids. These latter two forms of radiation (gamma rays and neutrons) are the most common in industrial measurement, with gamma rays used in through-vessel applications and neutrons typically used in backscatter applications.

[42] Beta particles are *not* orbital electrons, but rather than product of elementary particle decay in an atom's nucleus. These electrons are spontaneously generated and subsequently ejected from the nucleus of the atom.

Through-vessel and *backscatter* nuclear level instrument applications appear contrasted in these two illustrations:

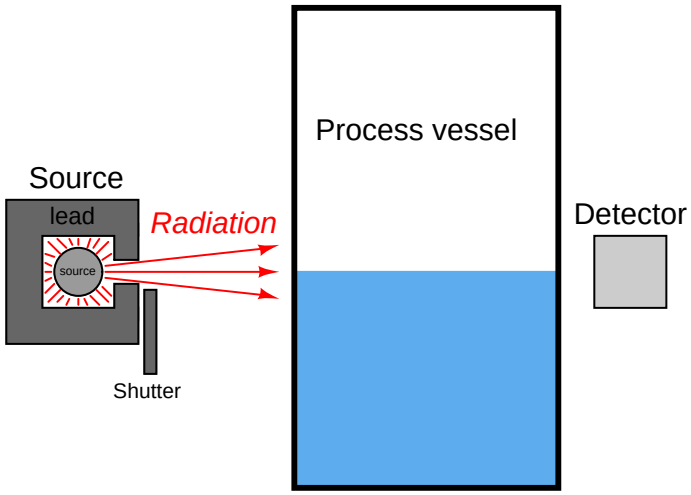


Nuclear radiation sources consist of radioactive samples contained in a shielded box. The sample itself is a small piece of radioactive substance encased in a double-wall stainless steel cladding, typically resembling a medicinal pill in size and shape. The specific type and quantity of radioactive source material depends on the nature and intensity of radiation required for the application. The basic rule here is that less is better: the smallest source capable of performing the measurement task is the best one for the application.

Common source types for gamma-ray applications are Cesium-137 and Cobalt-60. The numbers represent the *atomic mass* of each isotope: the sum total of protons and neutrons in the nucleus of each atom. These isotopes' nuclei are unstable, decaying over time to become different elements (Barium-137 and Nickel-60, respectively). Cobalt-60 has a relatively short half-life[43] of 5.3 years, whereas Cesium-137 has a much longer half-life of 30 years. This means radiation-based sensors using Cesium will be more stable over time (i.e. less calibration drift) than sensors using Cobalt. The trade-off is that Cobalt emits more powerful gamma rays than Cesium, which makes it better suited to applications where the radiation must penetrate thick process vessels or travel long distances (across wide process vessels).

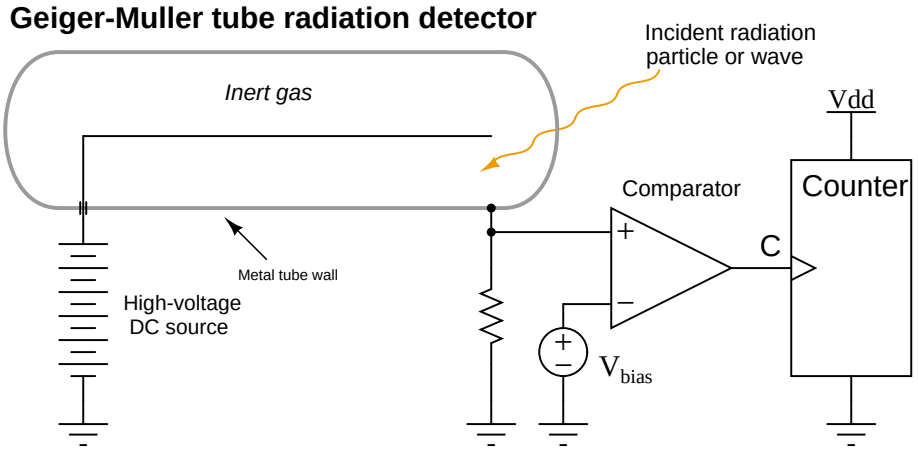
[43] The *half-life* of a radioactive substance is the amount of time it takes for one-half of the original quantity to experience radioactive decay. To illustrate, a 10-gram quantity consisting of 100% Cobalt-60 atoms will only contain 5 grams of Cobalt-60 after 5.3 years, and then only 2.5 grams of Cobalt-60 after another 5.3 years (10.6 years from the start), and so on. The actual mass of the sample does not change significantly over this time period because the Cobalt atoms have decayed into atoms of Nickel, which still have the same atomic mass value. However, the intensity of the gamma radiation emitted by the sample decreases over time, proportional to the percentage of Cobalt remaining therein.

One of the most effective methods of shielding against gamma ray radiation is with very dense substances such as lead or concrete. This is why the source boxes holding gamma-emitting radioactive pellets are lined with lead, so the radiation escapes only in the direction intended:



Radioactive sources naturally emit radiation, requiring no source of energy such as electricity to do their job. As such, they are "always-on" devices and may be locked out for testing and maintenance only by dropping a lead *shutter* over the "window" of the box. The lever actuating the shutter typically has provisions for lock-out/tag-out (LOTO) so a maintenance person may place a padlock on the lever and prevent anyone else from "turning on" the source during maintenance. For point-level (level switch) applications, the source shutter acts as a simple simulator for either a full vessel (in the case of a through-vessel installation) or an empty vessel (in the case of a backscatter installation). A full vessel may be simulated for neutron backscatter instruments by placing a sheet of plastic (or other hydrogen-rich substance) between the source box and the process vessel wall.

The detector for a radiation-based instrument is by far the most complex and expensive component of the system. Many different detector designs exist, the most common at the time of this writing being *ionization chambers* such as the Geiger-Muller (G-M) tube. In such devices, a thin metal wire centered in a metal cylinder sealed and filled with inert gas is energized with high voltage DC. Any ionizing radiation such as alpha, beta, or gamma radiation entering the tube causes gas molecules to ionize, allowing a pulse of electric current to travel between the wire and tube wall. A sensitive electronic circuit detects and counts these pulses, with a greater pulse rate corresponding to a greater intensity of detected radiation.



The following photograph shows an aluminum Geiger-Muller tube connected to a portable, battery-powered counter. This *Geiger counter* may be used as a piece of test equipment to measure radiation intensity while diagnosing problems in nuclear level measurement systems:



Geiger-style radiation detectors used as part of permanently-installed level measurement systems are housed in rugged housings, internally similar to the portable G-M tube shown in the photograph but designed for the rigors of continuous use in harsh industrial environments.

Neutron radiation is notoriously difficult to electronically detect, since neutrons are non-ionizing. Ionization tubes specifically made for neutron radiation detection are typically based on the Geiger-Muller design, but using tubes filled with special substances known to react with neutron radiation to produce (secondary) ionizing radiation. One example of such a detector is the so-called *fission chamber*, which is an ionization chamber lined with a fissile material such as uranium-235 (^{235}U). When a neutron enters the chamber and is captured by a fissile nucleus, that nucleus undergoes fission (splits into separate pieces) with a subsequent emission of gamma rays and charged particles, which are then detected by ionization in the chamber. Another variation on this theme is to fill an ionization tube with boron trifluoride gas. When a boron-10 (^{10}B) nucleus captures a neutron, it transmutates into lithium-7 (^{7}Li) and ejects an alpha particle and several beta particles, both of which cause detectable ionization in the chamber.

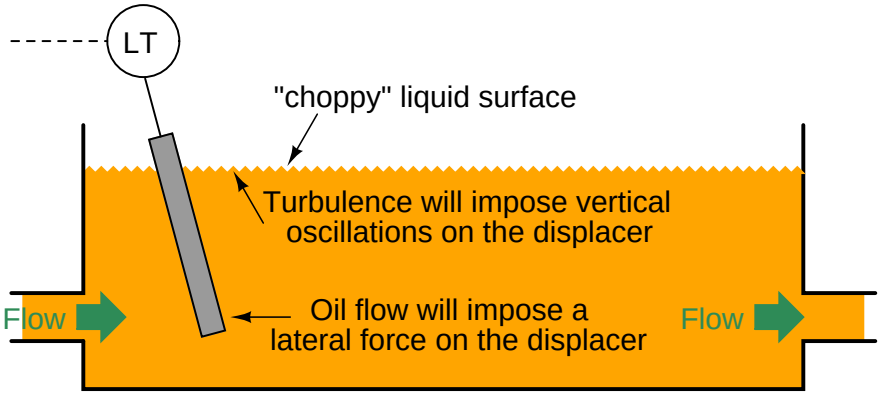
The accuracy of radiation-based level instruments varies with the stability of process fluid density, vessel wall coating, source decay rates, and detector drift. The multitude of error variables in radiation-based level measurement is one reason why they are more typically found as point-level (i.e. level switch) devices rather than continuous level (i.e. transmitter) measurement applications.

With their generally poor accuracy and the additional need for NRC (Nuclear Regulatory Commission) licensing to operate such instruments at an industrial facility, radiation instruments are typically used where no other instrument is practical. Examples include the level measurement of highly corrosive or toxic process fluids where penetrations into the vessel must be minimized and where piping requirements make weight-based measurement impractical (e.g. hydrocarbon/acid separators in alkylation processes in the oil refining industry), as well as processes where the internal conditions of the vessel are too physically violent for any instrument to survive (e.g. delayed coking vessels in the oil refining industry, where the coke is "drilled" out of the vessel by a high-pressure water jet).

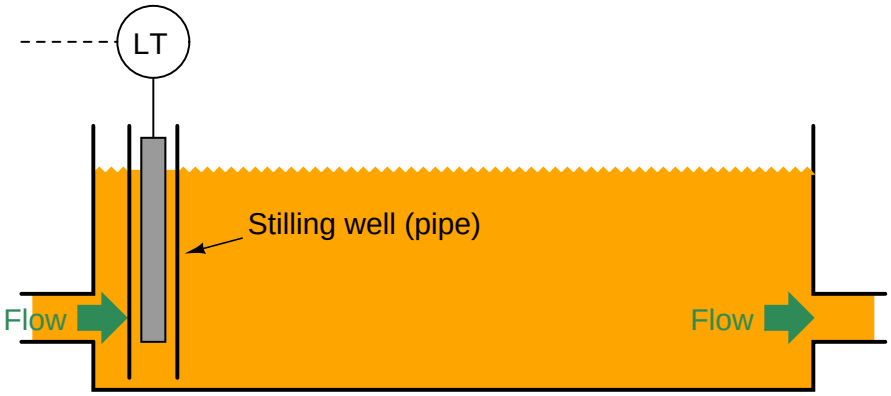
20.9 Level sensor accessories

Disturbances in the liquid tend to complicate liquid level measurement. These disturbances may result from liquid introduced into a vessel above the liquid level (splashing into the liquid's surface), the rotation of agitator paddles, and/or turbulent flows from mixing pumps. Any source of turbulence for the liquid surface (or liquid-liquid interface) is especially problematic for echo-type level sensors, which *only* sense interfaces between vapors and liquids, or liquids and liquids.

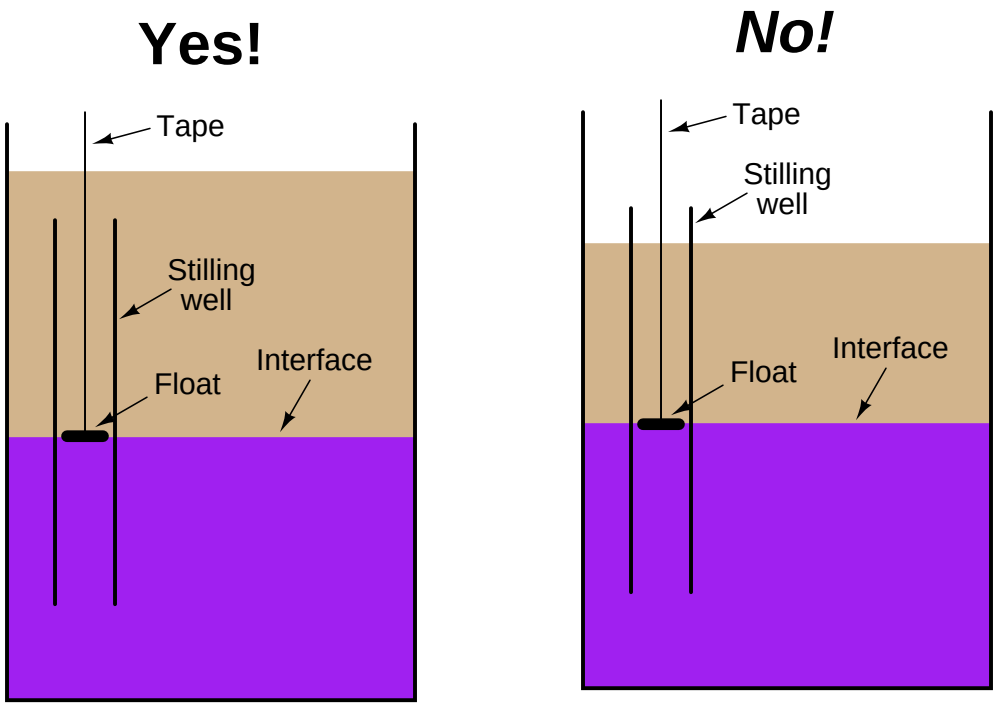
If it is not possible to eliminate disturbances inside the process vessel, a relatively simple accessory one may add to the process vessel is a vertical length of pipe called a *stilling well*. To understand the principle of a stilling well, first consider the application of a hydraulic oil reservoir where we wish to continuously measure oil level. The oil flow in and out of this reservoir will cause problems for the displacer element:



A section of vertical pipe installed in the reservoir around the displacer will serve as a shield to all the turbulence in the rest of the reservoir. The displacer element will no longer be subject to a horizontal blast of oil entering the reservoir, nor any wave action to make it bob up and down. This section of pipe *quiets*, or *stills*, the oil surrounding the displacer, making it easier to measure oil level:

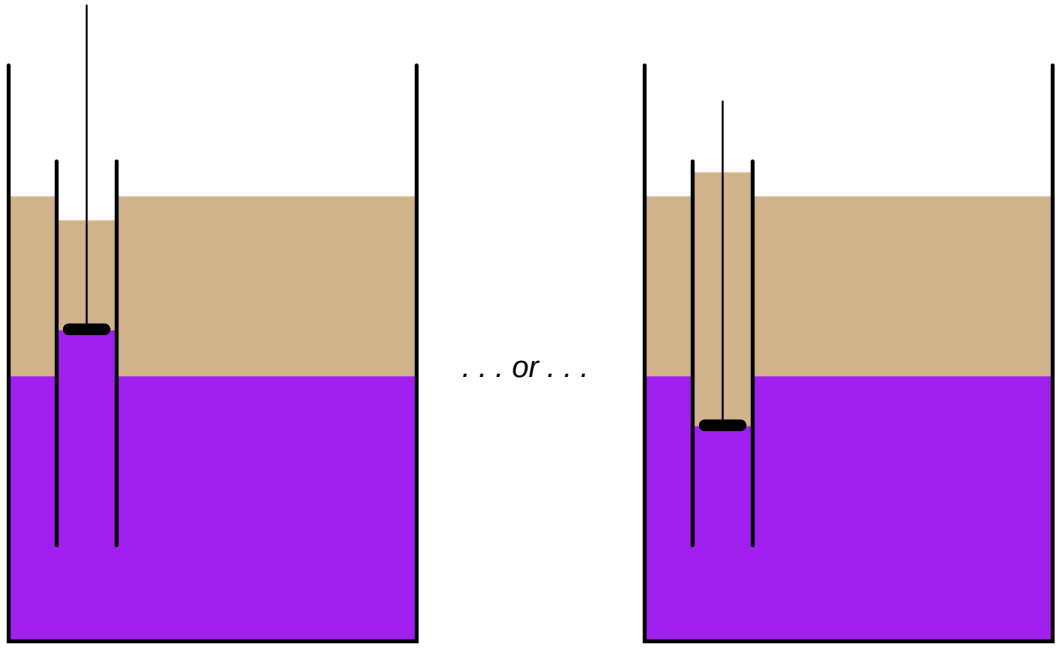


Stilling wells may be used in conjunction with many types of level instruments: floats, displacers, ultrasonic, radar, and laser to name a few. If the process application necessitates liquid-liquid interface measurement, however, the stilling well must be properly installed to ensure the interface level inside the well match the interface levels in the rest of the vessel. Consider this example of using a stilling well in conjunction with a tape-and-float system for interface measurement:



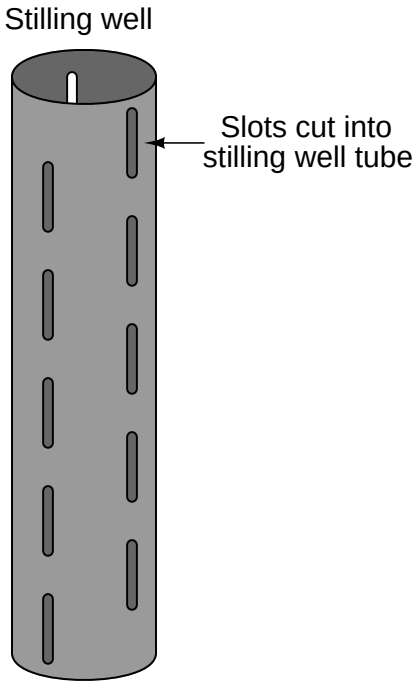
In the left-hand installation where the stilling well is completely submerged, the interface levels will always match. In the right-hand installation where the top of the stilling well extends above the total liquid level, however, the two levels may not always match.

This potential problem for the non-submerged stilling well is graphically illustrated here:



The problem here is analogous to what we see with sightglass-style level gauges: interfaces may be reliably indicated if and only if both ends of the sightglass are submerged (see section 20.1.2 beginning on page 1412 for an illustrated explanation of the problem).

If it is not possible or practical to ensure complete submersion of the stilling well, an alternative technique is to drill holes or cut slots in the well to allow interface levels to equalize inside and outside of the well tube:



Such equalization ports are commonly found as a standard design feature on coaxial probes for guided-wave radar level transmitters, where the outer tube of the coaxial transmission line acts as a sort of stilling well for the fluid. Coaxial probes are typically chosen for liquid-liquid interface radar measurement applications because they do the best job of preventing dispersion of the electromagnetic wave energy[44], but the "stilling well" property of a coaxial probe practically necessitates these equalization ports to ensure the interface level within the probe always matches the interface level in the rest of the vessel.

[44]So much of the incident power is lost as the radar signal partially reflects off the gas-liquid interface, then the liquid-liquid interface, then *again* through the gas-liquid interface on its return trip to the instrument that every care must be taken to ensure optimum received signal strength. While twin-lead probes have been applied in liquid-liquid interface measurement service, the coaxial probe design is still the best for maintaining radar signal integrity.

20.10 Review of fundamental principles

Shown here is a partial listing of principles applied in the subject matter of this chapter, given for the purpose of expanding the reader's view of this chapter's concepts and of their general interrelationships with concepts elsewhere in the book. Your abilities as a problem-solver and as a life-long learner will be greatly enhanced by mastering the applications of these principles to a wide variety of topics, the more varied the better.

- **Definition of pressure**: $P = \frac{F}{A}$ (pressure is the amount of force applied over a specified area by a fluid.
- **Pascal's principle**: changes in fluid pressure are transmitted evenly throughout an enclosed fluid volume. Relevant to pressure measurement, as fluid pressure in all parts of an enclosed system will experience the same changes in pressure.
- **Hydrostatic pressure**: fluids having substantial weight generate pressure proportional to their density and to their vertical height ($P = \gamma h$ and $P = \rho g h$). Relevant to pressure offsets generated in vertical spans of impulse or capillary tubing, causing a pressure instrument to register more or less pressure than that at the process vessel connection.
- **Archimedes' principle**: the buoyant force experienced by an object submerged in liquid is equal to the weight of the fluid that object displaces, which is equal to the volume displaced multiplied by the weight density of the fluid ($F_{buoyant} = \gamma V$). Relevant to displacer-type instruments, which work by sensing the buoyant force exerted on an object as liquid rises around it.
- **Time, velocity, and distance**: $x = vt$, describing the relationship between velocity (v), time of travel (t), and distance traveled (x). Relevant to all types of "echo" level instruments, where travel time of a wave is used to measure distance.
- **Transmission lines**: short-duration (pulsed) electrical signals travel along a cable at nearly the speed of light, reflecting off of any discontinuity along the cable. Relevant to guided-wave radar level-sensing instruments, where the waveguide serves the same purpose as a transmission line, and the fluid-fluid boundary constitutes a discontinuity (sudden change in electrical permittivity) causing a reflected signal to arise.
- **Ideal Gas Law**: $PV = nRT$, used to calculate corrections to gas permittivity. Relevant to the "gas phase effect" of radar level instruments.
- **Capacitance**: $C = \frac{\epsilon A}{d}$, capacitance being proportional to the area of two overlapping conductors (A), the permittivity of the insulating (dielectric) substance between them (ϵ), and the distance (d) separating the conductors. Relevant to capacitive level sensing, where changes in liquid level alter the effective area, permittivity, and/or distance.

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